ELEC3104: Digital Signal Processing (Guided Online Delivery)
Term 3. 2019

In day-to day life, there are many examples of real-world analogue signals that we encounter, including speech, music, video and biomedical signals. The objective of signal processing is to extract and modify useful information that is carried by these signals. Digital Signal Processing is the process of converting these signals into a digital stream and then analysing and mathematically manipulating them. ELEC3104 is an introductory course, which takes students through the fundamentals of Digital Signal Processing theory so that they can develop the techniques for solving discrete-time signal processing problems.

Course Learning Outcomes
At the end of the course, students should be able to:

1. Analyse linear time-invariant systems
2. Demonstrate competency in time and frequency domain analysis of signals and systems including transform methods
3. Design and analyse digital filters for a given specification
4. Implement a simple multi-rate system

Assessments

	Online Learning	Activities & Assessments
Week 1	Signals and Systems (CH1) & DSP Fundamentals (CH2)	**Face-to-face** group discussion (Monday 7-8pm; Room 102); submission and assessment of reflection sheet (2 marks); (Wednesday).
Week 2	DSP Fundamentals (CH2) & Discrete-Time Systems (CH3)	Online group discussion (Monday 7-8pm); submission and assessment of reflection sheet (2 marks); (Wednesday); 10 min individual online oral assessment (6 marks); (Friday: 6-8pm)
Week 3	Discrete-Time Systems (CH3) & Intro to z-Transform (CH4)	Online group discussion (Monday 7-8pm); submission and assessment of reflection sheet (2 marks); (Wednesday);
Week 4	Intro to z-Transform (CH4) & Intro to Digital Filters (CH5)	**NO** online group discussion on Monday 7th October; submission and assessment of reflection sheet (2 marks); (Wednesday); 10 min individual online oral assessment (6 marks); (Friday: 6-8pm)
Week 5	Tutorial-Labs 1 (group); Written Exam 1 (individual);	Face-to-face tutorial labs (Monday to Thursday: 5-8pm); class written exam 1 (15 marks); (Friday: 5-6pm) - Chapters 1 to 5
Week 6	Discrete-Time Fourier Transform (CH6) & Analog Filters (CH7)	Online group discussion (Monday 7-8pm); submission and assessment of reflection sheet (2 marks); (Wednesday);
Week 7	IIR Digital Filter Design (CH8)	Online group discussion (Monday 7-8pm); submission and assessment of reflection sheet (2 marks); (Wednesday); 10 min individual online oral assessment (6 marks); (Friday: 6-8pm)
Week 8	FIR Digital Filter Design (CH8)	Online group discussion (Monday 7-8pm); submission and assessment of reflection sheet (2 marks); (Wednesday);
Week 9	Multirate DSP (CH9) & DSP Applications	Online group discussion (Monday 7-8pm); submission and assessment of reflection sheet (2 marks); (Wednesday); 10 min individual online oral assessment (6 marks); (Friday: 6-8pm)
Week 10	Tutorial Labs 2 (group); Mini-Project (individual); Written Exam 2 (individual);	Face-to-face tutorial labs & mini-project (Monday to Thursday: 5-8pm); Mini-project assessment (individual, 20 marks); (Friday: 6-8pm) Mini-project report (28th November ,10 marks) class written exam 2 (15 marks) (Friday: 5-6pm) - Chapters 6 to 9

Overview

- This is an online course. All the teaching material is from online lectures every week.
- The expected workload is 12-15 hours per week.
- Reading additional text books will further enhance your learning experience.
- Group learning is encouraged.
- In this course we will use "Moodle" and Microsoft Teams: as online learning and teaching platforms
- One tutor will be allocated per 10 students, and there will be a total of 40 students in the course.
- Any email enquiry regarding the course should be made from your student email to your online tutor.
- There is no final exam for this online course.
- There are **no** tutorial-lab exemptions or rescheduling of face-to-face tutorial labs.
- If online format is not suitable for your learning preferences, then please do not enroll in this course.

Requirements

- Compulsory weekly online (video conference) group discussions (Monday 7:00pm to 8:00pm).
- Compulsory weekly submissions of reflection sheets on the content and discussion (Wednesdays).
- Compulsory 10 minutes individual online oral assessment (video conference) and feedback by tutors (Fridays between 6-8pm).
- Compulsory face-to-face tutorial labs (In Week 5, from 5:00pm to 8:00pm Monday to Thursday)
- Compulsory 1-hour class written exams 1 and 2 (Weeks 5 and 10: Fridays 5:00pm to 6:00pm)
- Compulsory face-to-face individual mini-project (Week 10: 5:00pm to 8:00pm Monday to Thursday)
- Compulsory individual mini-project assessment (Week 10: Friday 6:00pm to 8:00pm)
- Compulsory individual submission of a project report by Wednesday week 11, 28th November 2019.

Chapter 1

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Chapter 1: Signals and Systems

1.1 Introduction

The terms 'signals' and 'systems' are given various interpretations. For example, a **system** is an electric network consisting of resistors, capacitors, inductors and energy sources. **Signals** are various voltages and currents in the network. The signals are thus functions of time and they are related by a set of equations.

Example:

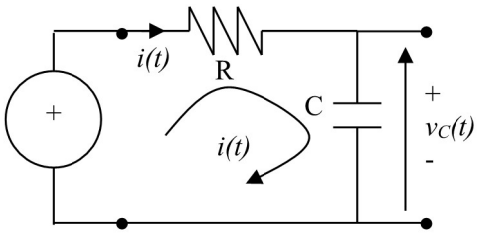


Figure 1.1: An electric circuit

The objective of system analysis is to determine the behaviour of the system subjected to a specific input or excitation. It is often convenient to represent a system schematically by means of a box as shown in Figure 1.2.

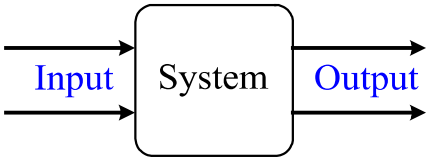


Figure 1.2: General representation of a system.

1.2 Signals

There are two types of signals:

(a) Continuous – time signals

(b) Discrete – time signals

In the case of a continuous-time signal, $x(t)$, the independent variable t is continuous and thus $x(t)$ is defined for all t (see Figure 1.3) where t is a continuos time -independent variable ($-\infty < t < \infty$)

On the other hand, discrete-time signals are defined only at discrete times and consequently the independent variable takes on only a discrete set of values (see Figure 1.3). A discrete-time signal is thus a sequence of numbers where n is a discrete time -independent variable (n = … -2, -1, 0, 1, 2,…)

Examples:

- A person's body temperature is a **continuous-time** signal.
- The prices of stocks printed in the daily newspapers are **discrete-time** signals.
- Voltages and currents are usually represented by **continuous-time** signals. They are also represented by **discrete-time** signals if they are specified only at a discrete set of values of t.

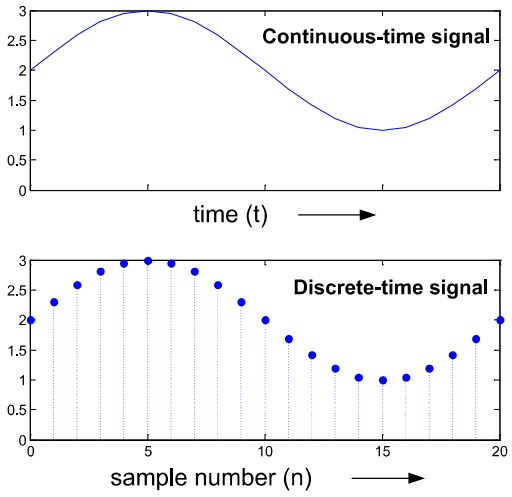


Figure 1.3: An example of continuous-time and discrete-time signals

1.2.1 Sampling

A discrete-time signal is often formed by sampling a continuous -time signal *x(t)*. If the samples are equidistant then

$$x[n] = x(t)\big|_{t=nT} = x(nT) \qquad (1.1)$$

Square brackets [ ] ⇒ Discrete time signals
Round Brackets ( ) ⇒ Continuous signals

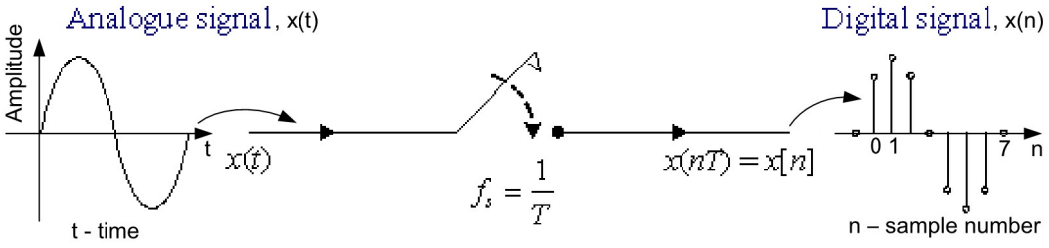


Figure 1.4 An example of acquiring discrete-time signal by sampling continuous-time signal

The constant T is the sampling interval or period and $f_s = 1/T$ is the sampling frequency.

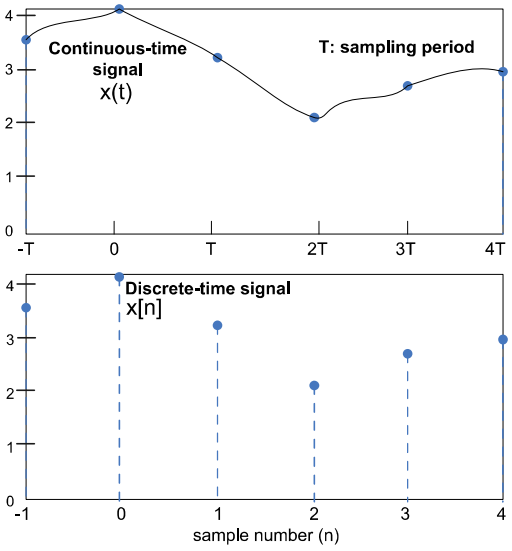


Figure 1.5: An example of acquiring discrete-time signals by sampling continuous-time signals.

$$x[n] = \{\ 3.5,\ \underset{\uparrow}{4},\ 3.25,\ \underset{\uparrow}{2},\ 2.5,\ \underset{\uparrow}{3.0}\ \}$$

n=-1 n=0 n=2 n=4

It is important to recognize that $x[n]$ is only defined for integer values of n. It is not correct to think of $x[n]$ as being zero for n not an integer, say n=1.5. $x[n]$ is simply undefined for non-integer values of n.

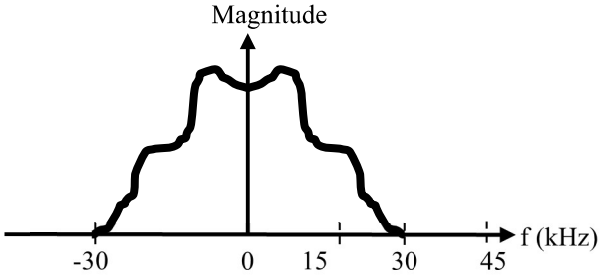
Sampling Theorem

If the highest frequency contained in an analogue signal $x(t)$ is f_{max} and the signal is sampled at a rate $f_s \geq 2\, f_{max}$ then $x(t)$ can be exactly recovered from its sample values using an interpolation function.

For example, audio CDs use a sampling rate $f_s = 44.1kHz$ for storage of the digital audio signal. This sampling frequency is slightly more than $2 \times f_{max}$ [f_{max} = 20kHz], which is generally accepted upper limit of human hearing and perception of music sounds.

Exercise:

What is the minimum sampling frequency, f_s, required to avoid aliasing when sampling a signal with the following spectrum?

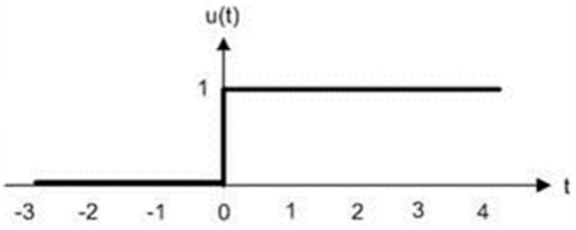


a) 30 kHz
b) 60 kHz
c) 45 kHz
d) none of the above

Unit step function

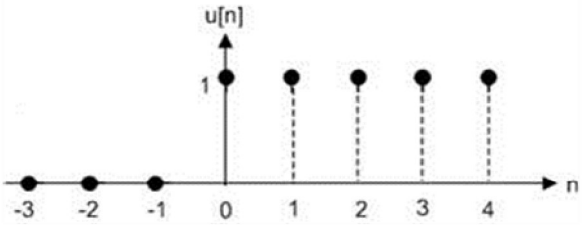
A continuous-time unit step function $u(t)$ is shown below:

$$u(t)=\begin{cases}1 & t\geq 0\\ 0 & t<0\end{cases} \qquad (1.2)$$

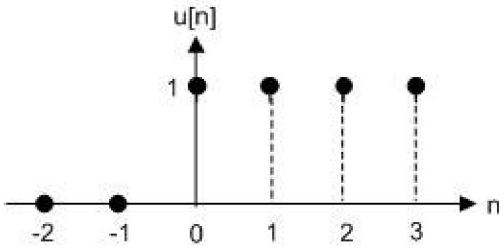


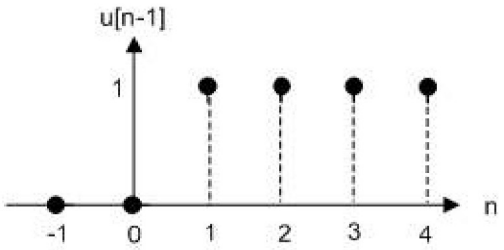
Note that the unit step is discontinuous at $t = 0$. Its samples $u[n] = u(t)|_{t=nT}$ form the discrete-time signal and defined by

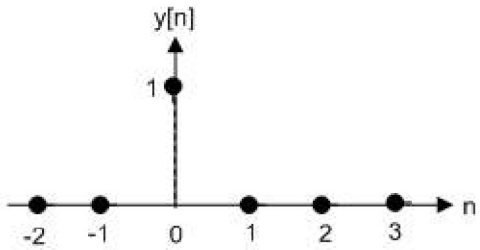
$$u[n]=\begin{cases}1 & n\geq 0\\ 0 & n<0\end{cases} \qquad (1.3)$$



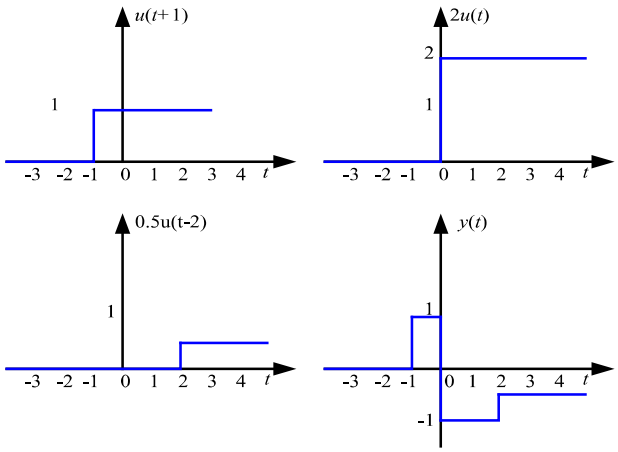
Example: Sketch the wave form: $y[n]=u[n]-u[n-1]$







Example: Sketch the waveform for $y(t) = u(t+1) - 2u(t) + 0.5u(t-2)$

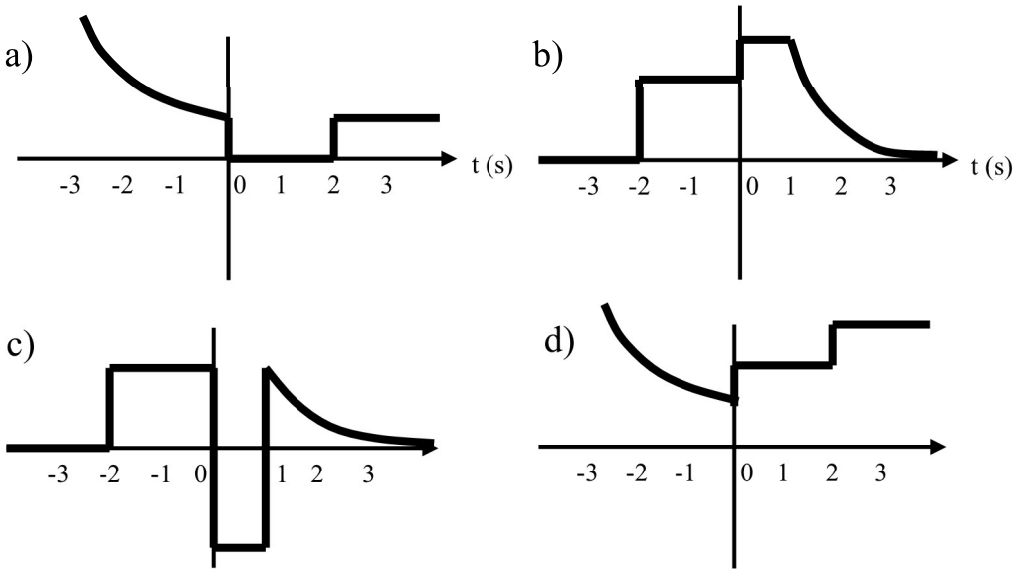


Exercise:

1) Sketch $x(t) = e^{-3t}[u(t) - u(t-2)]$

2) Which of the following depicts

$$y(t) = u(t+2) - 2u(t) + 2e^{-(t-1)}u(t-1)$$



Example:

Continuous-time Signal → Discrete-time signal

1. $x(t) = e^{at} \xrightarrow{t=nT} x[n] = e^{anT}$

   time (t)

   Sample number (n) [0,1,2,3,...]

   Sampling Period (T)

   $f_s = \frac{1}{T} Hz$ ← *sampling frequency*

2. $x(t) = 10e^{-t} - 5e^{-0.5t} \xrightarrow{t=nT} x[n] = 10^{-nT} - 5e^{-0.5nT}$

   sample number (n)

3. $x(t) = A\cos(\omega_a t) \xrightarrow{t=nT} x[n] = A\cos(\omega_a nT)$

   Analogue Frequency (ω_a) in radians

   $\omega_a = 2\pi f_a$

$$= A\cos(2\pi f_a n \frac{1}{f_s})$$

$$= A\cos(2\pi \frac{f_a}{f_s} n)$$

$$x[n] = A\cos(n\theta)$$

$$where \quad \theta = 2\pi \frac{f_a}{f_s} = \omega_a T$$

$$\theta : digital\ frequency(rad)$$

$$-\pi \le \theta \le \pi$$

Exercise: An analog signal $x(t) = 14\sin(5000\pi t)$ is sampled at a rate of 6 kHz. The resulting digital signal, $x[n]$, is given by:

a) $\sin[2\pi\frac{5}{6}n]$

b) $14\sin[\frac{5\pi}{6}n]$

c) $14\sin[1000\pi n]$

d) $14\sin[11000\pi n]$

Periodic Signals

An important class of signals is the periodic signals. A periodic continuous-time signal $x(t)$ has the property that there is a positive value of P for which

$$x(t) = x(t+P) \qquad (1.4)$$

for all values of t. In other words, a periodic signal has the property that is unchanged by a time shift of P. In this case we say $x(t)$ is periodic with period P.

Example

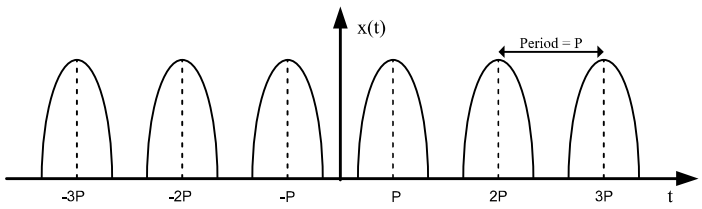
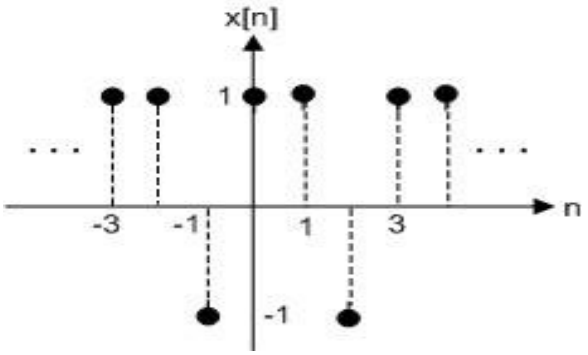


Figure 1.6: An example of a continuous-time periodic signal

Periodic signals are defined analogously in discrete time. A discrete-time signal $x[n]$ is periodic with period N, where N is a positive integer, if for all values of n.

$$x[n] = x[n+N]$$

Example:



x[n] with Period, N = 3 samples

1.2.2 Discrete-Time Sinusoidal Signals

A continuous-time sinusoidal signal is given by

$$x(t) = A\sin(\omega_a t) = A\sin(2\pi f_a t) \qquad (1.6)$$

f_a : *analogue frequency*

A discrete - time sinusoidal signal may be expressed as

$$\begin{aligned} x[n] &= x(t)|_{t=nT} = x(nT) \\ x[n] &= A\sin(n\omega_a T) \quad = A\sin(2\pi \frac{f_a}{f_s} n) \\ x[n] &\quad = A\sin(n\theta) \end{aligned} \qquad (1.7)$$

Sampling frequency: $f_s = \dfrac{1}{T}$

θ - Digital frequency: $\theta = 2\pi \dfrac{f_a}{f_s} = \omega_a T$ (1.8)

A discrete-time signal is said to be periodic with a period length N, if N is the smallest integer for which

$$\begin{aligned} x[n+N] &= x[n] \\ \therefore\ A\sin((n+N)\theta) &= A\sin(n\theta) \end{aligned}$$

which can only be satisfied for all n if

$N\theta = 2\pi k$ (where k is an arbitrary integer)

$$\therefore\ N = \frac{2\pi k}{\theta} = \frac{2\pi k}{2\pi \dfrac{f_a}{f_s}} \leftarrow \text{see eq. (1.8)}$$

$$N = \frac{f_s}{f_a} k \qquad (1.9)$$

For example, f_a = 1000Hz and f_s = 8000 Hz then $N=f_a / f_s=8$ samples (assume k is the smallest positive integer, i.e. k=1)
The following diagram depicts the cosine wave $x[n]=\cos\left(\frac{2\pi n}{12}\right)$ with period N = 12 samples.

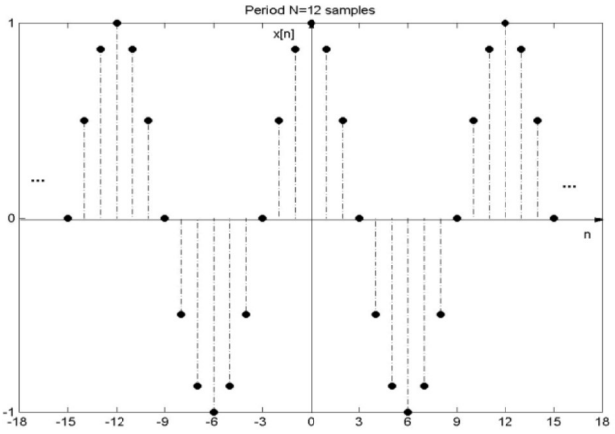


Figure 1.7 An example of a periodic sinusoidal sequence

Example:
Determine the fundamental period of $x[n]=10\cos\left(\frac{2\pi}{15}n+\frac{\pi}{3}\right)$

Digital frequency: $\theta = \frac{2\pi}{15}$

Recall (equation 1.9): N = $\frac{2\pi k}{\theta}$; k=1 is the smallest positive integer

$\therefore$ N = $\frac{2\pi\times1}{2\pi/15}$ = 15 samples

Example:
The sinusoidal signal x[n] has fundamental period N=10 samples. Determine the smallest θ for which x[n] is periodic:

$$\theta=\frac{2\pi k}{N}=\frac{2\pi}{10}k$$

Smallest value of θ is obtained when k = 1

$$\therefore \theta=\frac{2\pi}{10}=\frac{\pi}{5}radians/cycle$$

1.2.3 Discrete-time Exponential Signals

In discrete time, it is common practice to write a real exponential signal as

$$x[n] = c\alpha^n \qquad (1.10)$$

If c and α are real and if $|\alpha|>1$ the magnitude of the signal grows exponentially with n, while if $|\alpha|<1$ we have decaying exponential.

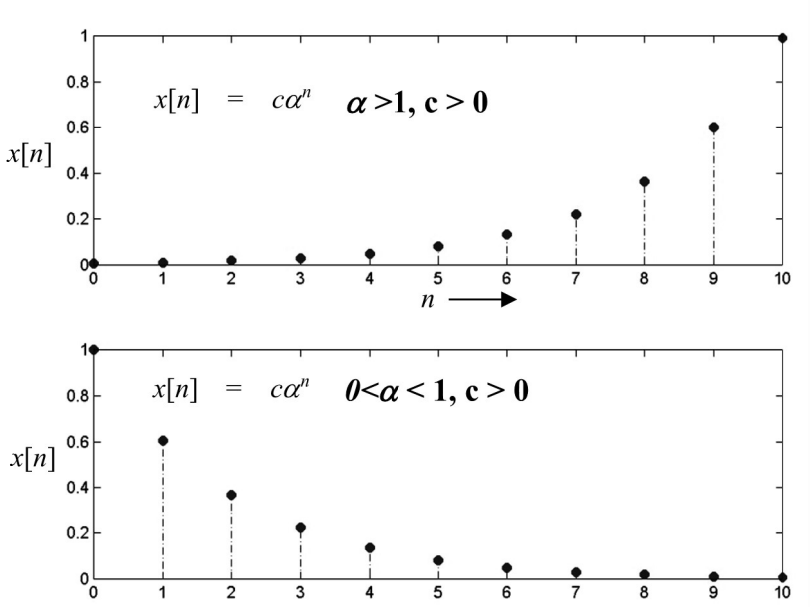
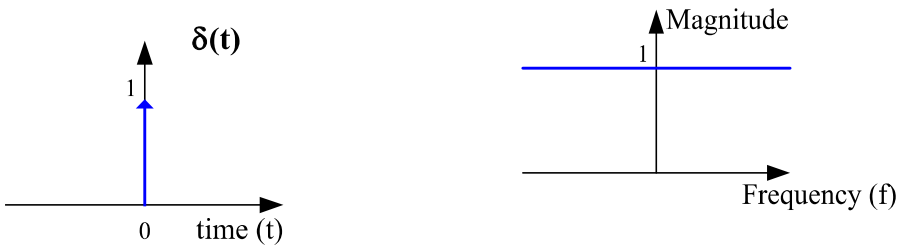


Figure 1.8: Examples of discrete-time exponential signals.

1.2.4 The Unit Impulse

An important concept in the theory of linear systems is the continuous time unit impulse function. This function, known also as the Dirac delta function is denoted by $\delta(t)$ and is represented graphically by a vertical arrow.

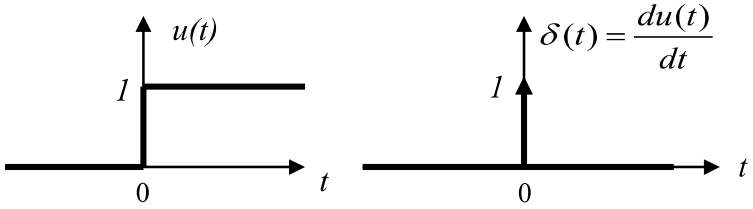


(a) (b)

Figure 1.9 Characteristics of impulse response function (a) in time domain and (b) in the frequency domain

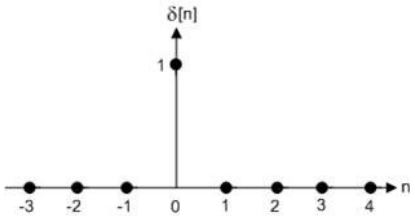
The impulse function $\delta(t)$ is the derivative of the step function $u(t)$.

$$\delta(t) = \frac{du(t)}{dt} \qquad (1.11)$$



The discrete-time unit impulse function $\delta[n]$ is defined in a manner similar to its continuous time counterpart. We also refer $\delta[n]$ as the unit sample.

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases} \qquad (1.12)$$



Exercise:

1) Which of the following signals is not periodic?

   a) x[n] = sin[πn/9]
   b) x[n] = sin[nπ^2]
   c) x[n] = cos[πn^2/15]
   d) x[n] = sin[πn/5 + π] + cos[πn/10 – π]

2) A discrete-time signal is given by x[n] = 3cos[**Error! Bookmark not defined.** $\frac{4\pi}{34}$ n + **Error! Bookmark not defined.** $\frac{\pi}{12}$]. The digital frequency (θ) and fundamental period (N) are given by:

   a) $\frac{4\pi}{34}$ and 4 samples
   b) **Error! Bookmark not defined.** $\frac{\pi}{12}$ and 34 samples
   c) $\frac{4\pi}{34}$ n and 12 samples

d) $\frac{2\pi}{17}$ and 17 samples

1.2.5 Simple Manipulations of Discrete-Time Signals

A signal $x[n]$ may be shifted in time by replacing the independent variable n by $n-k$ where k is an integer.

If $k>0 \Rightarrow$ the time shift results in a **delay** of the signal by k samples [ie. shifting a signal to the right]

If $k<0 \Rightarrow$ the time shift results in an **advance** of the signal by k samples. [ie. shifting a signal to the left]

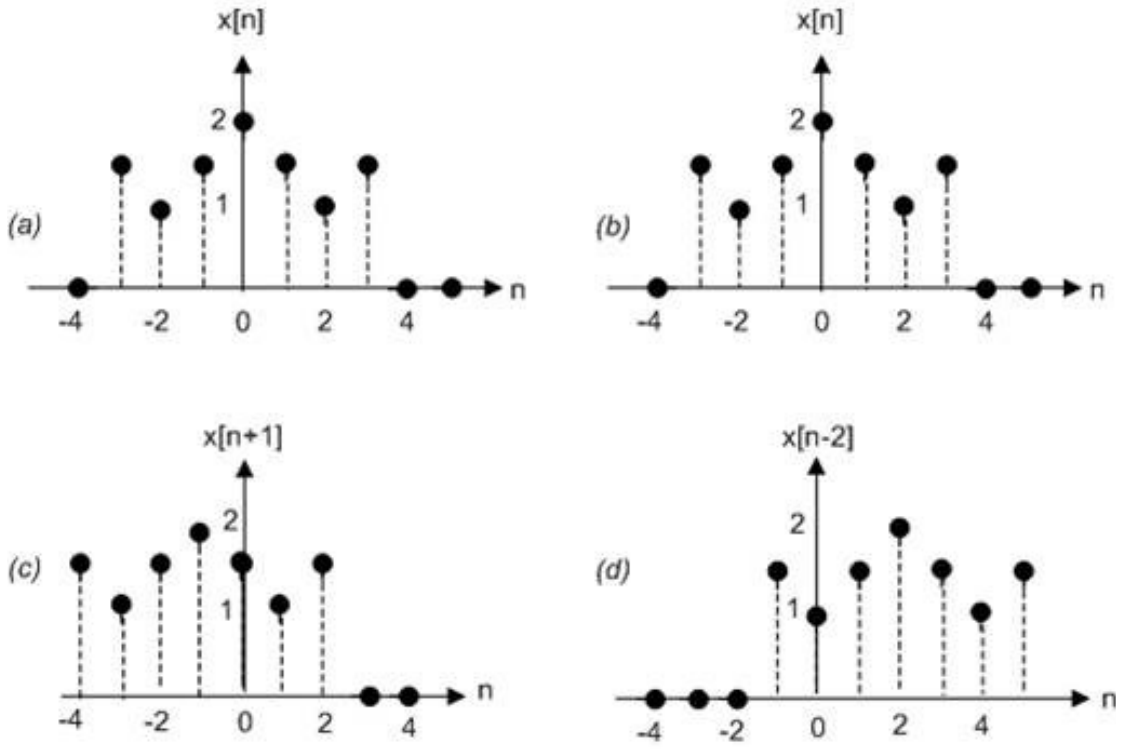


Figure 1.10: (a,b) Original signal x[n], (c) x[n] is advanced by 1 sample, (d) x[n] is delayed by 2 samples

1.3 Systems

A continuous-time system is one whose input $x(t)$ and output $y(t)$ are continuous time functions related by a rule as shown in Figure 1.11.

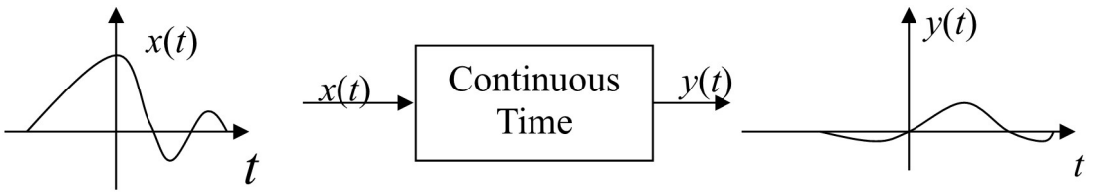


Figure 1.11: General representation of continuous-time systems.

A discrete system is one whose input $x[n]$ and output $y[n]$ are discrete time function related by a rule as shown in Figure 1.12.

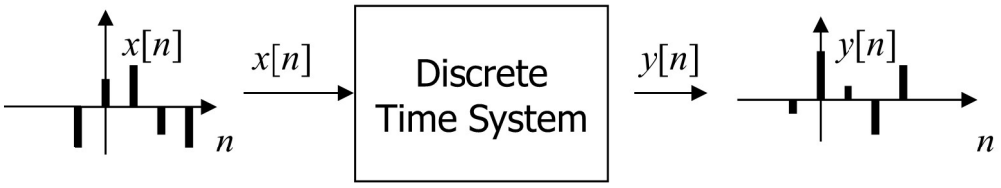
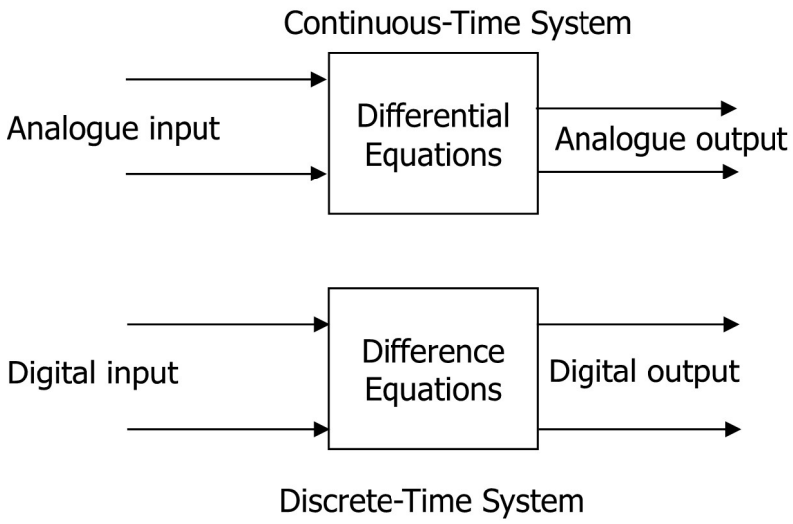


Figure 1.12: General representation of discrete-time systems.

An important mathematical distinction between continuous-time and discrete-time systems is the fact that the former are characterized by differential equations whereas the latter are characterized by difference equations.



Example: The RC circuit (known as a low-pass filter or an integrator) shown in Figure 1.13 is a continuous-time system.

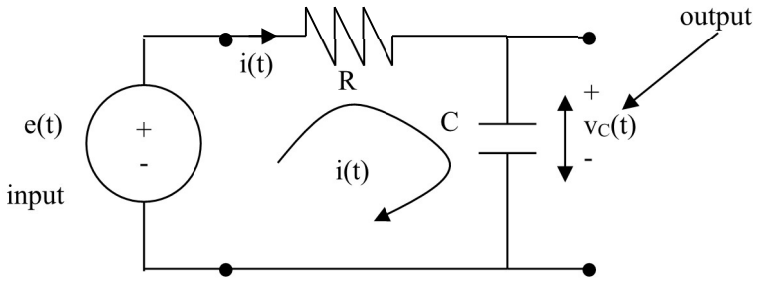


Figure 1.13: A diagram of RC circuit as an example of continuous-time systems.

If we regard $e(t)$ as the input signal and $v_C(t)$ as the output signal, we obtain using simple circuit analysis

$$\frac{dv_C(t)}{dt} + \frac{1}{RC} v_C(t) = \frac{1}{RC} e(t) \qquad (1.13)$$

From equation (1.13), a discrete -time system can be developed as follows: If the sampling period T is sufficiently small, (see figure 1.14)

$$\left.\frac{dv_C(t)}{dt}\right|_{t=nT} = \frac{v_C(nT) - v_C(nT-T)}{T} \qquad (1.14)$$

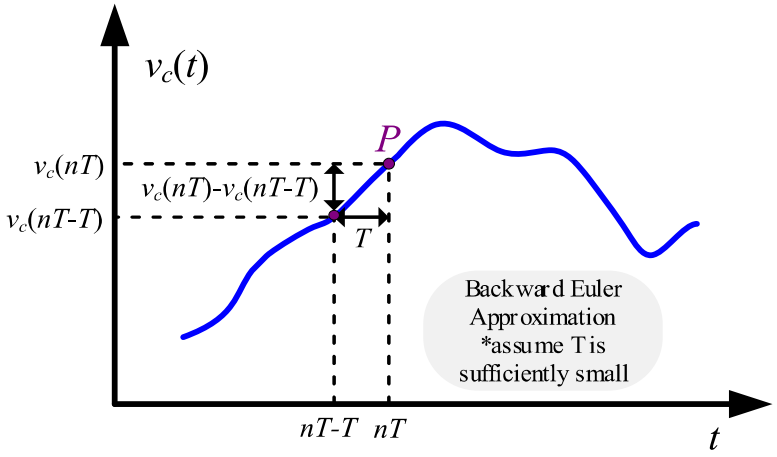


Figure 1.14: An approximation of discrete-time systems from the continuous-time systems.

By substituting equation (1.14) into (1.13) and replacing *t* by *nT*, we obtain:

$$\frac{v_C(nT)-v_C(nT-T)}{T}+\frac{1}{RC}v_C(nT)=\frac{1}{RC}e(nT)$$

The difference equation is:

$$\frac{v_C[n]-v_C[n-1]}{T}+\frac{1}{RC}v_C[n]=\frac{1}{RC}e[n]$$

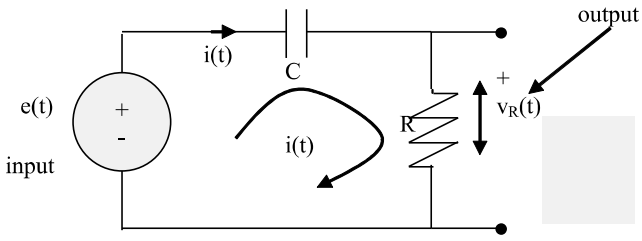
$$v_C[n]=\frac{RC}{RC+T}v_C[n-1]+\frac{T}{RC+T}e[n] \quad (1.10)$$

difference equation

output previous output input

Exercise:

The RC circuit, known as a high-pass filter or a differentiator, is shown below. Derive a difference equation relating the output and input of the circuit.



CHAPTER 1: PROBLEM SHEET 1

Q1. Sketch the following:

a) $x(t) = u(t-3) - u(t-5)$

b) $y[n] = u[n+3] - u[n-10]$

c) $x(t) = e^{2t}u(-t)$

d) $y[n] = u[-n]$

e) $h[n] = 2\delta[n+1] + 2\delta[n-1]$

f) $h[n] = u[n]$, $p[n] = h[-n]$; $q[n] = h[-1-n]$, $r[n] = h[1-n]$

Q2.

a) Consider a discrete-time sequence $x[n] = \cos\left[n\frac{\pi}{8} + \frac{\pi}{5}\right]$

Determine the fundamental period of $x[n]$. **Ans: 16 samples**

b)

i) Consider the sinusoidal signal

$x(t) = 10\sin(2\pi f_a t)$ *where* f_a -analogue frequency and t- time,

Write an equation for the discrete time signal $x[n]$. **Ans: x[n]=10 sin(nθ)**

ii) If f_a = 200 Hz and sampling frequency, f_s = 8000 Hz, determine the fundamental period of $x[n]$. **Ans: N=40**

End of Chapter 1

Chapter 2

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Chapter 2: Digital Signal Processing Fundamentals

2.1 Introduction

Digital Signal Processing (DSP) is a rapidly developing technology for scientists and engineers. In the 1990s the digital signal processing revolution started, both in terms of the consumer boom in digital audio, digital telecommunications and the wide used of technology in industry.

Due to the availability of low cost digital signal processors, manufacturers are producing plug-in DSP boards for PCs, together with high-level tools to control these boards. There are many areas where DSP technology is now being used and the current proliferation of such technology will open up further applications.

In the medical field, DSP systems are widely utilized for recording data analysis and the interpretation of ECG signals.

Audiologists and speech therapists are exposed to DSP systems for both testing a person's level of hearing and subsequently DSP hearing aid filtering.

The professional music industry uses spectrum analysers, digital filtering, sampling conversion filters etc and is one of the biggest users and exploiters of DSP technology.

In summary, DSP is applied in the area of control and power systems, biomedical engineering, instrumentation (test and measurement), automotive engineering, telecommunications, mobile communication, speech analysis and synthesis, audio and

video processing, seismic, radar and sonar processing and neural computing.

There are many advantages to using DSP techniques for variety of applications, these include:

- High reliability and reproducibility
- Flexibility and programmability
- The absence of component drift problem
- Compressed storage facility

DSP hardware allows for programmable operations. Through software, one can easily modify the signal processing functions to be performed by the hardware. For all these reasons, there has been vast growth in DSP theory & applications over the past decade.

2.2 Overview of a DSP System

An analogue signal processing system is shown in Figure 2.1, in which both the input signal and output signal are in analogue form

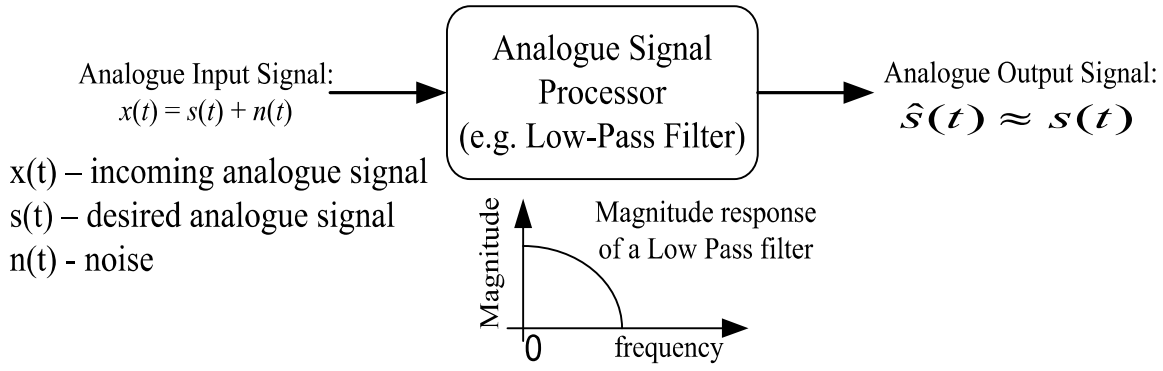


Figure 2.1 A general description of analogue systems whose input and output are in analogue form

A digital signal processing system in Figure 2.1 provides an alternative method for processing the analogue signal.

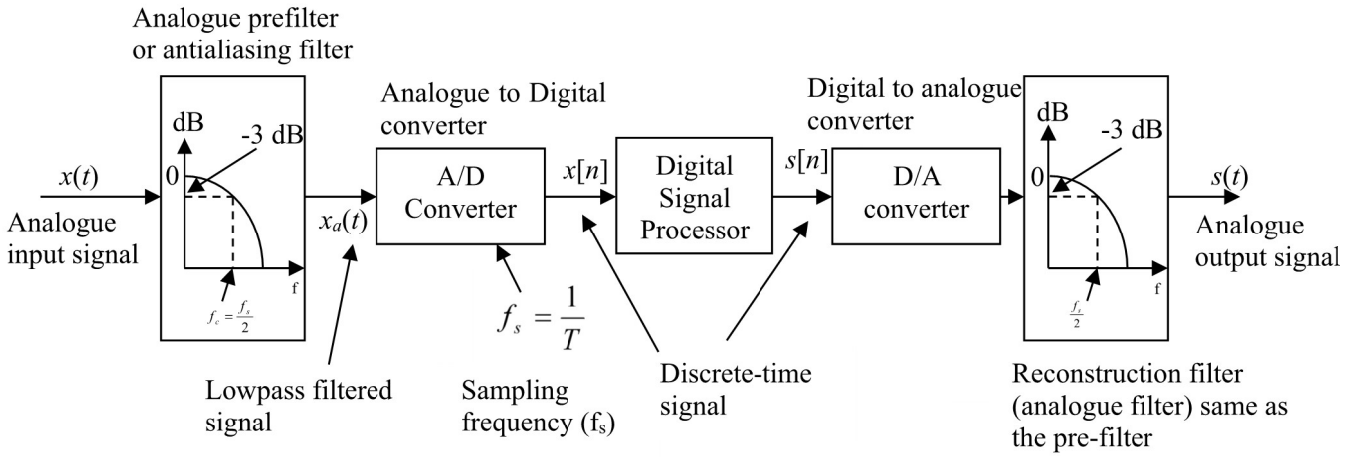


Figure 2.1: A general process of converting analogue signals into digital signals and back to analogue form.

Name	Function
Anti-aliasing filter ($f_c = \frac{f_s}{2}$)	To band-limit the analogue input signal prior to digitisation to reduce **aliasing** (see Section 2.7)
Analogue-to-digital converter	To convert analogue input signal into digital output signal by sampling ($f_s = \frac{1}{T}$)
Digital Signal Processor (heart of the system)	To process the digital signal according to the pre-defined rules
Digital-to-analogue converter	To convert the digital input signal into analogue output signal by interpolating
Reconstruction filter ($f_c = \frac{f_s}{2}$)	To smooth out the output of D/A converter To remove unwanted high frequency components

Table 2.1 Description of blocks contained in a general DSP process

The digital signal processor may implement one of the several DSP algorithms, for example digital filtering (low-pass filter) mapping the digital input signal $x[n]$ into digital output signal $s[n]$.

2.3 Analogue to Digital Conversion Process

Before any DSP algorithm can be performed, the signal must be in a digital form. The A/D conversion process involves the following steps:

- The signal (Band-limited) is first sampled, converting the analogue signal into a discrete-time signal
- The amplitude of each sample is quantised into one of 2^B levels (where B is the number of bits used to represent a sample in the A/D converter)
- The discrete amplitude levels are represented or encoded into distinct binary words each of length B bits.

A practical representation of the A/D conversion process is shown in Figure 2.3.

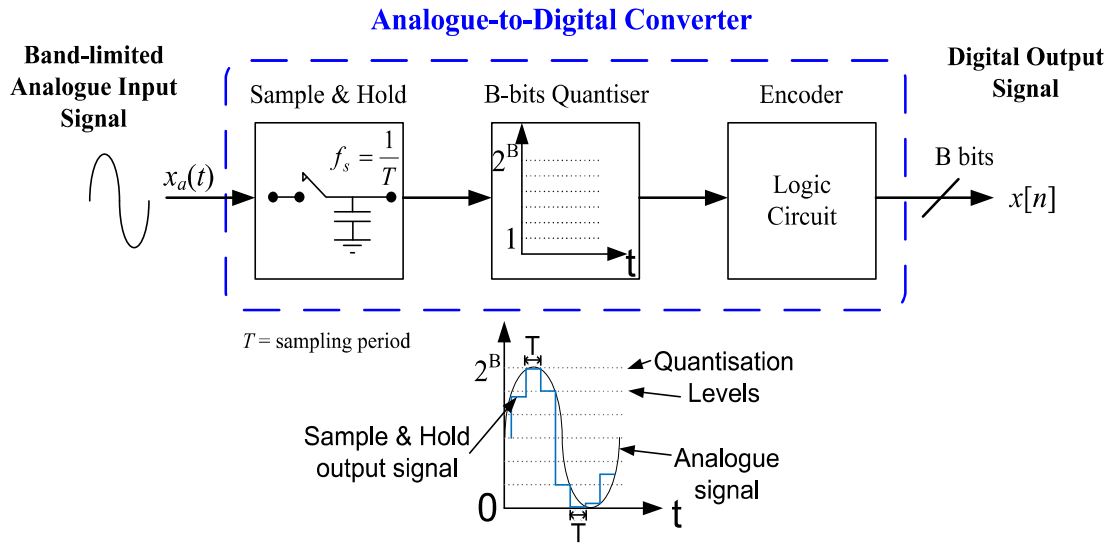
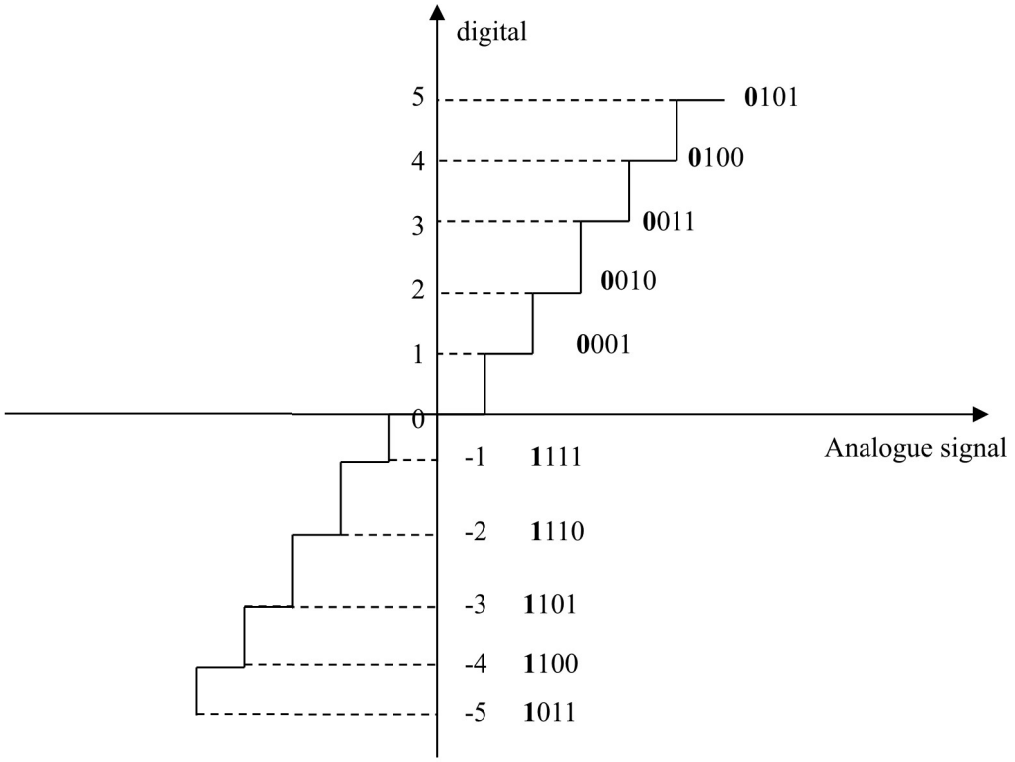


Figure 2.3 Analogue to digital conversion process

Sample and hold (S/H) takes a snapshot of the analogue signal every T sec and then holds that value constant for T secs until the next snapshot is obtained.

Example: 4-bit (B = 4) A/D converter (bipolar)



Input-output characteristic of 4-bit quantiser (linear) (two's complement notation)

2.4 Quantisation and Encoding

Before conversion to digital, the analogue sample is assigned one of 2^B values (see Figure 2.4). This process, termed quantization, introduces an error, which cannot be removed.

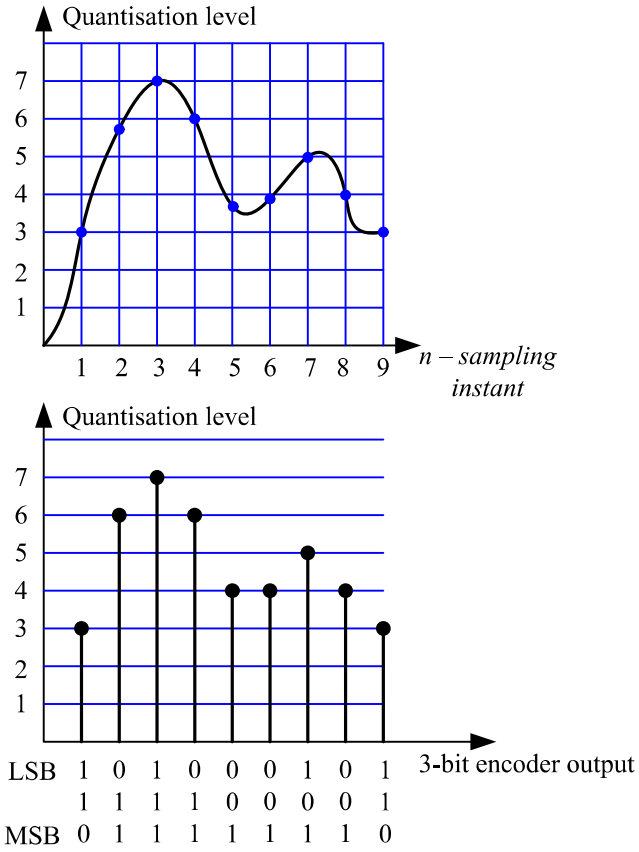
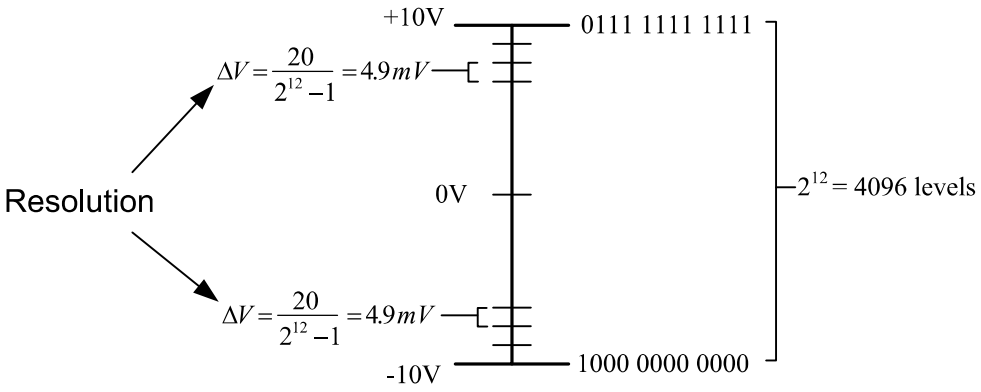


Figure 2.4: Quantisation of discrete-time signals

Example

A 12 bit A/D converter (bipolar) with an input voltage range of ±10V will have a least significant bit (LSB) of $\frac{20V}{2^{12}-1}mV = 4.9mV$ (resolution or quantisation step size, ΔV)



For a B-bit A/D converter, the number of quantisation level is 2^B, and the interval between levels, that is known as the quantisation step size or resolution (ΔV) is given by:

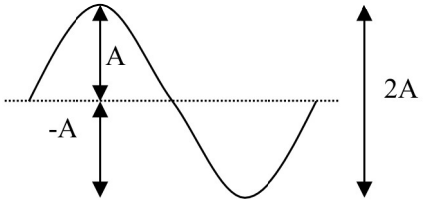
$$\therefore \quad \Delta V = \frac{V}{2^B - 1} \approx \frac{V}{2^B}$$

The approximation holds when when B is large (say B > 10 bits)

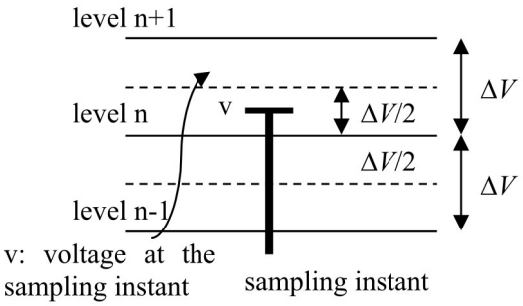
where ΔV = resolution, B = number of bits and V = peak-to-peak amplitude.

For a sine wave input of amplitude A, the quantisation step size becomes

$$\Delta V \approx \frac{2A}{2^B}$$



Quantisation Error (e):

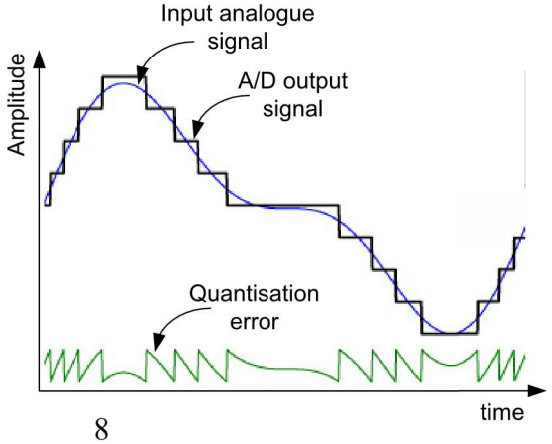


Quantisation error (e) $= \frac{\Delta V}{2}$

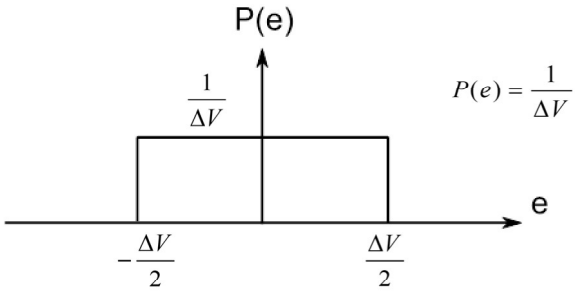
For an A/D converter with bipolar signal inputs, the quantisation error (e) is rounded up or down.$\pm$ $\Delta V/2$. The quantisation error ($e[n]$) for each sample, is normally assumed to be random and uniformly distributed in the interval $\pm$ $\Delta V/2$ with zero mean.

$$e[n] = A - A_q$$

Quantized amplitude — Actual amplitude



The probability density function of the error $P(e)$ has the form as shown below



The quantisation noise power or variance σ_e^2 is hence given by

$$\sigma_e^2 = \int_{-\frac{\Delta V}{2}}^{\frac{\Delta V}{2}} e^2 P(e) de$$

Probability of quantisation error is constant

$$P(e) = \frac{1}{\Delta V}$$

$$= \frac{1}{\Delta V} \int_{-\frac{\Delta V}{2}}^{\frac{\Delta V}{2}} e^2 de = \frac{1}{\Delta V} \left[\frac{e^3}{3} \right]_{-\frac{\Delta V}{2}}^{\frac{\Delta V}{2}}$$

Hence, the quantisation noise power

$$\sigma_e^2 = \frac{\Delta V^2}{12}$$ for uniform quantisation

(Note: Uniform quantisation - all steps (ΔV) are of equal size)

Example

Signal-to-quantisation noise power ratio (SQNR) is defined as

$$SQNR = \frac{P_S}{P_n}$$

(P_S: signal power; P_n: noise power)

$$P_S = \frac{1}{N}\sum_{n=1}^{N}\{x[n]\}^2 \quad and \quad P_n = \frac{1}{N}\sum_{n=1}^{N}\{e[n]\}^2$$

$$or \quad \sigma_e^2 = \frac{\Delta V^2}{12}$$

$$SQNR(dB) = 10\log\frac{P_S}{P_n} = 10\log\frac{\sum_{n=1}^{N} x^2[n]}{\sum_{n=1}^{N} e^2[n]}$$

The dynamic range, R, of the signal is defined as

$$R = \{x[n]\}_{\max} - \{x[n]\}_{\min}$$

The quantisation step size of resolution ΔV is defined as

$$\Delta V = \frac{R}{L}$$

(L: number of levels in the quantiser)

$$\therefore\ SQNR(dB) = 10\log\frac{P_S}{\frac{(R/L)^2}{12}} = 10\log\frac{12P_S L^2}{R^2}$$

$$\boxed{SQNR(dB) = 10\log P_S + 20\log L + 10\log 12 - 20\log R}$$

Example: For the sine wave input, the average signal power is $A^2/2$, i.e. $(A/\sqrt{2})^2$ rms value. The signal-to-quantisation noise power ratio (SQNR) in decibels is

$$SQNR = 10\log\left(\frac{\frac{A^2}{2}}{\frac{\Delta V^2}{12}}\right) = 10\log\left(\frac{\frac{A^2}{2}}{\frac{\left(2A/2^B\right)^2}{12}}\right) = 10\log\left(\frac{3\times 2^{2B}}{2}\right)$$

$$\boldsymbol{SQNR = 6.02B + 1.76 \quad dB}$$

The SQNR increases with the number of bits, B. In many DSP applications, an A/D converter resolution between 12 and 16 bits is adequate.

Number of Bits	Levels	SQNR
3	8	19.7 dB
4	16	25.3 dB
5	32	31.6 dB
6	64	37.7 dB
7	128	43.8 dB

Thus, the signal-to-quantisation noise ratio increases approximately 6dB for each bit.

Exercise: Show that the input signal x(t) to quantisation noise ratio of a linear A/D converter is given by

$$SQNR = 10\log P_s + 10.8 + 20\log L - 20\log R$$

where P_s is the signal power $\left[P_s = \frac{1}{N}\sum_{n=0}^{N-1} x^2[n]\right]$; L is the number of quantisation levels and R is the dynamic range of the input signal. Using the above equation, show that for a B-bit quantiser, $SQNR = 6.02B + 1.76\ (dB)$ if $x(t) = A\cos(2\pi ft)$

2.5 Continuous-Time Fourier Transform (FT)

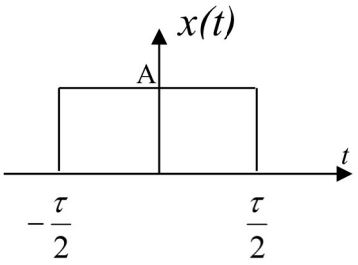
The Fourier transform for non-periodic continuous-time signals is a mathematical transformation to transform signals between time domain and frequency domain, which has many applications in engineering. Most signals of practical importance are non-periodic. The FT pairs is given by

$$\text{Fourier Transform (FT): } X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

$$\text{Inverse Fourier Transform (IFT): } x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}d\omega$$

The FT converts the time domain signal, $x(t)$, into its frequency domain representation, $X(\omega)$ and the IFT converts the frequency domain representation, $X(\omega)$, back into the time domain $x(t)$.

Example: Evaluate the Fourier transform of a rectangular pulse shown below:

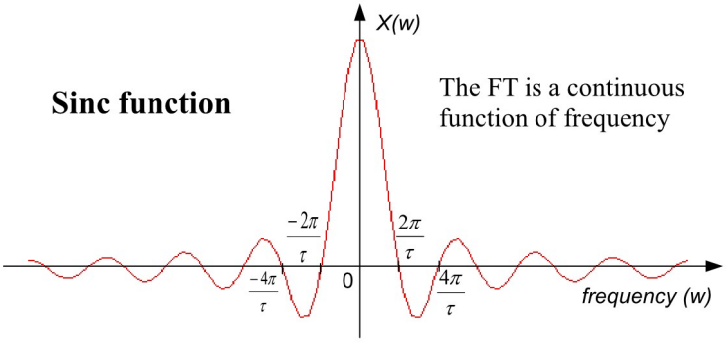


$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt = \int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} x(t)e^{-j\omega t}dt = -\frac{A}{j\omega}\left[e^{-\frac{j\omega\tau}{2}} - e^{\frac{j\omega\tau}{2}}\right]$$

$$X(\omega) = A\tau\frac{\sin\frac{\omega\tau}{2}}{\frac{\omega\tau}{2}}$$

Sinc function

Sinc function

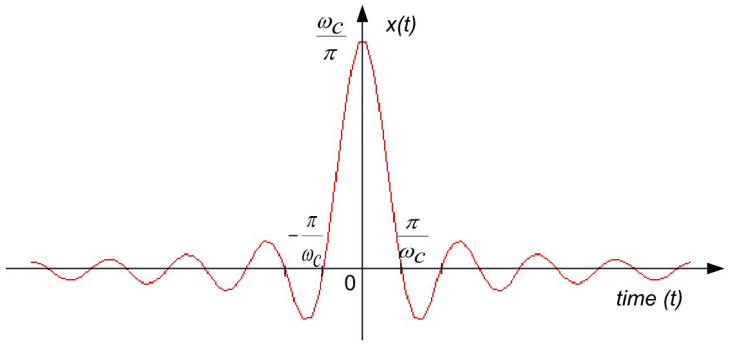


The FT is a continuous function of frequency

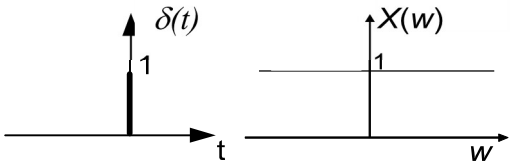
Example: Find the **inverse** Fourier transform of the rectangular spectrum shown below:

$$x(t)=\frac{1}{2\pi}\int_{-\omega_c}^{\omega_c}X(\omega)e^{j\omega t}d\omega;\quad x(t)=\frac{1}{2\pi}\int_{-\omega_c}^{\omega_c}1\cdot e^{j\omega t}d\omega$$

$$x(t)=\frac{1}{\pi t}\sin(\omega_C t)=\frac{\omega_C}{\pi}\frac{\sin\omega_C t}{\omega_C t}$$



Example: Find the Fourier Transform of $x(t) = \delta(t)$.



$$X(\omega)=\int_{-\infty}^{\infty}\delta(t)e^{-j\omega t}dt=1$$

Example:

$$FT\left\{e^{j\omega_1 t}\right\}=\int_{-\infty}^{\infty}e^{j\omega_1 t}e^{-j\omega t}dt=\int_{-\infty}^{\infty}e^{j(\omega_1-\omega)t}dt=2\pi\delta(\omega_1-\omega)$$

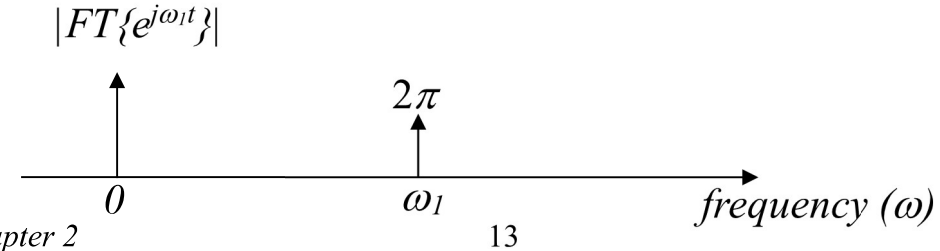
$$\boxed{FT\left\{e^{j\omega_1 t}\right\}=2\pi\delta(\omega-\omega_1)}$$

Using the properties:

$\int_{-\infty}^{\infty}e^{j\omega t}dt=2\pi\delta(\omega)$

and $\delta(\omega-\omega_1)=\delta(\omega_1-\omega)$

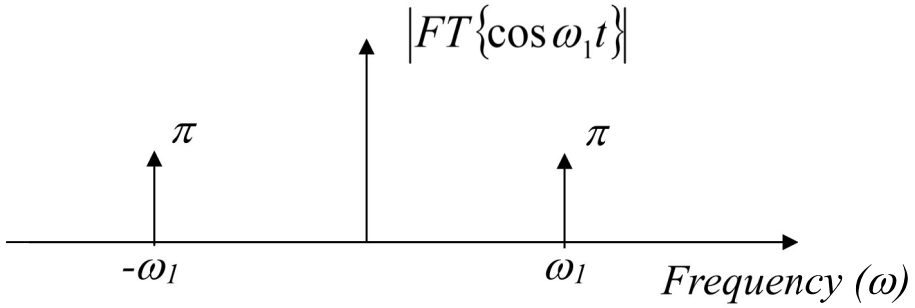
Hence, the magnitude spectrum of $e^{j\omega_1 t}$ is show below:



Example:

$$FT\{\cos\omega_1 t\} = FT\left\{\frac{e^{j\omega_1 t} + e^{-j\omega_1 t}}{2}\right\} = \frac{1}{2}FT\{e^{j\omega_1 t}\} + \frac{1}{2}FT\{e^{-j\omega_1 t}\}$$
$$= \pi\delta(\omega - \omega_1) + \pi\delta(\omega + \omega_1)$$

The magnitude spectrum of $\cos\omega_1 t$ is show below



Exercise:

Find the Fourier Transform of x(t)

$$x(t) = A\sin\omega t + A\cos 2\omega t$$

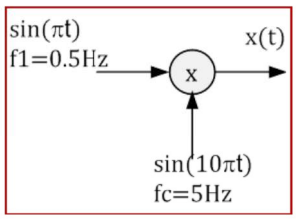
Sketch the magnitude spectrum of $x(t)$

Ans: $X(\omega) = A\pi[\delta(\omega - 2\omega_0) + \delta(\omega + 2\omega_0) - j\delta(\omega - \omega_0) + j\delta(\omega + \omega_0)]$

Multiplication of sinusoid

Consider a signal $x(t)$ which is a product of two sinusoids, as given by

$$x(t) = \sin(\pi t)\sin(10\pi t)$$

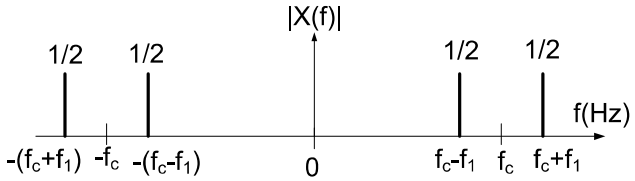


$x(t)$ can be re-written as

$$x(t) = \left(\frac{e^{j10\pi t} - e^{-j10\pi t}}{2j}\right)\left(\frac{e^{j\pi t} - e^{-j\pi t}}{2j}\right) = \frac{1}{4}\left(e^{-j10\pi t} - e^{j10\pi t}\right)\left(e^{-j\pi t} - e^{j\pi t}\right)$$

$$x(t) = \frac{1}{4}e^{j11\pi t} + \frac{1}{4}e^{-j11\pi t} - \frac{1}{4}e^{j9\pi t} - \frac{1}{4}e^{-j9\pi t}$$

From $x(t)$ it can be seen that there are four spectral compoments at frequencies f_c+f_1= 5.5Hz, f_c-f_1 = 4.5Hz, $-(f_c+f_1)$ = -4.5Hz, $-(f_c-f_1)$ =-5.5Hz, as shown below



Magnitude spectrum of x(t)

Exercise:

(a) Sketch the magnitude and phase responses of the following signal.

$$x(t) = \sin(2\pi t) + \cos(3\pi t)$$

(b) Find the Fourier Transform of $\sin\omega_1 t$ and draw the amplitude spectrum of $\sin\omega_1 t$.

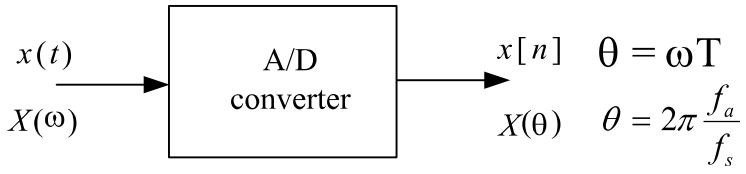
(c) Consider the amplitude modulated signal

$$x(t) = 5\cos(3\pi t)\cos(\pi t), \qquad t - \mu s$$

Determine its spectrum (ie. FT) and draw its amplitude spectrum.

2.6 Sampling of continuous-time signal

If a continuous-time signal $x(t)$ is sampled every T seconds, then at the output of the Analogue to Digital Converter, as shown below we obtain a discrete-time signal $x[n]$



where t = time; n = sample number; ω = analogue frequency; θ = digital frequency; $X(\theta)$ = digital spectrum; $X(\omega)$ = analogue spectrum.

The Discrete-Time Fourier Transform (DTFT) pair is given by

Discrete Time Fourier Transform (DTFT):	$X(\theta) = \sum_{n=-\infty}^{\infty} x[n] e^{-jn\theta}$
Inverse Discrete Time Fourier Transform (IDTFT):	$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\theta) e^{jn\theta} d\theta$ where $-\pi \le \theta \le \pi$

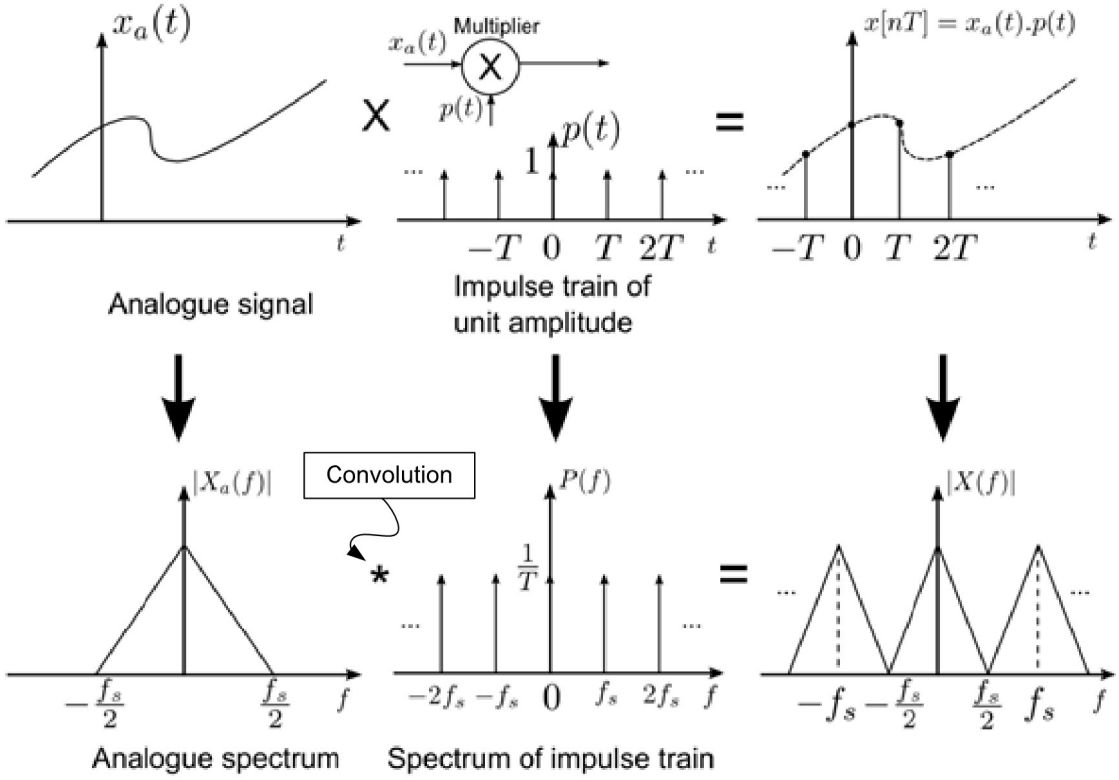
Using Inverse Fourier Transforms in Continuous-time and discrete-time domains, we can show that (see tutorial 2)

$$X(\theta) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(\omega + k\frac{2\pi}{T})$$

It is seen that $X(\theta)$ is periodic with period 2π. The digital spectrum is repetition of the bandlimited analogue spectrum.

2.6.1 The Ideal Sampling Operation

An analogue signal multiplied by a periodic impulse train results in a train of impulses that match the values of the analogue signal at the sampling instants.



Multiplication of the analogue signal and the ideal impulse train results in the convolution of their respective spectra.

The spectrum of the sampled signal $x[n]$ thus consists of replicas of $X_a(f)$ at multiples of the sampling rate f_s ($f_s = \frac{1}{T}$ or $w_s = \frac{2\pi}{T}$).

Exercise: Find the Discrete Time Fourier Transform of x[n]

x[n]=[1 0 0 1]

Ans: $X(\theta) = (2\cos 2\theta)e^{-j2\theta}$

2.7 Aliasing

Aliasing arises when a continuous-time signal is sampled at a rate that is insufficient to capture the changes in the signal. If aliasing occurs, the original continuous time signal cannot be recovered. The Nyquist sampling theorem states that the sampling frequency, f_s, should be at least twice the highest frequency, f_c, contained in the signal ($f_s \geq 2f_c$) to avoid aliasing.

Figure 2.5 illustrates the relationship between the digital spectrum $X(\theta)$ and the analogue spectrum $X(\omega)$ for the case $X(\omega) = 0$, $|\omega| \geq \frac{\pi}{T}$ or $|f| \geq \frac{f_s}{2}$.

Case 1: $X(\omega) = 0$, $|\omega| \geq \frac{\pi}{T}$ (sampling theorem holds)

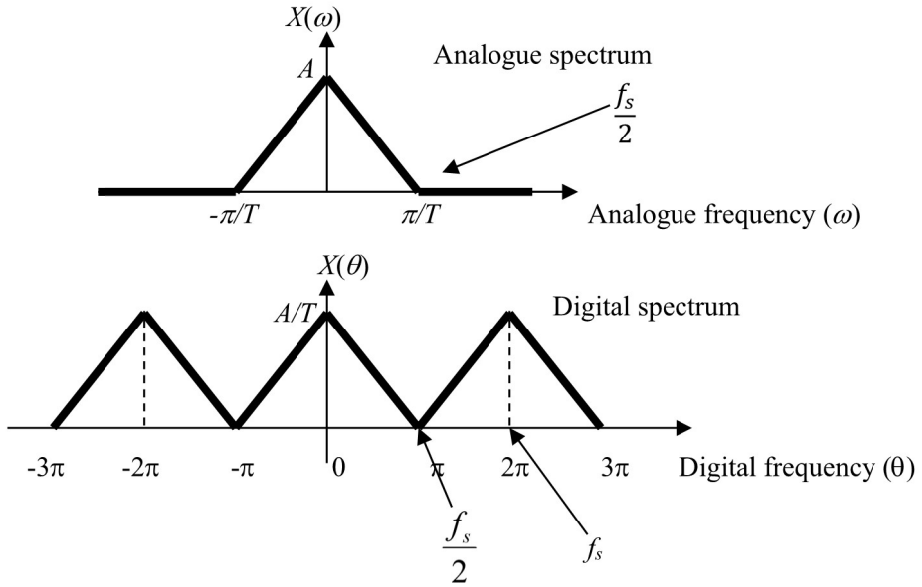


Figure 2.5: Above: Frequency response of an analogue signal. Below: Frequency response of the sampled analogue signal.

Note: $\omega = \frac{\pi}{T}$ corresponds to $\theta = \pi$ (or $f = \frac{f_s}{2}$)

The digital spectrum is the same as the original analogue spectrum and repeats at multiples of the sampling frequency f_s (refer to figure 2.6) as given by:

$$f_k = f_0 + kf_s$$

where k is an integer ($k = \pm 1, \pm 2, \pm 3, \ldots$) and f_0 is the frequency present in the fundamental region of the original analogue spectrum. It is clear that the frequency f_k is outside the fundamental frequency range $-\frac{f_s}{2} \le f_0 \le \frac{f_s}{2}$.

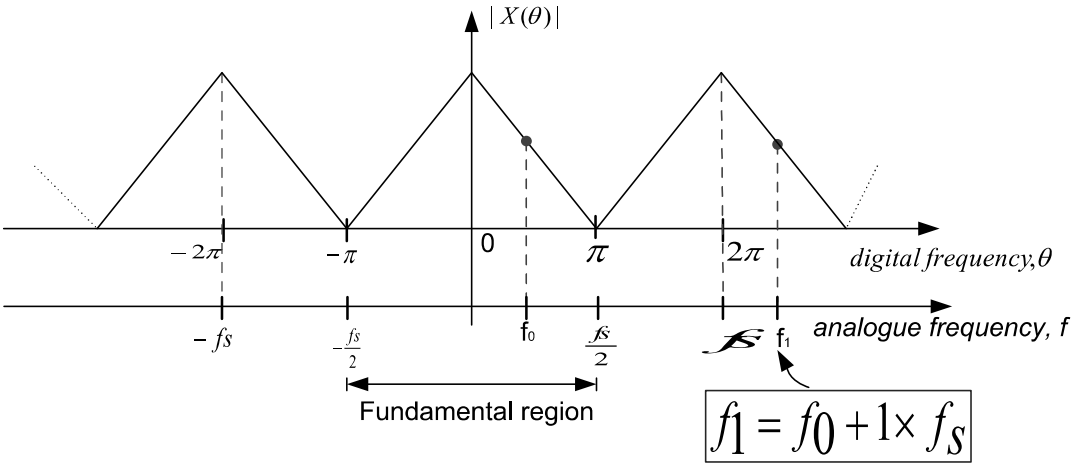


Figure 2.6

Case 2: $X(\omega) \neq 0$, $|\omega| > \frac{\pi}{T}$, but $X(\omega) = 0$, $|\omega| > \frac{3\pi}{2T}$

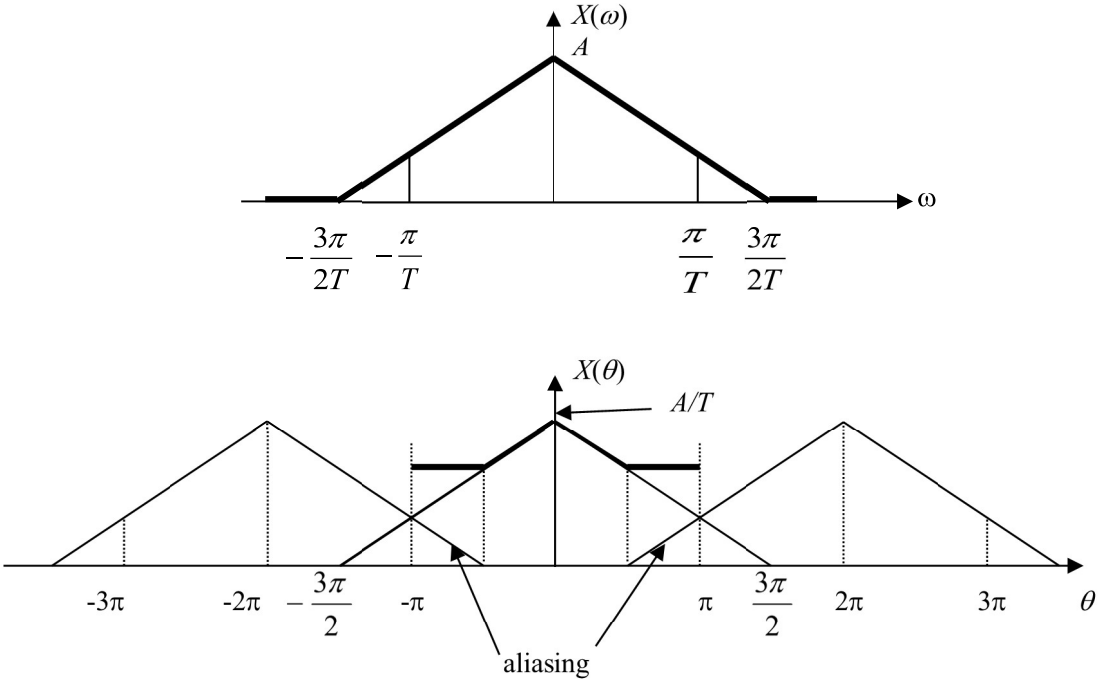


Figure 2.7: Above: Frequency response of an analogue signal whose highest frequency component is larger than the sampling frequency. Below: Frequency response of the sampled analogue signal. The overlapped region represents aliasing.

If the sampling frequency, f_s is not sufficiently high, the spectrum centred on f_s will fold over or alias into the base band frequencies (Figure 2.7). Aliasing can only be avoided if the analogue signal is band limited such that $X(\omega) = 0$, $|\omega| \geq \frac{\pi}{T} \Rightarrow |2\pi f| \geq \frac{\pi}{T} \Rightarrow |f| \geq \frac{f_s}{2}$. This results in the familiar sampling theorem.

Note:

f_s – sampling frequency, f_a – analog frequency

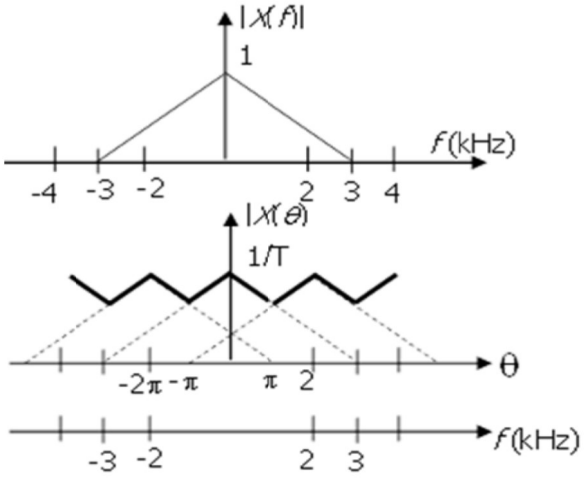
$\frac{f_s}{2}$ corresponds to $\theta = \pi$

$\frac{f_s}{2}$ is the highest frequency that can be represented uniquely with a sampling rate f_s

$\frac{fs}{2}$ is called half the sampling frequency or folding frequency.

Digital frequency, $\theta = \omega T = 2\pi \frac{f_a}{f_s}$

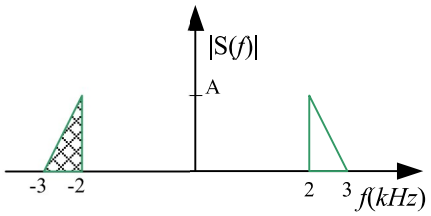
Example: Suppose $x(t)$ has the spectrum $X(f)$ as shown below. Sketch the digital spectrum $|X(\theta)|$ if the sampling frequency f_s = 2 kHz.



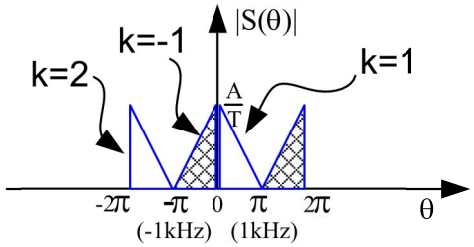
Example: Consider a signal, s(t), with spectrum satisfying the following equation

$$|S(f)| = \begin{cases} 3 - |f| & 2kHz < |f| < 3kHz \\ 0 & \text{otherwise} \end{cases}$$

f – frequency in kHz. The signal $s(t)$ is sampled uniformly with a sampling frequency of 2kHz. Sketch the digital spectrum $|S(\theta)|$, if it is sampled at a sampling frequency of 2kHz.

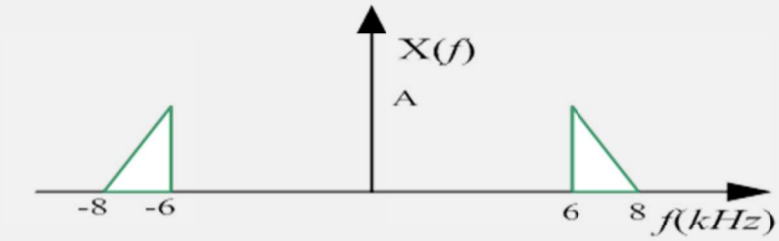


$$f_s = 2kHz, \quad f_k = f_o + kf_s$$
$$f_o = f_k - kf_s, \quad k = 0, \pm1, \pm2, \ldots$$
$$\text{For } k = 1: f_o = 2 - 1(2) = 0;\ f_o = 3 - 1(2) = 1$$
$$\text{For } k = -1: f_o = -2 + 1(2) = 0;\ f_o = -3 + 1(2) = -1$$



Exercise: An analogue signal $x(t)$ with a frequency spectrum shown below is sampled at a rate of 4 *kHz*. Determine the resulting digital spectrum.

$$X(\theta) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} X(f + kf_S)$$



Example: Consider the analogue signal

$x(t) = 3\cos 50\pi t + 10\sin 300\pi t - \cos 100\pi t$

What is the Nyquist rate for this signal?

The frequencies present in the signal above are

$f_1 = 25\text{Hz}; f_2 = 150\text{ Hz}; f_3 = 50\text{ Hz}$

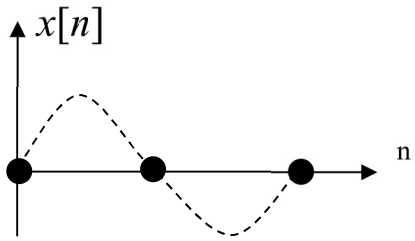
Hence, $f_{max} = 150\text{ Hz}$

$f_{sampling} > 2 f_{max} = 300\text{ Hz}$

The Nyquist rate is $f_N = 2 f_{max} = 300\text{ Hz}$.

Example : Consider $x(t) = 10\sin(300\pi t)$

$$f_s \geq 2 \times f = 300 \text{ Hz}$$
$$x[n] = 10\sin(300\pi nT)$$
$$= 10\sin\left(\frac{300\pi}{f_s} n\right)$$
$$= 10\sin(\pi n)$$



We are sampling the analogue sinusoid at its zero-crossing points and hence we miss the signal completely. The situation will not occur if the sinusoid is offset by some phase (here). In such case we have

$x(t) = 10\sin(300\pi t + \phi)$, where $f_s = 300$Hz.

$x[n] = 10\sin(\pi n + \phi)$

$\therefore \quad = 10[\sin(\pi n)\cos(\phi) + \cos(\pi n)\sin(\phi)]$ for n = 0,1,2,..

$= 10\cos(\pi n)\sin(\phi)$

Since $\cos(\pi n) = (-1)^n$, $\quad x[n] = (-1)^n 10\sin(\phi)$

If $\theta = 0$ or $\theta = \pi$, the samples of the sinusoid taken at the Nyquist rate are not all zero.

Example: Consider the analogue signal

$$x(t) = 3\cos(2000\pi t) + 5\sin(6000\pi t) + 10\cos(12000\pi t)$$

(a) What is the Nyquist rate for this signal?

The frequencies existing in the analogue signal are:

$f_1 = 1$ kHz; $f_2 = 3$ kHz; $f_3 = 6$ kHz

Thus $f_{max} = 6$ kHz and according to the sampling theorem,

$f_s > 2 f_{max} = 12$ kHz

The Nyquist rate is = 12 kHz.

(b) Assume now that we sample this signal $x(t)$ using a sampling rate f_s = 5 kHz (samples/sec). What is the discrete-time signal obtained after sampling?

First Method:

$f_s = 5000\text{Hz} \Rightarrow \frac{f_s}{2} = 2500$

$x(t) = 3\cos(2\pi \times 1000t) + 5\sin(2\pi \times 3000t) + 10\cos(2\pi \times 6000t)$

$$x[n] = 3\cos\left(2\pi\frac{1000}{5000}n\right) + 5\sin\left(2\pi\frac{3000}{5000}n\right) + 10\cos\left(2\pi\frac{6000}{5000}n\right)$$

$$= 3\cos\left(2\pi\left(\frac{1}{5}\right)n\right) + 5\sin\left(2\pi\frac{3}{5}n\right) + 10\cos\left(2\pi\left(\frac{6}{5}\right)n\right)$$

$$= 3\cos\left(2\pi\left(\frac{1}{5}\right)n\right) + 5\sin\left(2\pi\left(1-\frac{2}{5}\right)n\right) + 10\cos\left(2\pi\left(1+\frac{1}{5}\right)n\right)$$

$$= 3\cos\left(2\pi\left(\frac{1}{5}\right)n\right) + 5\sin\left(-2\pi\frac{2}{5}n\right) + 10\cos\left(2\pi\left(\frac{1}{5}\right)n\right)$$

$$x[n] = 13\cos\left(2\pi\left(\frac{1}{5}\right)n\right) - 5\sin\left(2\pi\left(\frac{2}{5}\right)n\right)$$

Second Method:

$$f_s = 5kHz \Rightarrow \frac{f_s}{2} = 2.5kHz$$

We have $f_k = f_0 + kf_s$

$f_0 = f_k - kf_s$ can be obtained by subtracting from f_k an integer multiple of f_s such that $-\frac{f_s}{2} \le f_0 \le \frac{f_s}{2}$.

The frequency f_1 = 1000 Hz is $< \frac{f_s}{2}$ (= 2500 Hz) and thus it is not affected by aliasing.

However, the other two frequencies f_2 & f_3 are above the folding frequency and they will be changed by the aliasing effect.

$f_2 = f_2 - 1\, f_s = 3000 - 5000 = -2$ kHz

$f_3 = f_3 - 1\, f_s = 6000 - 5000 = 1$ kHz

$$x[n] = 3\cos\left(2\pi\left(\frac{1000}{5000}\right)n\right) + 5\sin\left(2\pi\left(-\frac{2000}{5000}\right)n\right) + 10\cos\left(2\pi\left(\frac{1000}{5000}\right)n\right)$$

This is agreement with the result obtained before.

(c) What is the analogue signal $y(t)$ we can reconstruct from the samples if we use ideal interpolation?

Since only frequency components at 1 kHz and 2 kHz are present in the sampled signal, the analogue signal we can recover is,

$$y(t)=13\cos(2000\pi t)-5\sin(4000\pi t)$$

which is obviously different from the original signal $x(t)$.
The distortion of the original analogue signal was caused by the aliasing effect, due to the low sampling rate used.

Exercise: A digital communication link caries binary-coded words representing samples of an input signal

$$x(t) = 5\cos(600\pi t) + 4\cos(1800\pi t) .$$

The link is operated at 10,000 bits/s and each input sample is quantised into 1024 different voltage levels.

(i) What are the frequencies in the resulting discrete-time signal x(n)?

(ii) Determine the resulting discrete time signal x(n).

2.8 Digital-to-Analogue Conversion (D/A) – Signal recovery

The D/A conversion process is employed to convert the digital signal into an analogue form after it has been digitally processed. The reason for such conversion may be for example, to generate an audio signal to drive a loudspeaker or to sound an alarm. The D/A process is shown in Figure 2.8. A register is used to buffer the D/A's input to ensure that its output remains the same until the D/A is fed the next digital input.

Note: The inputs to the D/A are series of impulses, while the output of the DAC has a staircase shape as each impulse is held for a time T sec.

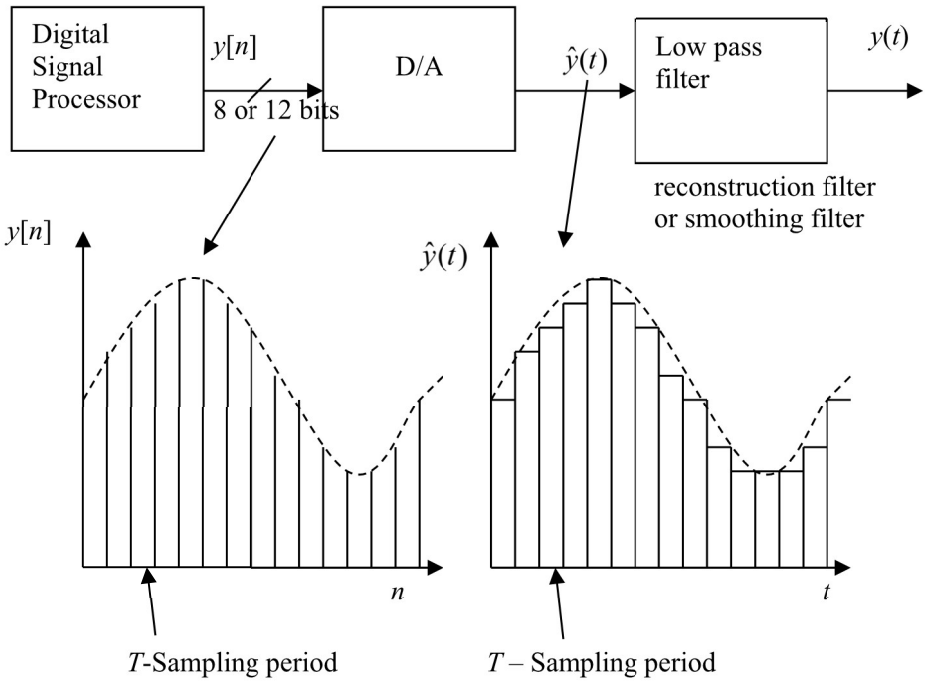


Figure 2.8: Conversion process from digital signals to analogue signals.

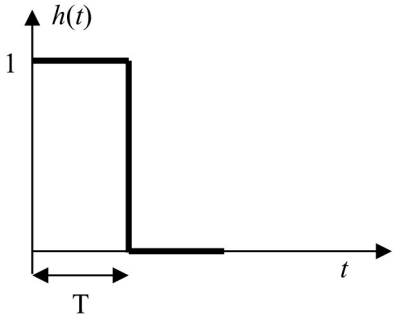
The D/A shown in Figure 2.8 is referred to as a zero-order hold.

By comparing its output $\hat{y}(t)$ and its input $y[n]$, it is evident that for each digital code fed into the D/A, its output is held for a time T. The result is the characteristic staircase shape at the D/A output.

The D/A output approximates the analogue signal by a series of rectangular pulses whose height is equal to the corresponding value of the signal pulse.

Just consider one pulse.

$$h(t) = \begin{cases} 1 & 0 \le t \le T \\ 0 & otherwise \end{cases}$$



The corresponding frequency response is

$$H(\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t}dt = \int_{0}^{T} e^{-j\omega t}dt = \left[\frac{e^{-j\omega t}}{-j\omega}\right]_0^T = \frac{T}{2} \quad e^{-\frac{j\omega T}{2}} \quad \frac{\sin\frac{\omega T}{2}}{\frac{\omega T}{2}}$$

The magnitude of $H(\omega)$ is plotted in Figure 2.9.

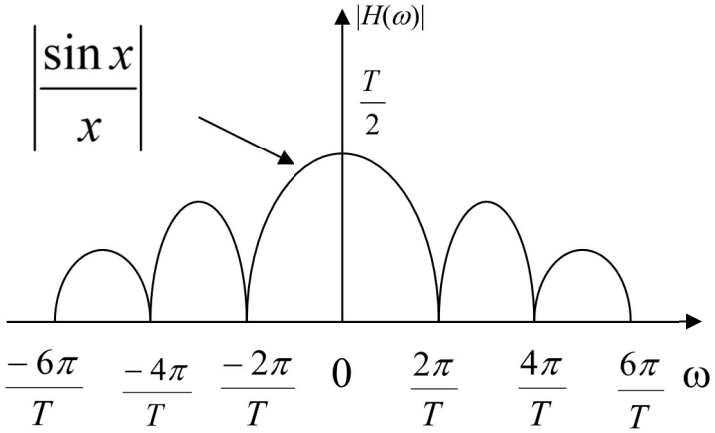
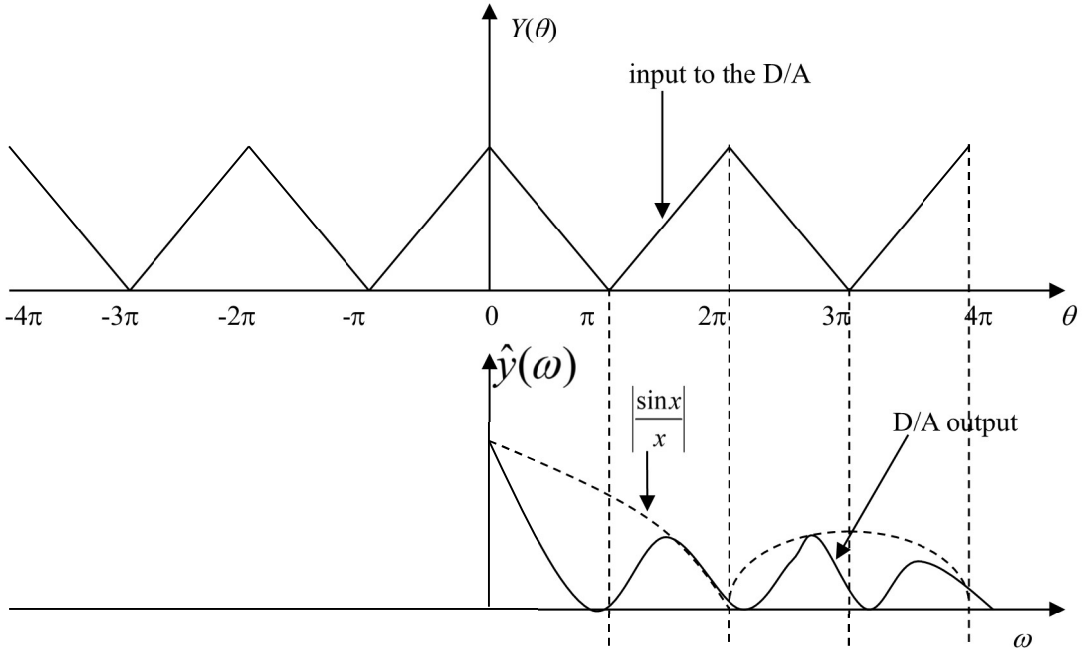


Figure 2.9: Magnitude response of a rectangular pulse.

In the frequency domain, the staircase action of the DAC introduces a type of distortion known as the $\frac{\sin x}{x}$ or aperture distortion, where $x = \frac{\omega T}{2}$.



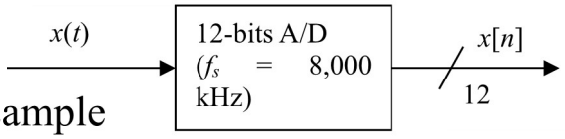
The amplitude of the output signal spectrum is multiplied by the $\frac{\sin x}{x}$ function, which acts like a lowpass filter, with the high frequencies heavily attenuated. The $\frac{\sin x}{x}$ effect is due to the holding action of the DAC and, in signal recovery, introduces an amplitude distortion. For a zero-order hold, the function $\frac{\sin x}{x}$ falls to about 4 dB at half the sampling frequency $\left(\frac{f_S}{2}\right)$ giving an average error of about 36.4%. Aperture error can be eliminated by equalization. In practice this can be achieved by first applying the signal, before converting it to analogue, through a digital filter whose amplitude-frequency response has a $\frac{x}{\sin x}$ shape.

2.8.1 Reconstruction Filter

The output of the D/A converter contains unwanted high frequency at multiples of the sampling frequency as well as the desired frequency components. The role of the output filter is to smooth out the steps in the D/A output thereby removing the unwanted high frequency components. In general, the requirements of the anti-imaging filter are similar to those of the anti-aliasing filter.

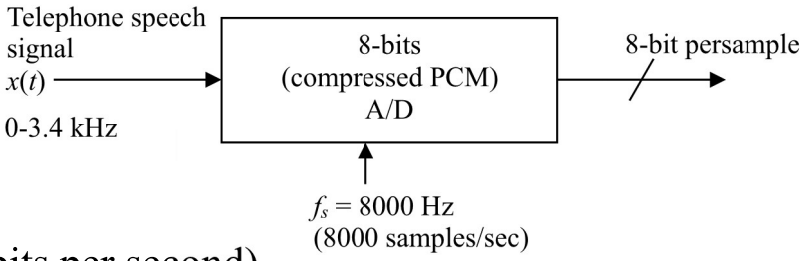
Note:

(a) bit rate = f_s × no of bits



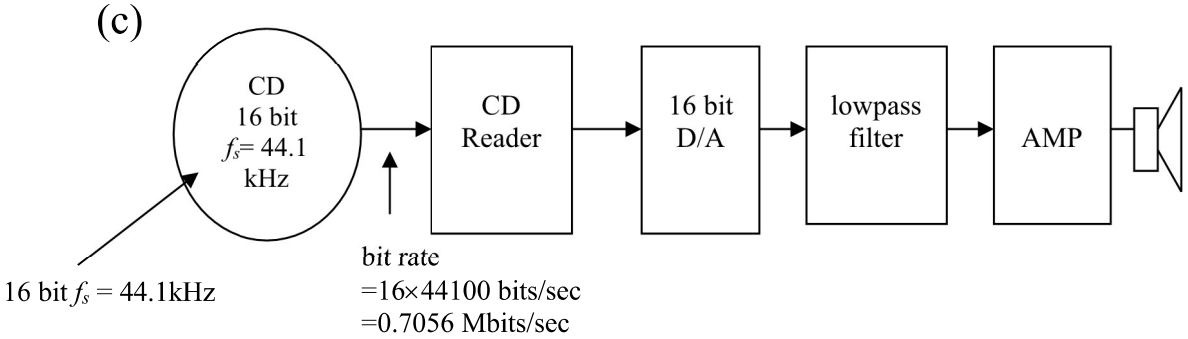
= 8000 samples/sec × 12 bits/sample
= 96000 bits/sec

(b) In the case of Pulse Code Modulation (PCM), speech signals are filtered to remove effectively all frequency components above 3.4 kHz and the sampling rate is 8000 samples per sec



Bit rate (bits per second)
= sampling frequency × bits/sample
= 8000 samples/second × 8 bits/sample
= 64,000 bits/sec

(c)



CHAPTER 2: PROBLEM SHEET 2

Q1) For a linear 16 bit A/D converter with an input signal range of ±4V, what is the minimum quantisation error? Ans: $1/2^{14}$

Q2) A sampled signal that varies between -2V and 2V is quantised using B bits. What value of B will ensure that the quantisation noise power is less than 25×10^{-6}? Ans: 8 bits

Q3) A sinusoidal signal with peak-to-peak amplitude of 5V is sampled at 50kHz with uniform quantisation. Find the minimum number of bits for the analogue to digital converter to achieve a SQNR of at least 92dB. State any assumptions made. Ans: $B = 15$

Q4) Show that the signal to quantisation noise ratio (SQNR) of a linear B-bit analogue-to-digital converter is given by

$$SQNR = 6.02B + 4.77 - 20\log\left(\frac{A}{\sigma_{sig}}\right)$$

where the input range of the A/D converter is ±A and the rms value of the input signal is σ_{sig}. Determine the SQNR if B is 16 bits and the input is a signal with an rms value of $\frac{A}{5}$. Ans: $SQNR = 6.02 \times 16 + 4.77 - 20log\frac{A}{A} \times 5$

Q5) An analogue signal $x(t) = \sin(480\pi t)+3\sin(720\pi t)$ is sampled 600 times per second.

(a) Determine the Nyguist sampling rate for $x(t)$.

(b) Determine the folding frequency (or half the sampling frequency).

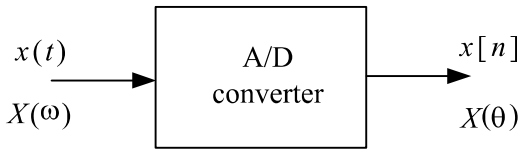
(c) What are the frequencies, in radians, in the resulting discrete time signal $x[n]$?

(d) If $x[n]$ is passed through an ideal D/A converter what is the reconstructed signal $y(t)$?

Q6) An analogue signal $x_a(t) = \cos(2\pi 500t)$ is sampled at a rate of f_s=4kHz. Determine the resulting analogue and digital magnitude spectra.

Q7) Find the Fourier Transform of $x(t) = \delta(t-a)$. *Hence,* show that $\int_{-\infty}^{\infty} e^{j\omega t}dt = 2\pi\delta(\omega)$.

Q8) If an analogue signal $x(t)$ is sampled every T seconds, then at the output of the Analogue to Digital Converter, as shown in a diagram below:



Using Inverse Fourier Transforms in Continuous time and discrete time domains, show that

$$X(\theta) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(\omega + k\frac{2\pi}{T}).$$

End of Chapter 2

Chapter 3

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Chapter 3: Discrete-Time Systems

3.1 Introduction

A discrete-time system is a device or algorithm that operates on a discrete-time signal called the input or excitation according to some well defined rule, to produce another discrete-time signal called the output or response.

We say that the input signal $x[n]$ is transformed by the system into a signal $y[n]$, and express the general relationship between $x[n]$ and $y[n]$ as

$$y[n] = H\{x[n]\} \qquad (3.1)$$

where the symbol H denotes the transformation or processing performed by the system on $x[n]$ to produce $y[n]$ (see Figure 3.1).

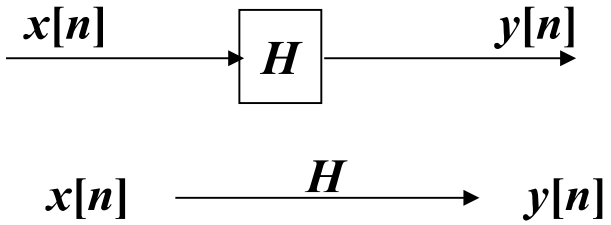


Figure 3.1: Block diagram representation of a discrete-time system

3.2 Block Diagram Representation

In order to introduce a block diagram representation of discrete-time systems, we need to define some basic blocks that can be interconnected to form complex systems.

3.2.1 An adder

A system that performs the addition of two signal sequences to form another sequence, which we denote as $y[n]$.

Note: It is not necessary to store either one of the sequences in order to perform the addition. In other words, the addition operation is **memoryless**.

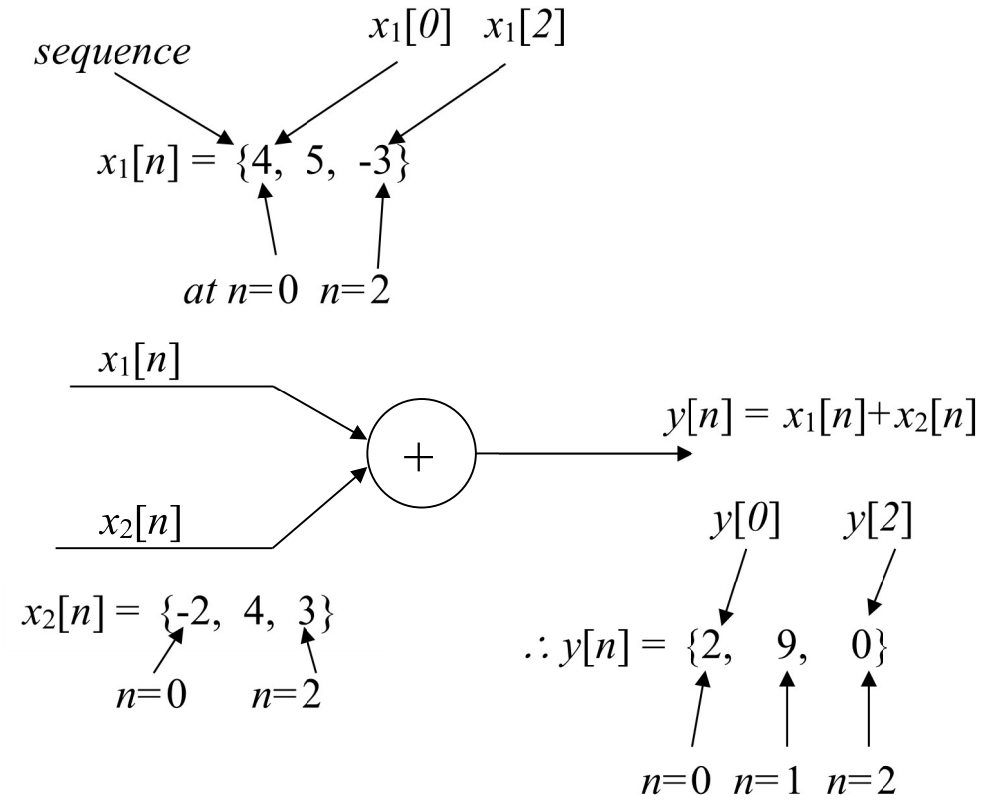


Figure 3.2: Block diagram representation of an adder, $x_1[n]$ and $x_2[n]$ denote discrete-time input signals and y[n] denote a discrete-time output signals.

3.2.2 A constant multiplier

This operation simply represents applying a scale factor on the input *x[n]*. Note that this operation is also **memoryless**.

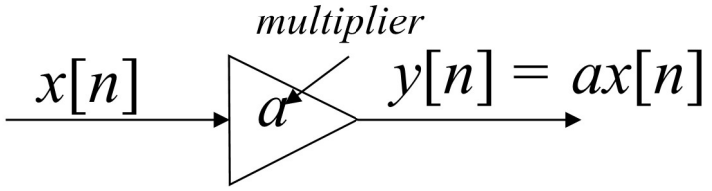
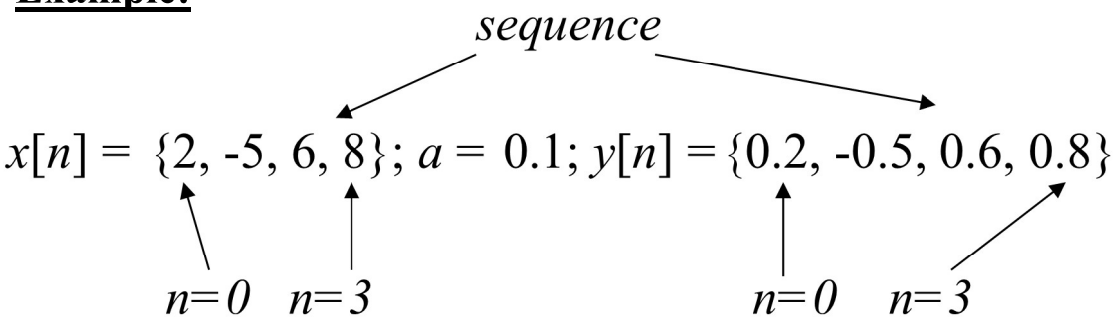


Figure 3.3: Block diagram representation of a multiplier. x[n] and y[n] denote discrete-time input and output signals respectively. 'a' denotes a scalar multiplier.

Example:



3.2.3 A Unit Delay Element

The unit delay is a special system that simply delays the signal passing through it by one sample.

If the input signal is $x[n]$, the output is $x[n\text{-}1]$. In fact, the sample $x[n\text{-}1]$ is stored in memory at time $n\text{-}1$ and it is recalled from memory at time n to form $y[n] = x[n\text{-}1]$.

Thus this basic building block requires memory. We use the symbol T or z^{-1} to denote the unit of delay.

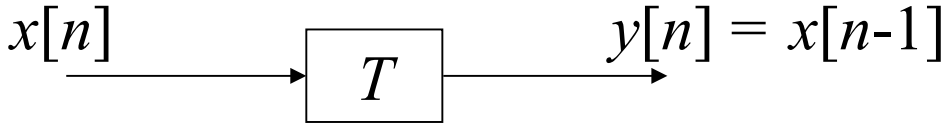
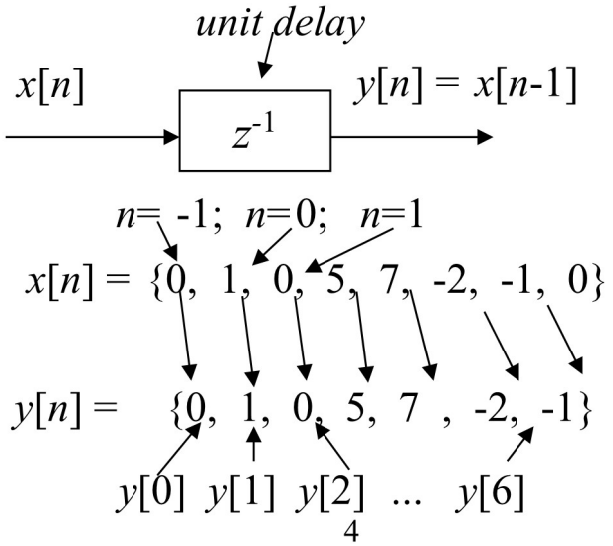


Figure 3.4: Block diagram representation of a unit delay.T denotes the sampling period.

Example:



$$\{ y[n] = x[n\text{-}1];\ y[0]=x[0\text{-}1]=x[\text{-}1];\ y[1] = x[1\text{-}1]=x[0];\ldots\}$$

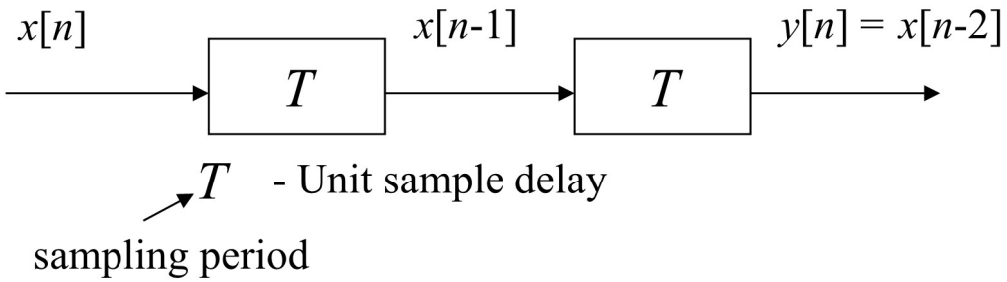
Note: Normally a combination of adders, multipliers and unit delays form a complex discrete-time system.

3.3 Difference Equations

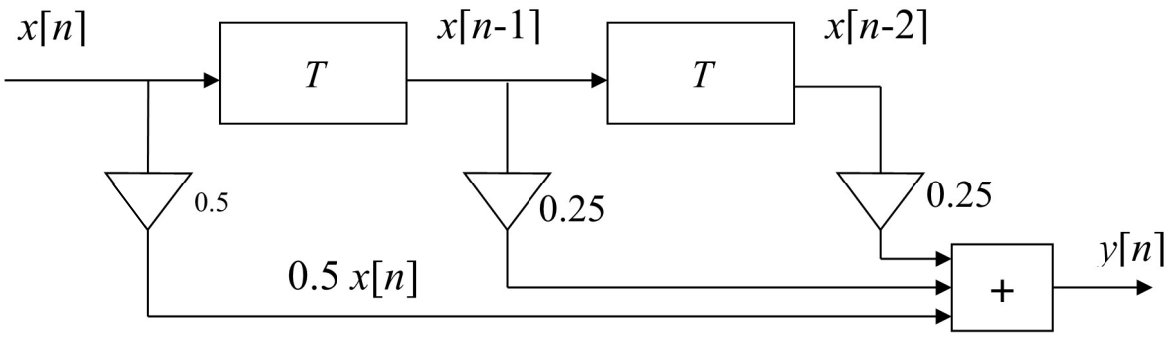
A discrete-time system consisting of combinations of adders, multipliers and unit delays can always be described by a set of difference equations. The equations would be ordinary algebraic equations if no delays were present.

Examples:

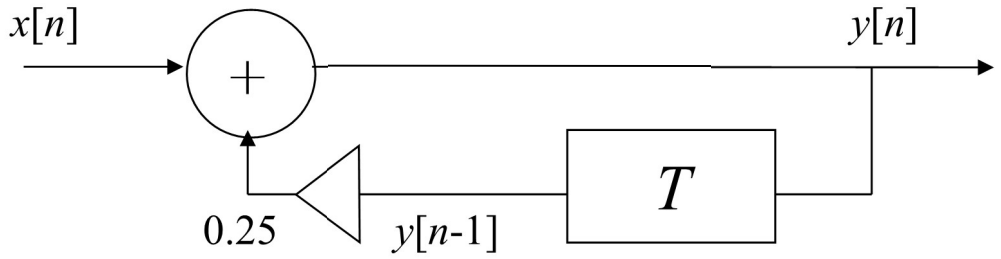
(a) $y[n] = x[n\text{-}2]$



(b) $y[n]=\frac{1}{2}x[n]+\frac{1}{4}x[n-1]+\frac{1}{4}x[n-2]$



(c) $y[n] = x[n] + 0.25y[n-1]$



Exercise:

Draw a system implementation for the difference equation as given below:

$$y[n] = x[n] + x[n-1] + x[n-2] + x[n-3]$$

Example $y[n] = a_0 x[n] + a_1 x[n-1] - b_1 y[n-1]$

Draw a system implementation for the above difference equation.

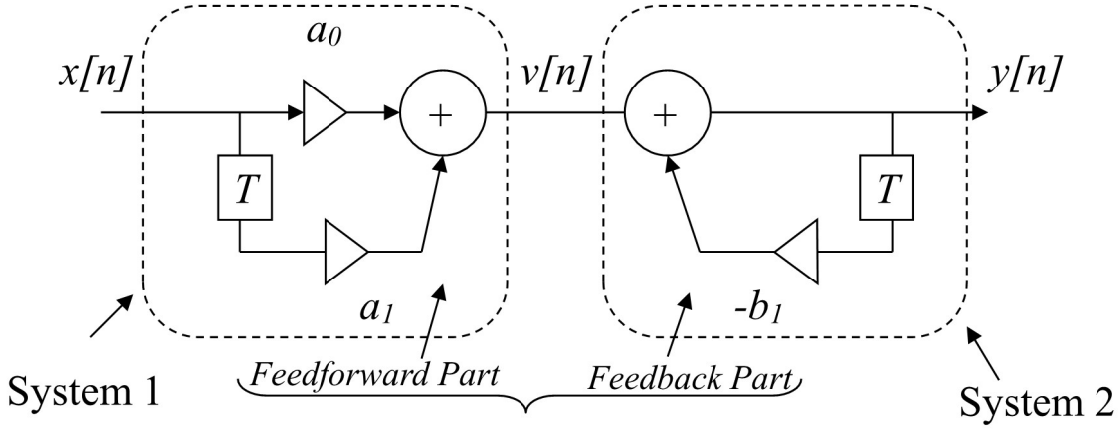


Figure 3.5: Direct Form I structure

We can write the above difference equation as a set of two equations

$$v[n] = a_0 x[n] + a_1 x[n-1] \quad \text{- system 1}$$
$$y[n] = v[n] - b_1 y[n-1] \quad \text{- system 2}$$

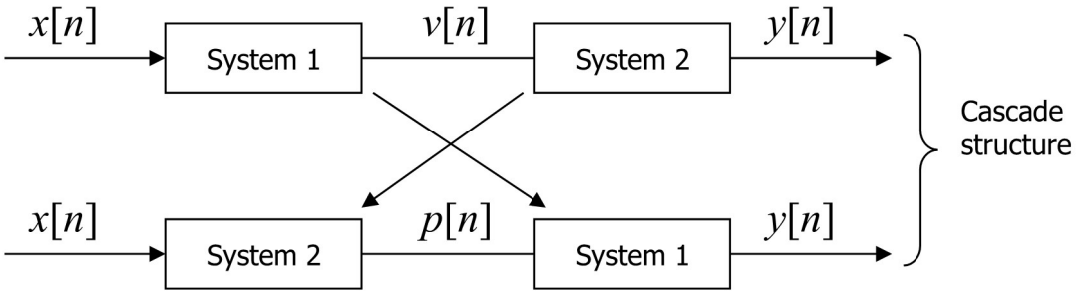


Figure 3.6: Two systems forming a cascade structure can be interchanged without affecting the final output signal.

Without changing the input-output relationship, we can reverse the ordering of the two systems in the cascade representation.

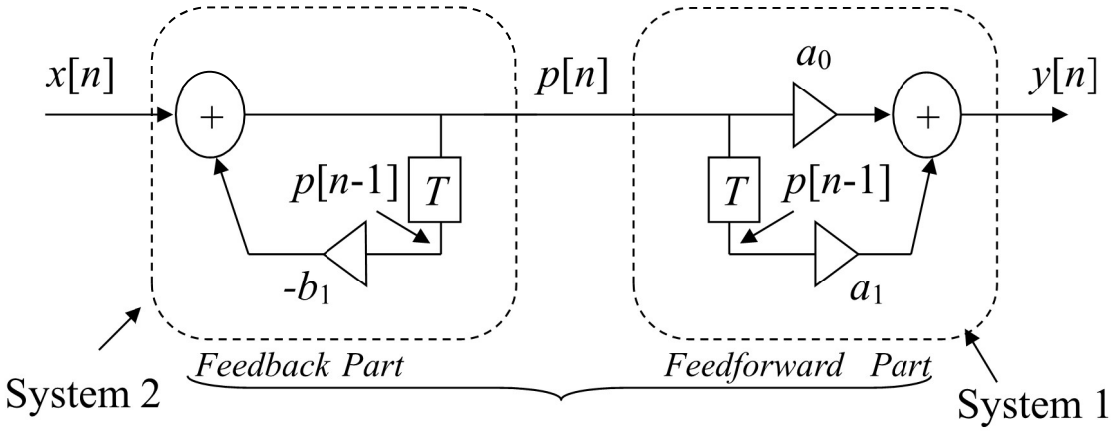


Figure 3.8: Direct Form II structure

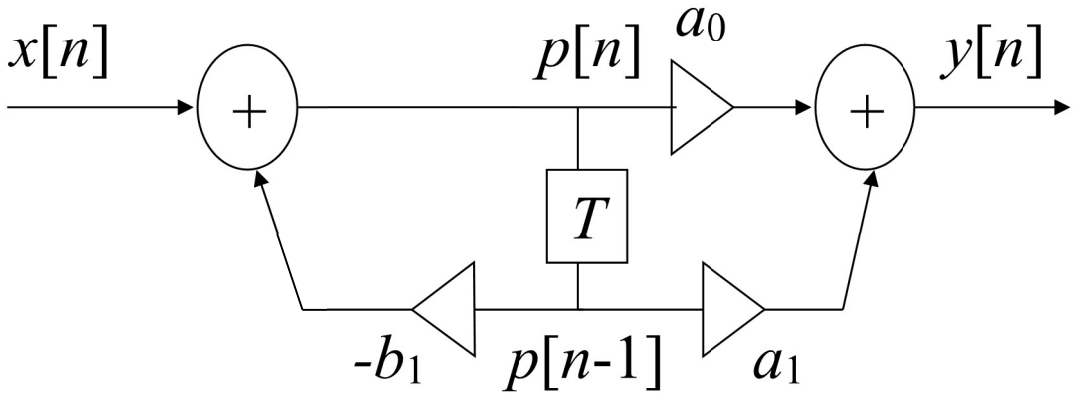


Figure 3.8: Canonic form.

There is no need for two delay operations; they can be combined into a single delay as shown in Figure 3.8. Since delay operations are implemented with memory in a computer, the implementation of Figure 3.8 would require less memory compared to the implementation of Figure 3.8.

It can be proven that both block diagrams Figure 3.5 and Figure 3.8/Figure 3.8 represent the same difference equation.

Proof:

$$y[n] = a_0 x[n] + a_1 x[n-1] - b_1 y[n-1] \qquad (3.2)$$

From Figure 3.8:

$$p[n] = x[n] - b_1 p[n-1] \qquad (3.2.a)$$

$$y[n] = a_0 p[n] + a_1 p[n-1] \qquad (3.2.b)$$

Substituting $n \rightarrow n\text{-}1$ in equation (3.2.b),

$$y[n-1] = a_0 p[n-1] + a_1 p[n-2] \qquad (3.2.c)$$

Multiplying equation (3.2.c) by b_1,

$$b_1 y[n-1] = a_0 b_1 p[n-1] + a_1 b_1 p[n-2] \qquad (3.2.d)$$

Adding equation (3.2.b) and (3.2.d),

$$y[n] + b_1 y[n-1] = \underbrace{a_0 p[n] + a_0 b_1 p[n-1]}_{a_0 x[n]} + \underbrace{a_1 p[n-1] + a_1 b_1 p[n-2]}_{a_1 x[n-1]}$$

Therefore,

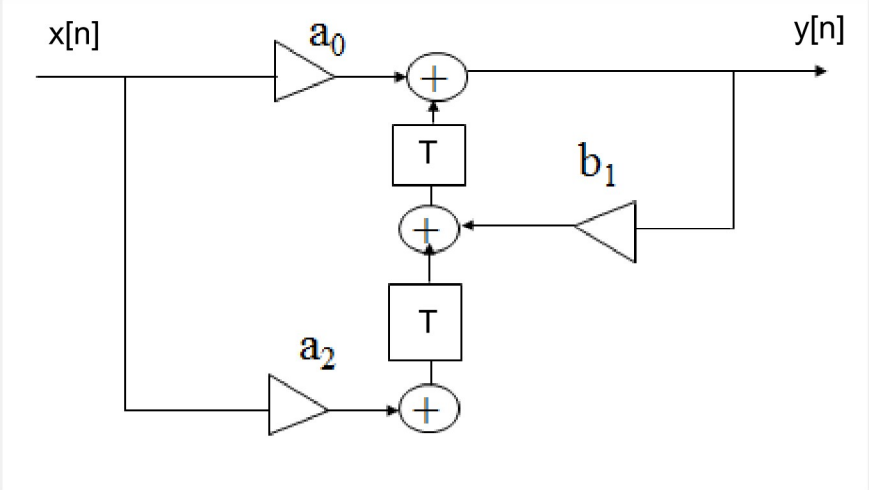
$$y[n] = a_0 x[n] + a_1 x[n-1] - b_1 y[n-1]$$

as in equation (3.2)

Exercise:

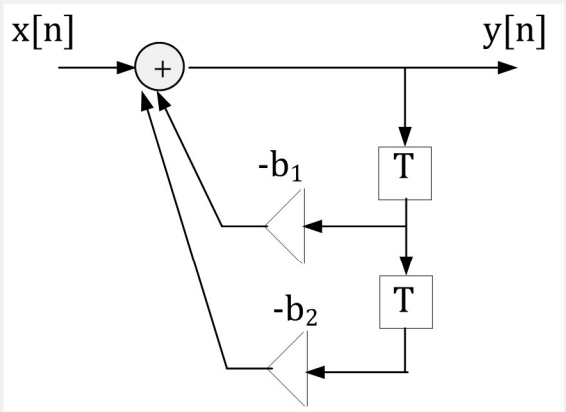
For the given systems shown below, write the difference equation.

a)



Ans: $y[n]=a_0x[n]+ a_2x[n-2]+ b_1y[n-1]$

b)



Ans: $y[n]=x[n]- b_1y[n-1]- b_2y[n-2]$

c) Draw a canonic structure for the difference equation given below:

$$y[n] = x[n] - a_2x[n-2] + b_1y[n-1] - b_2y[n-2]$$

3.4 Classification of Discrete-Time Systems

In the analysis as well as in the design of systems, it is desirable to classify the systems according to the general properties that they satisfy. For a system to possess a given property, the property must hold for every possible input signal to the system. If a property holds for some input signals but for others, the system does not possess the property.

General Categories are:

- Static systems
- Time - invariant systems
- Linear systems
- Causal systems
- Stable systems

3.4.1 Static systems

A discrete-time system is called static or memoryless if its output at any instant '*n*' depends at most on the input sample at the same time, but not on past or future samples of the input.

Example:

$$y[n] = ax[n]$$

$$y[n] = nx[n] + bx^3[n]$$

Both are static or memoryless.

On the other hand, the systems described by the following input-output relations, such as

$$y[n] = x[n] + 3x[n-1]$$

$$y[n] = \sum_{k=0}^{N} x[n-k]$$

are dynamic systems or system with memory.

3.4.2 Time-invariant systems

A time-invariant system is defined as follows:

$$x[n-n_0] \xrightarrow{H} y[n-n_0]$$

where $y[n] = H\{x[n]\}$.

Specifically, a system is time invariant if a time shift in the input signal results in an identical time shift in the output signal.

Example: Determine if the system is time variant or time invariant.

$$y[n] = H\{x[n]\} = nx[n] \tag{3.3}$$

The response of this system to $x[n-k]$ is

$$w[n] = nx[n-k]$$

Now if we delay $y[n]$ in (3.3) by k units in time, we obtain

$$\begin{aligned} y[n-k] &= (n-k)x[n-k] \\ &= nx[n-k] - kx[n-k] \end{aligned}$$

This system is time variant, since

$$y[n-k] \neq w[n]$$

Exercise:

Examine the time-invariant properties of the following systems:

a) y[n]= nx(n)
b) y[n]=|x[n] |

Ans: a) Time variant b) Time invariant

3.4.3 Linear Systems

A linear system is defined as follows:

$$a_1 x_1[n] + a_2 x_2[n] \xrightarrow{H} a_1 y_1[n] + a_2 y_2[n] \quad (3.4)$$

where a_1 and a_2 are arbitrary constants.

Example: Three sample averager

$$y[n] = \frac{1}{3}\{x[n+1] + x[n] + x[n-1]\} = H\{x[n]\}$$

$$x[n] \xrightarrow{H} y[n]$$

$$\begin{aligned} H\{ax_1[n] + bx_2[n]\} = \frac{1}{3}\{&ax_1[n+1] + bx_2[n+1] \\ &+ ax_1[n] + bx_2[n] \\ &+ ax_1[n-1] + bx_2[n-1]\} \\ = [&ay_1[n] + by_2[n]] \end{aligned}$$

The 3-sample averager is a linear system.

Example:

$$\begin{aligned} y[n] &= H\{x[n]\} = x^2[n] \\ H\{ax_1[n] + bx_2[n]\} &= \{ax_1[n] + bx_2[n]\}^2 \\ &= a^2 x_1^2[n] + b^2 x_2^2[n] + 2abx_1[n]x_2[n] \end{aligned}$$

which is not equal to $ax_1^2[n] + bx_2^2[n]$. This system is nonlinear.

Example:

$$\begin{aligned} y[n] &= nx[n] = H\{\ x[n]\ \} \\ H\{\ ax_1[n] + bx_2[n]\ \} &= anx_1[n] + bnx_2[n] \\ &= ay_1[n] + by_2[n] \end{aligned}$$

The system is linear.

3.4.4 Causal systems

A system is said to be causal if the output of the system at any time '*n*' depends only on present and past inputs, but does not depend on future inputs. If a system does not satisfy this definition, it is called *noncausal*. Such a system has an output that depends not only on present and past inputs but also on future inputs.

Example:

$$y[n] = x[n] - x[n-1] \rightarrow \text{Causal}$$
$$y[n] = ax[n] \rightarrow \text{Causal}$$
$$y[n] = x[n] + 3x[n+4] \rightarrow \text{Noncausal}$$
$$y[n] = x[-n] \rightarrow \text{Noncausal}$$

{Let n = -1 $\Rightarrow$ y[-1]= x [1], the output at n = -1 depends on the input at n = 1.}

Discrete - time sequence is called **causal** if it has zero values for $n<0$.

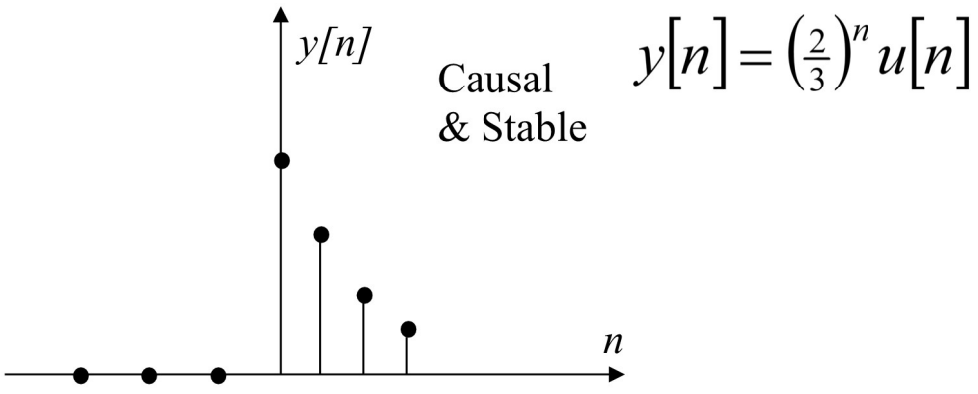


Figure 3.9: An example of causal discrete-time sequence.

3.4.5 Stable Systems

- A discrete signal $x[n]$ is bounded if there exists a finite M such that $|x[n]| < M$ for all n.

- A discrete-time system in Bounded Input-Bounded Output (**BIBO**) stable if every bounded input sequence $x[n]$ produced a bounded output sequence.

$$\text{If } x[n]_{\max} \le A, \text{ then } y[n]_{\max} \le B$$

Example:

The discrete-time system

$$y[n] = ny[n-1] + x[n], \quad n > 0$$

is at rest [i.e. $y[-1]=0$]. Check if the system is BIBO stable.

If $x[n]=u[n]$, then $|x[n]| \le 1$. But for this bounded input, the output is

$$\begin{aligned} n = 0 &\rightarrow y[0] = x[0] = 1 \\ n = 1 &\rightarrow y[1] = 1y[0] + x[1] = 2 \\ n = 2 &\rightarrow y[2] = 2y[1] + x[2] = 5 \\ &\vdots \qquad\qquad \vdots \end{aligned}$$

which is unbounded. Hence the system is unstable.

y[0]=1 → y[1]=2 → y[2]=5 → increasing

Exercise: A discrete-time system can be (1) Static or Dynamic, (2) Linear or nonlinear, (3) Time invariant or time varying, (4) Causal or noncausal, (5) Stable or unstable

Examine the following system with respect to the properties above.

(a) $y[n] = e^{a\,x[n]}$

(b) $y[n] = x[n] + nx[n+1]$

Ans: a) Static, nonlinear, time invariant, casual, stable
b) Dynamic, linear, time varying, noncasual, unstable

3.5 Linear Time-Invariant Discrete (LTD) Systems

3.5.1 Transformation of Discrete-Time signals

A discrete-time signal, $x[n]$ may be shifted in time (delayed or advanced) by replacing the variables n with n-k where $k > 0$ is an integer

$x[n-k]$ => $x[n]$ is delayed by k samples
$x[n+k]$ => $x[n]$ is advanced by k samples

For example consider a shifted version of the unit impulse function (see Figure 3.10). If we multiply an arbitrary signal $x[n]$ by this function, we obtain a signal that is zero everywhere, except at $n = k$.

$$\therefore\ y[n] = x[n] \cdot \delta[n-k] = x[k] \cdot \delta[n-k] \qquad (3.5)$$

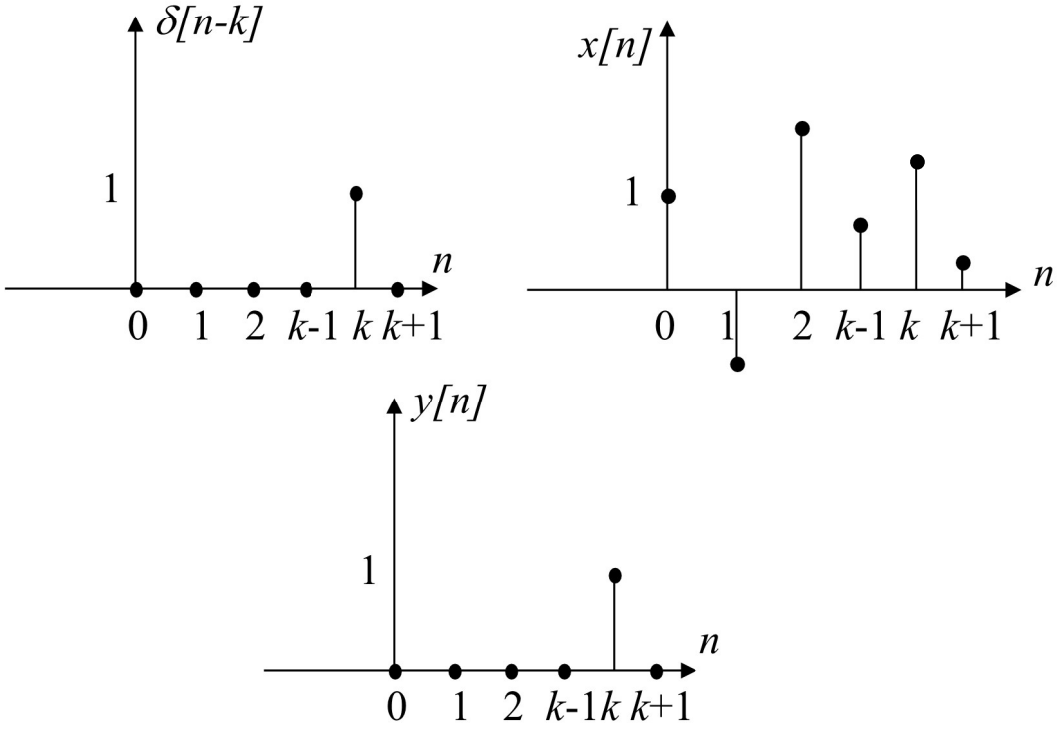
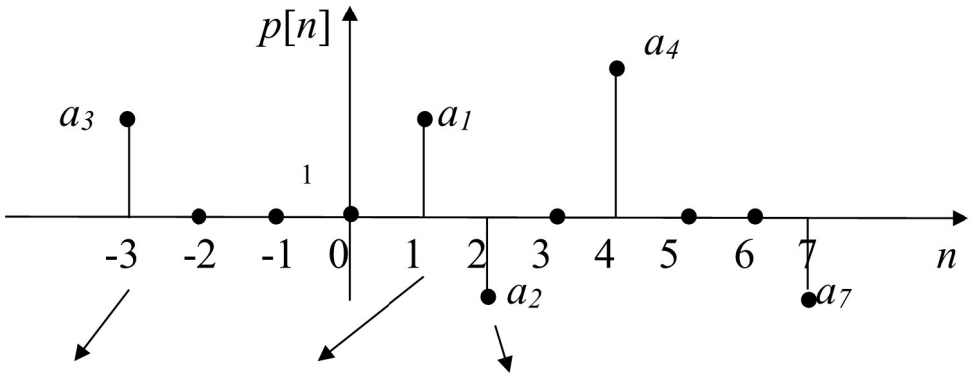


Figure 3.10: Multiplying a discrete-time signal, x[n], with a shifted unit impulse function, δ[n-k], produces a discrete-time signal whose sample is zero except at n=k.

An arbitrary sequence can then be expressed as a sum of scaled and delayed unit impulses.



$$p[n] = a_{-3}\delta[n+3] + a_1\delta[n-1] + a_2\delta[n-2] + a_4\delta[n-4] + a_7\delta[n-7]$$

Figure 3.11: An example of expressing arbitrary discrete-time sequences as a sum of scaled and delayed unit impulses.

More generally, the discrete-time sequence can be expressed according to

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \qquad (3.6)$$

For real time signals

$$x[n] = \sum_{k=0}^{\infty} x[k]\delta[n-k]$$

and for a real-time signal with a finite number of samples N.

$$x[n] = \sum_{k=0}^{N-1} x[k]\delta[n-k] \qquad (3.7)$$

If $x[n]$ has finite duration, the infinite sum in equation (3.7) may be replaced by a finite sum. That is if $x[n] \neq 0$ for $-N_2 \leq n \leq N_1$.

$$x[n] = \sum_{k=-N_2}^{N_1} x[k]\delta[n-k]$$

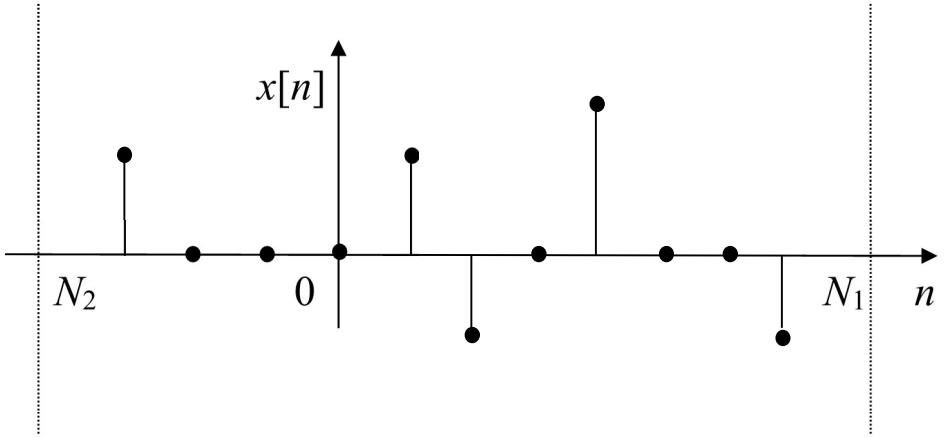


Figure 3.12: An example of finite duration discrete-time sequences.

Equation (3.7) is a special form of convolution. Generally, the convolution of two sequences $x[n]$ and $y[n]$ is defined as

$$x[n] * y[n] = \sum_{k=-\infty}^{\infty} x[k]y[n-k] = \sum_{k=-\infty}^{\infty} x[n-k]y[k] \quad \text{(convolution)} \tag{3.8}$$

Note that, convolution is commutative :

$$\text{i.e. } x[n] * y[n] = y[n] * x[n]$$

3.5.2 The impulse Response of a LTI system

For example consider the discrete-time system, *H*, shown in Figure 3.13.

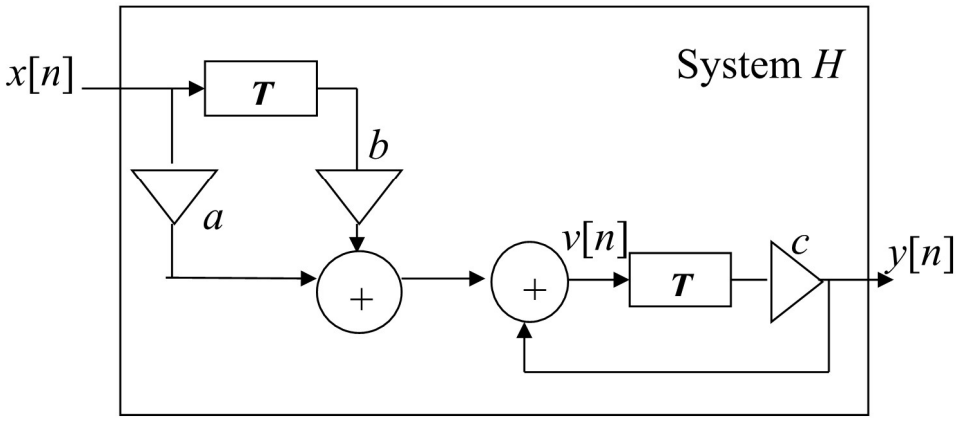


Figure 3.13: An example of discrete-time system, whose input and output are represented by x[n] and y[n], respectively.

Difference equation for the system *H*:

$$v[n] = ax[n] + bx[n-1] + y[n]$$

$$y[n] = cv[n-1] \qquad (3.9)$$

From the above difference equations, *y[n]* can be determined for a given input.

Let $x[n] = \delta[n]$ ***Unit impulse***

Assume $v[n] = 0$ for $n \leq 0 \Rightarrow y[n]$ is also initially zero for $n \leq 0$. Substituting $n = 0,1,2,...$ in equation (3.9), we obtain

$n = 0$ $\quad v[0]=ax[0]+bx[-1]+y[0]=a\cdot 1+b\cdot 0+0=a$
$\Rightarrow y[1] = a\cdot c$

$n = 1$ $\quad v[1] = ax[1] + bx[0] + y[1] = a\cdot 0 + b\cdot 1 + a\cdot c = b+ac$
$\Rightarrow y[2] = cv[1] = c(b+ac)$

$n = 2$ $\quad v[2] = ax[2] + bx[1] + y[2] = 0 + 0 + c(b+ac)$

$\Rightarrow y[3] = cv[2] = c^2(b+ac)$

...

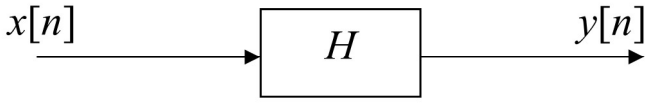
$n = n - 1$ $\quad v[n\text{-}1] = bc^{n-2} + ac^{n-1}$

$\therefore\ y[n] = c\,v[n\text{-}1] = bc^{n-1} + ac^n$

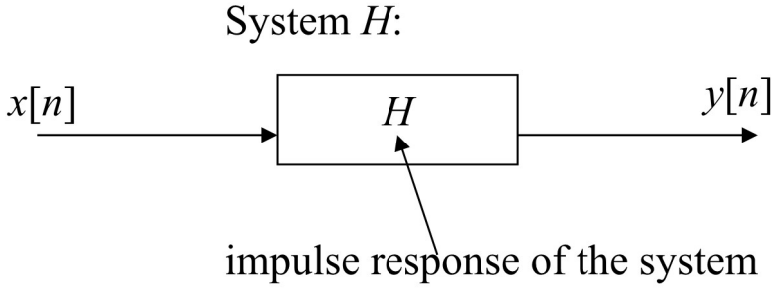
$$\therefore\ y[n]\big|_{x[n]=\delta[n]} = h[n] = bc^{n-1}u(n-2) + ac^n u(n-1)$$

Impulse response

The response $y[n] \overset{\Delta}{=} h[n]$ to an impulse excitation ($x[n] = \delta[n]$) is known as the impulse response and it is a very important characteristic of a discrete system.



If $x[n] = \delta[n]$, then $y[n] = h[n]$. The output tells us the system behaviour as the system is being hit by all input frequencies. $h[n]$ completely characterizes the system.The response to an arbitrary input signal $x[n]$ is the convolution of $x[n]$ with the impulse response of the system.



$$y[n] = x[n] * h[n] \qquad (3.10)$$

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$$y[n] = h[n] * x[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k] \qquad (3.11)$$

where $h[n]=H\{\delta[n]\}$.

3.5.3 Finite Impulse Response (FIR) System

If the impulse response of a LTI system is of finite duration, the system is said to be Finite Impulse Response (FIR).

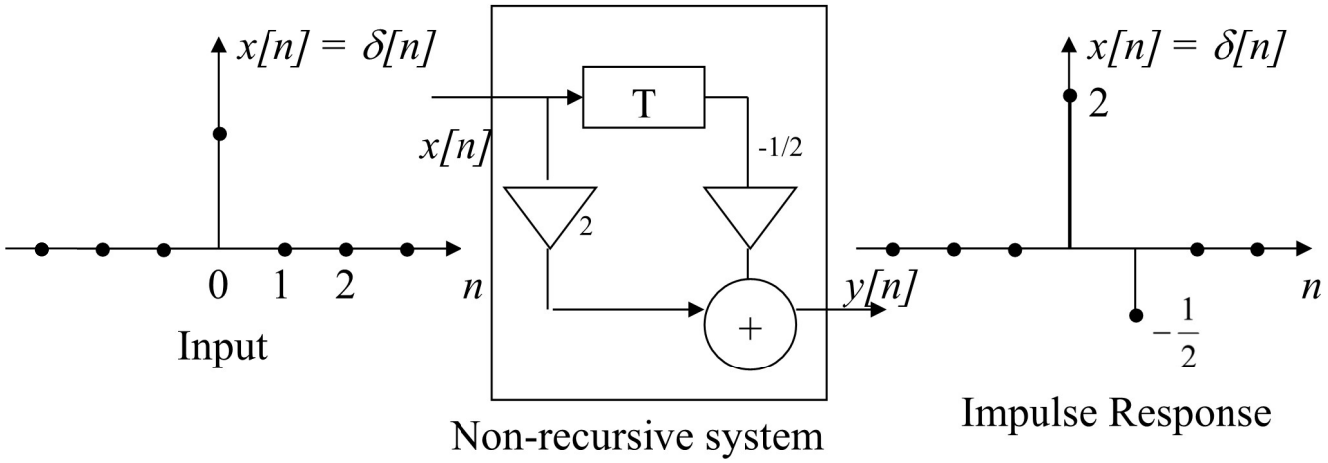


Figure 3.14: An example of LTI systems with finite impulse response.

$x[n] = \delta[n]y[n] = 2x[n] - 0.5x[n\text{-}1]$

$n = 0$ $\quad y[0] = h[0] = 2x[0] - 0.5x[-1] = 2$
$n = 1$ $\quad y[1] = h[1] = 2x[1] - 0.5x[0] = -0.5$
$n = 2$ $\quad y[2] = h[2] = 2x[2] - 0.5x[1] = 0$

$n = n$ $\quad 0$

3.5.4 Infinite Impulse response (IIR) system

If the impulse response of a linear time-invariant system is of infinite duration, the system is said to be an Infinite Impulse Response (IIR) system.

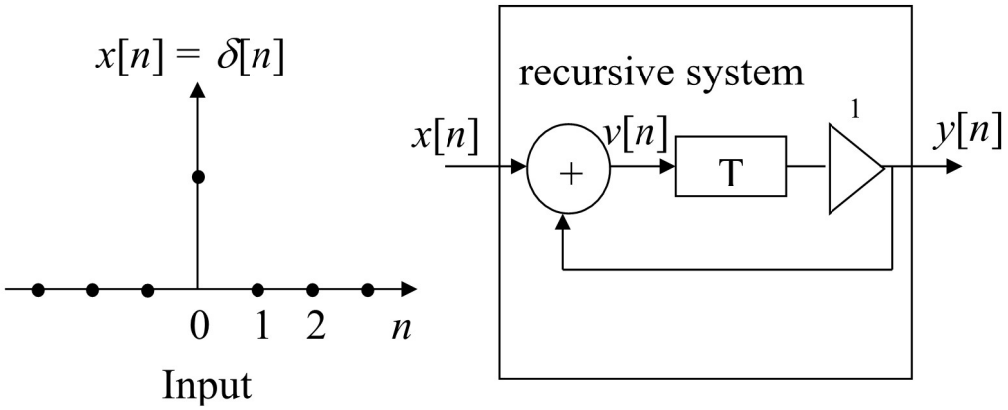


Figure 3.15: An example of discrete systems with infinite impulse response.

$$v[n] = x[n] + y[n]$$
$$y[n] = 1.v[n-1]$$

If $x[n] = \delta[n]$, calculate $h[n]$ for n=0,1,2,...

Example:

Find the impulse response $h[n]$ of the following first-order recursive system.

$$y[n] = \begin{cases} ay[n-1] + x[n] & n \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

To find $h[n]$, we let $x[n] = \delta[n]$ and apply the zero initial condition.

$n = 0, y[0] = h[0] = ay[-1] + \delta[0] = 1$

$n = 1, y[1] = h[1] = ay[0] + \delta[1] = a$

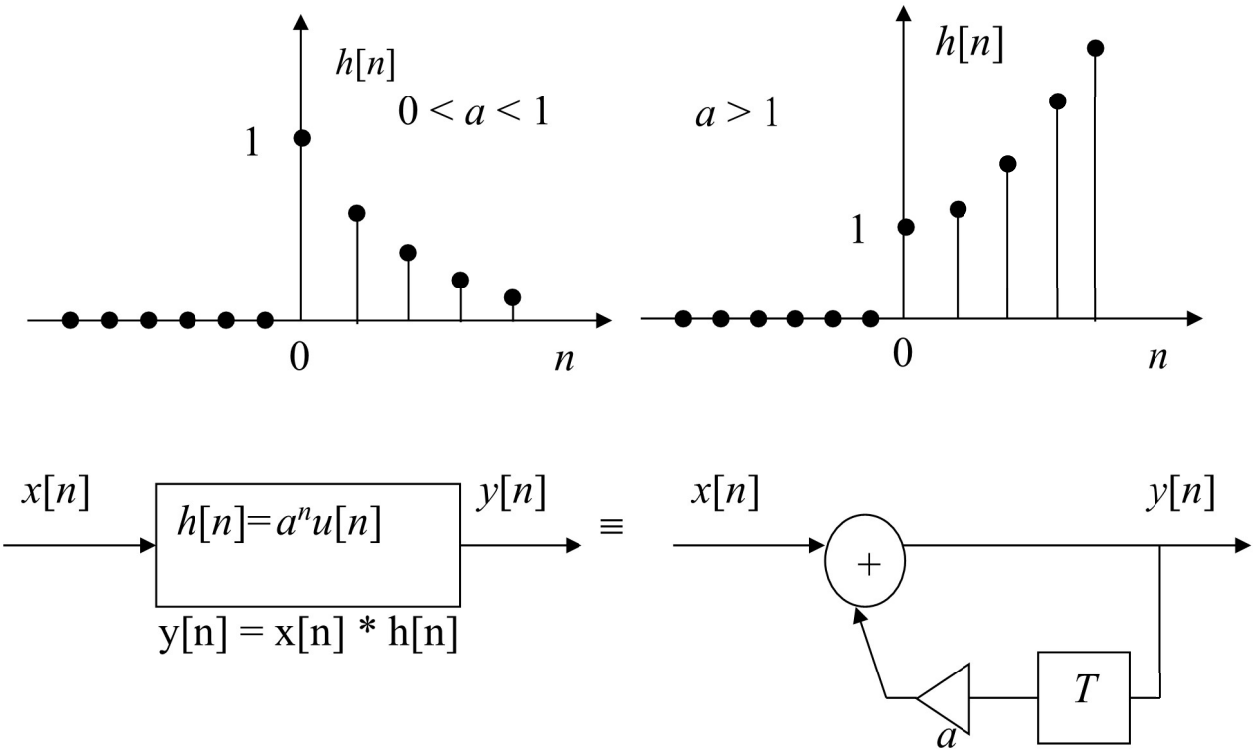
$n = 2, y[2] = h[2] = ay[1] + \delta[2] = a^2$

$\vdots$

$n = n, y[n] = h[n] = a^n \quad \text{for } n \geq 0$

$y[n] = h[n] = 0$ for $n < 0$, because $\delta[n]$ is zero for $n < 0$ and $y[-1] = 0$.

Hence, $h[n] = a^n u[n]$ for all n



Example:

Find the impulse response $h[n]$ of the following fourth order non-recursive system.

$$y[n] = a_0[n] + a_1 x[n-1] + a_2 x[n-2] + a_3 x[n-3] + a_4 x[n-4]$$

To find $h[n]$, we let $x[n] = \delta[n]$.

n=0 ➔ $h[0] = a_0\delta[0] + a_1\delta[-1] + a_2\delta[-2] + a_3\delta[-3] + a_4\delta[-4] = a_0$

n=1 ➔ $h[1] = a_0\delta[1] + a_1\delta[0] + a_2\delta[-1] + a_3\delta[-2] + a_4\delta[-3] = a_1$

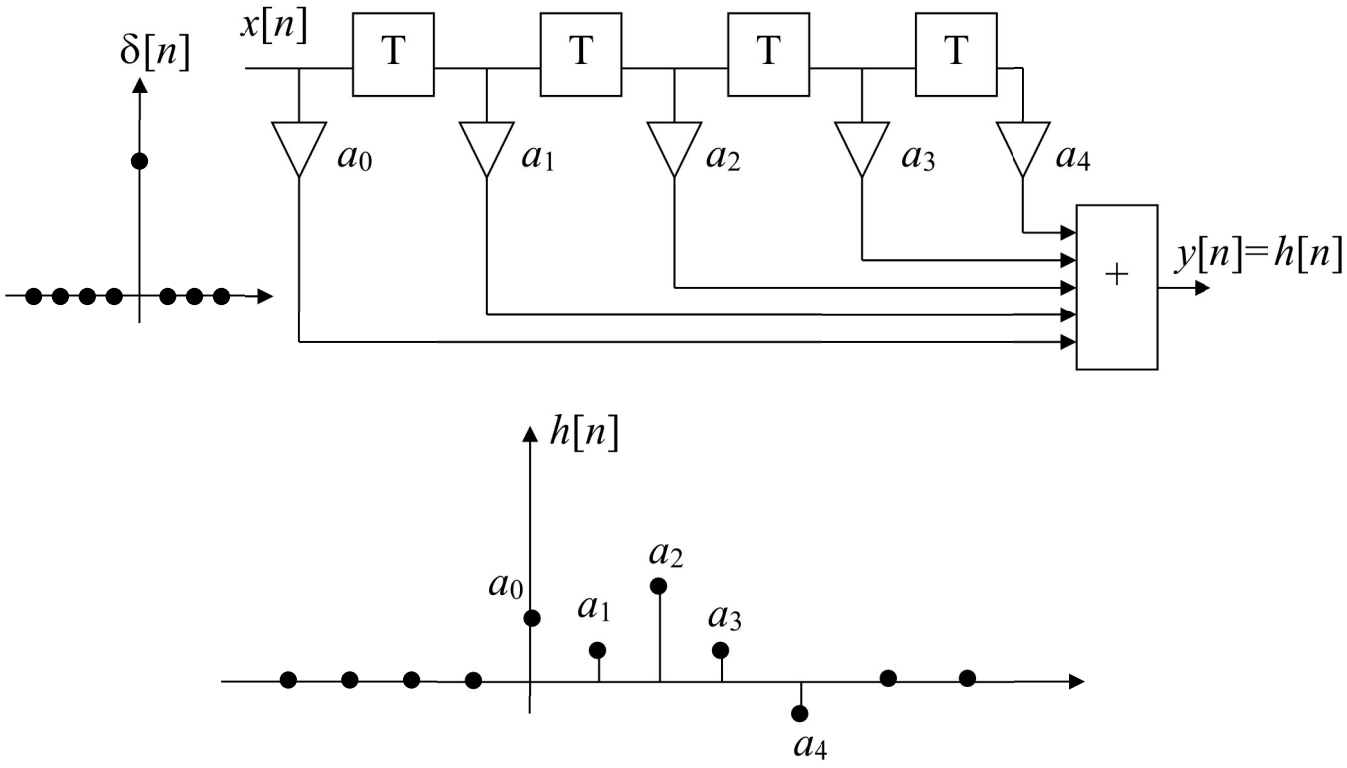
n=2 ➔ $h[2] = a_0\delta[2] + a_1\delta[1] + a_2\delta[0] + a_3\delta[-1] + a_4\delta[-2] = a_2$

n=3 ➔ $h[3] = a_0\delta[3] + a_1\delta[2] + a_2\delta[1] + a_3\delta[0] + a_4\delta[-1] = a_3$

n=4 ➔ $h[4] = 0 + 0 + 0 + 0 + a_4\delta[0] = a_4$

n=5 ➔ $h[5] = 0 + 0 + 0 + 0 + a_4\delta[1] = 0$

For $n \geq 5$, $h[n] = 0$, since the nonzero value of $\delta[n]$ has moved out of the memory of this system.

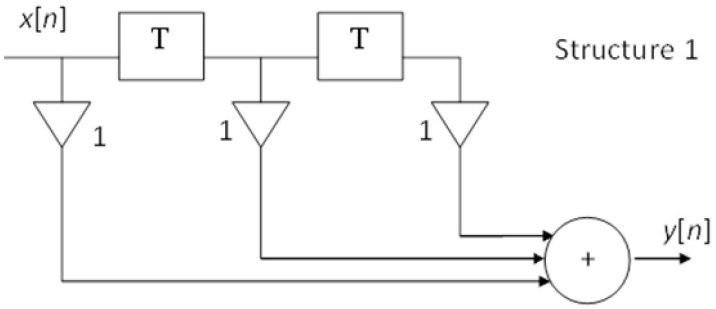


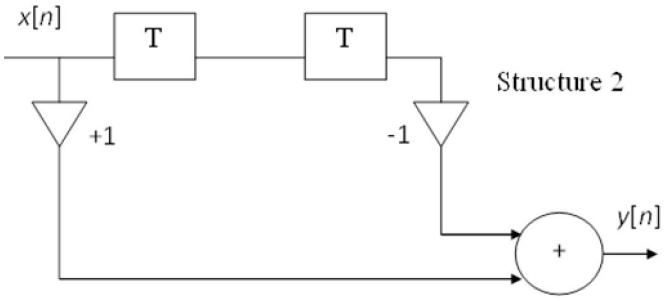
a_0, a_1, a_2, a_3 and a_4 are called coefficients (+ or -) or constants.

Example: Two structures are shown below:

(a) Write the difference equation

(b) Calculate the impulse response





Structure 1: $y[n] = x[n] + x[n\text{-}1] + x[n\text{-}2]$

$h[n]=y[n]|_{x[n]=\delta[n]}$

$h[n] = \delta[n] + \delta[n\text{-}1] + \delta[n\text{-}2]$

$h[0]=1;\ h[1] = 1;\ h[2] = 1$ and $h[n] = 0 \quad n \geq 3$

Structure 2: $y[n] = x[n]\text{-}x[n\text{-}2]$

$h[n] = \delta[n] - \delta[n\text{-}2]$

$h[0]=1;\ h[1] = 0;\ h[2] = -1$ and $h[n] = 0 \quad n \geq 3$

Exercise:

A difference equation for a particular filter is given by

y(n) = 0.12 x(n) – 0.1 x(n-2) + 0.82 x(n-3) – 0.1 x(n-4) + 0.12 x(n-6)

Find the impulse response of the above filter

Ans: h[n]=[0.12 0 -0.1 0.82 -0.1 0 0.12]

3.5.5 Convolution

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

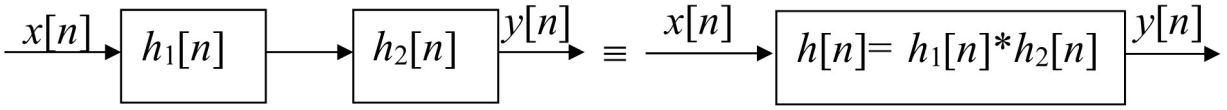
$$y[n] = h[n] * x[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

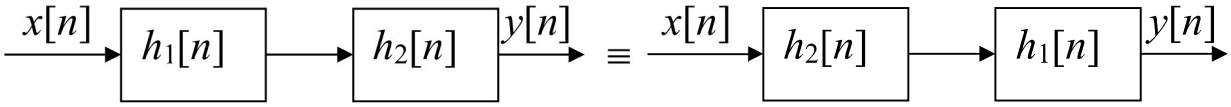
Commutative Law:

$$x[n] * h[n] = h[n] * x[n]$$

Associative Law:

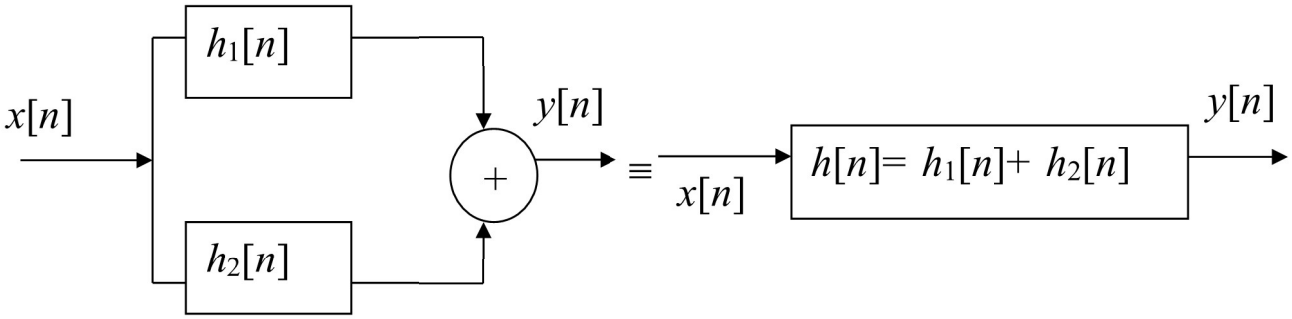
$$(x[n] * h_1[n]) * h_2[n] = x[n] * (h_1[n] * h_2[n])$$





Distributive Law:

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$



Convolution of Finite Sequence

The convolution of two finite-length sequences is also of finite length.

Find the convolution of $h[n]$ = [2, 5, 0, 4] and $x[n]$ = [4, 1, 3]; i.e., $y[n] = x[n] * h[n]$. Assume that both sequences start at n=0. The flipped sequence is $h[-i]$ = [4, 0, 5, 2]. Line up flipped sequence below x[n] to begin overlap and shift it successively, summing the product sequence to obtain the discrete convolution

0 shift

$x[n]$: 4 1 3
4 0 5 2 ← $h[-i]$
$y[0]$ = Sum of products = 8

1 shift

$x[n]$: 4 1 3
4 0 5 2 ← $h[1-i]$
$y[1]$ = (5x4)+(2x1) = 22

2 shifts

$x[n]$: 4 1 3
4 0 5 2 ← $h[2-i]$
$y[2]$ = 0 + 5 + 6 = 11

3 shifts

$x[n]$: 4 1 3
4 0 5 2 ← $h[3-i]$
$y[3]$ = 16 + 0 + 15 = 31

4 shifts

$x[n]$: 4 1 3
4 0 5 2 ← $h[4-i]$
$y[4]$ = 4 + 0 = 4

5 shifts

$x[n]$: 4 1 3
4 0 5 2 ← $h[5-i]$
$y[5]$ = 12

$$\therefore\ y[n] = x[n] * h[n] = [8 \quad 22 \quad 11 \quad 31 \quad 4 \quad 12]$$

Note: Length of $y[n]$ = Length of $x[n]$ + Length of $h[n]$ - 1
6 = 3 + 4 - 1

Exercise:

Compute the convolution $y[n] = x_1[n] * x_2[n]$ of the digital signals given by

$$x_1[n] = [1, -2, 1]$$

$$x_2[n] = \begin{cases} 1, & \text{for } 0 \le n \le 5 \\ 0, & \text{elsewhere} \end{cases}$$

3.6 Stability of Linear Time-Invariant Systems

An LTD system is stable if, and only if, the stability factor denoted by S, and defined by

$$S = \sum_{k=-\infty}^{\infty} |h[k]| < \infty \qquad (3.12)$$

is finite.

Let $x[n]$ be a bounded input sequence {i.e. $|x[n]| < M$ for all n, where M is a finite number}. We must show that the output is bounded when S is finite. To this end, we work again with the convolution formula.

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

If we take the absolute value of both sides of the above equation, we obtain

$$|y[n]| = \left| \sum_{k=-\infty}^{\infty} h[k]x[n-k] \right|$$

Now, the absolute value of the sum of terms is always less than or equal to the sum of the absolute values of the terms

$$|y[n]| \leq \sum_{k=-\infty}^{\infty} |h[k]| \quad |x[n-k]|$$

Since the input values are bounded, say by M, we have for all n:

$$|y[n]| \leq \quad M \sum_{k=-\infty}^{\infty} |h[k]| \quad \leq MS$$

Hence, since both M and S are finite, the output is also bounded. ie, a LTD system is stable if its impulse response is absolutely summable.

Example: Check the stability of the first-order recursive system shown below:

$$y[n] = ay[n-1] + x[n]$$

The impulse response of this system is:

$$h[n] = a^n u[n] \quad \text{for all formula}$$

$$S = \sum_{k=-\infty}^{\infty} |h[n]| = \sum_{k=-\infty}^{\infty} |a|^n$$

It is obvious that S is unbounded for $|a| \geq 1$, since then each term in the series is ≥ 1.

For $|a| < 1$, we can apply the infinite geometric sum formula, to find

$$S = \frac{1}{1-|a|} \quad \text{for } |a| < 1$$

Since S is finite for $|a| < 1$, the system is stable.

Exercise:

For the discrete time system given below, check if it is a linear time-invariant and BIBO stable. Assume $y(-1) = 0$.

$$y(n) = ny(n-1) + x(n) \qquad n \geq 0$$

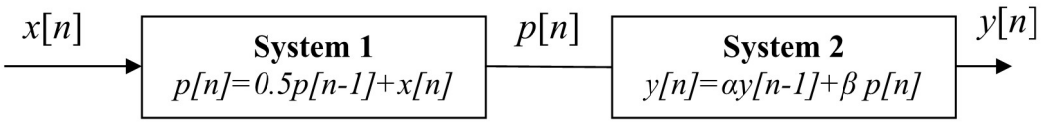
CHAPTER 3: PROBLEM SHEET 3

Q1) Draw the block diagrams of the following system in both Direct form I and Direct Form II.

$$w[n] = -0.5w[n-1] + 7x[n]$$
$$y[n] = 2w[n] - 4w[n-1]$$

(**Note:** $x[n]$ is the input and $y[n]$ is the output)

Q2) Consider the cascade of the following two systems as depicted below:

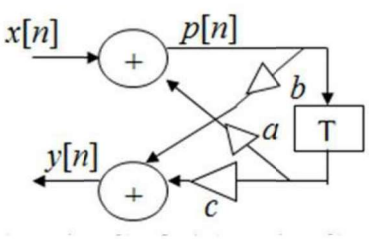


The difference equation relating $x[n]$ of $y[n]$ is:

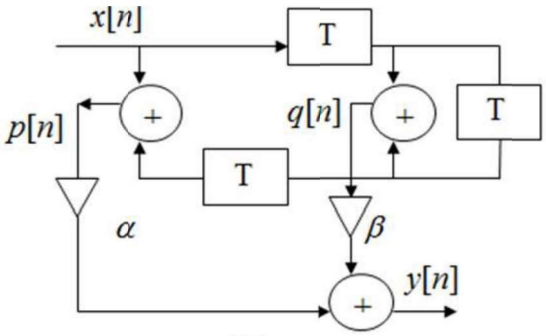
$$y[n] = x[n] + \tfrac{3}{4}y[n-1] - \tfrac{1}{8}y[n-2]$$

Determine α and β. Ans: α=1/4 and β=1

Q3) For the block diagram realisations given below, develop the relation between $y[n]$ and $x[n]$ in each case.



(a)



(b)

Ans: a) $y(n)-ay(n-1)=bx(n)+cx(n-1)$; b) $y(n)=\alpha x(n)+\beta x(n-1)+\beta x(n-2)+\alpha x(n-3)$

Q4) Draw a system implementation for each of the following difference equations:

a) $2y[n] + y[n-1] - 4y[n-3] = x[n] + 3x[n-5]$

b) $y[n] = x[n] - x[n-N]$

c) $y[n] = a_0x[n] + a_1x[n-1] + a_2x[n-2] - b_1y[n-1] - b_2y[n-2]$

Q5) A difference equation for a particular filter is given by

$$y[n] = 0.1x[n] - 0.1x[n-2] + 0.8x[n-3] + 0.1x[n-4] + 0.6x[n-6]$$

Find the impulse response of the above filter.

Ans: h[n]={0.1, 0, -0.1, 0.8, 0.1, 0, 0.6, 0, 0, 0...}

Q6) Show that the convolution of the two infinite duration sequences

$$p[n] = a^n u[n] \quad \text{and} \quad q[n] = b^n u[n]$$

for all n, where $u[n]$ is the unit step function and $a \neq b$, is given by

$$y[n] = \frac{a^{n+1} - b^{n+1}}{a - b}$$

Q7) Consider the system described by the difference equation

$$y[n] = ay[n-1] + bx[n]$$

Determine 'b' in terms of 'a' so that $\sum_{n=-\infty}^{\infty} h[n] = 1$ Ans:b=1-a

Q8) For each of the following systems, determine whether or not the system is (i) linear and (ii) time-invariant

a) $y[n] = \cos(x[n])$ Ans: Non-linear; Time invariant

b) $y[n] = x[n].\cos(0.2\pi n)$ Ans: Linear; Time variant

c) $y[n] = x[n] - x[n-1]$ Ans: Linear; Time invariant

d) $y[n] = |x[n]|$ Ans: Non-linear; Time invariant

e) $y[n] = x[n] + nx[n+1]$ Ans: linear; Time variant

f) $y[n] = \sum_{k=-\infty}^{n+1} x[k]$ Ans: Linear; Time invariant

g) Show that $y[n] = x[-n]$ is not a time-invariant system.

Q9)

a) Let $x[n]$ = {1 4 0 2} and $h[n]$ = {1 2 1}. Find their convolution (Both sequences start at n=0).

Ans:{1, 6, 9, 6, 4, 2}

b) Let $x[n]$ = {0.5 0.5 0.5} and $h[n]$ = {3 2 1}. Find their convolution (Both sequences start at n=0).

Ans:{1.5, 2.5, 3, 1.5, 0.5}

End of Chapter 3

Chapter 4

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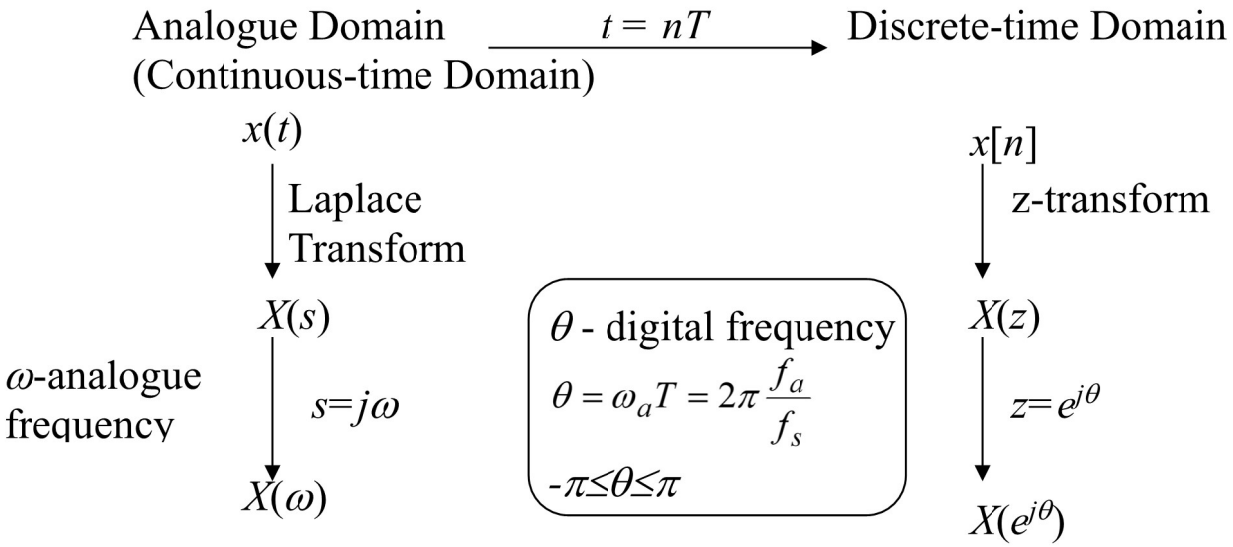
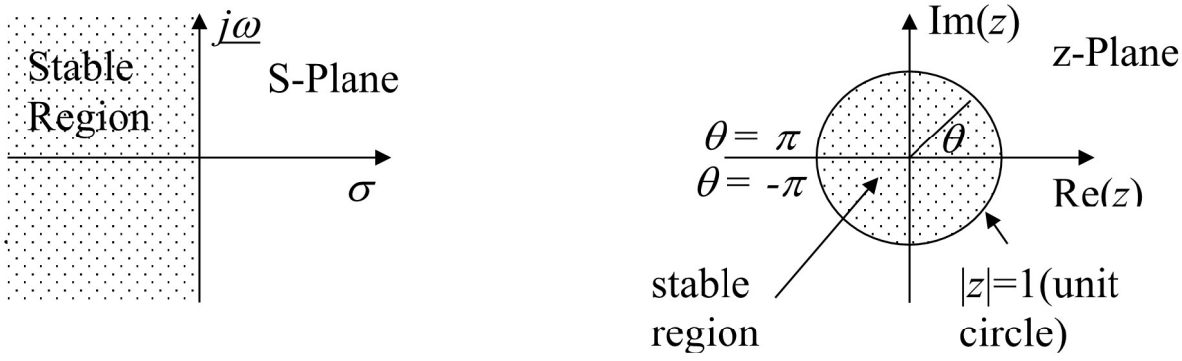
Chapter 4: Introduction to z-Transform





The primary role of the Laplace transform in engineering is transient and stability analysis of causal LTI system described by differential equations.

The primary roles of the z-transform are the study of system characteristics and derivation of computational structures for implementing discrete-time systems on computers. The transform is also used solve difference equations.

4.1 z-Transform

The z-transform of a discrete-time signal

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

two-sided transform

$z = re^{j\theta}$, θ = digital frequency
where z is a complex variable

In causal systems, $x[n]$ may be zero when $n < 0$

$$X(z) = \sum_{n=0}^{\infty} x[n] z^{-n}$$ one-sided transform

Clearly, the z-transform is a power series with an infinite number of terms and so may not converge for all values of z.

The region where the z-transform converges is known as the region of convergence (ROC) and in this region the values of $X(z)$ are finite.

The step sequence:

$$x[n] = \begin{cases} 1 & 0 \le n \le \infty \\ 0 & n < 0 \end{cases}$$

$$X(z) = \sum_{n=0}^{\infty} 1 \cdot z^{-n} = 1 + z^{-1} + z^{-2} + \ldots$$

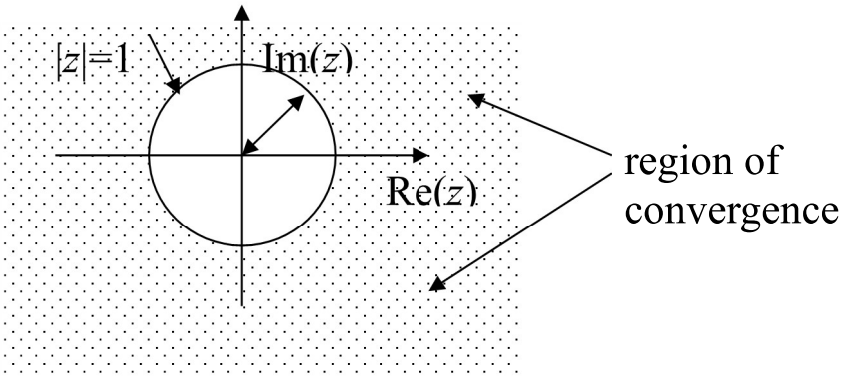
This is a geometric series with a common ratio of z^{-1}.

The series converges if $|z^{-1}| < 1$ or equivalently if $|z| > 1$

$$X(z) = 1 + z^{-1} + z^{-2} + \ldots = \frac{1}{1 - z^{-1}} = \frac{z}{z - 1}$$

In this case, the z-transform is valid everywhere outside a circle of unit radius whose centre is at the origin (see below)

$|z|=1$ is a circle of unit radius referred to as the 'unit circle'



$|z| > 1$ then $X(z)$ converges and $|z| < 1$ then it diverges

Let $z = 2$ $$X(z) = 1 + 2^{-1} + 2^{-2} + ... = \frac{1}{1-2^{-1}} = 2$$

Let $z = 0.5$ $$X(z) = 1 + \left(\tfrac{1}{2}\right)^{-1} + \left(\tfrac{1}{2}\right)^{-2} + \ldots = 1 + 2 + 4 + \ldots$$

So the Region Of Convergence (ROC) is seen to be bounded by the circle |z| = 1, the radius of the **pole** of $X(z)\left\{=\frac{z}{z-1}\right\}$

The delta sequence: $\delta[n]$

$$Z\{\delta[n]\} = \sum_{n=0}^{\infty} \delta[n] z^{-n} = 1$$

The geometric sequence: $x[n] = a^n$

$$X(z) = \sum_{n=0}^{\infty} a^n z^{-n} = \frac{1}{1-\frac{a}{z}} = \frac{z}{z-a} \qquad for \qquad \left|\frac{a}{z}\right| < 1$$

or equivalently, $|z| > |a|$.

when $a = 1$, $x[n] = 1$ for $n \geq 0$ ie. $x[n] = u[n]$

$$X(z) = \frac{z}{z-1} \qquad \text{ROC: } |z|>1$$

The complex exponential sequence: $x[n] = e^{jn\theta}$

$$z\left\{e^{j\theta n}\right\} = \sum_{n=0}^{\infty} e^{j\theta n} z^{-n} = \frac{1}{1-\frac{e^{j\theta}}{z}} = \frac{z}{z-e^{j\theta}} = \frac{z}{z-e^{j\theta}} \times \frac{z-e^{-j\theta}}{z-e^{-j\theta}}$$

$$= \frac{z(z-e^{-j\theta})}{(z-e^{j\theta})(z-e^{-j\theta})} = \frac{z(z-e^{-j\theta})}{z^2-(e^{j\theta}+e^{-j\theta})z+1}$$

$$= \frac{z(z-\cos\theta+j\sin\theta)}{z^2-2z\cos\theta+1}$$

$$Z\{e^{j\theta n}\} = \frac{z(z-\cos\theta)}{z^2-2z\cos\theta+1} + j\frac{z\sin\theta}{z^2-2z\cos\theta+1}$$

$$Z\{\cos\theta n + j\sin\theta n\} = \frac{z(z-\cos\theta)}{z^2-2z\cos\theta+1} + j\frac{z\sin\theta}{z^2-2z\cos\theta+1}$$

Exploiting the linearity property

$$\therefore Z\{\cos n\theta\} = \frac{z(z-\cos\theta)}{z^2-2z\cos\theta+1}$$

$$Z\{\sin n\theta\} = \frac{z\sin\theta}{z^2-2z\cos\theta+1}$$

4.2 Properties of z-Transform

4.2.1 Linearity

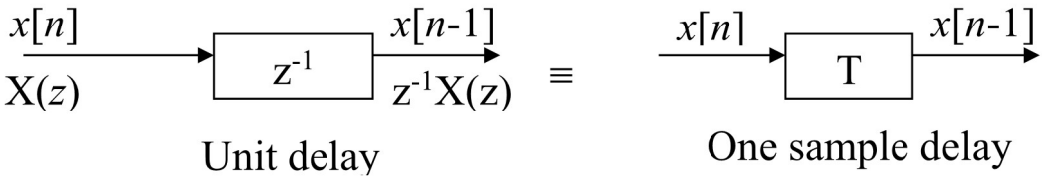
$$ax[n]+by[n] \xleftrightarrow{Z} aX(z)+bY(z)$$

4.2.2 Shifting property (Delay theorem)

$$x[n-k] \xleftrightarrow{Z} z^{-k}X(z)$$

A very important property of the z-transform is the delay theorem.

eg:
$$Z\{x[n-1]\} = z^{-1}X(z)$$
$$Z\{x[n-2]\} = z^{-2}X(z)$$



Unit delay One sample delay

4.2.3 Time reversal

$$x[-n] \xleftrightarrow{z} X\left(\frac{1}{z}\right) \quad or \quad X\left(z^{-1}\right)$$

Example: Find the z-transform of $x[n] = u[-n]$.

$$u[n] \xleftrightarrow{z} \frac{z}{z-1}; \quad u[-n] \xleftrightarrow{z} \frac{z^{-1}}{z^{-1}-1} = \frac{1}{1-z}$$

ROC: $|z| < 1$

4.2.4 Multiplication by exponential sequence

$$a^n x[n] \xleftrightarrow{z} X(a^{-1}z)$$

As a special case if $x[n]$ is multiplied by $e^{jn\theta}$

$$e^{j\theta n} x[n] \xleftrightarrow{z} X(e^{-j\theta}z)$$

4.2.5 Differentiation in the z-domain

$$nx[n] \xleftrightarrow{z} -z\frac{d}{dz}X(z)$$

Example:

Find the z-transform, G(z), if $g[n] = \begin{cases} na^n & n \geq 0 \\ 0 & n < 0 \end{cases}$

where a is a constant

$$g[n] = na^n \quad let\ x[n] = a^n \ \Rightarrow \ X(z) = \frac{z}{z-a}$$

$$G(z) = -z\frac{dX}{dz} \quad \leftarrow \quad \frac{dX(z)}{dz} = \frac{-a}{(z-a)^2}$$

$$G(z) = -z\frac{-a}{z^2(1-az^{-1})^2} = \frac{az^{-1}}{(1-az^{-1})^2}$$

4.2.6 Discrete convolution

If $y[n] = x[n] * h[n]$
then Y(z) = X(z) . H(z)

$$Z\{x[n]*h[n]\} = Z\left\{\sum_{k=-\infty}^{\infty} u[k]x[k]\cdot u[n-k]h[n-k]\right\}$$

$$= \sum_{n=0}^{\infty}\left\{\sum_{k=-\infty}^{\infty} u[k]x[k]\cdot u[n-k]h[n-k]\right\}z^{-n}$$

$$= \sum_{k=-\infty}^{\infty} u[k]x[k]\left\{\sum_{n=0}^{\infty} u[n-k]h[n-k]z^{-n}\right\} \quad Let \quad m = n-k$$

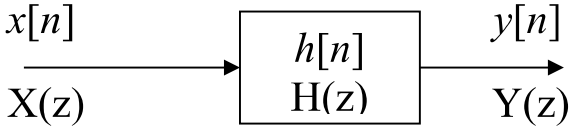
$$= \sum_{k=-\infty}^{\infty} u[k]x[k]\left\{\sum_{m=-k}^{\infty} u[m]h[m]z^{-m-k}\right\}$$

$$= \sum_{k=-\infty}^{\infty} u[k]x[k]z^{-k}\left\{\sum_{m=-k}^{\infty} u[m]h[m]z^{-m}\right\}$$

$$= \sum_{k=0}^{\infty} x[k]z^{-k}\left\{\sum_{m=0}^{\infty} h[m]z^{-m}\right\}$$

$$X(z) \qquad H(z)$$

$$= X(z)\cdot H(z)$$



$y[n] = x[n] * h[n]$
$Y(z) = X(z) \cdot H(z)$

Example: Concept of the transfer function

$$y[n] = a_0 x[n] + a_1 x[n-1] + a_2 x[n-2] + b_1 y[n-1] + b_2 y[n-2]$$

Take the z-transform of both sides:

$$Y(z) = a_0 X(z) + a_1 X(z) z^{-1} + a_2 X(z) z^{-2} + b_1 Y(z) z^{-1} + b_2 Y(z) z^{-2}$$

$$Y(z)\left[1 - b_1 z^{-1} - b_2 z^{-2}\right] = X(z)\left[a_0 + a_1 z^{-1} + a_2 z^{-2}\right]$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2}}{1 - b_1 z^{-1} - b_2 z^{-2}}$$

Example: Find the difference-equation of the following transfer function:

$$H(z) = \frac{5z + 2}{z^2 + 3z + 2}$$

First rewrite $H(z)$ as a ratio of polynomials in z^{-1}

$$\frac{Y(z)}{X(z)} = H(z) = \frac{5z^{-1} + 2z^{-2}}{1 + 3z^{-1} + 2z^{-2}}$$

$$Y(z) + 3Y(z) z^{-1} + 2Y(z) z^{-2} = 5X(z) z^{-1} + 2X(z) z^{-2}$$

Take inverse z-transform

$$y[n] + 3y[n-1] + 2y[n-2] = 5x[n-1] + 2x[n-2]$$

$$y[n] = 5x[n-1] + 2x[n-2] - 3y[n-1] - 2y[n-2]$$

Example:

If $x[n] = u[n] - u[n\text{-}10]$, find $X(z)$.

$$X(z) = \sum_{n=0}^{9} (1) z^{-n} = \frac{1 - z^{-10}}{1 - z^{-1}}$$

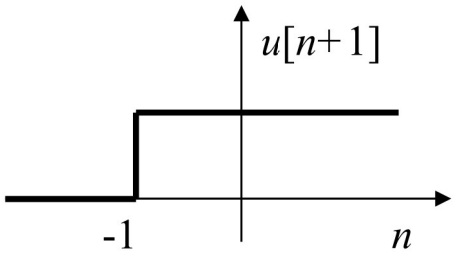
Example: If $y[n] = -\alpha^n u[-n-1]$, find Y(z).

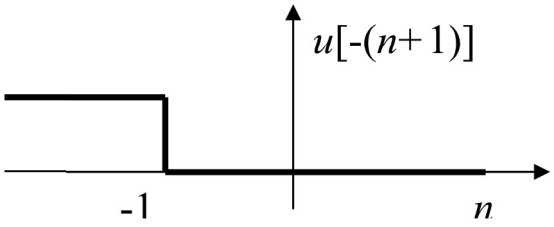
$$Y(z) = \sum_{n=-\infty}^{\infty} -\alpha^n u[-n-1] z^{-n}$$

$$= -\sum_{n=-\infty}^{-1} \left(\frac{\alpha}{z}\right)^n$$

$$= -\sum_{k=1}^{\infty} \left(\frac{z}{\alpha}\right)^k$$

$$= 1 - \sum_{k=0}^{\infty} \left(\frac{z}{\alpha}\right)^k$$



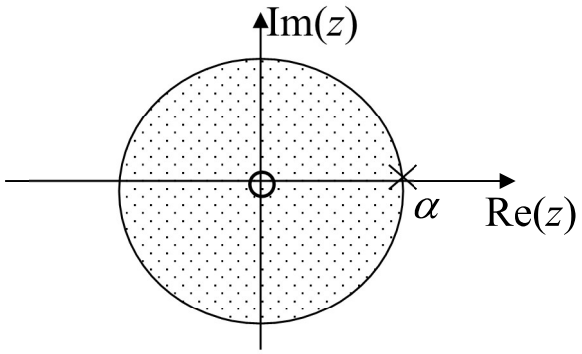


The sum converges provided $\left|\frac{z}{\alpha}\right| < 1$ ie. $|z| < |\alpha|$

$$Y(z) = 1 - \frac{1}{1 - z\alpha^{-1}} \qquad |z| < |\alpha|$$

$$= \frac{z}{z - \alpha} \qquad |z| < |\alpha|$$

Depict the ROC and pole and zero locations in the z-plane



Example: Discrete convolution

Compute the convolution $y[n]$ of the digital signals given by
$x_1[n] = [1, -2, 1]$;
$x_2[n] = 1$ for $0 \le n \le 5$,
$x_2[n] = 0$ elsewhere

$$y[n] = x_1[n] * x_2[n] \Rightarrow Y(z) = X_1(z) \cdot X_2(z)$$

$$X_1(z) = 1 - 2z^{-1} + z^{-2}$$

$$X_2(z) = 1 + z^{-1} + z^{-1} + z^{-3} + z^{-4} + z^{-5}$$

$$Y(z) = X_1(z) \cdot X_2(z)$$

$$= 1 - z^{-1} - z^{-6} + z^{-7}$$

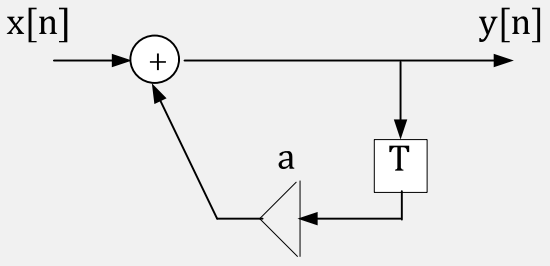
$\downarrow$ Inverse z-transform

$$y[n] = [1, -1, 0, 0, 0, 0, -1, 1]$$

Exercise: Compute the convolution $y[n]$ of the digital signals given by $x_1[n] = [2, 0, 1]$ and $x_2[n] = [1\ 0\ 0\ -1]$.

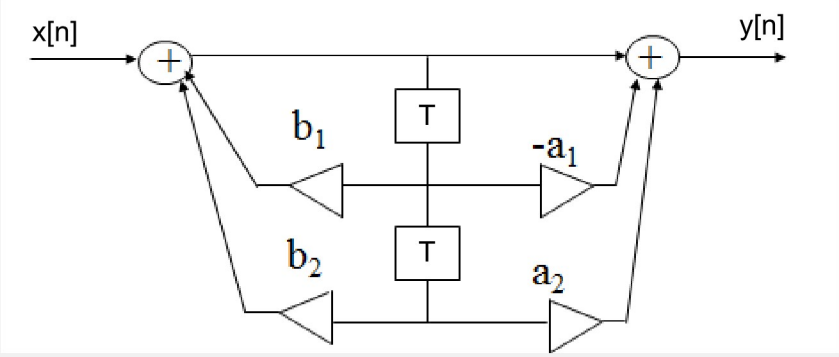
Ans: y[n]= [2 0 1 -2 0 -1]

Exercise: Determine the transer function $H(z)$:



Ans: $H(z) = \dfrac{1}{1 - az^{-1}}$

Exercise: Determine the transer function $H(z)$:



Ans: $H(z)=\dfrac{1-a_1z^{-1}-a_2z^{-2}}{1-b_1z^{-1}-b_2z^{-2}}$

Basic z-Transforms

Signal	Transform	ROC
$\delta[n]$	1	all z
$U[n]$	$\dfrac{1}{1-z^{-1}}$	$\|z\| > 1$
$\alpha^n\, u[n]$	$\dfrac{1}{1-\alpha z^{-1}}$	$\|z\| > \|\alpha\|$
$n\alpha^n\, u[n]$	$\dfrac{\alpha z^{-1}}{(1-\alpha z^{-1})^2}$	$\|z\| > \|\alpha\|$
$\cos(n\theta)\, u[n]$	$\dfrac{1-z^{-1}\cos\theta}{1-z^{-1}2\cos\theta+z^{-2}}$	$\|z\|>1$
$\sin(n\theta)\, u[n]$	$\dfrac{z^{-1}\sin\theta}{1-z^{-1}2\cos\theta+z^{-2}}$	$\|z\| > 1$
$r^n\cos(n\theta)\, u[n]$	$\dfrac{1-z^{-1}r\cos\theta}{1-z^{-1}2r\cos\theta+r^2z^{-2}}$	$\|z\| > r$
$r^n\sin(n\theta)\, u[n]$	$\dfrac{1-z^{-1}r\sin\theta}{1-z^{-1}2r\cos\theta+r^2z^{-2}}$	$\|z\|>r$

z-Transform Properties

Signal	Transform
$x[n]$	$X(z)$
$ax[n] + by[n]$	$aX(z) + bY(z)$
$x[n-k]$	$z^{-k}X(z)$
$a^n x[n]$	$X\left(\frac{z}{a}\right)$
$x[-n]$	$X\left(\frac{1}{z}\right)$
$x[n]*y[n]$	$X(z)\cdot Y(z)$
$nx[n]$	$-z\frac{d}{dz}X(z)$

Note: $S_N = 1 + z^{-1} + z^{-2} + \ldots + z^{-(N-1)} = \frac{1 - z^{-N}}{1 - z^{-1}}$

If $N \to \infty$ $S_\infty = \frac{1}{1 - z^{-1}}$, $|z^{-1}| < 1$

4.3 Inverse z-Transform

- Partial fraction method

The inverse z-transform allows us to recover the discrete-time sequence.

$$x[n] = z^{-1}[X(z)]$$

where $X(z)$ is the z-transform of $x[n]$.

Example: Find $x[n]$ for the following:

$$X(z)=\frac{z^{-1}}{1-0.25z^{-1}-0.375z^{-2}}$$

$$=\frac{z}{z^2-0.25z-0.375}=\frac{z}{(z-0.75)(z+0.5)}=z\left[\frac{A}{z-0.75}+\frac{B}{z+0.5}\right]$$

$$=z\left[\frac{\left(\frac{4}{5}\right)}{z-0.75}+\frac{\left(-\frac{4}{5}\right)}{z+0.5}\right]=\left[\frac{\left(\frac{4}{5}\right)z}{z-0.75}-\frac{\left(\frac{4}{5}\right)z}{z+0.5}\right]$$

$$x[n]=\frac{4}{5}Z^{-1}\left[\frac{z}{z-0.75}\right]-\frac{4}{5}Z^{-1}\left[\frac{z}{z+0.5}\right]$$

$$=\frac{4}{5}\left[(0.75)^n-(-0.5)^n\right],n>0$$

- Power series method
- Residue method - evaluating the contour integral

Exercise:

(a) Find h[n] for the following system and show that the system is stable if $|a|<1$:

$$H(z)=\frac{1}{1+az^{-1}}-\frac{az^{-1}}{1+az^{-1}}$$

(b) Find the impulse response h[n] for the following system:

$$H\ (z)=\frac{1-\frac{1}{2}z^{-1}}{\left[1-\frac{1}{4}z^{-1}\right]\left[1+\frac{1}{4}z^{-1}\right]}$$

4.3.1 Relationship between the z-transform and Laplace transform.

If we let $z = e^{sT}$, then $z = e^{(\sigma + j\omega)T}$ (T is the sampling period)

$$z = e^{\sigma T} \cdot e^{j\omega T} = e^{\sigma T} \cdot e^{j\theta} \qquad -\pi \le \theta \le \pi$$

Thus $|z| = e^{\sigma T}$ and $\angle z = \omega T = 2\pi \frac{f}{f_S} = \theta$ (θ = digital frequency)

As ω varies from -∞ to ∞ the s-plane is mapped to the z-plane as shown in Figure 4.1.

The entire jω axis in the s-plane is mapped onto the unit circle. The left-hand s-plane is mapped to the inside of the unit circle and the right-hand s-plane maps to the outside of the unit circle.

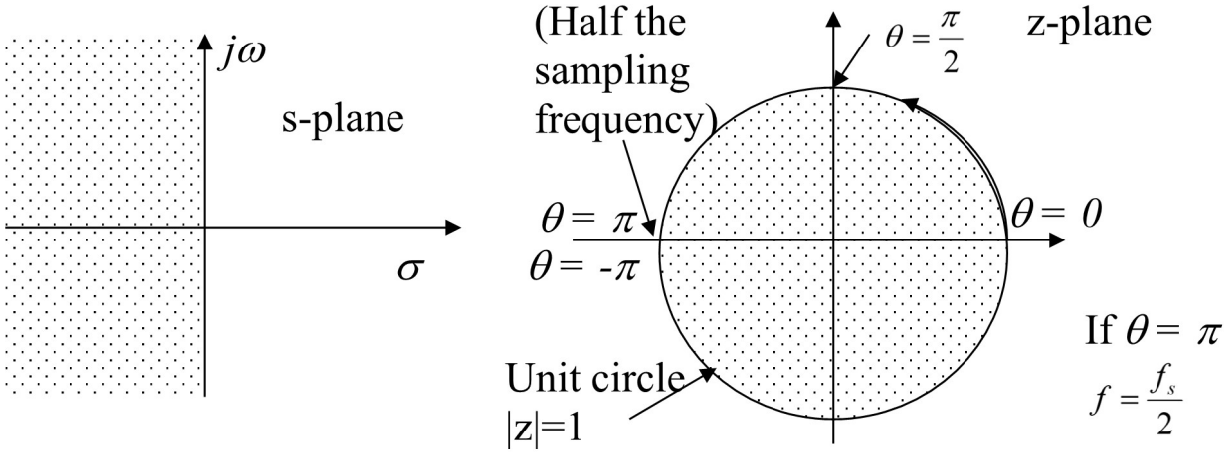


Figure 4.1: Mapping of the s-plane to the z-plane.

In terms of frequency response, the jω axis is the most important in the s-plane. In this case $\sigma = 0$ and the frequency points in the s-plane are related to points on the z-plane unit circle by $z = e^{\sigma T}.e^{j\omega T} = 1\, e^{j\theta}$

$$z = e^{j\theta}$$

4.4 Frequency Response Estimation

There are many instances when it is necessary to evaluate the frequency response of discrete-time systems. The frequency response of a system can be readily obtained from its z-transforms. For example, if we set $z = e^{j\theta}$, that is evaluate the z-transform around the unit circle, we obtain the Fourier Transform of the system.

$$H(\theta) = H(z)\big|_{z=e^{j\theta}} \text{ , } -\pi \le \theta \le \pi$$

Fourier transform of the discrete-time system

Example:

$$H(z) = \frac{1}{1-az^{-1}}, \; 0 < a < 1, \text{ say } a = 0.6$$

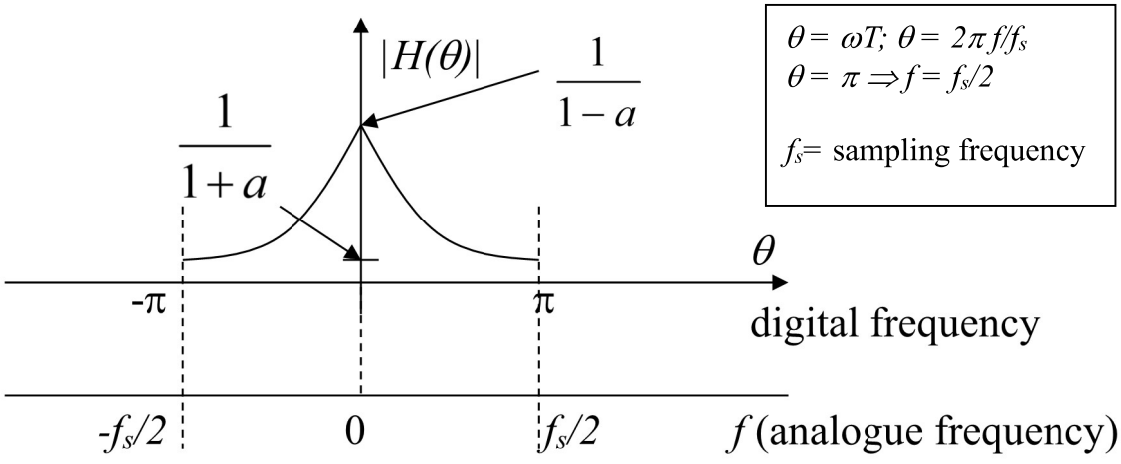
Find $H(\theta)$. {$H(\theta)$-the frequency response}

$$H(\theta) = H(z)\big|_{z=e^{j\theta}} \qquad -\pi \le \theta \le \pi$$

Fourier transform of the discrete-time system.

$$H(\theta) = H(z)\big|_{z=e^{j\theta}} = \frac{1}{1-ae^{-j\theta}} = \frac{1}{(1-a\cos\theta)+ja\sin\theta}$$

$$|H(\theta)| = \frac{1}{\sqrt{(1-a\cos\theta)^2+(a\sin\theta)^2}} = \frac{1}{\sqrt{1-2a\cos\theta+a^2}}$$



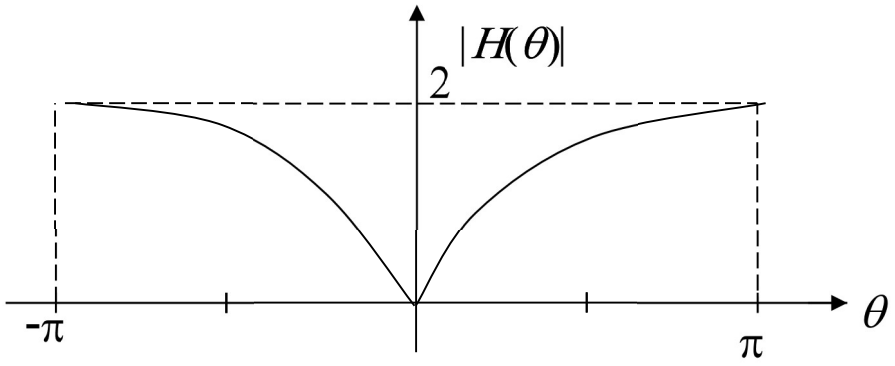
$\theta = \omega T$; $\theta = 2\pi f/f_s$
$\theta = \pi \Rightarrow f = f_s/2$

f_s= sampling frequency

Example: Find $H(\theta)$ if $H(z)=1-z^{-1}$

$$H(\theta)=1-e^{-j\theta}$$

$$|H(\theta)|=\sqrt{(1-\cos\theta)^2+(-\sin\theta)^2}=\sqrt{2-2\cos\theta}$$



Exercise:

Find $H(\theta)$ if $H(z)=\frac{1-a}{2}\frac{1+z^{-1}}{1-az^{-1}}$ where $0< a <1$.

Ans: $H(\theta)=\frac{1-a}{2}\frac{1+\cos\theta-j\sin\theta}{1-a\cos\theta+ja\sin\theta}$

Exercise: Find H(z) if

$$H(\theta)=\frac{0.5\cos\theta+j0.5\sin\theta}{1+\cos 2\theta-j\sin 2\theta}$$

Ans: $H(z)=\frac{0.5z}{1+z^{-2}}$

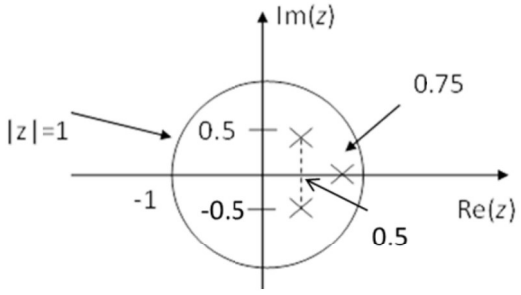
4.4 Pole-Zero Description of Discrete-Time Systems

The **zeros** of a z-transform *H(z)* are the values of *z* for which *H(z)=0*. The **poles** of a z-transform are the values of *z* for which *H(z)=∞* . If *H(z)* is a rational function , then

$$H(z)=\frac{Y(z)}{X(z)}=\frac{a_0+a_1z^{-1}+...+a_Mz^{-M}}{1+b_1z^{-1}+.....+b_Lz^{-L}}$$

$$=a_0\frac{(z-z_1)(z-z_2).....(z-z_M)}{(z-p_1)(z-p_2).....(z-p_L)}$$

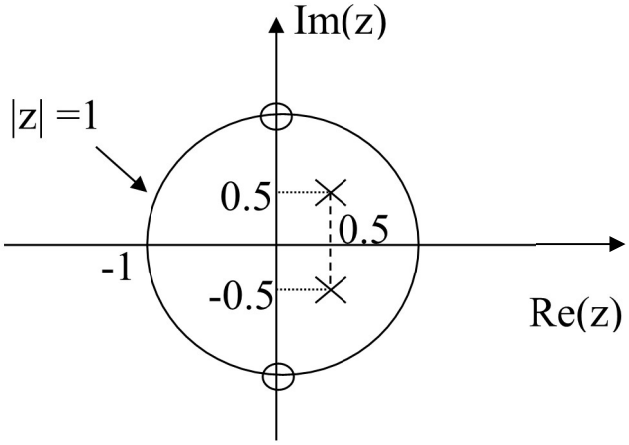
The complex quantities (or may be real) z_1, z_2, z_3 .... are called zeros of *H(z)* and the complex quantities (or may be real) p_1, p_2, p_3 … are called the poles of *H(z)*. We thus see that *H(z)* is completely determined , except for the constant a_0, by the values of poles and zeros. The information contained in the z-transform can be conveniently displayed as a poles-zero diagram (see figure below)



In the diagram, 'X' marks the position of a pole and 'O' denotes the position of a zero. The poles are located at $z = 0.5 \pm 0.5j$ and $z = 0.75$; a single zero is at $z = -1$. An important feature of the pole-zero diagram is the unit circle $|z|=1$. The pole-zero diagram provides an insight into the properties of a given discrete-time system. From the locations of the poles and zeros we can infer the frequency response of the system as well as its degree of stability.

For a stable system, all the poles must lie inside the unit circle. Zeros may lie inside, on, or outside the unit circle.

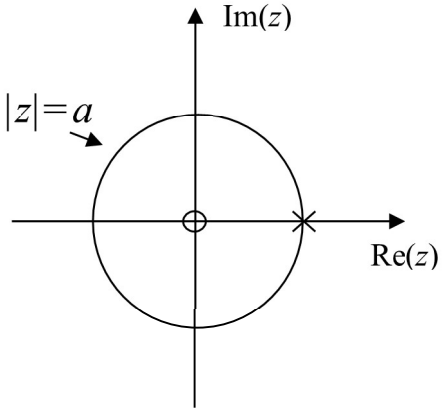
Example: Determine the transfer function *H(z)* of a discrete-time system with the pole-zero diagram shown below:



$$H(z) = \frac{K(z - j1)(z + j1)}{(z - 0.5 - 0.5j)(z - 0.5 + 0.5j)}$$

$$= \frac{K(1 + z^{-2})}{1 - z^{-1} - 0.5z^{-2}}$$

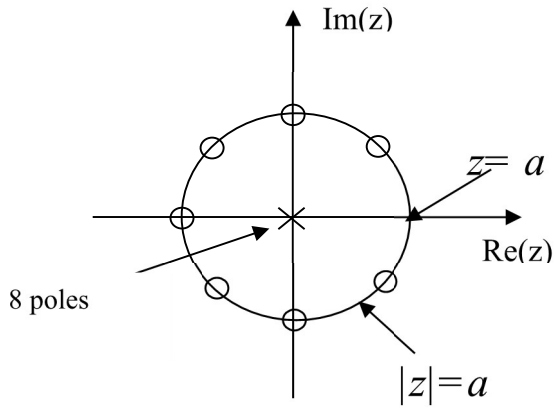
Example: Determine the pole-zero plot: $H(z) = \dfrac{z}{z - a}$



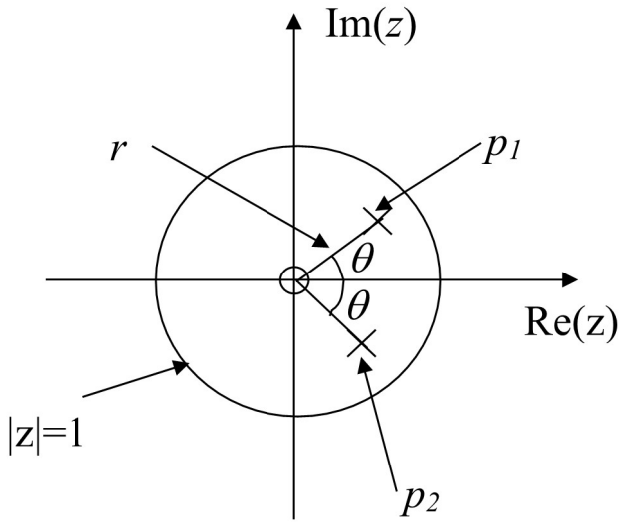
Example: Determine the pole-zero plot:

$$H(z) = \frac{z^M - a^M}{z^{M-1}(z - a)} = \frac{(z - z_1)(z - z_2).....(z - M)}{z^{M-1}}$$

The zero z = *a* cancels the pole at z = *a*. Thus, *H(z)* has *M*-1 zeros and *M*-1 poles as shown in the diagram below for *M* = 8.



Consider a system, *H(z)* with two complex conjugate poles in the z-plane :



$$p_1 = re^{j\theta}$$
$$p_2 = re^{-j\theta}$$
$$z_1 = 0 \quad (zero)$$
} Poles

A typical transfer function might be:

$$H(z) = \frac{z}{\left(z - re^{j\theta}\right)\left(z - re^{-j\theta}\right)} = z\left[\frac{A}{z - re^{j\theta}} + \frac{B}{z - re^{-j\theta}}\right]$$

$$= z\left[\frac{\frac{1}{2jr\sin\theta}}{z - re^{j\theta}} - \frac{\frac{1}{2jr\sin\theta}}{z - re^{-j\theta}}\right]$$

$$H(z) = \frac{1}{j(2r\sin\theta)}\left[\frac{1}{1 - re^{j\theta}z^{-1}} - \frac{1}{1 - re^{-j\theta}z^{-1}}\right]$$

$$h(n) = \frac{1}{j2r\sin\theta}\left[\left(re^{j\theta}\right)^n - \left(re^{-j\theta}\right)^n\right]$$

$$= \frac{r^n}{j2r\sin\theta}\left[e^{j\theta n} - e^{-j\theta n}\right]$$

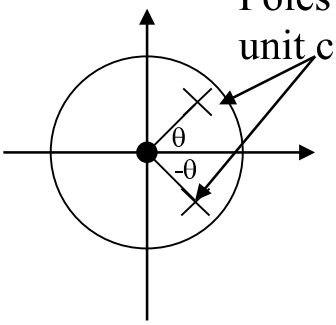
$$h(n) = \frac{1}{\sin\theta}\left[r^{n-1}\sin n\theta\right]$$ ← θ = frequency of oscillation

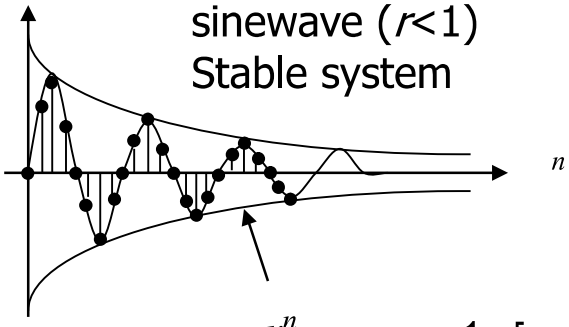
This is the impulse response of the 2nd order system with complex poles. We note that the impulse response will decay away to zero provided r is less than one. [$r < 1$]

Recall that *r* is also the distance from the origin in the z-plane to the poles p_1 or p_2, so that system will be stable if the poles in the z-plane lie inside the unit circle.

(a)**Example:**

Poles inside unit circle



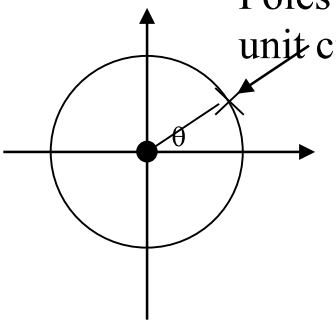


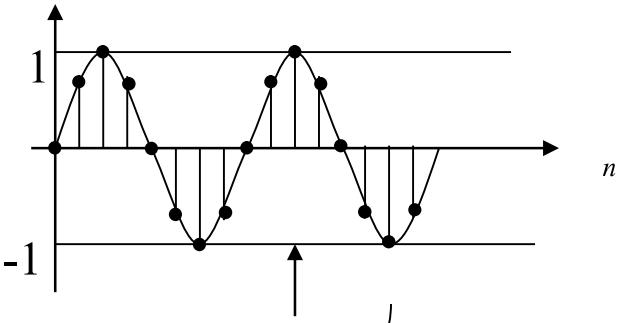
Exponential decay sinewave ($r<1$)
Stable system

r^n

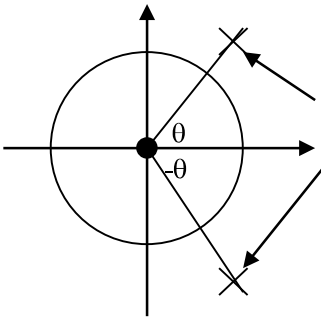
$$h(n)=\frac{1}{\sin\theta}\left[r^{n-1}\sin n\theta\right]$$

Poles on the unit circle

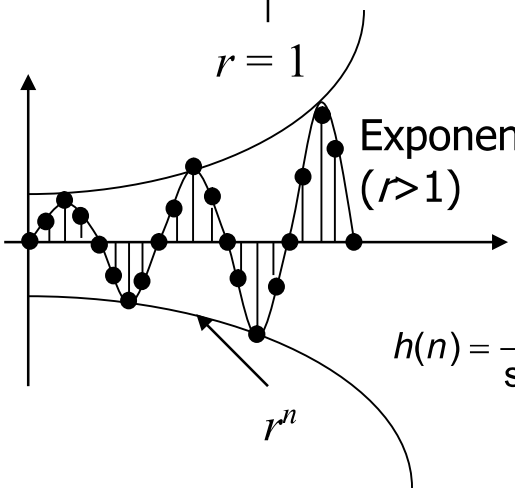




$r=1$



Poles outside unit circle

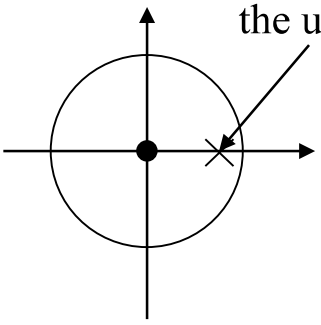


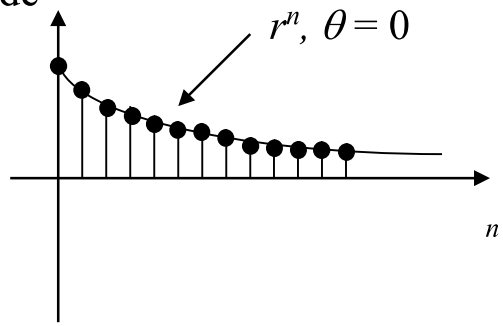
Exponential increasing ($r>1$)

$$h(n)=\frac{1}{\sin\theta}\left[r^{n-1}\sin n\theta\right]$$

r^n

One real Pole inside the unit circle





$r^n,\ \theta=0$

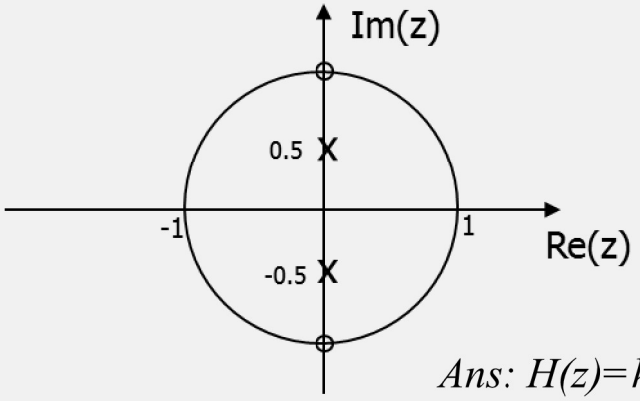
Note:

- A system that is both stable and causal must have all its poles inside that unit circle within the z-plane.

- We cannot have a pole outside the unit circle, since the inverse transform of a pole located outside the circle will contribute either a right sided increasing exponential term, which is not stable, or a left-sided decaying exponential term that is not causal.

Exercise:

(a) Determine the transfer function H(z) of a discrete-time system with the pole-zero map given below.

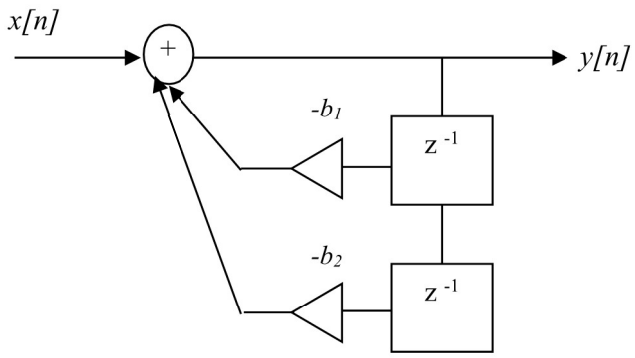


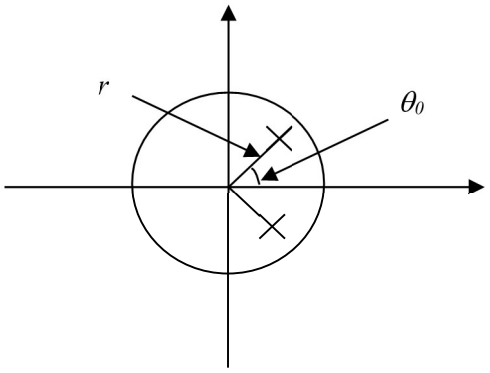
Ans: $H(z)=k(z^2+1)/(z^2+0.25)$

(b) Sketch the pole zero plot corresponding to the transfer function given below.

$$H(z) = \frac{z^2 + 1}{z^2 - z + 1/2}$$

4.5 A Second Order Resonant System (Complex Poles)





$$p_1 = re^{j\theta_0} = r\cos\theta_0 + jr\sin\theta_0$$

$$p_2 = re^{-j\theta_0} = r\cos\theta_0 - jr\sin\theta_0$$

$$H(z) = \frac{1}{1 + b_1 z^{-1} + b_2 z^{-2}} = \frac{z^2}{z^2 + b_1 z + b_2} \qquad \text{(A)}$$

All pole system has poles only (without counting the zeros at the origin)

$$H(z) = \frac{z^2}{z^2 + b_1 z^{-1} + b_2} = \frac{z^2}{(z - p_1)(z - p_2)} = \frac{z^2}{(z - re^{j\theta_0})(z - re^{-j\theta_0})}$$

$$H(z) = \frac{z^2}{z^2 - r(e^{j\theta_0} + e^{-j\theta_0})z + r^2} = \frac{z^2}{z^2 - 2r\cos\theta_0 z + r^2} \qquad \text{(B)}$$

Comparing (A) and (B), we obtain

$$\boldsymbol{b_1 = -2r\cos\theta_0} \quad \boldsymbol{b_2 = r^2} \quad \therefore \boldsymbol{\cos\theta_0 = \frac{-b_1}{2\sqrt{b_2}}} \qquad \theta_0 = \frac{2\pi f_0}{f_s}$$

θ_0 = resonant frequency

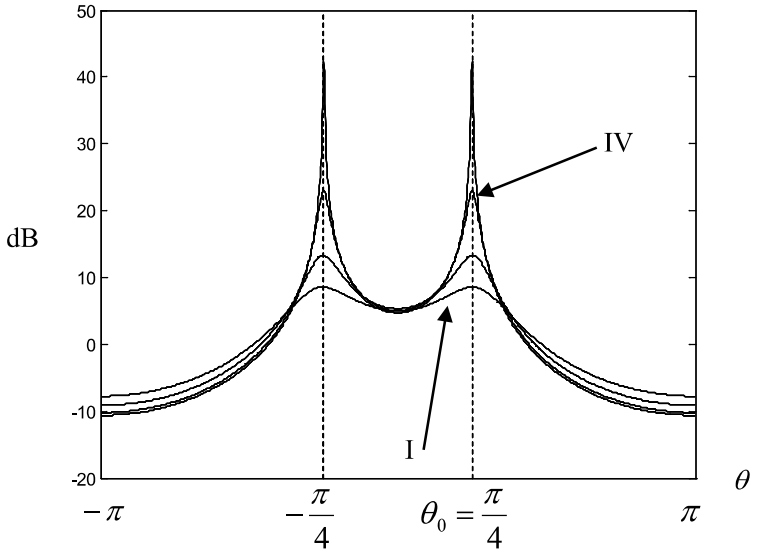
We can derive $H(\theta)$ and the magnitude from $H(z) = \dfrac{1}{1 + b_1 z^{-1} + b_2 z^{-2}}$

	b_1	$b_2 = r^2$
I	-0.94	0.5
II	-1.16	0.7
III	-1.34	0.9
IV	-1.41	0.99

$$b_2 = r^2$$

$$b_1 = -2r\cos(\theta_0)$$

$$\theta_0 = \cos^{-1}\left[\frac{-b_1}{2\sqrt{b_2}}\right]$$

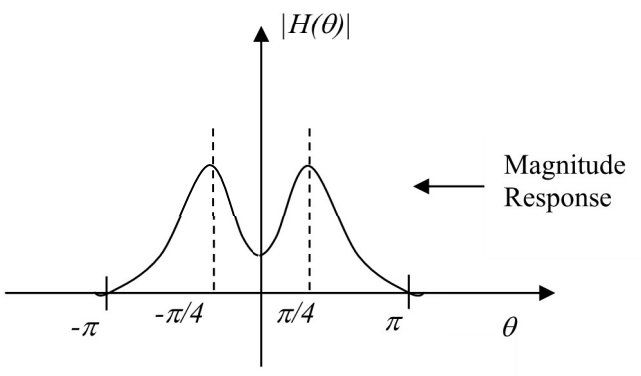


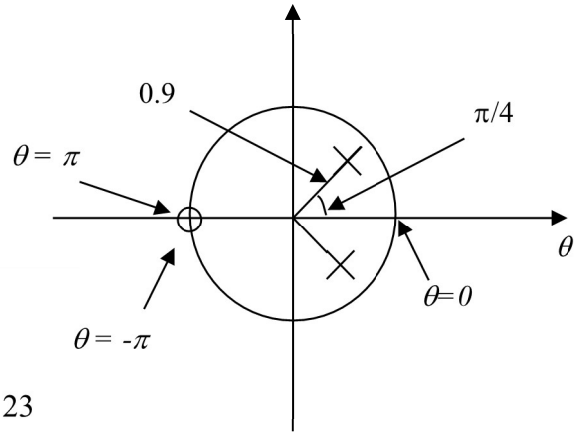
Example: Sketch the magnitude response for the system having the transfer function

$$H(z) = \frac{1 + z^{-1}}{(1 - 0.9e^{j\frac{\pi}{4}}z^{-1})(1 - 0.9e^{-j\frac{\pi}{4}}z^{-1})}$$

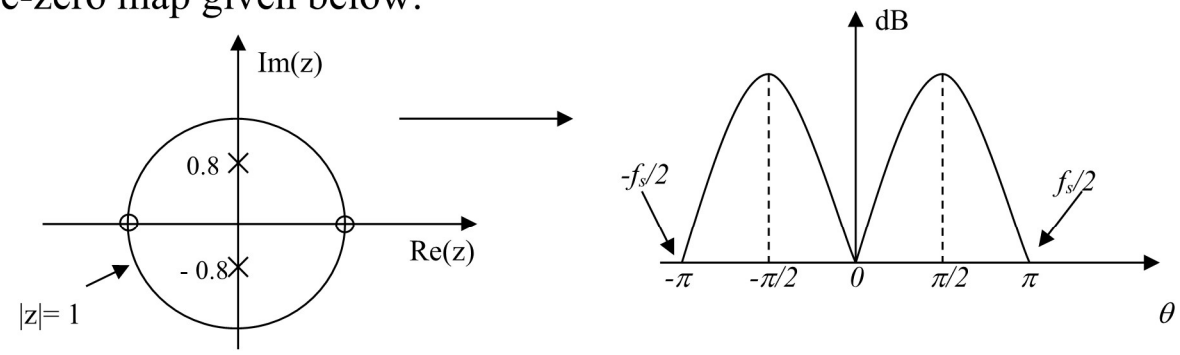
The system has a zero at $z = -1$ & poles at $z = 0.9e^{-j\frac{\pi}{4}}$

$\therefore$ Magnitude response will be zero at $\theta = \pi$ and large at $\theta_0 = \pm \pi/4$ because the poles are close to the unit circle.

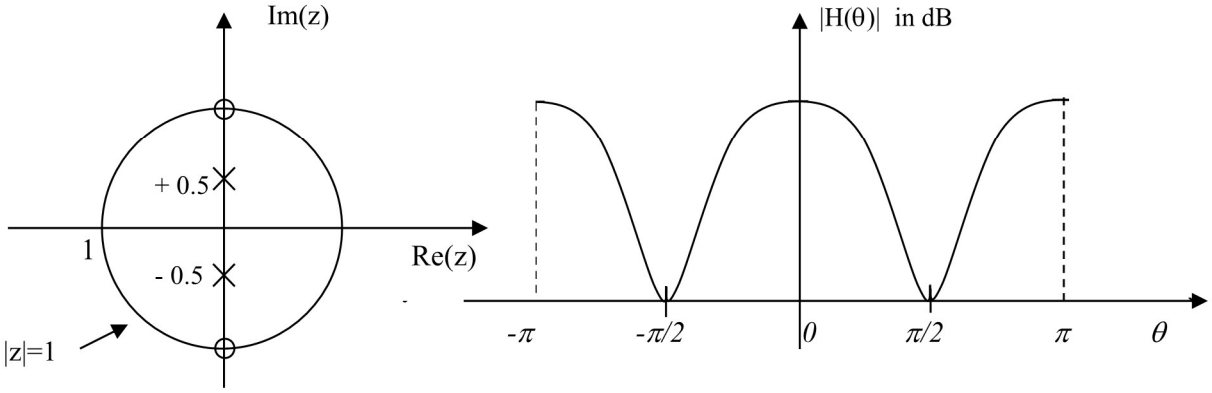




Example: Sketch the approximate magnitude response from the pole-zero map given below:



Example: Sketch an approximate magnitude response from the pole-zero map given below:

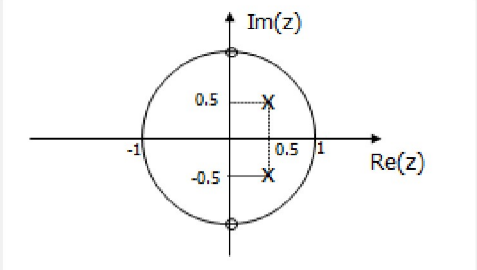


Exercise:

(a) Determine and sketch the approximate magnitude response for the following system:

$$H(z)=\frac{1-a}{1-az^{-1}} \qquad 0<a<1 \qquad Ans:\left|H(\theta)\right|=\frac{1-a}{\sqrt{1+a^2-2a\cos\theta}}$$

b) Sketch an approximate magnitude response from the pole-zero map given below:



CHAPTER 4: PROBLEM SHEET 4

Q1) Find the inverse of each of the following z-transforms:

a) $X(z)=\dfrac{1}{1-\left(\frac{1}{2}\right)z^{-1}}+\dfrac{3}{1-\frac{1}{3}z^{-1}},|z|>\dfrac{1}{2}$

Ans: $x[n]=\left(\frac{1}{2}\right)^n u[n]+3\left(\frac{1}{3}\right)^n u[n] \quad n\geq 0$

b) $X(z)=\dfrac{1}{1+3z^{-1}+2z^{-2}} \qquad |z|>2$

Ans: $x[n]=2(-2)^n u[n]-(-1)^n u[n] \quad n\geq 0$

Q2)

Show that the following systems are equivalent.

a) $y[n]=0.2y[n-1]+x[n]-0.3x[n-1]+0.02x[n-2]$

b) $y[n]=x[n]-0.1x[n-1]$

Q3) Determine and sketch the approximate magnitude response for the following filter:

$$H(z)=\frac{1-a}{1-az^{-1}} \qquad 0<a<1$$

Ans:

$$|H(\theta)|=\frac{1-a}{\sqrt{1-2a\cos\theta+a^2}}$$

Q4) Consider the system.

$$H(z)=\frac{1-2z^{-1}+2z^{-2}-z^{-3}}{(1-z^{-1})(1-0.5z^{-1})(1-0.2z^{-1})}, \qquad \text{ROC: } \boldsymbol{0.5<|z|<1}$$

i) Sketch the pole-zero pattern. Is this system stable?

Ans: Stable

ii) Determine the impulse response of the system.

Ans: $h[n]=10\delta(n)+\left\{5\left(\frac{1}{2}\right)^n-14\left(\frac{1}{5}\right)^n\right\}u(n)$

End of Chapter 4

Chapter 5

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Chapter 5: Introduction to Digital Filters

5.1 Introduction

There are two types of digital filters:

1. Recursive (there is at least one feedback path in the filter)
2. Non-recursive (no feedback paths)

A linear time invariant (LTI) discrete system described by the following equation is commonly called a digital filter:

$$y[n] = \sum_{k=0}^{M} a_k x[n-k] - \sum_{k=1}^{L} b_k y[n-k] \qquad (5.1)$$

Feed forward — Feedback

where $x[n]$ is the input signal, $y[n]$ is the output signal. a_0, a_1, a_2, ..., a_M; b_1, b_2, b_3, ..., b_L are constants (filter coefficients). These coefficients determine the characteristics of the system.

when $b_k = 0$ the filter is said to be non-recursive type
when $b_k \neq 0$ recursive type.

5.1.1 Non recursive digital filters (FIR)

If $b_k = 0$, then the calculation of *y[n]* does not require the use of previously calculated samples of the output (see equation (5.1)).

$$y[n]=\sum_{k=0}^{M} a_k x[n-k]= a_0 x[n]+ a_1 x[n-1]+\cdots+ a_M x[n-k]$$

This is recognised as a convolution sum.

$$y[n]=\sum_{k=0}^{M} h[k]x[n-k]$$

Therefore the impulse response is identical to the coefficients, that is,

$$h[n]=\begin{cases} a_n & 0\le n\le M \\ 0 & otherwise \end{cases}$$

$$y[n]=\sum_{k=0}^{M} a_k x[n-k]= a_0 x[n]+ a_1 x[n-1]+\cdots+ a_M x[n-k]$$

h_0 h_1 h_M

Any filter that has an impulse response of finite duration is called Finite Impulse Response (FIR) filter.

Example:

$$y[n] = a_0 x[n]+ a_1 x[n-1]+ a_2 x[n-2]$$

$$H(z)=\frac{Y(z)}{X(z)}= a_0 + a_1 z^{-1} + a_2 z^{-2}$$

This is a (non-recursive) second order FIR filter

A property of the FIR filter is that it will always be stable.

(a) Stability requires that there should be no poles outside the unit circle. This condition is automatically satisfied since there are no poles at all outside the origin. (In fact, all poles are located at the origin.)
(b) Another property of non-recursive filter is that we *can* make filters with exactly linear phase characteristics

Note: The ability to have an exactly linear phase response is one of the most important properties of a LTD system (filter). When a signal passes through a filter, it is modified in amplitude and/or phase. The nature and extent of the modification of the signal is dependent on the amplitude and phase characteristics of the filter.

The **phase delay** or **group delay** of the filter provides a useful measure of how the filter modified the phase characteristic of the signal. If we consider a signal that consists of several frequency components (eg. speech waveform) the phase delay of the filter is the amount of time delay each frequency component of the signal suffers in going through the filter.

$$phase_delay(T_p) = -\frac{\phi(\theta)}{\theta} \qquad (5.2)$$

Note: T_p = the negative of the phase angle divided by frequency

The group delay on the other hand is the average time delay the composite signal suffers at each frequency as it passes from the input to the output of the filter.

$$group_delay(T_g) = -\frac{d\phi(\theta)}{d\theta} \qquad (5.3)$$

Note: T_g = the negative of the derivative of the phase with respect to frequency]

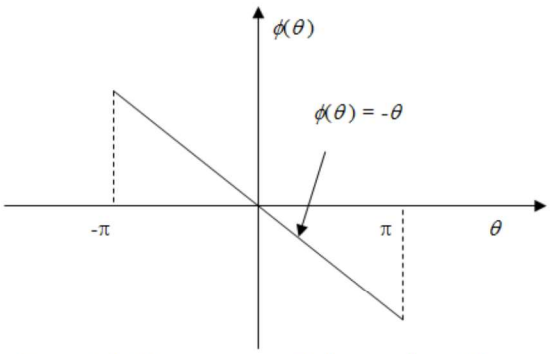
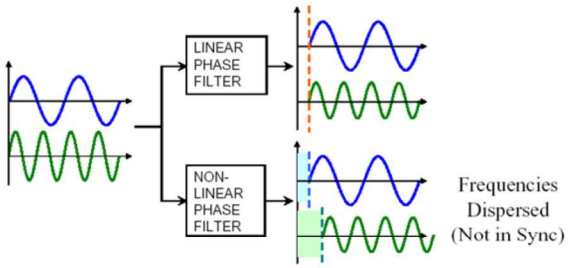


Figure 5.1: Phase response of a linear phase filter



A diagram comparing the performance of a linear phase filter and a non-linear phase filter

A *constant group delay* means that signal components at different frequencies receive the same delay in the filter.

A *linear phase filter* gives same time delay to all frequency components of the input signal. A filter with a nonlinear phase characteristic will cause a phase distortion in the signal that passes through it.

This is because the frequency components in the signal will each be delayed by an amount not proportional to frequency, thereby altering their harmonic relationship. Such a distortion is undesirable in many applications, for example music, video etc.

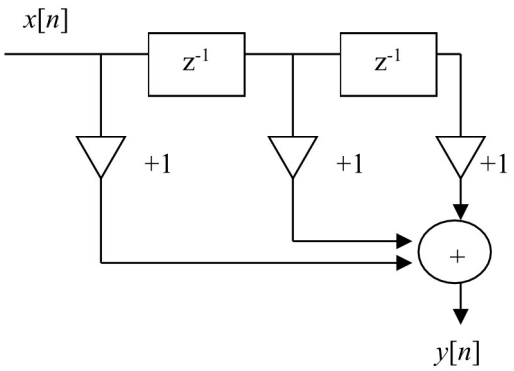
A filter is said to have a linear phase response if its phase response satisfies one of the following relationships:

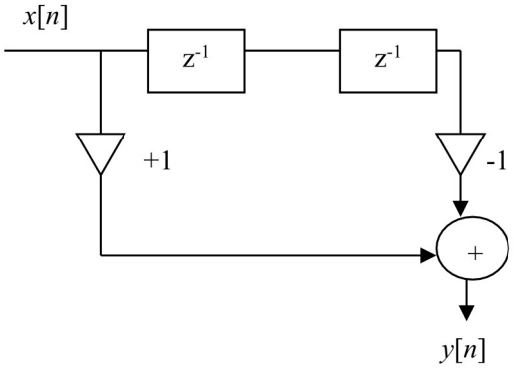
$$\phi(\theta) = -a\theta \qquad (5.4)$$
$$\phi(\theta) = b - a\theta$$

where '*a*' and '*b*' are constants.

Example:

Two filter structures are shown below. Show that both filters have linear phase.





$$y[n] = x[n] + x[n-1] + x[n-2]$$

$$H(z) = 1 + z^{-1} + z^{-2}$$

$$H(\theta) = 1 + e^{-j\theta} + e^{-j2\theta}$$

$$= e^{-j\theta}\left(e^{j\theta} + e^{-j\theta} + 1\right)$$

$$= e^{-j\theta}(1 + 2\cos\theta)$$

phase: $\phi(\theta) = -\theta$

linear phase

$$y[n] = x[n] - x[n-2]$$

$$H(z) = 1 - z^{-2}$$

$$H(\theta) = 1 - e^{-j2\theta}$$

$$= 2je^{-j\theta}\left(\frac{e^{j\theta} - e^{-j\theta}}{2j}\right)$$

$$= 2e^{j\frac{\pi}{2}}e^{-j\theta}\sin\theta$$

$$= e^{j\left(\frac{\pi}{2}-\theta\right)}(2\sin\theta)$$

phase: $\phi(\theta) = \pi/2 - \theta$

linear phase

Pole-zero Patterns of Linear Phase Filters

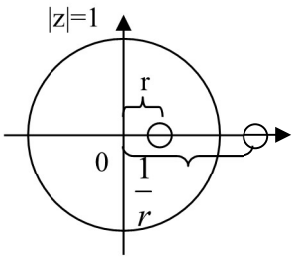
- Linear phase filters provide constant delay with no amplitude distortion.

- An FIR filter with its impulse response $h[n]$ symmetric about the midpoint is endowed with linear phase and constant delay.

- For a linear phase FIR filter, the poles must lie at the origin (z=0).

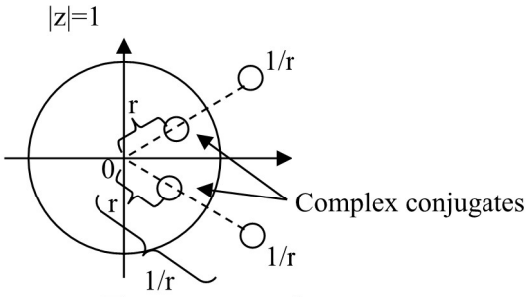
- Impulse responses that are symmetric about the origin (i.e. n=0) require $h[n] = \pm h[-n]$ and hence $H(z) = \pm H\left(\frac{1}{z}\right)$

- The zeros of a linear-phase filter must occur in reciprocal pairs and exhibit conjugate reciprocal symmetry as shown below.



For a real zero

A zero on the real axis is paired with just its reciprocal



For a complex zero

Each zero not on the real axis is paired with its reciprocal and its conjugate.

- Zeros at z=1 or at z=-1 can occur singly, because they form their own reciprocal and their own conjugate.

- If there are no zeros at z=1, a linear-phase impulse response is always even symmetric about its midpoint.

- For odd symmetry about the mid-point, there must be an odd number of zeros at z=1.

- The frequency response of linear phase filter may be written as $H(\theta) = e^{j\alpha\theta} A(\theta)$ (for even symmetry) or $H(\theta) = je^{j\alpha\theta} A(\theta) = e^{j\left(\alpha\theta+\frac{\pi}{2}\right)} A(\theta)$ (for odd symmetry).

Example:

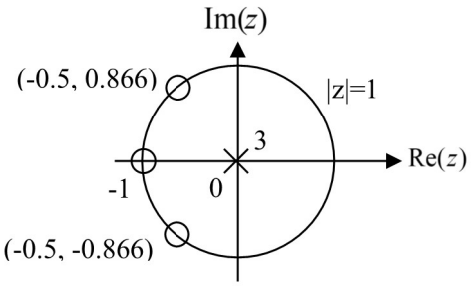
Is this a linear phase filter? Sketch the pole-zero plot.

$$H(z) = 1 + 2z^{-1} + 2z^{-2} + z^{-3}$$

n=0 n=1 n=2 n=3

$$\therefore\ h[n] = \{1, 2, 2, 1\}$$

$$H(z) = \frac{z^3 + 2z^2 + 2z + 1}{z^3}$$



All of its poles are at z=0;

Its zeros are at z=-1 and z=-0.5±j0.866

The real zero at z=-1 can occur singly

Complex conjugate pair of zeros lie on the unit circle

Ans: Linear phase filter, with an even symmetry impulse response *h[n]* about n=1.5 (mid point)

Example:

$$H(z) = z^2 + 4.25 + z^{-2}$$

$$\therefore\ h[n] = \{1, 0, 4.25, 0, 1\}$$

n=-2 n=-1 n=0 n=1 n=2

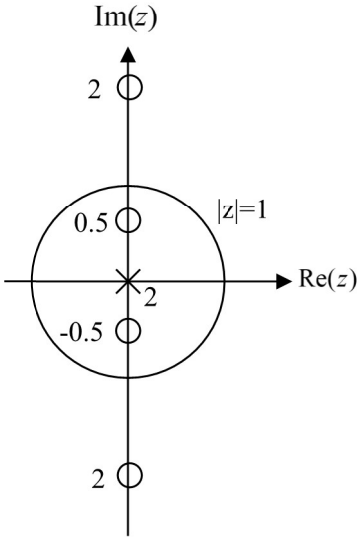
Is $H(z)$ linear phase? Sketch the pole-zero plot of $H(z)$.

Since $h[n]$ is even symmetric about n=0 with $h[n]$=$h[-n]$, we have $H(z)$=$H(1/z)$

$$H(z) = z^2 + 4.25 + z^{-2}$$

$$= \frac{z^4 + 4.25z^2 + 1}{z^2}$$

$$= \frac{(z + j0.5)(z - j0.5)(z + j2)(z - j2)}{z^2}$$



The four zeros at $z = j0.5$, $z = j2$, $z = -j0.5$ and $z = -j2$ are conjugate receiprocal symmetry. All poles are at $z = 0$.

Ans: Linear phase filter

Exercise:

$H(z) = 1-2z^{-1}+z^{-2}$

Is H(z) linear phase? Sketch the pole-zero plot of H(z).

Four types of linear phase filters:

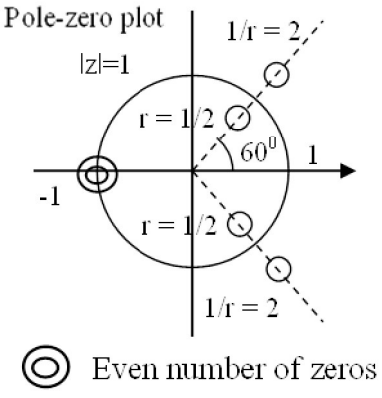
Type 1: Impulse response has even symmetry and odd length [Must have even number of zeros at z=-1 (if present) and z=1 (if present)]

Type 2: Impulse response has even symmetry and even length [Must have an odd number of zeros at z=-1 (if present) and even number of zeros at z=1 (if present)]

Type 3: Impulse response has odd symmetry and odd length. [Must have an odd number of zeros at z=-1 and odd number of zeros at z=1]

Type 4: Impulse response has odd symmetry and even length [Must have an odd number of zeros at z=1. The number of other zeros, if present at z=-1, must be even]

Example: Find all of the zero locations of a type 1 linear-phase sequence if it is known that there is a zero at $z = \frac{1}{2}e^{j\frac{\pi}{3}}$ and a zero at $z = -1$.



Due to conjugate reciprocal symmetry, the zero at $z = \frac{1}{2}e^{j\frac{\pi}{3}}$ implies we have a zero at $z = 2e^{j\frac{\pi}{3}}$, $z = \frac{1}{2}e^{-j\frac{\pi}{3}}$ and $z = 2e^{-j\frac{\pi}{3}}$. For a type 1 sequence, the number of zeros at z=-1 must be even, so there must be another zero at z=-1. Thus, there are 6 zeros.

5.1.2 Recursive digital filter (IIR)

Every recursive digital filter must contain at least one closed loop. Each closed loop contains at least one delay element.

$$y[n]=\sum_{k=0}^{M} a_k x[n-k]-\sum_{k=1}^{L} b_k y[n-k]$$

For recursive digital filters $b_k \neq 0$. Let $a_0 = a_0$, $a_k = 0$ for $k > 1$ and $b_1 = b_1$ & $b_k = 0$ for $k > 1$.

$$y[n] = a_0 x[n]- b_1 y[n-1]$$

$$H(z)=\frac{a_0}{1+b_1 z^{-1}}$$

IIR filter

A recursive filter is an infinite impulse response filter (IIR).

Example: 2nd order FIR filter

Transfer function	Description
$H_1(z) = a_0 + a_1 z^{-1} + a_2 z^{-2}$	all-zero second order FIR filter (excluding poles at origin)
$H_2(z) = \dfrac{1}{1 + b_1 z^{-1} + b_2 z^{-2}}$	all-pole second order IIR filter (excluding zeros at origin)
$H_3(z) = \dfrac{1 + a_1 z^{-1} + a_2 z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}}$	pole-zero second order IIR filter

Note: Poles and zeros can be real or imaginary

Example: Consider the 1st order IIR filter

$$H(z) = \frac{1}{1 - az^{-1}} \qquad 0 < a < 1$$

The difference equation is: $y[n] = x[n] + ay[n\text{-}1]$.

The DC gain of $H(z)$ can be obtained by substituting $\theta = 0$.

$$H(\theta)|_{\theta=0} = \frac{1}{1 - ae^{-j(0)}} = \frac{1}{1 - a}$$

If DC gain is undesirable, introduce a constant gain factor of (1-a), so that $H(z)$ becomes

$$H(z) = \frac{1 - a}{1 - az^{-1}} \qquad \text{DC gain} = 1$$

$$\therefore \; y[n] = (1\text{-}a)x[n] + ay[n\text{-}1]$$

Example: Consider a lowpass filter

$$y[n] = ay[n\text{-}1] + bx[n], \qquad 0 < a < 1$$

(i) Determine 'b' so that $|H(0)| = 1$.
(ii) Determine the 3dB bandwidth (here) for the normalised filter in part (i).

(i) $Y(z) = aY(z)\,z^{-1} + b\,X(z) \Rightarrow H(z) = \dfrac{b}{1 - az^{-1}}$

$$H(z) = \frac{b}{1 - ae^{-j\theta}} \qquad \text{we have } |H(0)| = 1$$

$$H(0) = \frac{b}{1 - a} = 1 \Rightarrow b = 1 - a$$

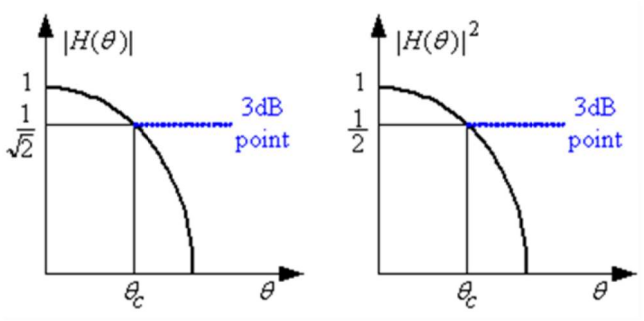
$$H(\theta)=\frac{1-a}{1-ae^{-j\theta}}=\frac{1-a}{(1-a\cos\theta)+ja\sin\theta}$$

$$|H(\theta)|=\frac{1-a}{\sqrt{(1-a\cos\theta)^2+(a\sin\theta)^2}}=\frac{1-a}{\sqrt{1+a^2-2a\cos\theta}}$$

Second Method:

$$\left\{|H(\theta)|^2=H(\theta)\cdot H^*(\theta)=\frac{1-a}{1-ae^{-j\theta}}\cdot\frac{1-a}{1-ae^{j\theta}}=\frac{(1-a)^2}{1-2a\cos\theta+a^2}\right\}$$

$$\therefore\ |H(\theta)|=\frac{1-a}{\sqrt{1-2a\cos\theta+a^2}}\qquad |H(0)|^2=1\therefore|H(\theta)|^2{}_{\theta=\theta_c}=\frac{1}{2}|H(0)|^2$$



$$\frac{1}{2}=\frac{(1-a)^2}{1+a^2-2a\cos\theta_c}$$

$$\theta_c=\cos^{-1}\left[\frac{-a^2+4a-1}{2a}\right]$$

Example: Consider a filter described by

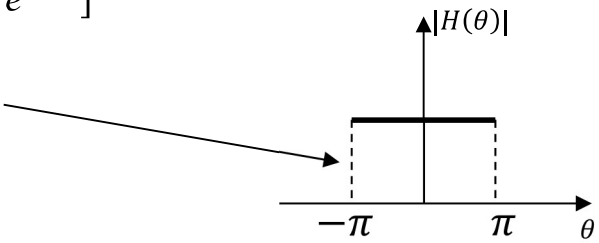
$$H(z)=\frac{a+cz^2}{c+az^2}$$ where a & c are constants.

Show that the magnitude response $|H(\theta)|$ is unity for all θ.

$$|H(\theta)|^2=H(\theta)\cdot H^*(\theta)=\frac{c+ae^{-j2\theta}}{a+ce^{-j2\theta}}\cdot\frac{c+ae^{j2\theta}}{a+ce^{j2\theta}}$$

$$=\frac{c^2+a^2+ac[e^{j2\theta}+e^{-j2\theta}]}{a^2+c^2+ac[e^{j2\theta}+e^{-j2\theta}]}=1$$

This is an **all-pass** filter.



Exercise: Determine the magnitude response of the following filter and show that it has an all-pass characteristic.

$$H(z)=\frac{1-(1/a)z^{-1}}{((1/a)-z^{-1})} \qquad |a|<1$$

5.2 Digital Filter Realisation

$$y[n]=\sum_{k=0}^{M} a_k x[n-k]-\sum_{k=1}^{L} b_k y[n-k]$$

$$Y(z)=\sum_{k=0}^{M} a_k X(z)z^{-k}-\sum_{k=1}^{L} b_k Y(z)z^{-k}$$

$$\therefore\ H(z)=\frac{Y(z)}{X(z)}=\frac{\sum_{k=0}^{M} a_k z^{-k}}{1+\sum_{k=1}^{L} b_k z^{-k}}$$

Structure I : $$H(z)=\underbrace{\sum_{k=0}^{M} a_k z^{-k}}_{zeros}\cdot\underbrace{\frac{1}{1+\sum_{k=1}^{L} b_k z^{-k}}}_{poles}$$

Structure II : $$H(z)=\underbrace{\frac{1}{1+\sum_{k=1}^{L} b_k z^{-k}}}_{poles}\cdot\underbrace{\sum_{k=0}^{M} a_k z^{-k}}_{zeros}$$

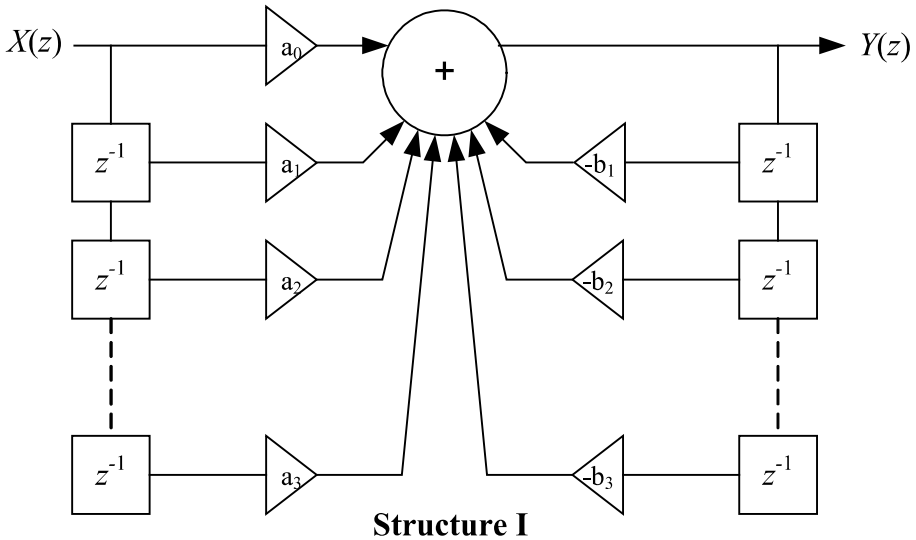


Figure 5.1 Structure I or Direct Form I

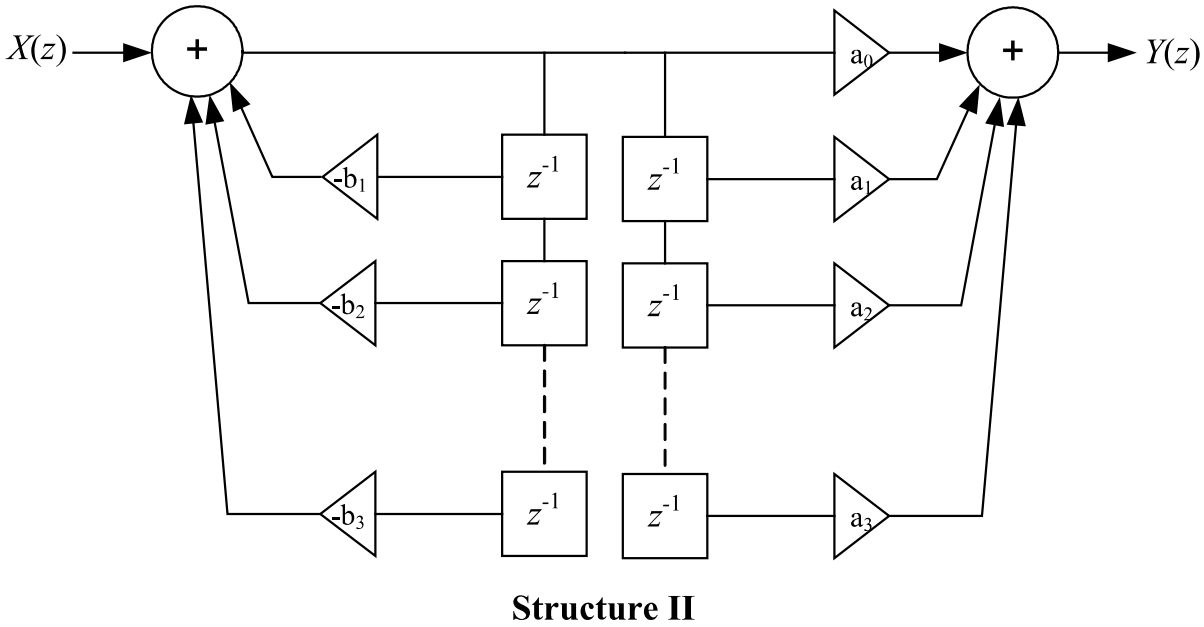


Figure 5.2 Structure II or Direct Form II

In the case when $L = M$, we have canonic form realisation.

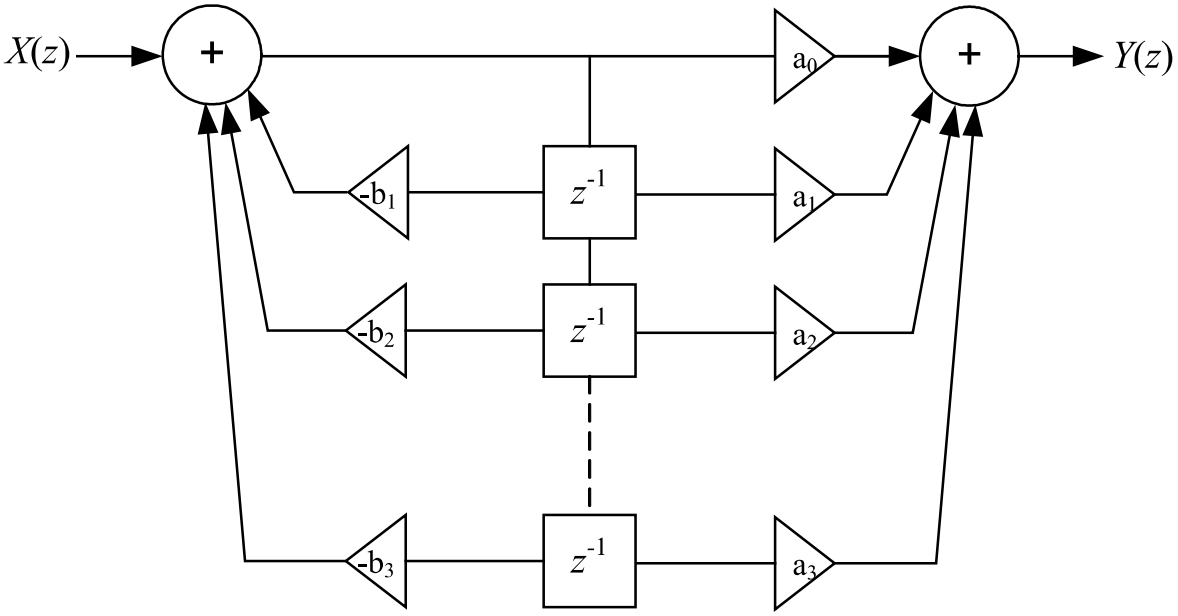


***Figure 5.3**: Canonic form*

A discrete-time filter is said to be canonic if it contains the minimum numbers of delay elements necessary to realise the associated frequency response.

5.2.1 Parallel realisation

$$H(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + \cdots + a_M z^{-M}}{1 + b_1 z^{-1} + b_2 z^{-2} + \cdots + b_L z^{-L}}$$

$$= \underbrace{\sum_{i=1}^{k} H_i(z) = H_1(z) + H_2(z) + H_3(z) + \ldots + H_k(z)}_{parallel_structure}$$

(use partial fraction to obtain $H_i(z)$)

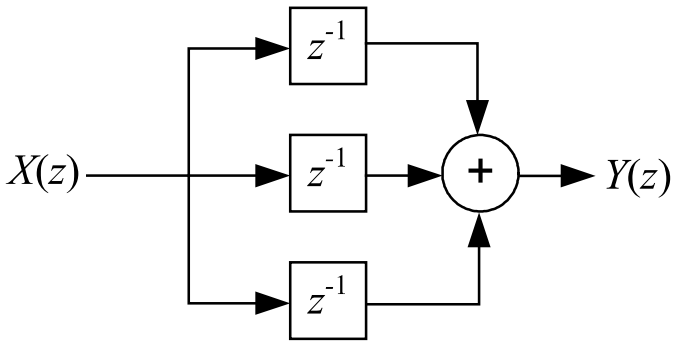


Figure 5.4 Parallel structure

Example: Draw a parallel realisation of a third order system $H(z)$ as given below:

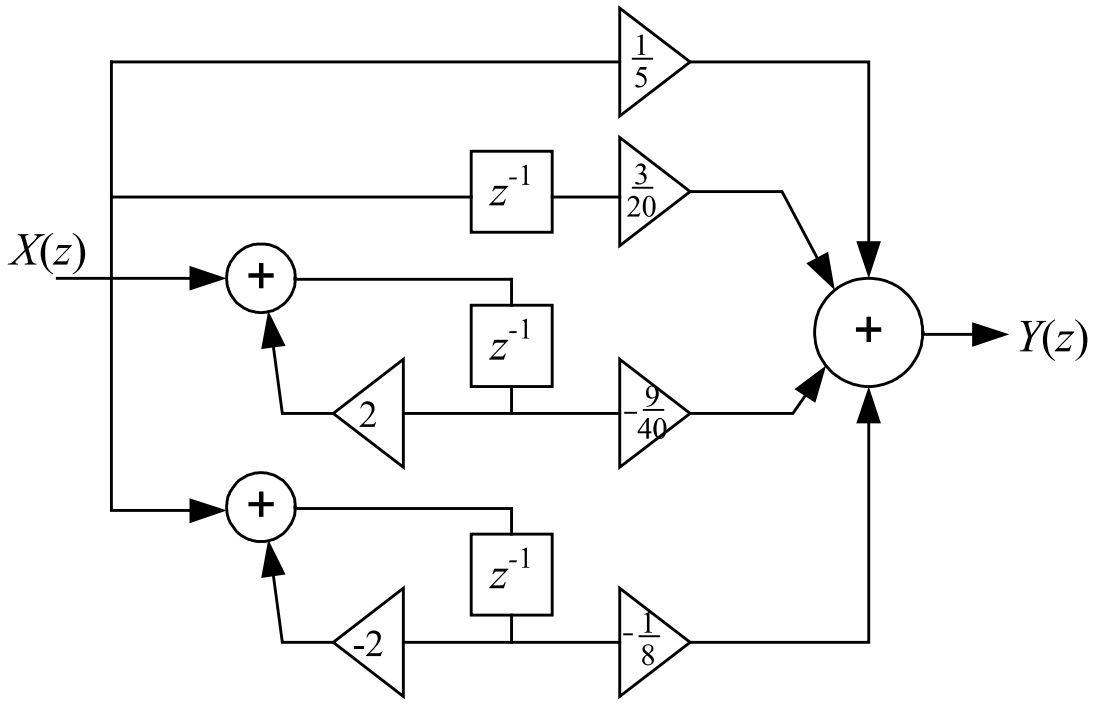
$$H(z) = \frac{z^3 - z^2 - 5z - 3}{5z(z^2 - 4)}$$

By using a partial fraction decomposition, $H(z)$ becomes:

$$H(z) = A + \frac{B}{z} + \frac{C}{z-2} + \frac{D}{z+2}$$

$$H(z) = \frac{1}{5} + \frac{3}{20z} - \frac{9}{40(z-2)} - \frac{1}{8(z+2)}$$

$$H(z) = \frac{1}{5} + \frac{3}{20}z^{-1} - \frac{9z^{-1}}{40(1-2z^{-1})} - \frac{z^{-1}}{8(1+2z^{-1})}$$



5.2.2 Cascade realisation

$$H(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + \cdots + a_M z^{-M}}{1 + b_1 z^{-1} + b_2 z^{-2} + \cdots + b_L z^{-L}}$$

$$H(z) = \prod_{i=1}^{k} \hat{H}_i(z) = \underbrace{\hat{H}_1(z) \cdot \hat{H}_2(z) \cdot \hat{H}_3(z) \ldots \hat{H}_k(z)}_{cascade_structure}$$

(Product of lower order transfer function ie. 1st or 2nd order sections). The cascade structure is the most popular form

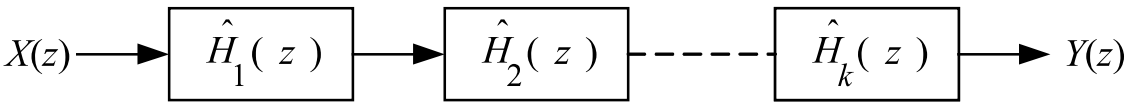


Figure 5.6: Cascade structure

Example: A parallel realisation of a third order system $H(z)$ is given by

$$H(z)=\frac{23+40z^{-1}+36z^{-2}+19z^{-3}}{(2+z^{-1})(5+2z^{-1}+3z^{-2})}$$

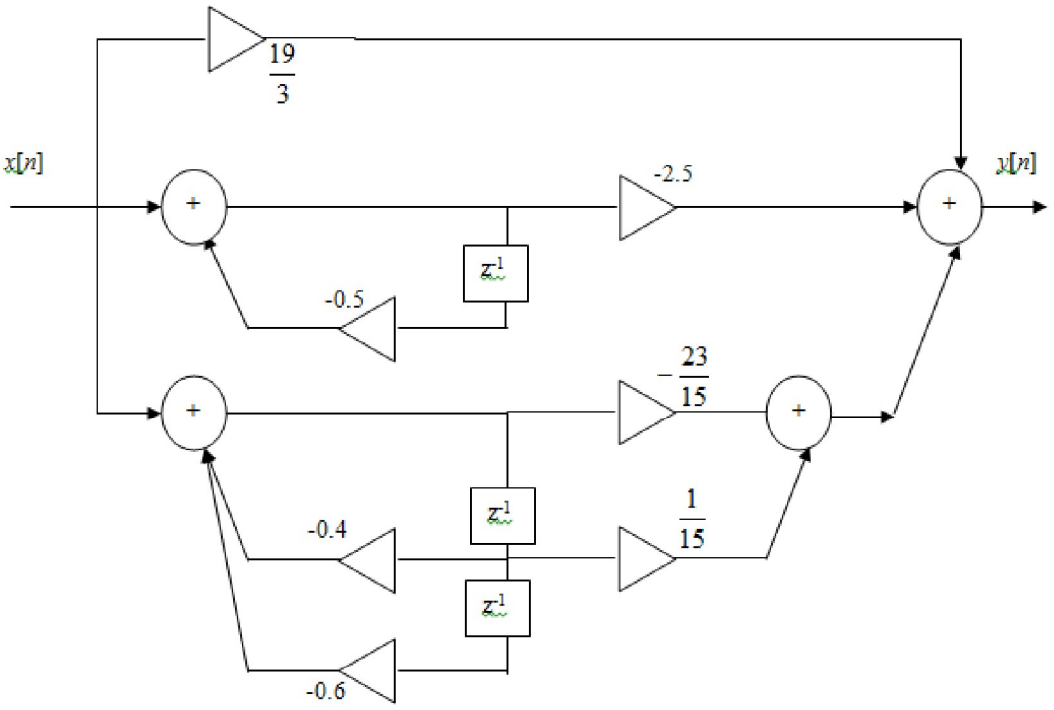
$$=A+\frac{B}{2+z^{-1}}+\frac{C-Dz^{-1}}{5+2z^{-1}+3z^{-2}}$$

$$H_0(z) \qquad H_1(z) \qquad H_2(z)$$

$$H(z)=\frac{19}{3}-\frac{5}{2+z^{-1}}-\frac{1}{3}\frac{(23-z^{-1})}{5+2z^{-1}+3z^{-2}}$$

$$H(z)=\frac{19}{3}-\frac{5}{2}\frac{1}{1+0.5z^{-1}}-\frac{1}{3}\frac{(23-z^{-1})}{5(1+0.4z^{-1}+0.6z^{-2})}$$

$$=\frac{19}{3}+\frac{(-2.5)}{1+0.5z^{-1}}+\frac{(-\frac{23}{15}+\frac{1}{15}z^{-1})}{1+0.4z^{-1}+0.6z^{-2}}$$



Example: A cascade realisation of a third-order system is given by

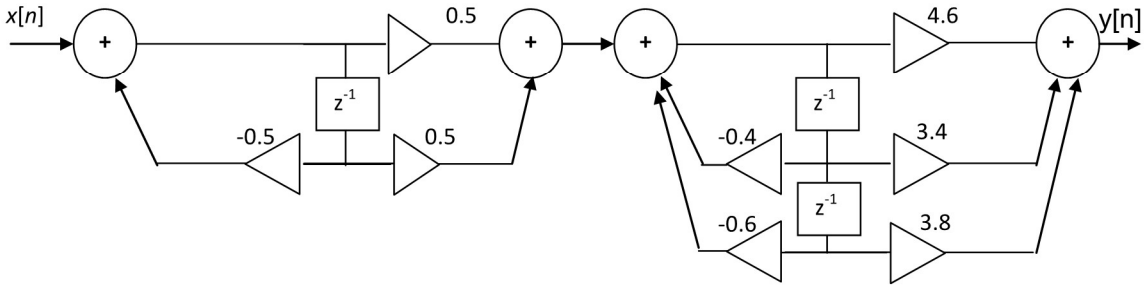
$$H(z)=\frac{23+40z^{-1}+36z^{-2}+19z^{-3}}{10+9z^{-1}+8z^{-2}+3z^{-3}}$$

$$=\frac{(1+z^{-1})}{(2+z^{-1})}\frac{23+17z^{-1}+19z^{-2}}{5+2z^{-1}+3z^{-2}}$$

$$=\frac{(1+z^{-1})}{2(1+0.5z^{-1})}\frac{23+17z^{-1}+19z^{-2}}{5(1+0.4z^{-1}+0.6z^{-2})}$$

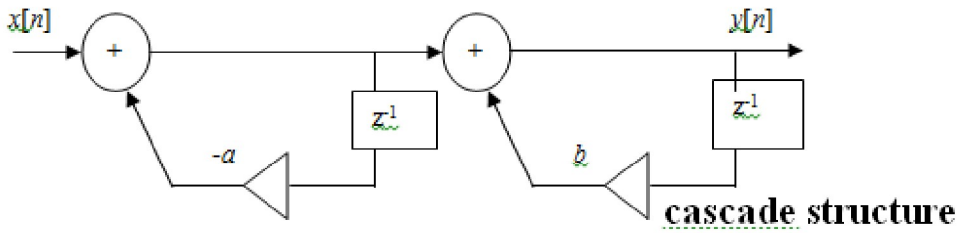
$$=\frac{(0.5+0.5z^{-1})}{1+0.5z^{-1}}\frac{4.6+3.4z^{-1}+3.8z^{-2}}{1+0.4z^{-1}+0.6z^{-2}}$$

$$=\left(\frac{(0.5+0.5z^{-1})}{1+0.5z^{-1}}\right)\left(\frac{4.6+3.4z^{-1}+3.8z^{-2}}{1+0.4z^{-1}+0.6z^{-2}}\right)$$

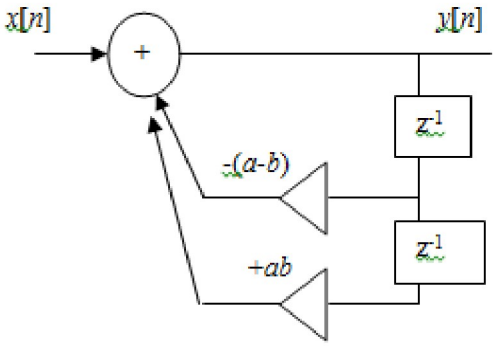


Example: Implement the following system in the cascade, direct form II and parallel structures. All coefficients are real.

(a) $H(z)=\dfrac{1}{(1+az^{-1})(1-bz^{-1})}$

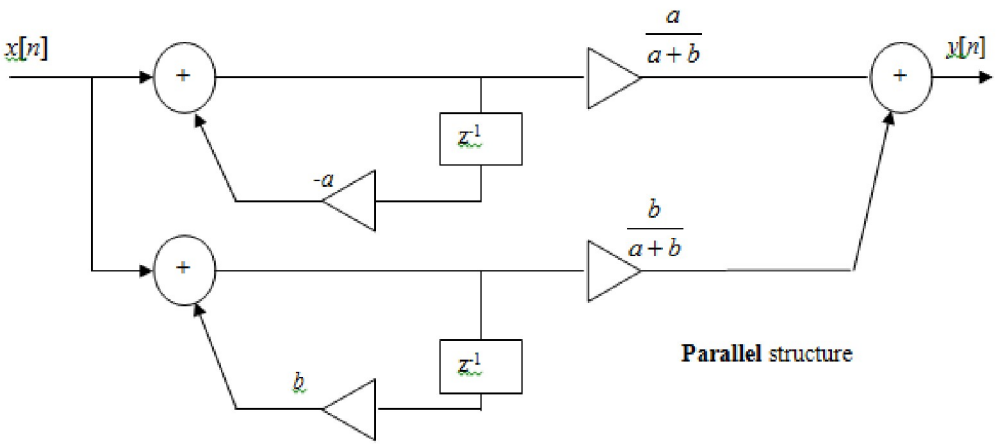


$$H(z)=\frac{1}{1+(a-b)z^{-1}-abz^{-2}}$$



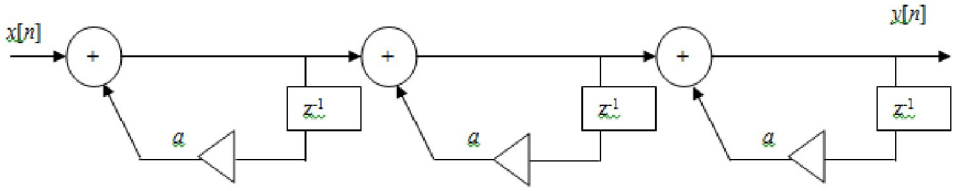
Direct form II

$$H(z)=\frac{A}{1+az^{-1}}+\frac{B}{1-bz^{-1}}=\frac{\frac{a}{a+b}}{1+az^{-1}}+\frac{\frac{b}{a+b}}{1-bz^{-1}}$$



Parallel structure

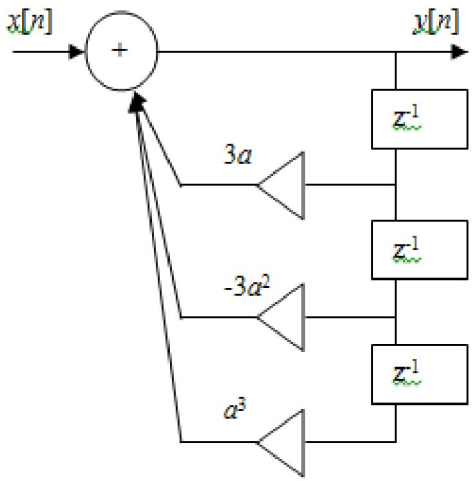
b) $H(z)=\frac{1}{(1-az^{-1})^3}$



cascade structure

No parallel structure exists because partial fraction expansion cannot be performed.

$$H(z)=\frac{1}{1-3az^{-1}+3a^2z^{-2}-a^3z^{-3}}$$



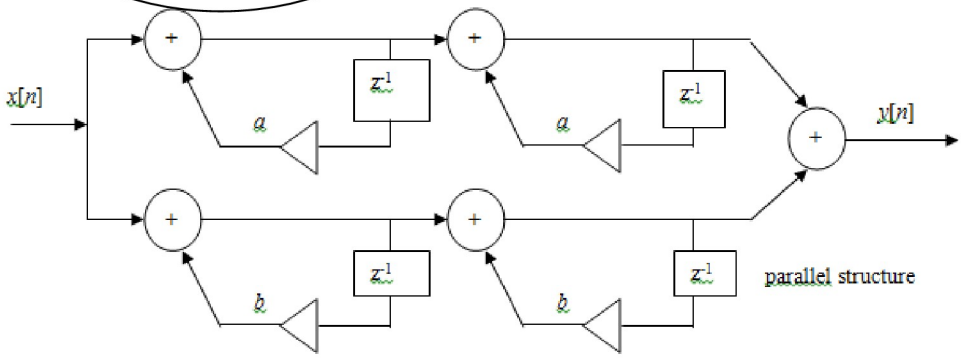
Direct Form II

(c)

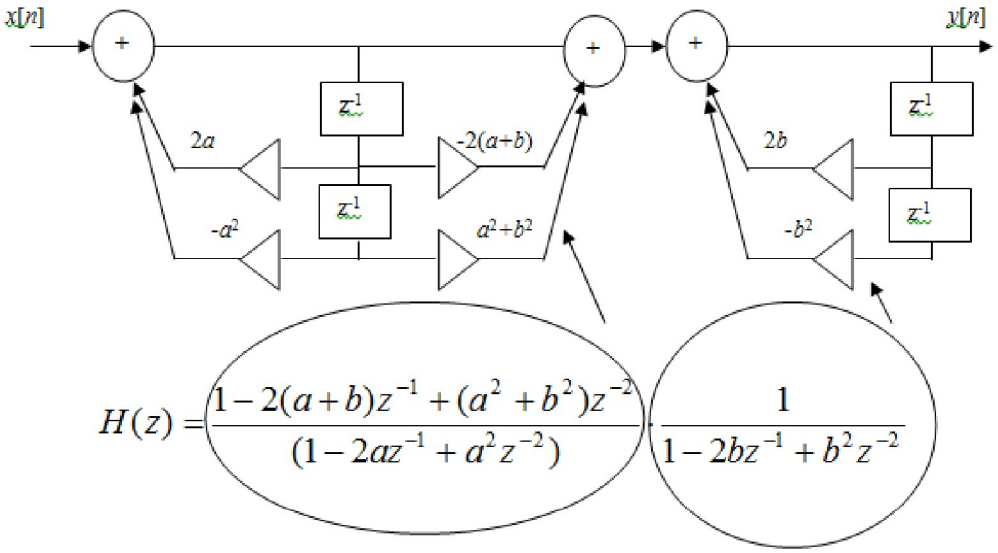
$$H(z)=\frac{1}{(1-za^{-1})^2}+\frac{1}{(1-bz^{-1})^2} \quad \leftarrow \quad parallel$$

$$=\frac{1-2(a+b)z^{-1}+(a^2+b^2)z^{-1}}{(1-2az^{-1}+a^2z^{-2})(1-2bz^{-1}+b^2z^{-2})} \quad \leftarrow \quad cascade$$

$$H(z)=\frac{1}{(1-za^{-1})(1-za^{-1})}+\frac{1}{(1-bz^{-1})(1-bz^{-1})}$$



parallel structure



Exercise:

A first-order digital filter has a transfer function given by:

$$H(z) = \frac{1}{4}\frac{1+z^{-1}}{1-0.5z^{-1}}$$

Draw the following realisations of H(z):
(a) Cascade; (b) Parallel; (c) Canonic.

5.3 Magnitude and Phase Response

- We can show that the magnitude response is an **even** function of frequency.
- The phase response is an **odd** function of frequency.

Example: Calculate the magnitude and phase response of the 3-sample averager given by

$$h[n]=\begin{cases}\frac{1}{3} & -1\le n\le 1\\ 0 & otherwise\end{cases}$$

$$H(z)=\sum_{n=-\infty}^{\infty}h(n)z^{-n}=\sum_{n=-1}^{1}h(n)z^{-n}=\frac{1}{3}z^{-1}+\frac{1}{3}z^{0}+\frac{1}{3}z^{1}$$

$$H(z)=\frac{1}{3}\left[z^{-1}+1+z^{1}\right]$$

$$H(\theta)=H(z)\big|_{z=e^{j\theta}}=\frac{1}{3}\left[e^{-j\theta}+1+e^{j\theta}\right]=\left[1+e^{j\theta}+e^{-j\theta}\right]\cdot\frac{1}{3}$$

$$H(\theta)=\frac{1}{3}\left[1+2\cos\theta\right]$$

Precautions must be taken when determining the phase response of a filter having a real-valued transfer function H(θ), because negative real values produce an additional phase of π radians.

For example, let us consider the following linear-phase form of the transfer function

$$H(\theta)=e^{-jk\theta}B(\theta)$$

real-valued function of θ that can take positive and negative values.

$$H(\theta)=B(\theta)\cos(-k\theta)+B(\theta)j\sin(-k\theta)$$
$$H(\theta)=B(\theta)\cos(k\theta)-jB(\theta)\sin(k\theta)$$

Let phase angle be ϕ:

$$\tan\phi=-\frac{B(\theta)\sin(k\theta)}{B(\theta)\cos(k\theta)}=-\tan(k\theta)$$

$\tan\phi = \tan(-k\theta) \therefore \phi = -k\theta$ or $\phi(\theta) = -k\theta$

phase angle

The phase function $\phi(\theta)$ includes linear phase term and also accommodates for the sign changes in $B(\theta)$. Since -1 can be expressed as $e^{\pm j\pi}$, phase jumps of $\pm\pi$ will occur at frequencies where $B(\theta)$ changes sign.

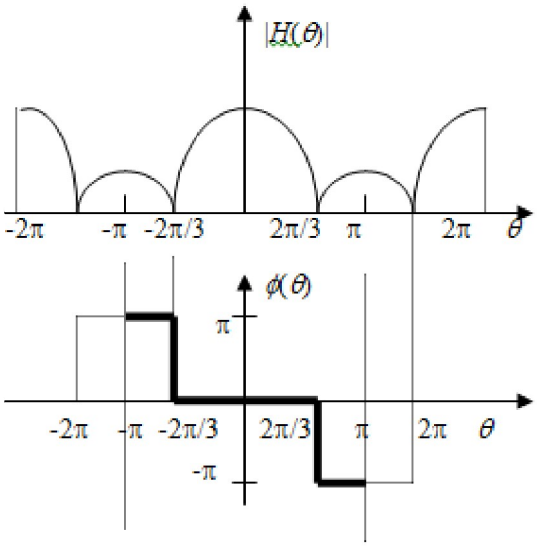
If $B(\theta) > 0$, then $\phi(\theta) = -k\theta$.
If $B(\theta) < 0$, then $\phi(\theta) = -k\theta \pm \pi$..

Let us get back to our example of the 3-sample averager

$$H(\theta) = \frac{1}{3}[1+2\cos\theta] \Rightarrow |H(\theta)| = \left|\frac{1}{3}[1+2\cos\theta]\right|$$

$$\begin{cases} \phi(\theta) = 0 & H(\theta) > 0 \qquad -\frac{2\pi}{3} < \theta < \frac{2\pi}{3} \\ \phi(\theta) = 0 \pm \pi & H(\theta) < 0 \qquad -\pi \le \theta \le -\frac{2\pi}{3} \text{ and } \frac{2\pi}{3} < \theta < \pi \end{cases}$$

The appropriate sign of π must be chosen to make $\phi(\theta)$ an odd function of frequency.



Even function

Odd function

Example: Find the magnitude and phase response of the following: $h(0)=\frac{1}{2},\quad h(1)=1,\quad h(2)=\frac{1}{2},\quad h(n)=0,$ otherwise.

$$H(z)=\frac{1}{2}z^{0}+1z^{-1}+\frac{1}{2}z^{-2}=\frac{1}{2}+z^{-1}+\frac{1}{2}z^{-2}$$

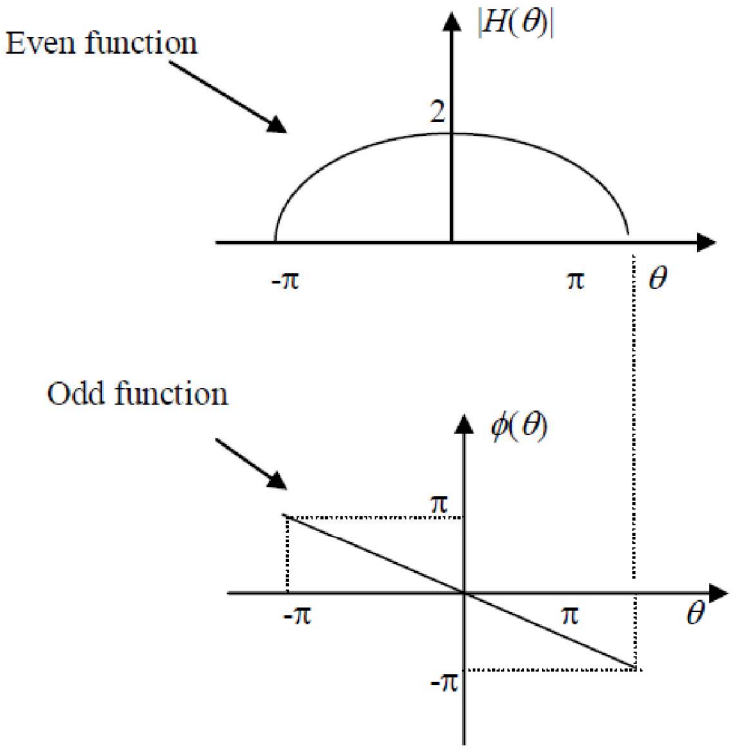
$$H(\theta)=\frac{1}{2}+e^{-j\theta}+\frac{1}{2}e^{-j2\theta}$$

$$=e^{-j\theta}\left[\frac{1}{2}e^{j\theta}+1+\frac{1}{2}e^{-j\theta}\right]$$

$$H(\theta)=e^{-j\theta}\underbrace{[1+\cos\theta]}_{B(\theta)}$$

$$\phi(\theta)=-\theta$$

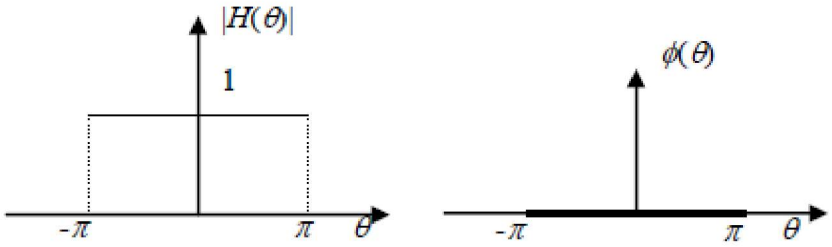
The amplitude function is never negative (therefore there is no phase jumps of $\pm\pi$)



Example:

Case 1:

$$\left.\begin{array}{ll}\delta(n)=1 & n=0\\ \delta(n)=0 & otherwise\end{array}\right\} \quad case\,1$$

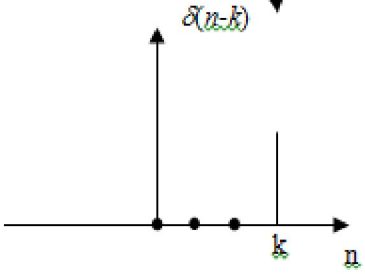


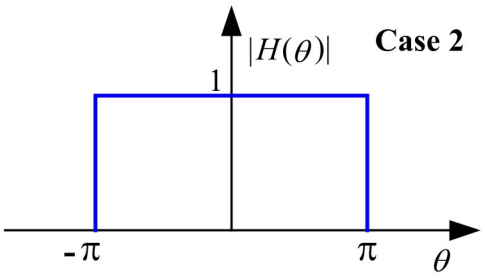
Case 2:

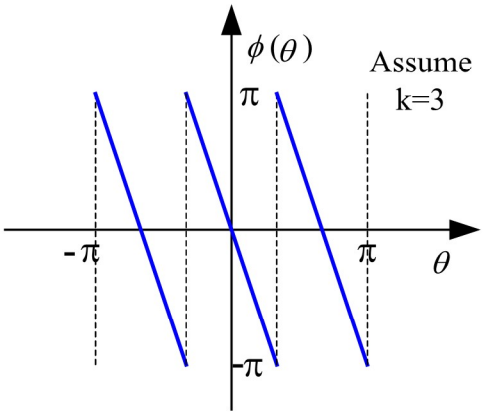
$h[n] = \delta[n-k]$

$H(z) = 1z^{-k}$

$H(\theta) = 1e^{-j\theta k}$







$B(\theta) = 1$

$\phi(\theta) = -\theta k$

Note: When phase exceeds ±π range a jump of ±2π is needed to bring the phase back into ±π range.

Phase Jumps: From the previous examples, we note that there are two occasions for which the phase function experiences discontinuities or jumps.

1. A jump of ±2π occurs to maintain the phase function within the principal value range of [-π and π]
2. A jump of ± π occurs when $B(\theta)$ undergoes a change of sign

The sign of the phase jump is chosen such that the resulting phase function is odd and, after the jump, lies in the range [-π and π].

Example: Magnitude and phase response of causal 3-sample average.

$$h[n]=\begin{cases}\frac{1}{3} & \text{for} \quad 0\le n\le 2\\ 0 & \text{otherwise}\end{cases}$$

$$H(z)=\frac{1}{3}+\frac{1}{3}z^{-1}+\frac{1}{3}z^{-2}\Rightarrow H(\theta)=\frac{1}{3}+\frac{1}{3}e^{-j\theta}+\frac{1}{3}e^{-j2\theta}$$

$$=\frac{1}{3}e^{-j\theta}\left[e^{j\theta}+1+e^{-j\theta}\right]$$

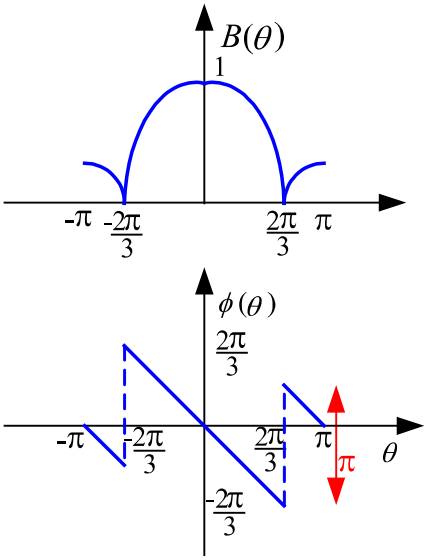
$$\therefore\ H(\theta)=e^{-j\theta}\underbrace{\frac{1}{3}[1+2\cos\theta]}_{B(\theta)}$$

$$\phi(\theta)=-\theta;\qquad |H(\theta)|=|B(\theta)|=\frac{1}{3}[1+2\cos\theta]$$

$$\phi(\theta)=-\theta \qquad B(\theta)>0 \qquad -\frac{2\pi}{3}<\theta<\frac{2\pi}{3}$$

$$=-\theta\pm\pi \qquad B(\theta)<0 \qquad -\pi<\theta<-\frac{2\pi}{3}$$

$$\frac{2\pi}{3}<\theta<\pi$$



Phase is undefined at points $|H(\theta)| = 0$ or B(θ)=0.

Example: Determine and sketch the magnitude and phase response of 1st order recursive filter (IIR filter)

$$y[n] = x[n] + ay[n-1]$$

$$\frac{Y(z)}{X(z)} = H(z) = \frac{1}{1-az^{-1}} \qquad H(\theta) = \frac{1}{1-ae^{-j\theta}}$$

$$H(\theta) = \frac{1}{1-ae^{-j\theta}} \cdot \frac{1-ae^{j\theta}}{1-ae^{j\theta}} = \frac{1-a\cos\theta}{1-2a\cos\theta+a^2} - \frac{ja\sin\theta}{1-2a\cos\theta+a^2}$$

$$\tan\phi = \frac{-\dfrac{a\sin\theta}{1-2a\cos\theta+a^2}}{\dfrac{1-a\cos\theta}{1-2a\cos\theta+a^2}} = \frac{-a\sin\theta}{1-a\cos\theta}$$

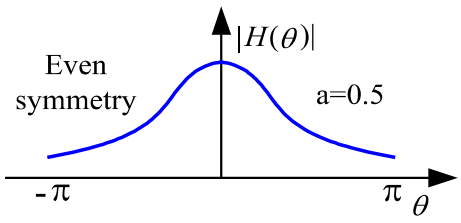
$$\phi(\theta) = \tan^{-1}\left[\frac{-a\sin\theta}{1-a\cos\theta}\right] \qquad \textit{phase}$$

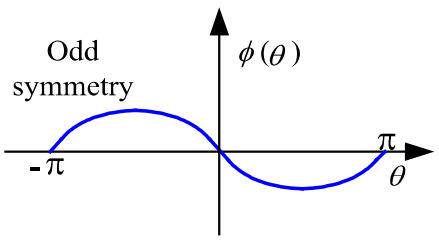
Magnitude:

$|H(\theta)|^2 = H(\theta)\cdot H^*(\theta)$ ($H^*(\theta)$ is the complex conjugate of $H(\theta)$)

$$= \frac{1}{1-ae^{-j\theta}} \cdot \frac{1}{1-ae^{j\theta}} \qquad 0 < a < 1$$

$$\boxed{|H(\theta)| = \frac{1}{1-2a\cos\theta+a^2}}$$





Example:

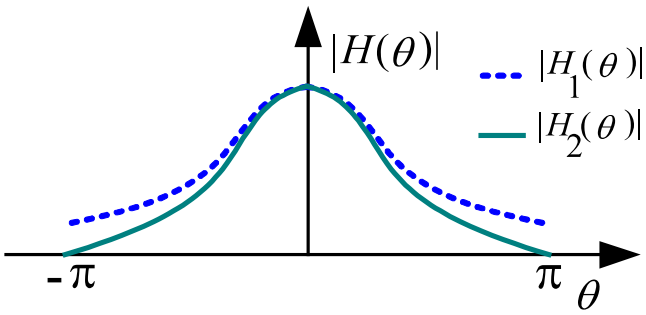
(a) $H_1(z) = \dfrac{k_0}{1 - az^{-1}} \Rightarrow \dfrac{1-a}{1-az^{-1}}$ (Low pass filter)

The gain k_0 can be selected as $1 - a$, so that the filter has unity gain at $\theta = 0$.

(b) $H_2(z) = k_0 \dfrac{1+z^{-1}}{1-az^{-1}}$

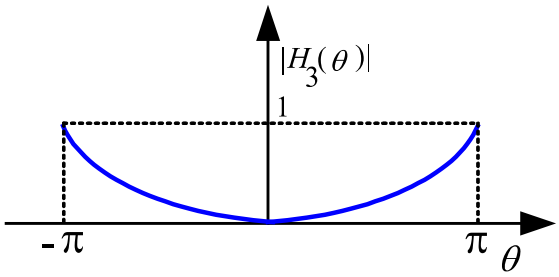
In this case, $k_0 = \dfrac{1-a}{2}$ (for unity gain at $\theta = 0$).

The addition of a zero at $z = -1$ further attenuates the response of the filter at high frequencies



(c) We can obtain simple highpass filters by reflecting (folding) the pole-zero locations of the lowpass filters about the imaginary axis in the z-plane.

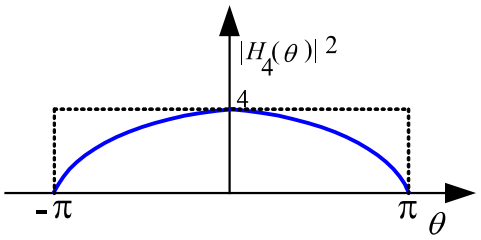
$H_3(z) = \dfrac{1-a}{2} \cdot \dfrac{1-z^{-1}}{1+az^{-1}}$ high pass filter



(d)

$$H_4(z) = 1 + z^{-1}$$
$$y[n] = x[n] + x[n-1]$$ lowpass filter

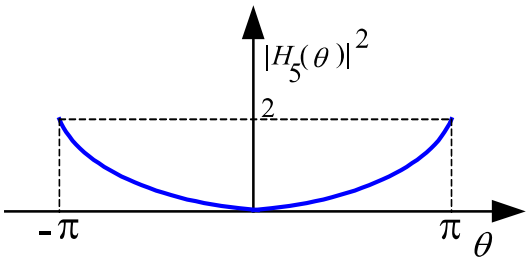
$$|H_4(\theta)|^2 = 2(1+\cos\theta)$$



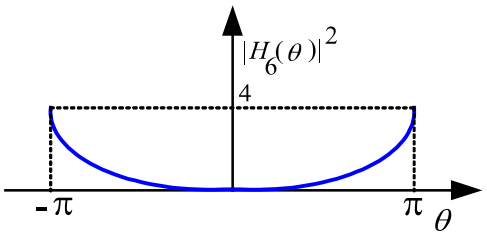
(e)

$$H_5(z) = 1 - z^{-1}$$
$$y[n] = x[n] - x[n-1]$$ high pass filter

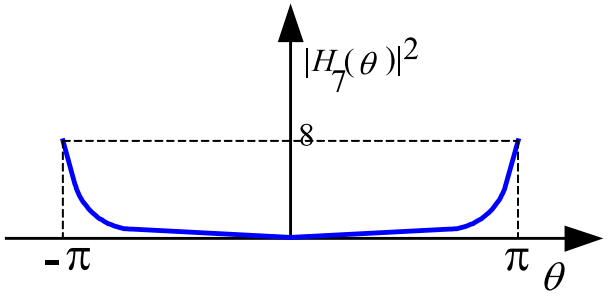
$$|H_5(\theta)|^2 = 2(1-\cos\theta)$$



(f) $H_6(z) = (1 - z^{-1})^2$



(g) $H_7(z) = (1 - z^{-1})^3$

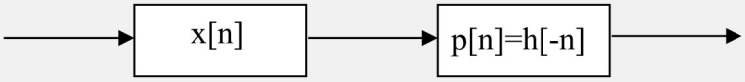


Exercise:
(a) Calculate and Sketch the Magnitude and Phase responses of the following FIR filter

$$y[n] = \tfrac{1}{4}\left[x[n] - 2x[n-1] + x[n-2]\right]$$

(b) Show that if x[n] is a real valued signal, the structure shown below has a zero-phase response; where h[n] is the impulse response of the filter H(z) given by,

$$H(z) = \left(\frac{1}{2} + z^{-1} + \frac{1}{2}z^{-2}\right)$$

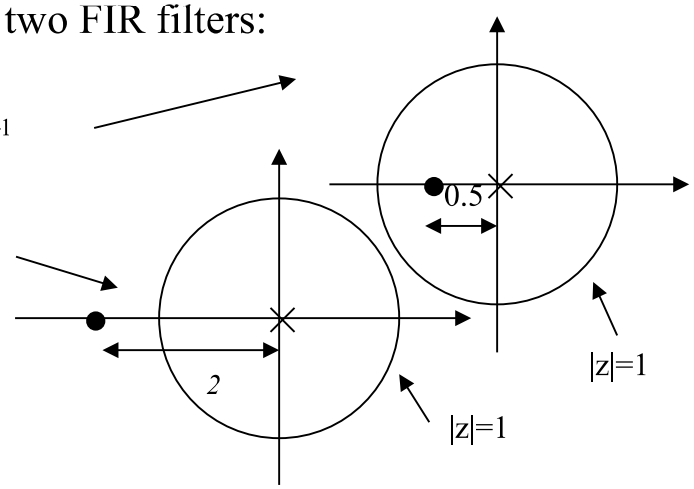


5.4 Minimum, Maximum and Mixed phase systems

Let us consider two FIR filters:

$$H_1(z) = 1 + \frac{1}{2}z^{-1}$$

$$H_2(z) = \frac{1}{2} + z^{-1}$$



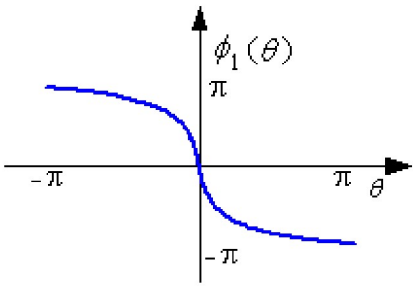
$H_2(z)$ is the reverse of the system $H_1(z)$. This is due to the reciprocal relationship between the zeros of $H_1(z)$ and $H_2(z)$.

$$H_1(\theta) = 1 + \frac{1}{2}e^{-j\theta} \;\&\; H_2(\theta) = \frac{1}{2} + e^{-j\theta}$$

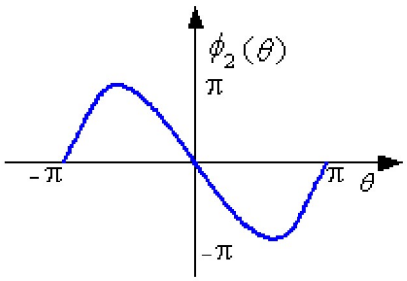
$$|H_1(\theta)| = |H_2(\theta)| = \sqrt{\frac{5}{4} + \cos\theta}$$

The magnitude characteristics for the two filters are identical because the roots of $H_1(z)$ and $H_2(z)$ are reciprocal.

$$\phi_1(\theta) = \tan^{-1}\frac{-\sin\theta}{2+\cos\theta}$$



$$\phi_2(\theta) = \tan^{-1}\frac{-\sin\theta}{\frac{1}{2}+\cos\theta}$$



Note: If we reflect a zero with magnitude $|z| = \rho$ that is inside the unit circle into a zero with magnitude $|z| = \frac{1}{\rho}$ outside the unit circle the magnitude characteristic of the system is unaltered, but the phase response changes.

We observe that the phase characters $\phi_1(\theta)$ begins at zero phase at frequency $\theta = 0$ and terminates at zero phase at the frequency $\omega = \pi$. Hence the net phase change.

$$\boxed{\phi_1(\pi) - \phi_1(0) = 0}$$ ← Minimum phase filter

On the other hand, the phase characteristic for the filter with the zero outside the unit circle undergoes a net phase change

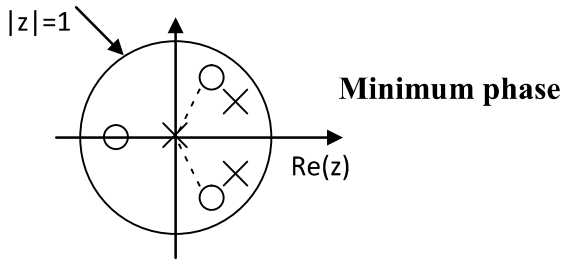
$$\boxed{\phi_2(\pi) - \phi_2(0) = -\pi \text{ radians}}$$

As a consequence of these different phase characteristics, we call the first filter a **minimum-phase** system and the second system is called a **maximum-phase** system.

If a filter with M zeros has some of its zeros inside the unit circle and the remaining outside the unit circle, it is called a **mixed-phase** system. A minimum-phase property of FIR filter carries over to IIR filter. Let us consider

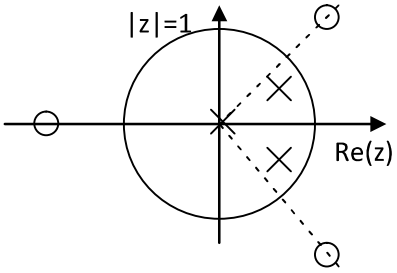
$$H(z) = \frac{B(z)}{A(z)}$$

is called minimum phase if all its poles and zeros are inside the unit circle.



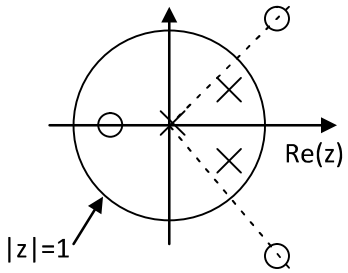
Minimum phase

If all the zeros lie outside the unit circle, the system is called maximum phase.



Maximum phase

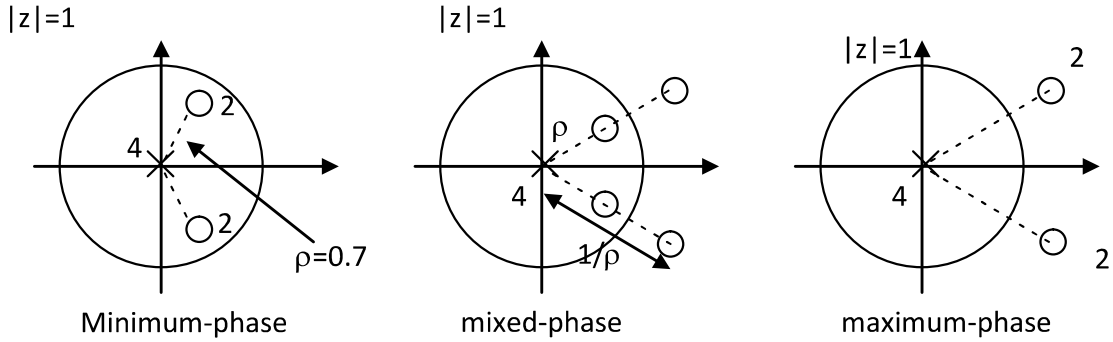
If zeros lie both inside and outside the unit circle, the system is called mixed-phase.



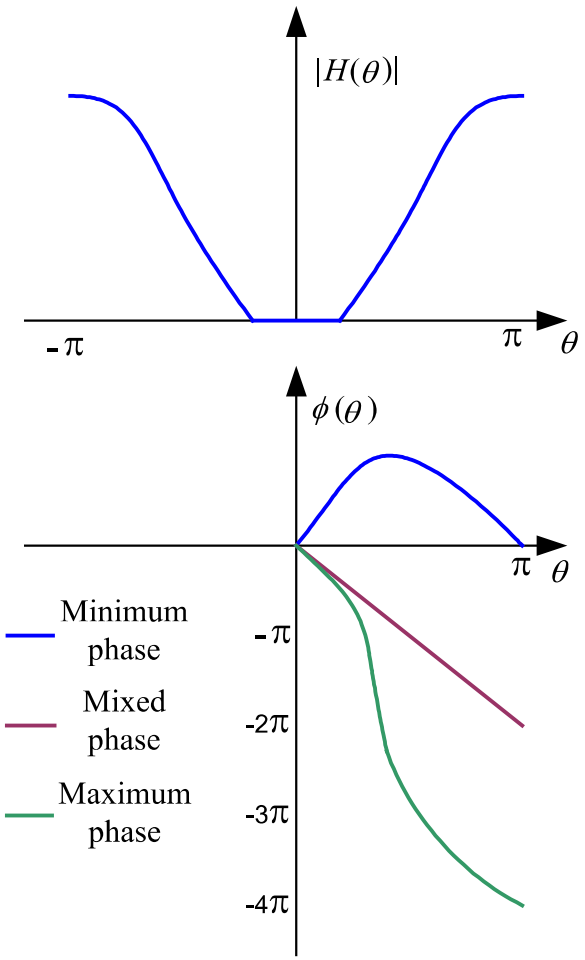
Mixed phase

Note: For a given magnitude response, the minimum-phase system is the causal system that has the smallest magnitude phase at every frequency (θ). That is, in the set of causal and stable filters having the same magnitude response, the minimum-phase response exhibits the smallest deviation from zero phase.

Example: Consider a fourth-order all-zero filter containing a double complex conjugate set of zeros located at $z = 0.7e^{\pm j\frac{\pi}{4}}$. The minimum-phase, mixed phase and maximum phase system pole-zero patterns having identical magnitude response are shown below.



The magnitude response and the phase response of the three systems are shown below: The minimum-phase system seems to have the phase with the smallest deviation from zero at each frequency.



*Mixed-phase is linear in this case

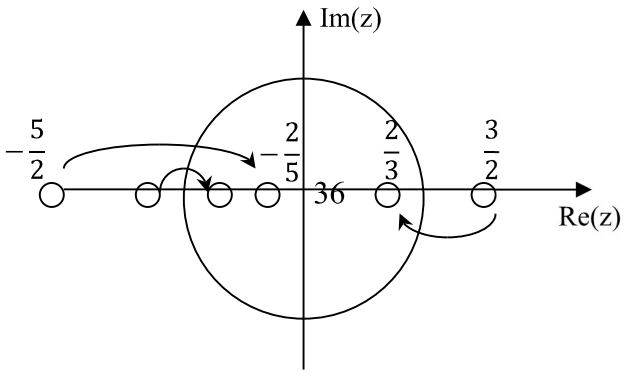
Example: A third order FIR filter has a transfer function $G(z)$ given by: $G(z) = (6 - z^{-1} - 12z^{-2})(2 + 5z^{-1})$

From $G(z)$, determine the transfer function of an FIR filter whose magnitude response is identical to that of $G(z)$ and has a minimum phase response.

$$G(z) = (2 - 3z^{-1})(3 + 4z^{-1})(2 + 5z^{-1})$$

$$G(z) = 12(1 - \frac{3}{2}z^{-1})(1 + \frac{4}{3}z^{-1})(1 + \frac{5}{2}z^{-1})$$

> 1



$-\frac{4}{3}$ $-\frac{3}{4}$

The minimum phase filter $P(z) = k(1-\frac{2}{3}z^{-1})(1+\frac{3}{4}z^{-1})(1+\frac{2}{5}z^{-1})$

Exercise:

The transfer function of a minimum-phase FIR filter is given by

$$G(z) = \left(-3 + 2z^{-1}\right)\left(4 + 3z^{-1}\right)\left(5 + 2z^{-1}\right)$$

From G(z) determine a transfer function of an FIR filter whose magnitude response is identical to that of G(z).

5.5 All-Pass Filters

An all-pass filter is one whose magnitude response is constant for all frequencies, but whose phase response is not identically zero.

[The simplest example of an all-pass filter is a pure delay system with system function $H(z) = z^{-k}$]

A more interesting all-pass filter is one that is described by

$$H(z) = \frac{a_L + a_{L-1}z^{-1} + \cdots + a_1 z^{-L+1} + a_0 z^{-L}}{1 + a_1 z^{-1} + \cdots + a_L z^{-L}},$$

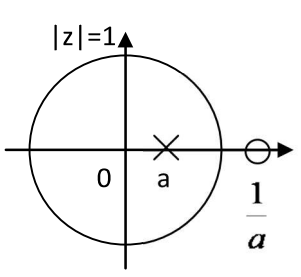
where $a_0 = 1$ and all coefficients are real. If we define the polynomial $A(z)$ as

$$A(z) = \sum_{k=0}^{L} a_k z^{-k} \qquad a_0 = 1$$

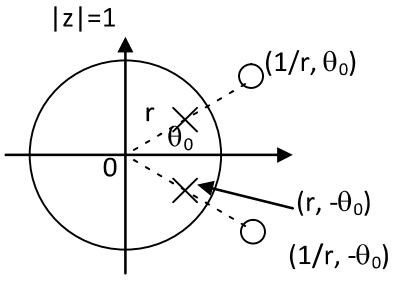
$$\therefore H(z) = z^{-L}\frac{A(z^{-1})}{A(z)} \Rightarrow |H(\theta)|^2 = H(z)\cdot H(z^{-1})\big|_{z=e^{j\theta}} = 1$$

i.e. all pass filter.

Furthermore, if $|z_0|$ is the modulus of a pole of $H(z)$, then $1/|z_0|$ is the modulus of a zero of $H(z)$ {i.e. the modulus of poles and zeros are reciprocals of one another}. The figure shown below illustrates typical pole-zero patterns for a single-pole, single-zero filter and a two-pole, two-zero filter.



All-pass



All pass filter

$$H(z)=\frac{1-\frac{1}{a}z^{-1}}{1-az^{-1}} \qquad |a|<1 \text{ for stability}$$

We can easily show that the magnitude response is constant.

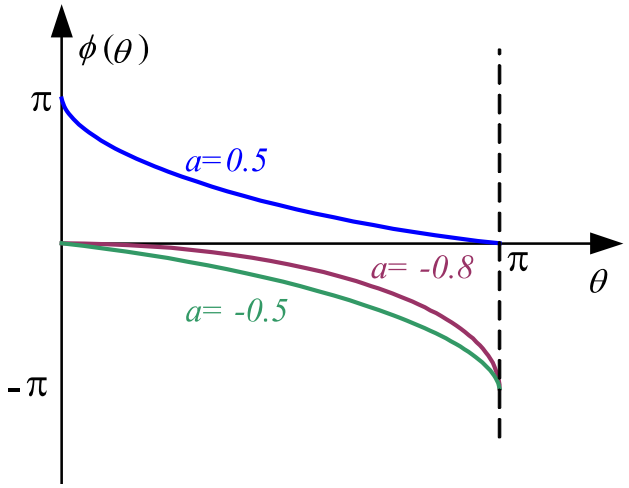
$$|H(\theta)|^2=|H(\theta)\cdot H^*(\theta)=H(z)\cdot H(z^{-1})|_{z=e^{-j\theta}}$$

$$=\frac{1-\frac{1}{a}e^{-j\theta}}{1-ae^{-j\theta}}\cdot\frac{1-\frac{1}{a}e^{j\theta}}{1-ae^{j\theta}}=\frac{1-\frac{2}{a}\cos\theta+\frac{1}{a^2}}{1-2a\cos\theta+a^2}=a^2$$

Phase response:

$$H(\theta)=\frac{1-\frac{1}{a}e^{-j\theta}}{1-ae^{-j\theta}}\cdot\frac{1-ae^{j\theta}}{1-ae^{j\theta}}=\frac{2-(a+a^{-1})\cos\theta-j(a-a^{-1})\sin\theta}{1-2a\cos\theta+a^2}$$

$$\therefore\ \phi(\theta)=\tan^{-1}\left[\frac{-(a-a^{-1})\sin\theta}{2-(a+a^{-1})\cos\theta}\right]$$



When $0 < a < 1$, the zero lies on the positive real axis. The phase over $0 \le \theta \le \pi$ is positive, at $\theta = 0$ it is equal to π and decreases until $\omega = \pi$, where it is zero.

When $-1 < a < 0$, the zero lies on the negative real axis. The phase over $0 \le \theta \le \pi$ is negative, starting at 0 for $\theta = 0$ and decreases to $-\pi$ at $\omega = \pi$.

Example: Show that the following transfer function $H_1(z)$ can be obtained using a parallel connection of two all-pass filters.

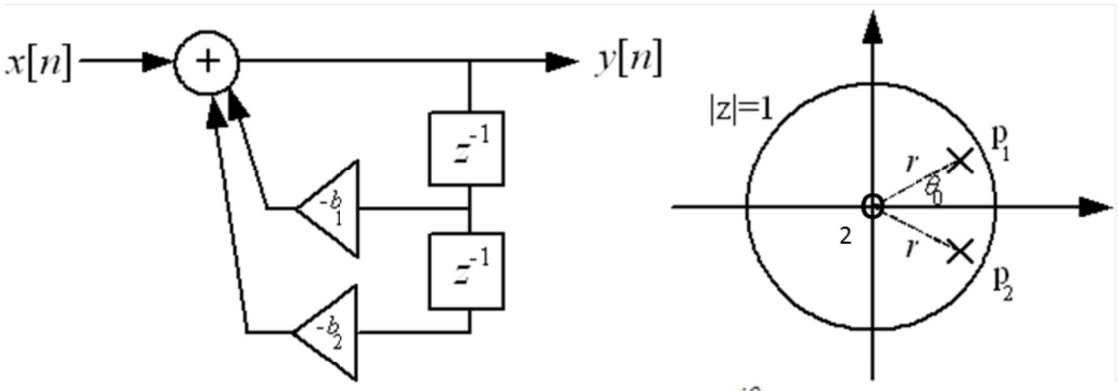
$$H_1(z) = \frac{10 + 6z^{-1}}{3 + z^{-1}}$$

$$H_1(z) = \frac{9 + 3z^{-1} + 1 + 3z^{-1}}{3 + z^{-1}}$$

$$= 3 + \frac{1 + 3z^{-1}}{3 + z^{-1}}$$

$$H_1(z) = \underbrace{3}_{All-pass\ filter} + \underbrace{\frac{1 + 3z^{-1}}{3 + z^{-1}}}_{All-pass\ filter}$$

5.6 A second Order Resonant Filter



$$p_1 = re^{j\theta_0} = r\cos\theta_0 + jr\sin\theta_0$$

$$p_2 = re^{-j\theta_0} = r\cos\theta_0 - jr\sin\theta_0$$

$$H(z) = \frac{1}{1 + b_1 z^{-1} + b_2 z^{-2}} = \frac{z^2}{z^2 + b_1 z + b_2} \qquad \text{(A)}$$

All pole systems has poles only (without counting the zeros as the origin)

$$H(z) = \frac{z^2}{z^2 + b_1 z^{-1} + b_2} = \frac{z^2}{(z - p_1)(z - p_2)} = \frac{z^2}{(z - re^{j\theta_0})(z - re^{-j\theta_0})}$$

$$H(z) = \frac{z^2}{z^2 - r(e^{j\theta_0} + e^{-j\theta_0})z + r^2} = \frac{z^2}{z^2 - 2r\cos\theta_0 z + r^2} \qquad \text{(B)}$$

Comparing (A) and (B), we obtain

$b_1 = -2r\cos\theta_0$ and $b_2 = r^2$

$$\therefore cos\theta_0 = -\frac{b_1}{2\sqrt{b_2}} \qquad \theta_0 = \frac{2\pi f_0}{f_s}$$

θ_0= resonant frequency

Exercise:

An all poles system H(z) has two poles $p_{1,2} = 0.8e^{\pm j\frac{\pi}{3}}$. Determine the transfer function of H(z) and sketch pole-zero plot of H(z).

5.7 Stability of a second-order filter

Consider a two-pole resonant filter given by

$$H(z) = \frac{z^2}{z^2 + b_1 z + b_2} = \frac{1}{1 + b_1 z^{-1} + b_2 z^{-2}}$$

b_1 & b_2 are coefficients

This system has two zeros at the origin and poles at

$$p_1, p_2 = -\frac{b_1}{2} \pm \frac{\sqrt{b_1^{\,2} - 4b_2}}{2}$$

- The filter is stable if the poles lies inside the unit circle i.e.
  $|p_1| < 1$ & $|p_2| < 1$

- For stability $b_2 < 1$. If $b_2 = 1$ then the system is an oscillator (Marginally stable)

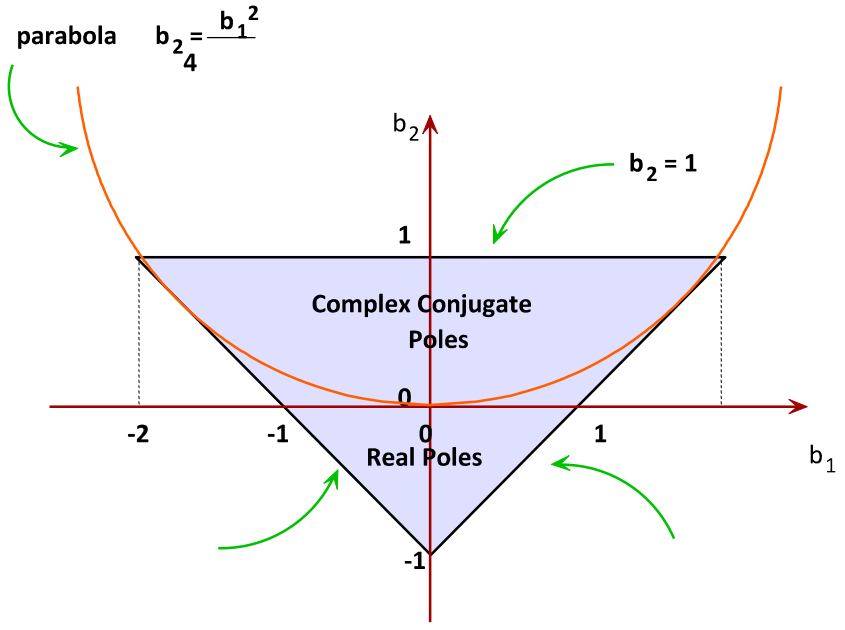
- Assume that the poles are complex

  ie. $b_1^2 - 4b_2 < 0 \Rightarrow b_1^2 < 4b_2$ and $b_1 < \pm 2\sqrt{b_2}$ $\quad b_2 > 0$

- If $b_1^2 - 4b_2 \geq 0$ then we get real roots.

The stability conditions define a region in the coefficient plane (b_1, b_2) which is in the form of a triangle (see below)

The system is only stable if and only if the point (b_1, b_2) lies inside the **stability triangle**.



Stability Triangle

If the two poles are real then they must have a value between -1 and 1 for the system to be stable.

$$-1 < \frac{-b_1 \pm \sqrt{b_1^2 - 4b_2}}{2} < 1$$

$$-2 + b_1 < \pm\sqrt{b_1^2 - 4b_2} < 2 + b_1$$

$$\therefore -2 + b_1 < -\sqrt{b_1^2 - 4b_2} \text{ and } \sqrt{b_1^2 - 4b_2} < 2 + b_1$$

$$(-2 + b_1)^2 > b_1^2 - 4b_2 \text{ and } b_1^2 - 4b_2 < (2 + b_1)^2$$

$$b_1 - b_2 - 1 < 0 \text{ and } b_1 + b_2 + 1 > 0$$

- The region below the parabola ($b_1^2 > 4b_2$) corresponds to real and distinct poles.
- The points on the parabola ($b_1^2 = 4b_2$) result in real and equal (double) poles.
- The points above the parabola correspond to complex-conjugate poles.

Exerise:

(a) Determine the stability region (triangle) for following the causal system

$$H(z) = \frac{1}{1 + b_1 z^{-1} + b_2 z^{-2}}$$

Your answer should include, sketch and equations of the boundaries of the region. Also, indicate the region where complex conjugate poles exist.

(b) The transfer function of a second order digital filter is given by $H(z) = \frac{1}{1 + b_1 z^{-1} + b_2 z^{-2}}$

Assuming $b_1^2 - 4b_2 < 0$ show that the resonant frequency (θ_0) of this filter is given by $\theta_0 = \cos^{-1}\left[\frac{-b_1}{2\sqrt{b_2}}\right]$

5.8 Digital Oscillators

A digital oscillator can be made using a second order discrete-time system, by using appropriate coefficients. A difference equation for an oscillating system is given by

$$p[n] = A\cos(n\theta)$$

From the table of z-transforms we know that the z-transform of $p[n]$ above is

$$P(z) = \frac{1 - \cos\theta z^{-1}}{1 - 2\cos\theta z^{-1} + z^{-2}}$$

Let $P(z) = \dfrac{Y(z)}{X(z)} = \dfrac{1 - \cos\theta z^{-1}}{1 - 2\cos\theta z^{-1} + z^{-2}}$,

Taking inverse z-transform on both sides, we obtain

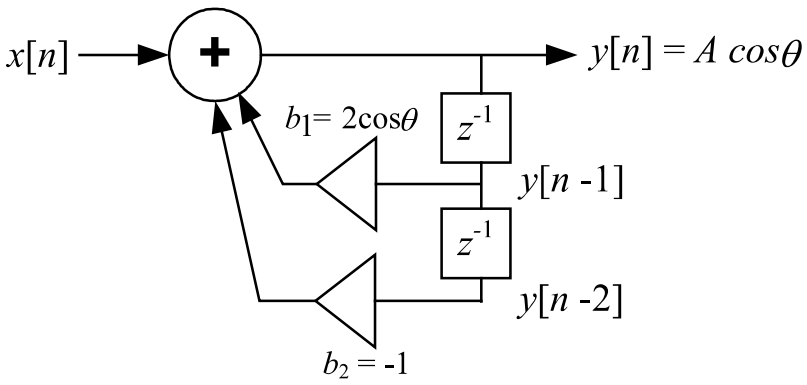
$$y[n] - 2\cos\theta y[n-1] + y[n-2] = x[n] - \cos\theta x[n-1]$$

An oscillator has no input. i.e., $x[n] = 0$ and $x[n-1] = 0$

So the equation of the digital oscillator becomes

$$y[n] = 2\cos\theta y[n-1] - y[n-2]$$

and its structure is shown below:



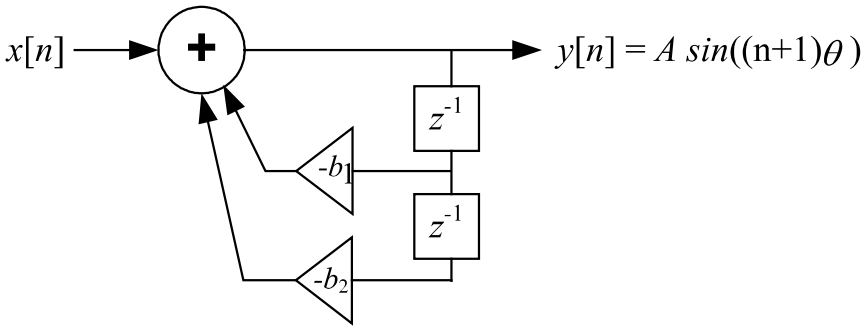
To obtain $y[n] = A\cos(n\theta)$, use the following initial conditions:

$$y[0] = A\cos(0.\theta) = A$$
$$y[-1] = A\cos(-1.\theta) = A\cos\theta$$

The frequency can be tuned by changing the coefficient b_1 (b_2 is a constant). The resonant frequency θ of the oscillator is,

$$\cos\theta = \frac{-b_1}{2\sqrt{b_2}} = -\frac{b_1}{2} \text{ (For an oscillator } b_2 = 1)$$

Example: A digital sinusoidal oscillator is shown below.



(a) Assuming θ_0 is the resonant frequency of the digital oscillator, find the values of b_1 and b_2 for sustaining the oscillation.

$$H(z) = \frac{K}{1 + b_1 z^{-1} + b_2 z^{-2}} = \frac{K}{(z - re^{j\theta_0})(z - re^{-j\theta_0})} = \frac{K}{z^2 - r(e^{j\theta_0} + e^{-j\theta_0})z + r^2}$$

$$= \frac{K}{z^2 - 2r\cos\theta_0 z + r^2}$$

$$\therefore b_1 = -2r\cos\theta_0; \qquad b_2 = r^2$$

For oscillation $b_2 = 1 \Rightarrow r = 1 \qquad \therefore b_1 = -2\cos\theta_0$

(b) Write the difference equation for the above figure. Assuming

$$x[n] = (A\sin\theta_0)\delta[n], \text{ and } y(-1) = y(-2) = 0.$$

Show, by analysing the difference equation, that the application of an impulse at $n = 0$ serves the purpose of beginning the sinusoidal oscillation, and prove that the oscillation is self-sustaining thereafter.

$$y[n] = -b_1 y[n-1] - y[n-2] + x[n]$$
$$y[n] = 2\cos\theta_0 y[n-1] - y[n-2] + A\sin\theta_0\delta[n]$$

$n = 0$

$$y[0] = 2\cos\theta_0\, y[-1] - y[-2] + A\sin\theta_0\,\delta[0]$$

$y[-1]$ → 0, $y[-2]$ → 0, $\delta[0]$ → 0

$$y[0] = A\sin\theta_0$$

$n = 1$

$$y[1] = 2\cos\theta_0\, y[0] - y[-1] + A\sin\theta_0\,\delta[1]$$

$y[0]$ → 0, $y[-1]$ → 0

$$y[1] = 2\cos\theta_0\, A\sin\theta_0 = A\sin 2\theta_0$$

$n = 2$

$$\begin{aligned} y[2] &= 2\cos\theta_0\, y[1] - y[0] + A\sin\theta_0\delta[2] \\ &= 2\cos\theta_0\, A\sin 2\theta_0 - A\sin\theta_0 \\ &= 2A\cos\theta_0\,[2\sin\theta_0\cos\theta_0] - A\sin\theta_0 \\ &= A\sin\theta_0\,[4\cos^2\theta_0 - 1] = A[3\sin\theta_0 - 4\sin^3\theta_0] \end{aligned}$$

where $\sin 3\theta_0 = 3\sin\theta_0 - 4\sin^3\theta_0$

$$y[2] = A\sin 3\theta_0 \text{ and so forth}$$

(c) By setting the input to zero and under certain initial conditions, sinusoidal oscillation can be obtained using the structure shown above. Find these initial conditions.

$$y[n] = 2\cos\theta y[n-1] - y[n-2] + x[n]$$

($x[n] = 0$ for an oscillator)

$n = 0 \qquad y[0] = 2\cos\theta_0\, y[-1] - y[-2]$

for oscillation, $y[-1] = 0 \Rightarrow$ no cosine terms

$y[0] = -y[-2] \Rightarrow y[-2] = -A\sin\theta_0$ (sine term is required)
$y[0] = 0-(-A\sin\theta_0) = A\sin\theta_0$

Initial conditions: $y[-1] = 0; y[-2] = -A\sin\theta_o$

Exercise:

The transfer function of a second order digital oscillator with an oscillation frequency of θ_0 is given by

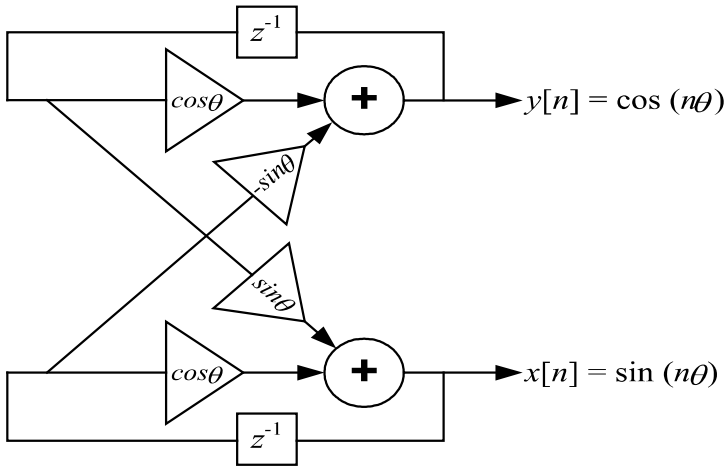
$$H(z) = \frac{z}{z^2 - b_1 z + 1}; b_1 = 2\cos\theta_0$$

Show that the impulse response corresponding to the system function H(z) is given by

$$h(n) = \frac{\sin(n\theta_0)\quad u(n)}{\sin(\theta_0)}$$

5.8.1 Sine and cosine oscillators

Sinusoidal oscillators can be used to deliver the carrier in modulators. In modulation schemes, both sines and cosines oscillators are needed. A structure that delivers sines and cosines simultaneously is shown below:



A simultaneous sine and cosine oscillator

Proof:

Trigonometric equation for $cos(n+1)\theta$ is:

$$\cos((n+1)\theta) = \cos(n\theta)\cos(\theta) - \sin(n\theta)\sin(\theta)$$

Let $y[n] = \cos(n\theta)$ and $x[n] = \sin(n\theta)$

$$\therefore\ y[n+1] = \cos(\theta)\, y[n] - \sin(\theta)\, x[n]$$

Replace n by n-1: $y[n] = \cos(\theta)\, y[n-1] - \sin(\theta)\, x[n-1]$ (A)

Similarly

$$\sin((n+1)\theta) = \sin(\theta)\cos(n\theta) + \sin(n\theta)\cos(\theta)$$
$$\therefore\ x[n+1] = \sin(\theta)\, y[n] + x[n]\cos(\theta)$$

Replace $n \rightarrow n$-1

$$x[n] = \sin(\theta)\, y[n-1] + x[n-1]\cos(\theta) \qquad \text{(B)}$$

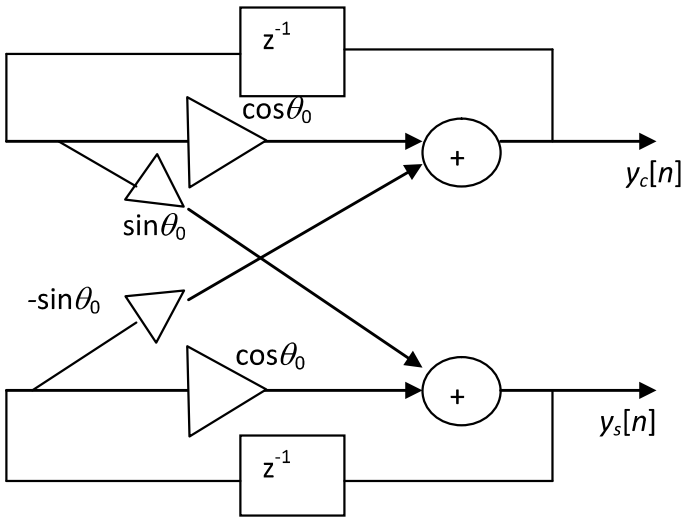
Using equations A & B above, the structure shown above can be obtained.

Example: An oscillator is given by the following coupled difference equations expressed in matrix form.

$$\begin{bmatrix} y_c[n] \\ y_s[n] \end{bmatrix} = \begin{bmatrix} \cos\theta_0 & -\sin\theta_0 \\ \sin\theta_0 & \cos\theta_0 \end{bmatrix} \begin{bmatrix} y_c[n-1] \\ y_s[n-1] \end{bmatrix}$$

Draw the structure for the realisation of this oscillator, where θ_0 is the oscillation frequency. If the initial conditions $y_c[-1] = A\cos\theta_0$ and $y_s[-1] = -A\sin\theta_0$, obtain the outputs $y_c[n]$ and $y_s[n]$ using the above difference equations.

$$y_c[n] = \cos\theta_0 y_c[n-1] - \sin\theta_0 y_s[n-1]$$
$$y_s[n] = \sin\theta_0 y_c[n-1] + \cos\theta_0 y_s[n-1]$$



n=0 $\quad y_s[0] = \sin\theta_0 (A\cos\theta_0) + \cos\theta_0 (-A\sin\theta_0) = 0$

n=0 $\quad y_c[0] = \cos\theta_0 (A\cos\theta_0) - \sin\theta_0 (-A\sin\theta_0) = \boldsymbol{A}$

n=1 $\quad y_c[1] = \cos\theta_0 . A - \sin\theta_0 . 0 = \boldsymbol{A\cos\theta_0}$

n=1 $\quad y_s[1] = A\sin\theta_0 + 0 = A\sin\theta_0$

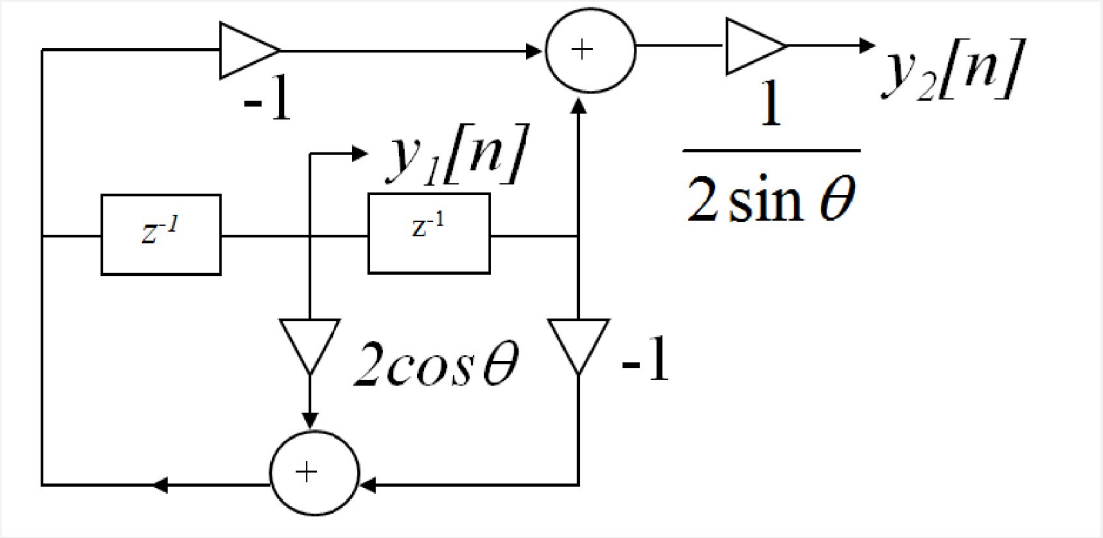
n=2 $\quad y_c[2] = \cos\theta_0\, y_c[1] - \sin\theta_0\, y_s[1]$

$= \cos\theta_0\, A\cos\theta_0 - \sin\theta_0\, A\sin\theta_0 = \boldsymbol{A\cos(2\theta_0)}$

$n = n$ $\quad y_c[n] = \boldsymbol{A\cos(n\theta_0)}$

similarly $\quad y_s[n] = A\sin(n\theta_0)$

Exercise: For the structure shown below, write down the appropriate difference equations and hence state the function of this structure.



CHAPTER 5: PROBLEM SHEET 5

Q1)

a) Show that both digital filters given below have the same magnitude response:

i) $y[n]=\sum_{i=-m}^{m} c_i x[n-i]$ ii) $y[n]=\sum_{i=-m}^{m} c_i x[n-m-i]$ i)

$y[n]$ = output; $x[n]$ = input; c_i = coefficients

b) Compute the 3dB bandwidth of the following filters

$$H_1(z)=\frac{1-a}{1-az^{-1}}; H_2(z)=\frac{1-a}{2}\frac{1+z^{-1}}{1-az^{-1}}, 0<a<1$$

Which filter has a smaller 3dB bandwidth?

Ans: $\theta_{c_1}=\cos^{-1}\left[\frac{4a-a^2-1}{2a}\right]$; $\theta_{c_2}=\cos^{-1}\left[\frac{2a}{1+a^2}\right]$; $\therefore \theta_{c_2}<\theta_{c_1} \Rightarrow 2^{nd}\ filter.$

c) Find the magnitude response for the system function $H(z)$ and comment on your result

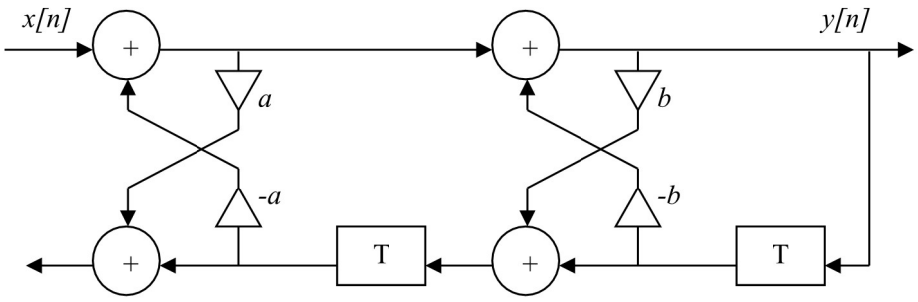
$$H(z)=\frac{1+3z^2}{3+z^2}$$

Draw the canonic realization of the system $H(z)$.

Ans: $|H(\theta)| = 1$ allpass filter

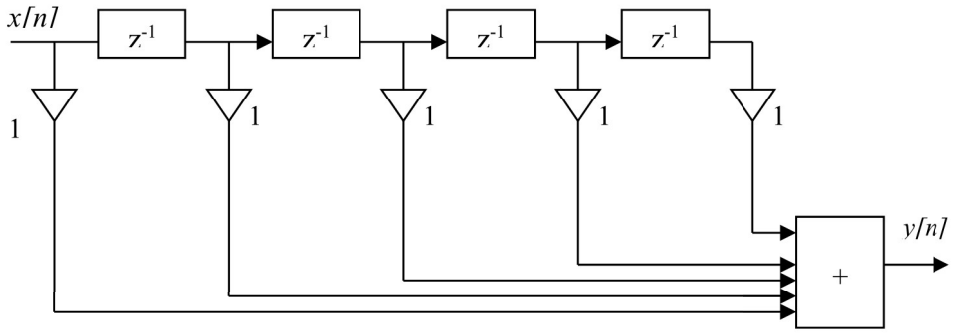
Q2)

a) Determine the frequency response $H(\theta)$ of the lattice filter



$$Ans: H(\theta)=\frac{1}{1+b(a+1)e^{-j\theta}+ae^{-j2\theta}}$$

b) Show that the filter structure shown below has a linear phase characteristic equation given by: $\phi(\theta) = -2\theta$.



Q3)

Consider the following causal IIR transfer function:

$$H(z) = \frac{2z^3 - 4z^2 + 9}{(z-3)(z^2 + z + 0.5)}$$

Is $H(z)$ a stable function? If it is not stable, find a stable transfer function $G(z)$ such that $|G(\theta)| = |H(\theta)|$. Is there any transfer function having the same magnitude response as $H(z)$?

Ans: unstable,

$$G(z) = \frac{2z^3 - 4z^2 + 9}{(1-3z)(z^2 + z + 0.5)} = H(z)A(z)$$

$$|G(z)| = |H(\theta)||A(\theta)|, A(z) = \frac{1-3z}{z-3} \quad \{Allpass\ filter\}$$

Q4)

a) A two-pole lowpass filter has the system function

$$H(z) = \frac{k_0}{(1 - b_1 z^{-1})^2}$$

Determine the values of k_0 and b_1 such that the frequency response $H(\theta)$ satisfy the conditions

$$H(0) = 1; \quad |H(\theta)|^2_{\theta=\frac{\pi}{4}} = \frac{1}{2}$$

Ans: k=0.46, b_1 = 0.32

b) A third order FIR filter has a transfer function $G(z)$ given by

$$G(z) = 30(1 - \frac{3}{2}z^{-1})(\frac{4}{3} + z^{-1})(1 + \frac{5}{2}z^{-1})$$

From $G(z)$, determine the transfer function of an FIR filter whose magnitude response is identical to that of $G(z)$ and has a maximum phase response.

Q5)

Obtain a parallel realization for the following $H(z)$

$$H(z) = \frac{z^2 - 4z + 3}{z(z + 0.5)^2}$$

Implement the parallel realization of *H(z)* which you have obtained

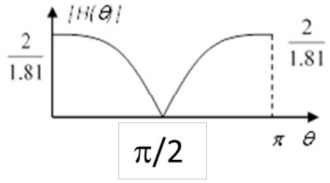
Ans: $H(z) = 12z^{-1} + \dfrac{-11z^{-1} - 16z^{-2}}{1 + z^{-1} + \dfrac{1}{4}z^{-2}}$

Q6) A notch filter is given by

$$H(z) = \frac{1 + z^{-2}}{1 + 0.8z^{-2}}$$

Determine the frequency response at DC, $\frac{f_s}{4}$ and $\frac{f_s}{2}$ (f_s – sampling frequency). Sketch the frequency response in the interval $0 \le f \le \frac{f_s}{2}$

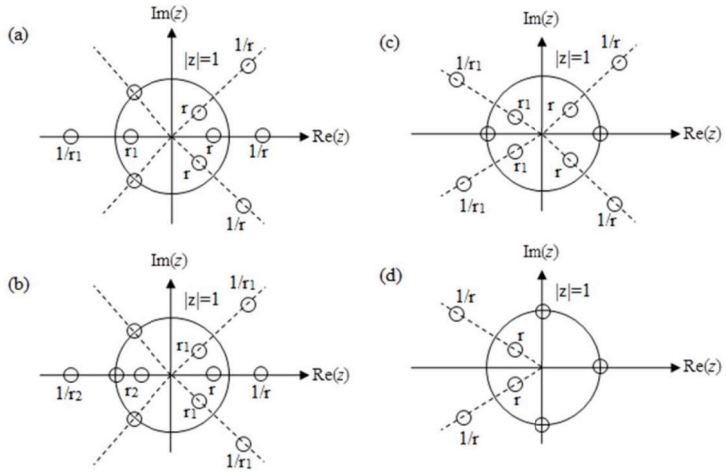
Ans:



Q7) Find all of the zeros of type 2 linear-phase sequence, if it is known that there are zeros at $z = \frac{1}{4}e^{j\frac{\pi}{6}}$ and $z = 1$.

Ans: $z = \frac{1}{4}e^{j\frac{\pi}{6}}$, $z = \frac{1}{4}e^{-j\frac{\pi}{6}}$, $z = 4e^{j\frac{\pi}{6}}$, $z = 4e^{-j\frac{\pi}{6}}$

Q8) Typical plots of zeros for linear-phase systems are shown below. From the pole-zero plot, identify the sequence type



Q9) Determine and sketch the magnitude and phase response of the following filters:

(i) $y[n] = \frac{1}{2}(x[n] - x[n-1])$

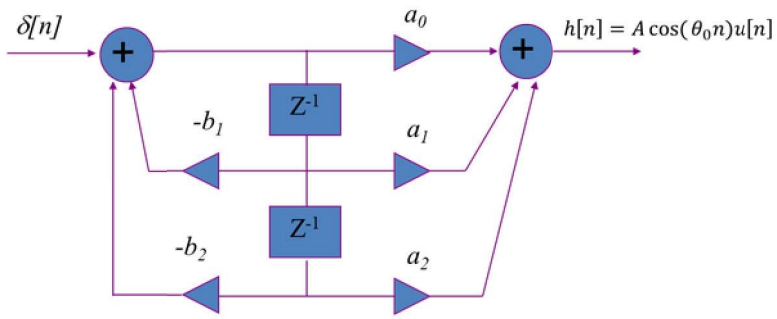
(ii) $y[n] = x[n] - x[n-8]$

(iii) $y[n] = x[n-4]$

Q10) The block diagram shows a digital oscillator which is initialized by an impulse as shown in the figure below. The desired unit impulse response is:

$$h[n] = A\cos(\theta_0 n)u[n]$$

Assuming θ_0 is the resonant frequency of the digital oscillator and writing appropriate equations find the values of a_0, a_1, a_2, b_1 and $b2$.



End of Chapter 5

Chapter 6

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Chapter 6: Discrete-Time Fourier Transform (DTFT)

The Fourier Transform is a very important tool in many aspects of engineering. In the case of digital signal processing, it allows signals to be represented in terms of their constituent frequency components, i.e. as signals in the 'frequency domain'. In some cases this means signal processing operations can be executed on frequency domain rather than time domain signals, which can have a number of advantages.

Before proceeding, it is worth noting that in signal processing what is referred to as 'the Fourier transform' may in fact be one of four possible types of Fourier transform, illustrated in Table 1. The difference between each type of transform is whether the time domain signal is continuous (i.e. analog) or discrete (i.e. digital), and whether its frequency domain representation is continuous (i.e. a function of real numbers ω or θ) or discrete (i.e. a function of integers n or k)

Table 1. Types of Fourier transforms for signal processing

		Frequency domain signal	
		Continuous	Discrete $X[k]$
Time domain signal	Continuous $x(t)$	Fourier transform $X(\omega)$	Fourier series X_n
	Discrete $x[n]$	Discrete time Fourier transform (DTFT) $X(\theta)$	Discrete Fourier Transform (DFT) $X[k]$

Fourier Transform of an arbitrary discrete signal is also called Discrete Time Fourier Transform (DTFT).

For discrete-time signals, the Fourier transform pair is defined by

$$X(\theta) = \sum_{n=-\infty}^{\infty} x[n] e^{-jn\theta}$$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\theta) e^{jn\theta} d\theta$$

$$x[n] \xleftrightarrow{\text{DTFT}} X(\theta)$$

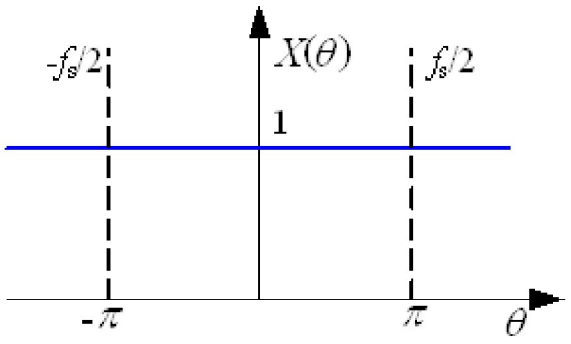
where θ is the digital frequency or relative frequency ($-\pi \leq \theta \leq \pi$) and $\theta = \omega T$. $X(\theta)$ is periodic with period 2π. (ie. $X(\theta + 2\pi) = X(\theta)$). Note that although $x[n]$ is discrete, $X(\theta)$ is a continuous function of frequency θ (hence the integral in the inverse transform).

Note: The Fourier transform of an analogue signal is not periodic.

Example: A unit impulse signal $x[n]$ is transformed into its frequency domain counterpart using the DTFT as follows:

$$x[n] = \delta[n]$$

$$X(\theta) = \sum_{n=-\infty}^{\infty} \delta[n] e^{-jn\theta} = 1$$



Example:

$$x[n]=\begin{cases}0 & n<0\\ a^n & n\geq 0,|a|<1\end{cases}$$

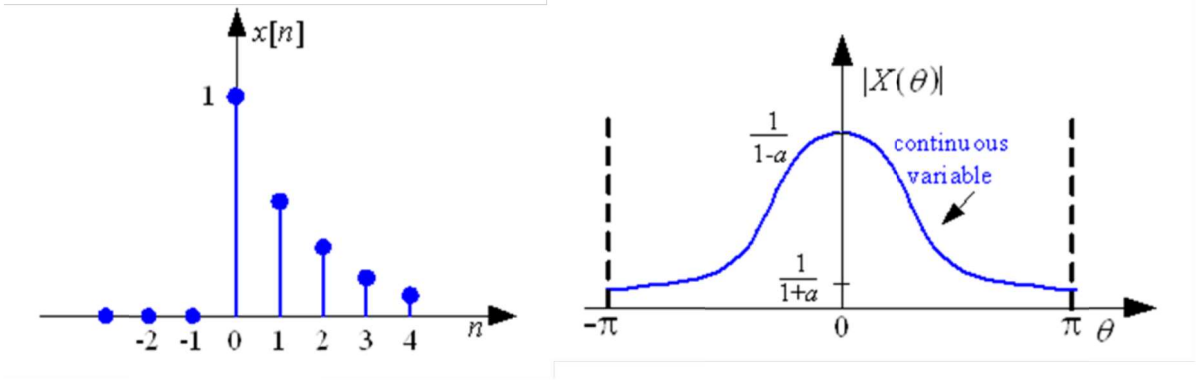
$$X(\theta)=\sum_{n=0}^{\infty}a^n e^{-jn\theta}=\sum_{n=0}^{\infty}(ae^{-j\theta})^n$$

$$=\frac{1}{1-ae^{-j\theta}}$$

$$X(\theta)=\frac{1}{(1-a\cos\theta)+ja(\sin\theta)}$$

$$\therefore |X(\theta)|=\frac{1}{\sqrt{1-2a\cos\theta+a^2}}$$

$$\angle X(\theta)\overset{\Delta}{=}\phi(\theta)=-\tan^{-1}\left(\frac{a\sin\theta}{1-a\cos\theta}\right)$$



Note: The above $X(\theta)$ can be obtained using the z-transform as well.

$$x[n]=a^n u[n]\Rightarrow X(z)=\frac{1}{1-az^{-1}}$$

$$X(\theta)=X(z)\big|_{z=e^{j\theta}}=\frac{1}{1-ae^{-j\theta}}$$

6.1 Properties of the DTFT

$$x[n] \overset{\text{DTFT}}{\rightarrow} X(\theta)$$
$$x_1[n] \overset{\text{DTFT}}{\rightarrow} X_1(\theta)$$
$$x_2[n] \overset{\text{DTFT}}{\rightarrow} X_2(\theta)$$

Linearity: $ax_1[n]+bx_2[n] \overset{\text{DTFT}}{\leftrightarrow} aX_1(\theta)+bX_2(\theta)$

Time Shift: $x[n-k] \overset{\text{DTFT}}{\leftrightarrow} e^{-j\theta k}X(\theta)$

Proof: $\left\{ x[n-k] \overset{Z}{\leftrightarrow} z^{-k}X(z) \right\}$

$z^{-k}X(z) = e^{-j\theta k}X(\theta)$ as we define $X(e^{j\theta}) \equiv X(\theta)$

Frequency Shift:

$x[n]e^{jn\theta_0} \overset{\text{DTFT}}{\leftrightarrow} X(\theta-\theta_0)$ where $X(\theta-\theta_0) \equiv X(e^{j(\theta-\theta_0)})$

Frequency Differentiation: $nx[n] \overset{DTFT}{\leftrightarrow} j\frac{d}{d\theta}X(\theta)$

Convolution: $x_1[n] * x_2[n] \leftrightarrow X_1(\theta)X_2(\theta)$

Multiplication: $x_1[n]x_2[n] \leftrightarrow X_1(\theta) * X_2(\theta)$

Parseval's Theorem:

$$\sum_{n=-\infty}^{\infty} x_1[n]x_2^*[n] \leftrightarrow \frac{1}{2\pi}\int_{-\pi}^{\pi} X_1(\theta) * X_2(\theta)d\theta$$

Exercise:
The frequency response of an ideal high pass filter, in the fundamental interval $-\pi \le \theta \le \pi$ is given by

$$H(\theta) = \begin{cases} 1 & \theta_C \le |\theta| \le \pi \\ 0 & 0 \le |\theta| < \theta_C \end{cases}$$

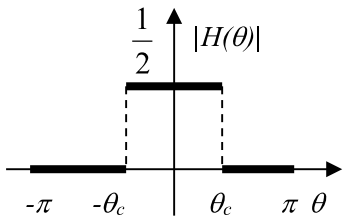
a) Find the impulse response of the ideal high pass filter
b) Sketch the impulse response for $\theta_C = \frac{\pi}{2}$

Example: The frequency response of an ideal low pass filter, in the fundamental interval $-\pi \le \theta \le \pi$ is given by

$$H(\theta) = \begin{cases} \frac{1}{2} & 0 \le |\theta| \le \theta_C \\ 0 & \theta_C < |\theta| \le \pi \end{cases}$$

(a) Find the impulse response of the ideal low pass filter.
(b) Sketch the impulse response for $\theta_C = \frac{\pi}{4}$.

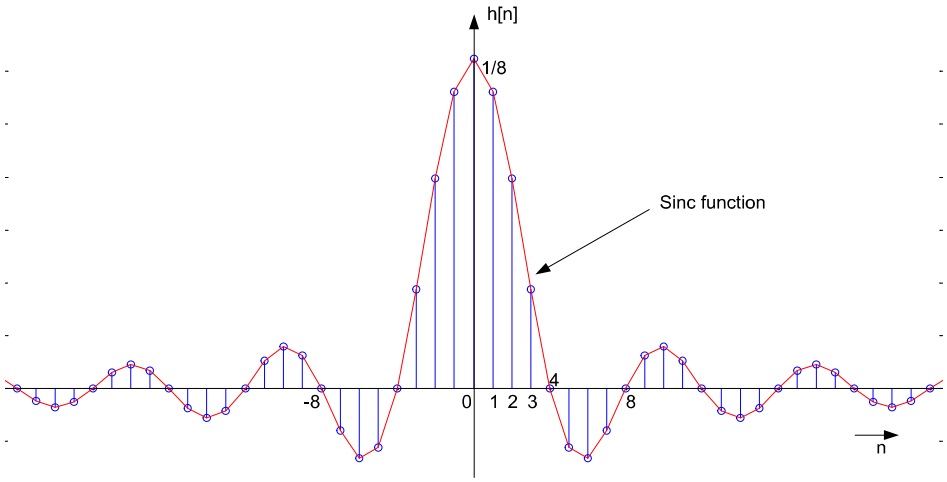
(a)



$$h[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(\theta)e^{jn\theta}d\theta = \frac{1}{2\pi}\int_{-\theta_C}^{\theta_C}\frac{1}{2}e^{jn\theta}d\theta = \frac{1}{4\pi}\frac{e^{j\theta_C n} - e^{-j\theta_C n}}{jn}$$

$$= \frac{1}{2\pi}\frac{\sin\theta_c n}{n}$$

(b) $\theta_c = \frac{\pi}{4}$ $\qquad h(n) = \frac{1}{2\pi}\frac{\sin\frac{n\pi}{4}}{n} = \frac{1}{8}\frac{\sin\frac{n\pi}{4}}{\frac{n\pi}{4}}$

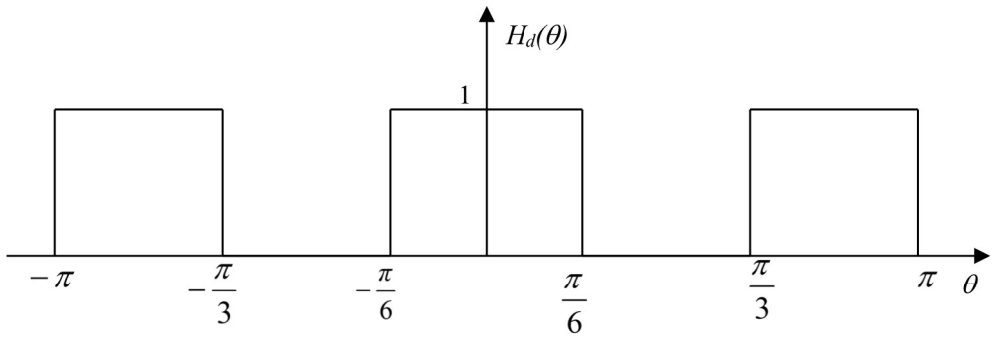


Example: The frequency response of a band stop filter is given by

$$H_d(\theta)=\begin{cases}1 & |\theta|\le\dfrac{\pi}{6}\\ 0 & \dfrac{\pi}{6}<|\theta|<\dfrac{\pi}{3}\\ 1 & \dfrac{\pi}{3}\le|\theta|\le\pi\end{cases}$$

Show that the impulse response $h_d[n]$ of the band stop filter is given by

$$h_d[n]=\delta[n]-\frac{\sin\left(\frac{\pi}{3}n\right)}{\pi n}+\frac{\sin\left(\frac{\pi}{6}n\right)}{\pi n}$$



$$h_d[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\theta)e^{j\theta n}d\theta = \frac{1}{2\pi}\left\{\int_{-\pi}^{-\frac{\pi}{3}} e^{j\theta n}d\theta + \int_{-\frac{\pi}{6}}^{\pi/6} e^{j\theta n}d\theta + \int_{\frac{\pi}{3}}^{\pi} e^{j\theta n}d\theta\right\}$$

$$= \frac{1}{2\pi}\left\{\left[\frac{e^{j\theta n}}{jn}\right]_{-\pi}^{-\frac{\pi}{3}} + \left[\frac{e^{j\theta n}}{jn}\right]_{-\frac{\pi}{6}}^{\frac{\pi}{6}} + \left[\frac{e^{j\theta n}}{jn}\right]_{\frac{\pi}{3}}^{\pi}\right\}$$

$$= \frac{1}{2\pi}\left\{\frac{e^{-j\frac{\pi}{3}n} - e^{-j\pi n}}{jn} + \frac{e^{j\frac{\pi}{6}n} - e^{-j\frac{\pi}{6}n}}{jn} + \frac{e^{j\pi n} - e^{-j\frac{\pi}{3}n}}{nj}\right\}$$

$$= \frac{\sin \pi n}{\pi n} + \frac{\sin \frac{\pi}{6}n}{\pi n} - \frac{\sin \frac{\pi}{3}n}{\pi n}$$

where $\frac{\sin \pi n}{\pi n} = 1 \qquad n = 0$ and zero elsewhere

$$h_d[n] = \delta[n] - \frac{\sin \frac{\pi}{3}n}{\pi n} + \frac{\sin \frac{\pi}{6}n}{\pi n}$$

6.2 The Discrete Fourier Transform (DFT)

The importance of the discrete Fourier transform (DFT) for practical applications is hard to overstate. If you've ever seen a real time spectrum of just about anything, chances are it was generated using a DFT. The only practical alternative to the DFT is a bank of analog filters (e.g. as used in a graphic equalizer), however with the availability of cheap processing power, and the fast algorithms for DFT computation, spectrum analysis is a viable component in many present-day systems.

$x[n] \xleftrightarrow{DTFT} X(\theta)$ DTFT

$x[n] \xleftrightarrow{DFT} X[k]$ DFT {DFT mapping $x[n]$ to another sequence}

The DFT denoted by $X[k]$ (complex valued sequence), is obtained by sampling the Discrete Time Fourier Transform $X(\theta)$ at a finite number of frequency points

This sampling is conventionally performed at equally spaced points over the period extending over $-\pi \le \theta \le \pi$.

The DFT allows us to determine the frequency content of a signal, that is, to perform spectral analysis.

The DFT plays a central role in the implementation of a variety of digital signal processing algorithms, as a result of the existence of the efficient algorithm for the Fast Fourier Transform (FFT).

(1) Fourier transform of a discrete signal (DTFT) is

$$X(\theta) = \sum_{n=-\infty}^{\infty} x[n] e^{-jn\theta}, \quad -\pi \le \theta \le \pi$$

(2) $X(\theta) = X(z)\,|_{z=e^{j\theta}}$

{Evaluate z-transform on the unit circle}

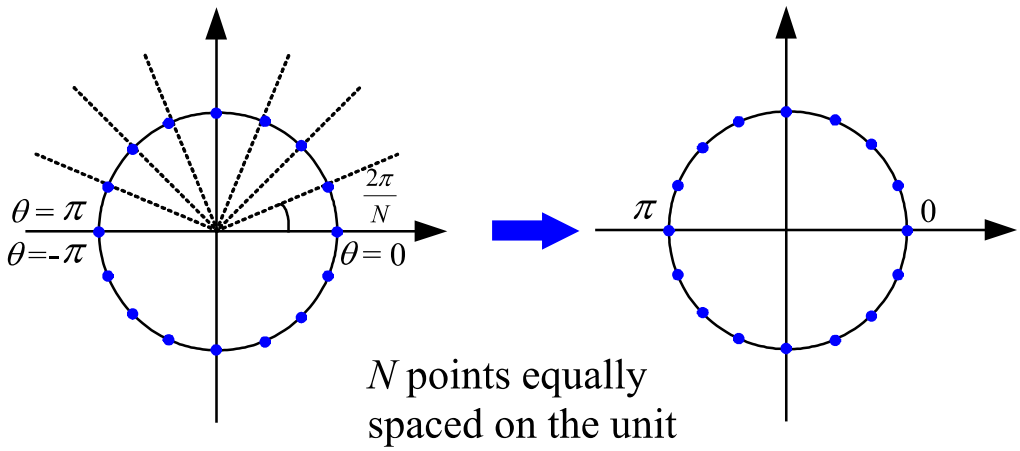
(3) Discrete Fourier Transform (DFT) or N-point DFT

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-jn\left(\frac{2\pi k}{N}\right)}$$

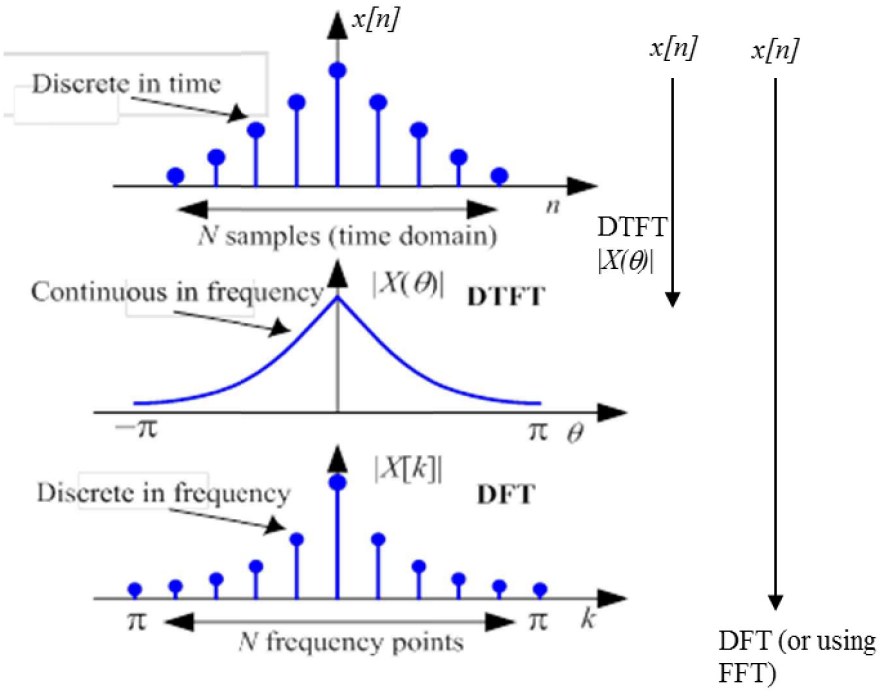
i.e. sampling $X(\theta)$ at N equally spaced interval, $k = 0,1,2,\ldots N\text{-}1$.
Note: One of the most important properties of the DFT is implied periodicity.

Number of time samples = N Number of frequency samples (k) = N

The DFT corresponds to sampling the z-transform of $X(z)$ at N-points equally spaced in angle around the unit circle.



Note:



- When the DFT of a block of N samples is calculated, the assumption is that the original signal actually repeats itself periodically, with period N.
- Clearly, for all real signals this will not be true.
- Even for artificial signals such as a pure sinusoid this will only be true if N is a multiple of the period of the sinusoid.
- This "windowing" process introduces a slight distortion into the frequency representation of the signal being analysed.

DFT calculation

$$X[k]=\sum_{n=0}^{N-1} x[n]\,e^{-jn\frac{2\pi k}{N}} \qquad k=0,1,2,3,....N$$

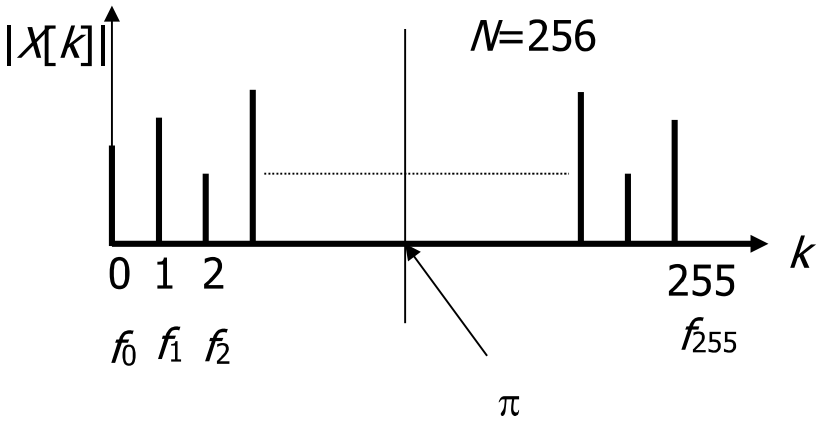
$$X[k]=\sum_{n=0}^{N-1} x[n]\cos\left(\frac{2\pi}{N}nk\right) - j\sum_{n=0}^{N-1} x[n]\sin\left(\frac{2\pi}{N}nk\right)$$

$$X[k]=A[k]-B[k]$$

$$|X[k]|=\sqrt{A[k]^2+B[k]^2} \qquad k=0,1,2,3,....N-1$$

$$\Delta\theta=\frac{2\pi}{N} \qquad \Delta f=\frac{f_S}{N}$$

$f_k = k(fs/N) = k\,\Delta f$ $(k=0, 1, 2, ...N\text{-}1)$



The frequency resolution (Δf) can be made as small as desired by increasing the value of N (window size being analysed)

Example: Let f_s = 8000 Hz, number of sample N = 1000

Frequency resolution $\Delta f = \frac{f_s}{N} = \frac{8000}{1000} = 8Hz$

$f_0 = 0; f_1 = 8\text{Hz}, f_2 = 16$ Hz, …. $f_{999} = 8000\text{Hz}$

$$X[k]=\sum_{0}^{999} x[n]e^{-jn\left(\frac{2\pi k}{1000}\right)}$$

$k = 0,1,2,3, \ldots$

Example: A speech signal is sampled at a rate of 20000 samples/sec. A sequence of length (N) 1024 samples is selected and the 1024-point DFT is computed.

(1) What is the time duration of segment of speech?

Duration = no of samples × sampling period.

= 1024 (1/20000) = 51.2 ms

(2) What is the frequency resolution (spacing in Hz) between the DFT values.

$$\text{Re}\,solution = \frac{f_s}{N} = \frac{20000}{1024} = 19.531Hz$$

Example: Consider the finite length sequence.

$$x[n] = \delta[n] + 0.2\delta[n-2]$$

Find the N-point DFT of $x[n]$ for $N = 50$

Solution: $X(z) = 1 + 0.2\, z^{-2} \Rightarrow X(\theta) = 1 + 0.2\, e^{-j2\theta}$

The N-point DFT is obtained by evaluating $X(\theta)$ at N points equally spaced around the unit circle.

$$\theta = \frac{2\pi k}{N}$$

Therefore the DFT is given by:

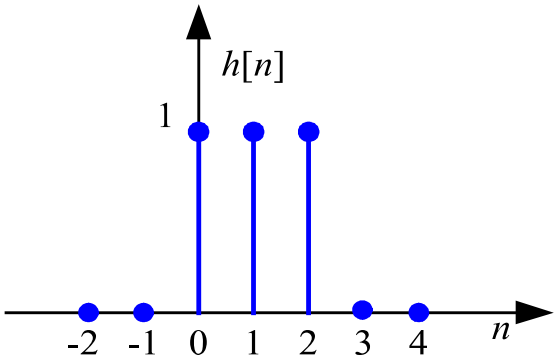
$$X[k] = 1 + 0.2e^{-j\left(\frac{2\pi k}{N}\right)2}$$

$$= 1 + 0.2e^{-j\left(\frac{2\pi k}{25}\right)}$$

k = 0, 1, 2, …. 49

Example: Find the DFT of the three-sample averager.

$$h[n]=\begin{cases}\frac{1}{3} & 0\le n\le 2\\ 0 & \text{otherwise}\end{cases}$$



$$H[k]=\sum_{n=0}^{N-1}h[n]e^{-jn(\frac{2\pi}{N}k)}=\sum_{n=0}^{2}\frac{1}{3}e^{-jn(\frac{2\pi}{N}k)}$$

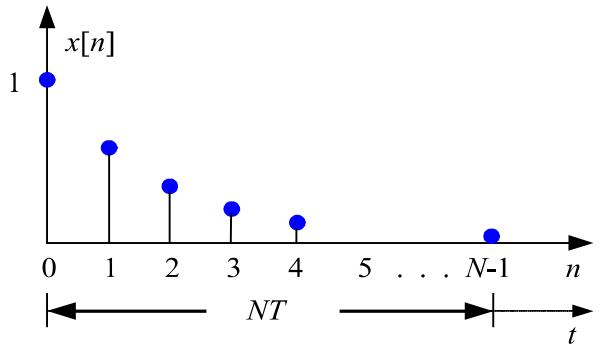
$$H[k]=e^{-j(\frac{2\pi}{N}k)}\frac{(1+2\cos(\frac{2\pi k}{N}))}{3}$$

$$|H[k]|=\frac{1}{3}\left|[1+2\cos(\frac{2\pi k}{N})]\right| \qquad \text{where } k=0,1,2$$

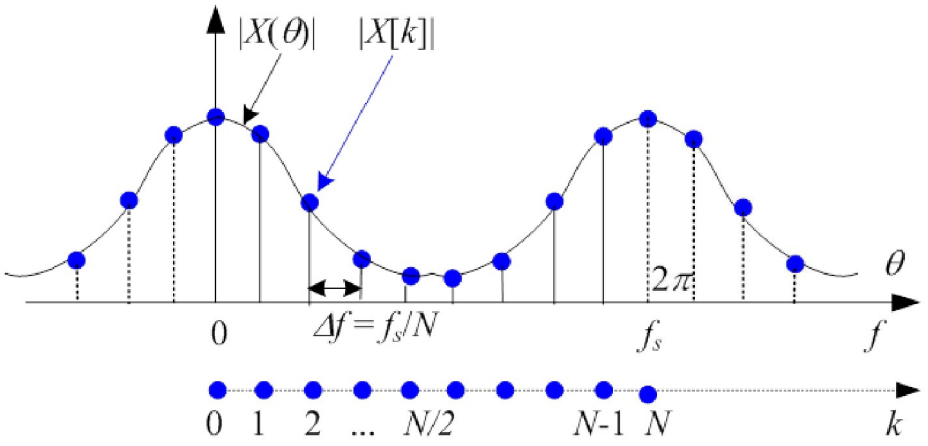
Exercise: Compute the N-point DFT $H[k]$ of the sequence $h[n]$. Show that when $N = 8$ the value of $H(2) = -1$.

$$h[n]=\begin{cases}1 & -3\le n\le 3\\ 0 & \text{otherwise}\end{cases}$$

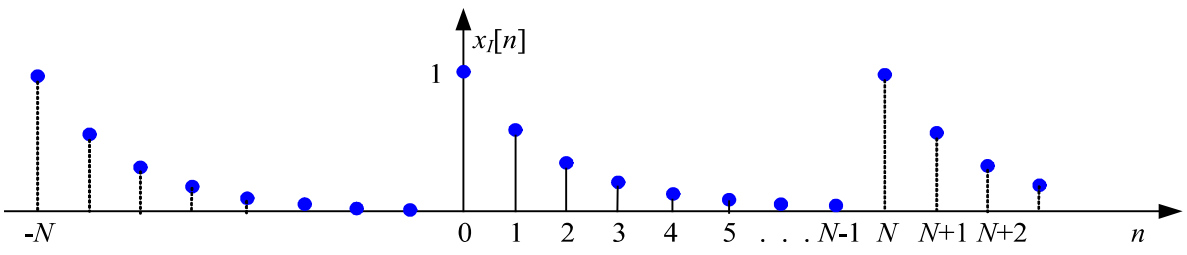
Example: Find the DTFT and DFT of a truncated exponential sequence $x_a(nT) = a^{nT}$, $n = 0,1,\ldots, N-1$, $|a| < 1$



This has DTFT $X(e^{j\theta}) = \dfrac{1 - a^{NT} e^{-j\theta N}}{1 - a^{T} e^{-j\theta}}$



and DFT $X[k] = \dfrac{1 - a^{NT}}{1 - a^{T} e^{-j\frac{2\pi k}{N}}}$. Note also that the inverse DFT $x_I[n]$ (below) is identical to $x[n]$ for $n = 0, 1, \ldots, N-1$, but outside this region creates periodic repetitions of the truncated $x[n] = x_a(nT)$.



6.2.1 Conjugate Symmetry

The DFT of a real sequence possesses conjugate symmetry about the origin with

$$X[-k] = X^*[k]$$

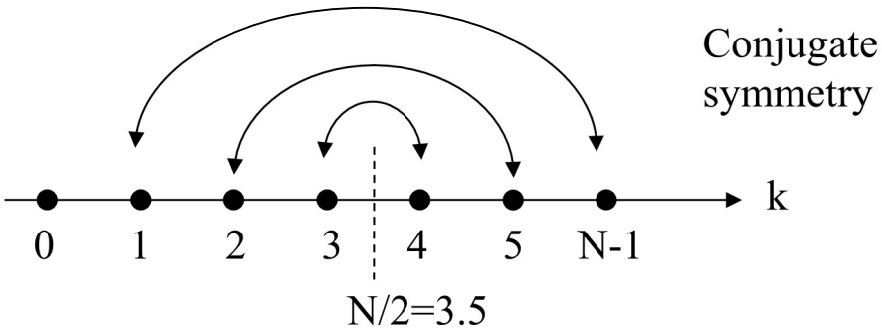
Since the DFT is periodic,

$$X[-k] = X[N-k]$$

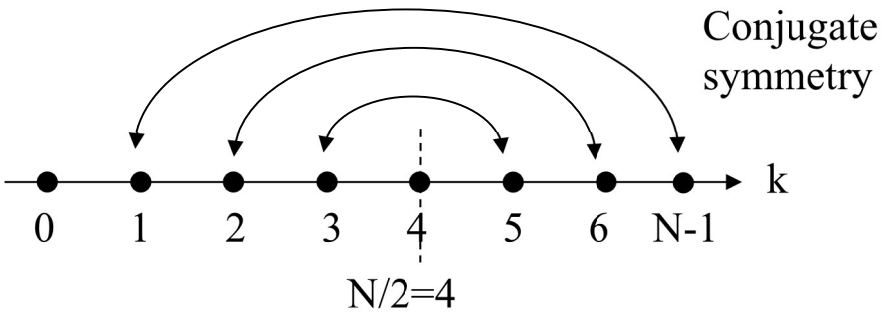
This implies conjugate symmetry about the index $k = \frac{N}{2}$

$$\therefore\ X[-k] = X^*[k] = X[k-K]$$

N is odd (N=7)



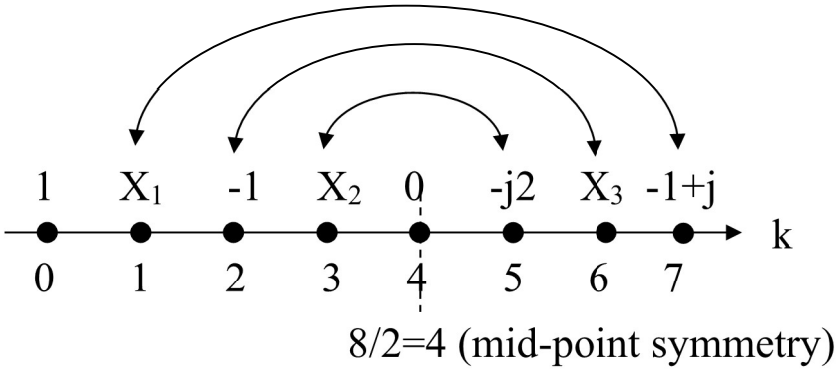
N is even (N=8)



Example: The DFT of a real signal is

$$\{1, X_1, -1, X_2, 0, -j2, X_3, -1+j\}$$

Find X_1, X_2 and X_3.



$$X_1 = x^*[1] = -1 - j$$
$$X_2 = x^*[5] = j2$$
$$X_3 = x^*[2] = -1$$

Note: 8 frequency points = 8 time samples (when IDFT is applied)

6.2.2 Parseval's Theorem

DFT is an energy-conserving transformation that allows to find the signal energy from the signal or its spectrum. Therefore, the sum of squares of the signal samples is related to the sum of squares of the magnitude of the DFT samples.

Example:

The DFT of a real signal is $\{1, X_1, -1, X_2, -7, -j2, X_3, -1+j\}$. What is the signal energy?

k=0 k=7

X_1, X_2 and X_3 can be found first using conjugate symmetry property

$$\text{Signal Energy} = \frac{1}{8}\sum |X[k]|^2$$

$$= \frac{1}{8}\left[1^2 + (\sqrt{2})^2 + (-1)^2 + 2^2 + 7^2 + 2^2 + 1^2 + \left(\sqrt{2}\right)^2\right]$$

$$= \frac{1}{8}[1+2+1+4+49+4+1+2]$$

$$= 8$$

Exercise:

a) Determine the 4-point DFT of a signal x[n]=[1 -1 -1 1].
b) Find the energy of x[n] using the information in time domain.
c) Find the energy of x[n] using its spectrum. Show that it is same as answer in question (b).

6.3 Inverse Discrete Fourier Transform (IDFT)

The inverse DFT equation comes directly from the Inverse Fourier Transform equation

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X(k)e^{jn\frac{2\pi}{N}k} \quad 0 \le n \le N\text{-}1$$

However, the signal which it produces will be a periodic signal repeating every N samples

Example:

If $X[k] = \{4, -j2, 0, j2\}$, find its IDFT
N=4

$$\therefore\ x[n] = \frac{1}{4}\sum_{k=0}^{3} X[k]e^{j\frac{2\pi kn}{4}}$$

$$= \frac{1}{4}\sum_{k=0}^{3} X[k]e^{j\frac{\pi kn}{2}}$$

$$n = 0 \quad x[0] = \frac{1}{4}[4 - j2 + 0 + j2] = 1$$

$$n = 1 \quad x[1] = \frac{1}{4}[4 - j2e^{j\frac{\pi}{2}} + 0 + j2e^{j\frac{3\pi}{2}}] = 2$$

$$n = 2 \quad x[2] = \frac{1}{4}[4 - j2e^{j\pi} + 0 + j2e^{j3\pi}] = 1$$

$$n = 3 \quad x[3] = \frac{1}{4}[4 - j2e^{j\frac{3\pi}{2}} + 0 + j2e^{j\frac{9\pi}{2}}] = 0$$

The IDFT is $x[n]=\{1, 2, 1, 0\}$

n=0

Use of DFT

- Clearly, the DFT is useful in that it allows the spectral content of a signal to be determined
- Additionally, once in the frequency domain the DFT of a signal can be processed in order to "filter" or "alter" the signal is some desired fashion
- The IDFT can then be used to regenerate the processed signal

Note: If any window, other than rectangular, has been used then DFT and IDFT blocks must overlap by 50%, if perfect reconstruction is required

6.4 Padding with Zeros and Frequency Resolution

$$DFT: \quad X[k] = \sum_{n=0}^{N-1} x[n] e^{-jn\frac{2\pi k}{N}} \qquad k = 0,1,2,3,....N$$

- The frequency resolution depends directly on the number of time samples N, i.e. the longer the sequence $x[n]$, the more frequency points in the DFT domain (frequency domain).

- To obtain more points in the DFT sequence, we can always increase the duration of *x[n]* by adding additional zero-valued elements. This procedure is called padding with zeros.

- These zero-valued elements contribute nothing to the sum in the above equation, but act to decrease the frequency spacing ($2\pi/N$).

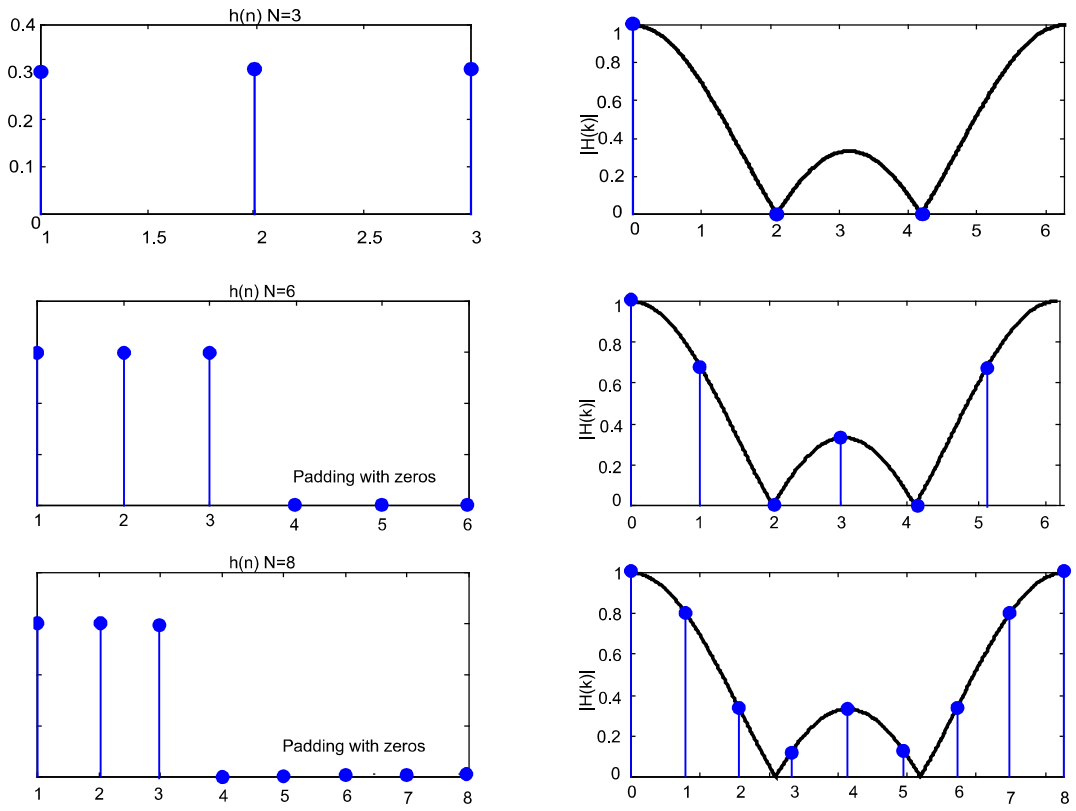
- The zero padding gives us a high-density spectrum and provided a better displayed version for plotting. But it does not give us a high resolution spectrum because no new information is added to the signal. Only additional zeros are added in the data.

Example: The three sample averager can be transformed using both the DTFT and the DFT. Because the DTFT is determined for all $n \in [-\infty,\infty]$, it is unaffected by the window length N.

On the other hand, adding zeros to the three sample averager gives a different view of the magnitude spectrum, depending on the length N.

Lengths of $N = 3$, 6 and 8 are shown below. Note that although the frequencies at which the DFT is evaluated are different in each case, the underlying spectrum (given by the DTFT) is exactly the same.

Three Sample Averager



Fast Fourier Transform

- The Fast Fourier Transform (FFT) is simply a mathematical technique to accelerate the calculation of the DFT. It was developed by Cooley and Tukey (1965) requires N to be a power of 2.

- Typically, if the DFT is calculated for a block of $2n$ samples e.g. 512 or 1024 samples (N) it would make the calculation of the DFT quite demanding.

- The FFT simply uses repetition and redundancy in the calculation to speed it up.

- The FFT is simply a TECHNIQUE to calculate the DFT, NOT a different transform.

CHAPTER 6: PROBLEM SHEET 6

Q1) A speech signal is sampled with a sampling period of 125ms. A frame of 256 samples is selected and a 256 point DFT is computed. What is the spacing between the DFT values in Hz. Ans: 0.03125Hz

Q2) Compute the N-point DFT, *H[k]* of the sequence *h[n]*. Show that when $N = 3$ the value of $H(3) = 3/5$

$$h[n] = \begin{cases} \frac{1}{5} & -1 \le n \le 1 \\ 0 & \text{otherwise} \end{cases}$$

Ans: $H[k] = \frac{1}{5}(2\cos\left(\frac{2\pi k}{3}\right) + 1)$

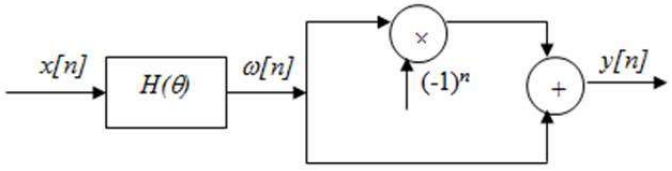
Q3) A 5 kHz sinusoidal signal is sampled at 40 kHz and 128 samples are collected and used to compute the 128-point DFT of the signal. What is the time duration in seconds of the collected samples? At what DFT indices do we expect to see any peaks in the spectrum?
Ans: 16 and 112

Q4) Consider the finite length sequence: $x[n] = \delta[n] + 0.2\delta[n-2]$
Write the equation for the *N*-point DFT of *x[n]* for $N = 25$

Ans: $X[k] = 1 + 0.2e^{\frac{j2\pi k}{25}}$

Q5) For the system in the figure below, sketch the output *y[n]* when the input *x[n]* is $\delta[n]$ and $H(\theta)$ is an ideal lowpass filter as follows:

$$H(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{2} \\ 0 & \frac{\pi}{2} < |\theta| \le \pi \end{cases}$$



Ans: $\omega(n) = \frac{1}{\pi}\left(\frac{\sin\frac{n\pi}{2}}{n}\right)$; $y(n) = \frac{1}{2}\left[(-1)^n + 1\right]\frac{\sin\frac{n\pi}{2}}{\frac{n\pi}{2}}$

Q6) Let x[n]={1, 2, 1, 0}. With N=4, compute the 4-point DFT.

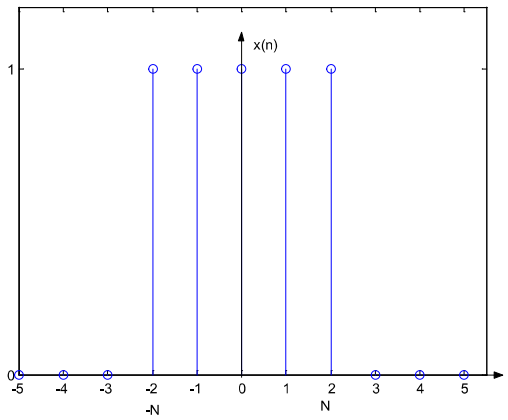
n=0

Ans:$X[k]$={4, -*j*2, 0, *j*2}

Q7) Find the 8-point DFT of $x[n]$ using conjugate symmetry property of $X[k]$. x[n]={1, 1, 0, 0, 0, 0, 0, 0}.

Ans: $X[k]$={2, 1.707-*j*0.707, 1-*j*, 0.293-*j*0.707, 0, 0.293+*j*0.707, 1+*j*, 1.707+*j*0.707}

Q8)



If $x[n] = \begin{cases} 1 & |n| \leq N \\ 0 & |n| > N \end{cases}$

Determine *X(θ)*.

$$ans: X(\theta) = \frac{\sin\left\{\frac{(2N+1)\theta}{2}\right\}}{\sin\left(\frac{\theta}{2}\right)}$$

End of Chapter 6

Chapter 7

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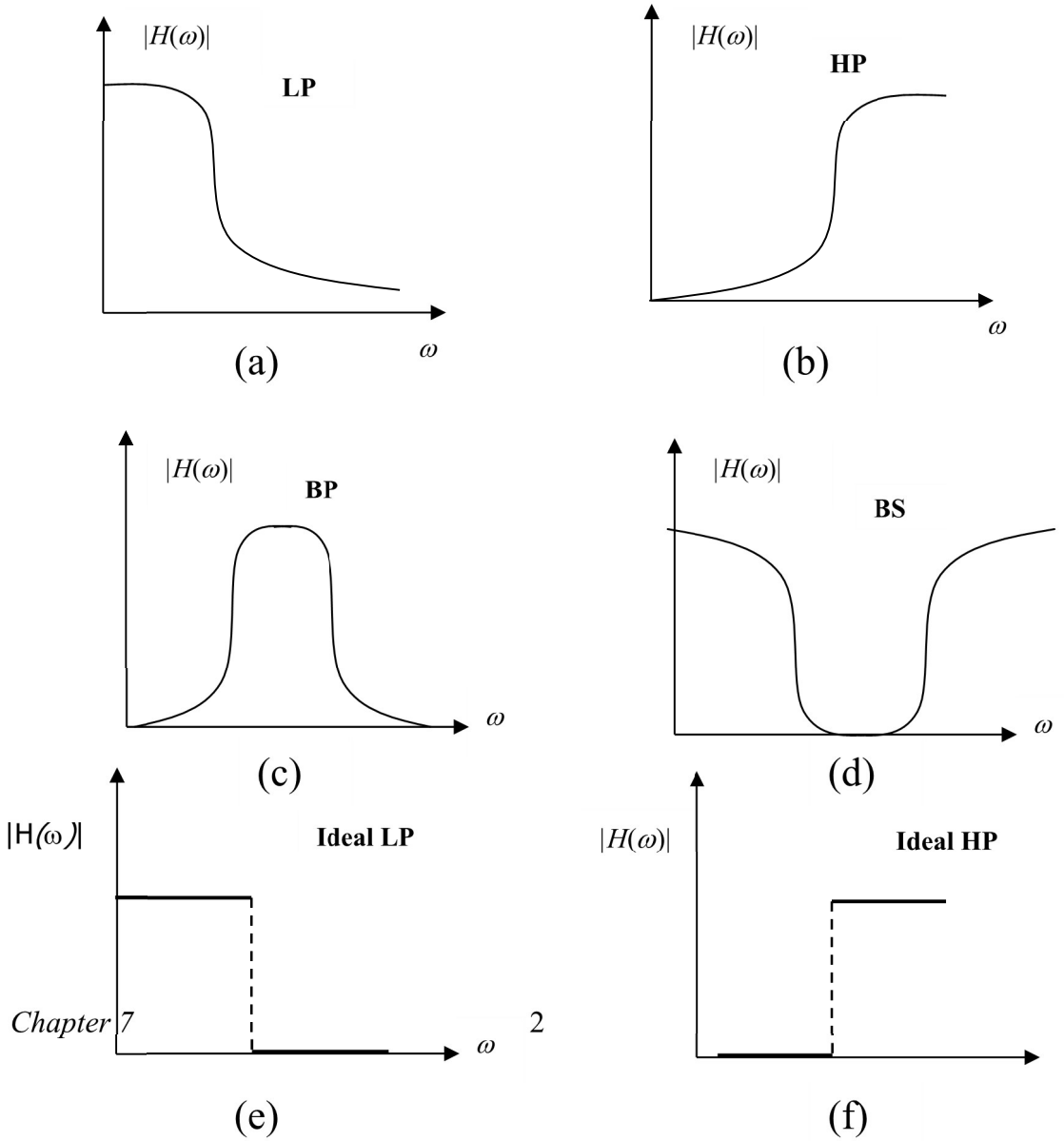
7.1 Introduction

A very important approach to the design of digital filters is to apply a transformation to an existing analogue filter.

The commonly required types of frequency filter fall into four main groups:

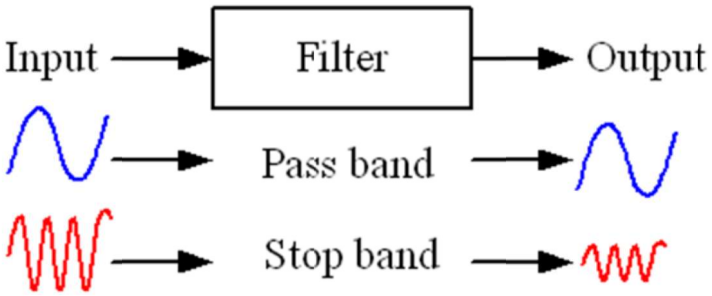
- Low-pass (LP)
- High-pass (HP)
- Band-pass (BP)
- Band-stop (BS)

The frequency responses of these filters are shown in Figure **Figure 7.1** (a to d). Also shown are the frequency responses for the **ideal** LP, HP, BP and BS filters.



The analogue filter design procedure normally begins with a specification of the frequency response for the filter describing how the filter reacts to sinusoid inputs.

If an input sinusoid is not attenuated or attenuated less than a specific tolerance as it goes through the system, it is said to be in a **pass band** of the filter.



If it is attenuated more than a specified value it is said to be stopped and within the **stop band** of the filter.

Input sinusoids with neither a little nor a large amount of attenuation are said to be in the **transition band**. A typical frequency response is shown in **Figure 7.2**, showing the pass band, transition band and stop band.

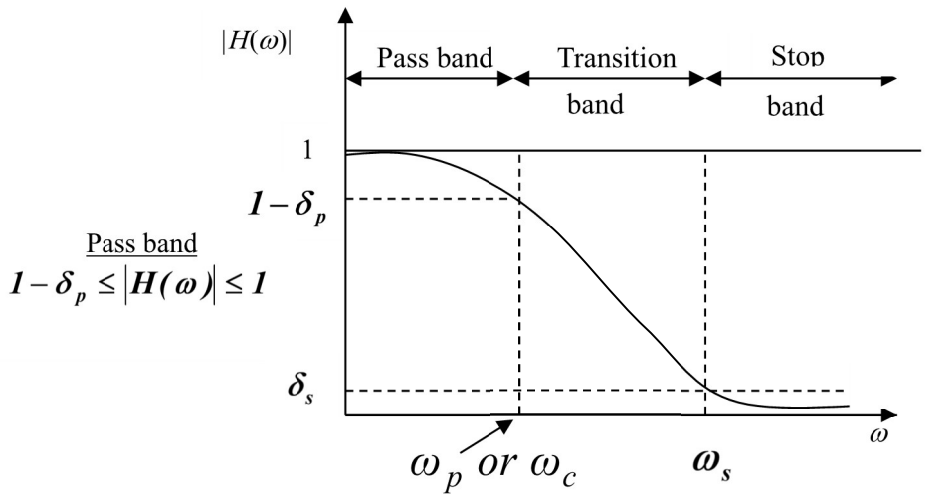


***Figure** Error! No text of specified style in document..**2**: Low-pass filter frequency response*

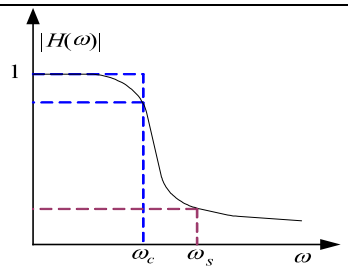
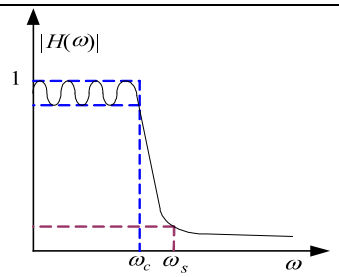
The filter with this type of frequency response is called a low-pass filter as it passes all frequencies less than a certain value ω_p (or ω_c), called the cut-off frequency and attenuates or stops all frequencies past ω_s, the stop band frequency.

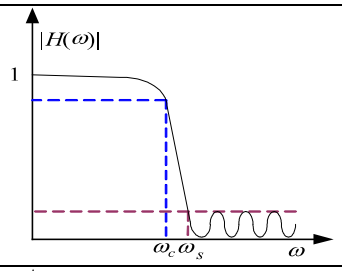
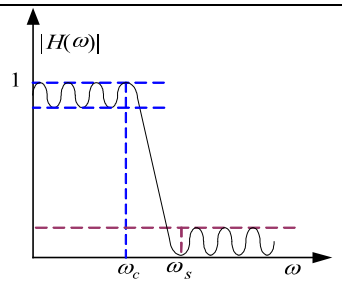
The analogue low pass filters are commonly used as **anti-aliasing** filters that are applied to continuous-time signals before A/D conversion and also used as reconstruction filters.

Note: The technique used to design analogue filters is to specify a prototype low-pass filter function which is normalised to provide a Cut-off frequency (ω_c) at 1 *rad/sec* and then apply transformations to achieve the actual desired cut-off frequencies and filter type.

Therefore, prime consideration will be given to low-pass filter design using the following:

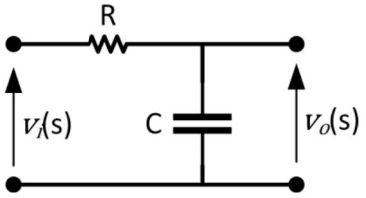
- Butterworth low-pass filter
- Chebyshev I low-pass filter
- Chebyshev II low-pass filter
- Elliptic low-pass filter
- Bessel low-pass filter

Name	**Property**	
Butterworth	Maximally flat in passband Stopband attenuation relatively poor	
Chebyshev I	Good match with ideal filter characteristic Maximal ripple in passband	

Chebyshev II	Good match with ideal filter characteristic Maximal ripple in stopband	
Elliptic	Minimal transition width Ripple in both passband and stopband	
Bessel	Maximally constant group delay $\{\tau_g = -\frac{d\phi(\theta)}{\theta}\}$	$\phi(\theta) = -a\theta$ *or* $\phi(\theta) = b - a\theta$

7.2 Passive RC filters

- Passive 1st order low-pass filter



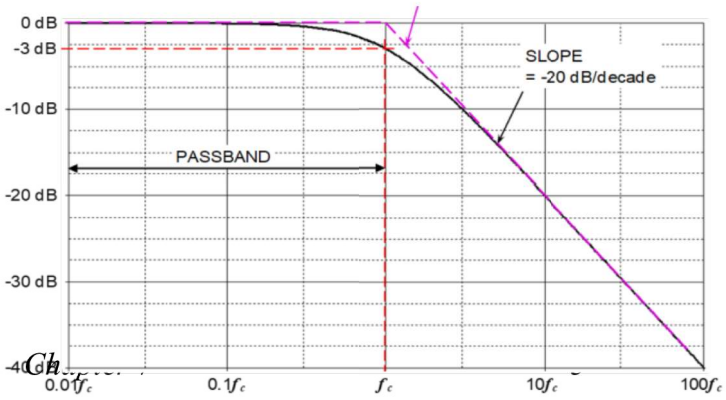
$$H(s) = \frac{v_o(s)}{v_i(s)} = \frac{\omega_c}{s + \omega_c} \quad \text{where } \omega_c = \frac{1}{RC}$$

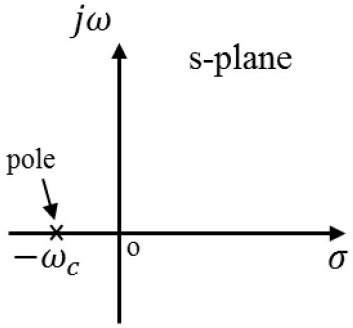
$$H(j\omega) = \frac{1}{1 + j\frac{\omega}{\omega_c}}$$

$$|H(j\omega)|_{dB} = 20\ log\left[\frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^2}}\right]$$

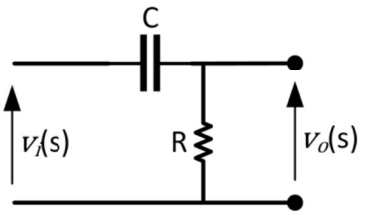
Note

- s - Complex frequency
- $s = j\omega$ sinusoid
- $s = j\omega + \sigma$ Exponential sinusoid
- $s = 0$ dc

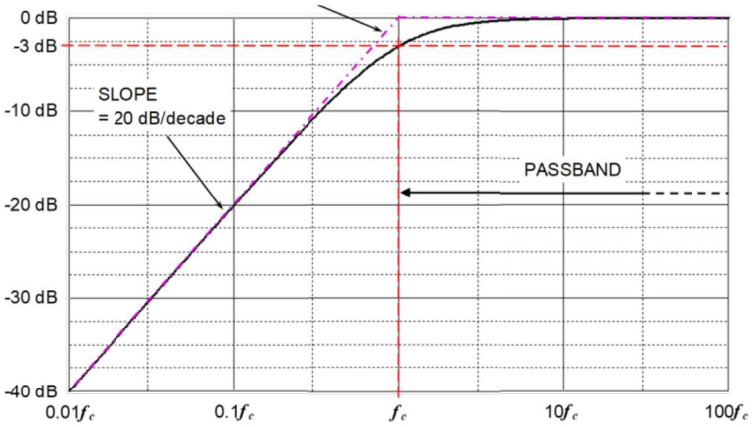


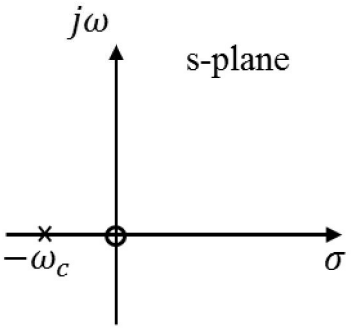


- Passive 1st order high-pass filter



$$H(s) = \frac{v_o(s)}{v_i(s)} = \frac{s}{s + \omega_c} \quad \text{where } \omega_c = \frac{1}{RC}$$





Note

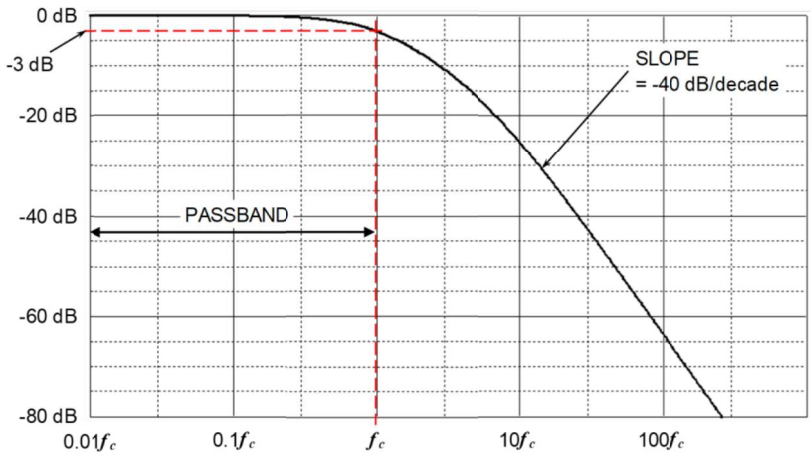
1st order LPF	1st order HPF
$H(s) = \frac{\omega_c}{s + \omega_c}$	$H(s) = \frac{s}{s + \omega_c}$

- Passive 2nd order low-pass filter

$$H(s) = \frac{{\omega_o}^2}{s^2 + b_o s + {\omega_o}^2} \qquad b_o = \frac{\omega_o}{Q}$$

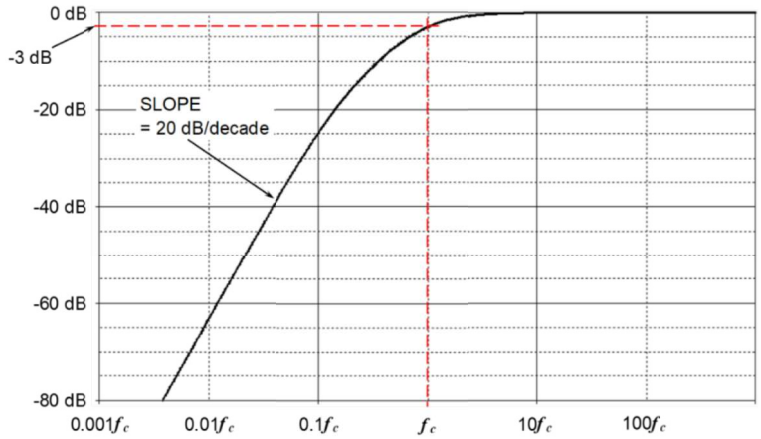
ω_o ← Undamped resonant frequency

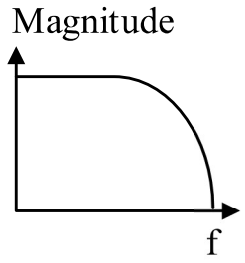
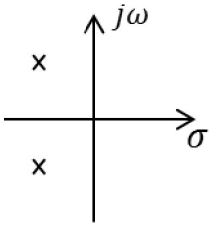
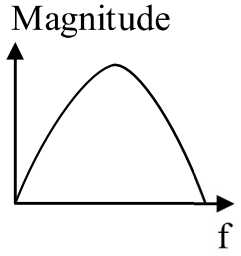
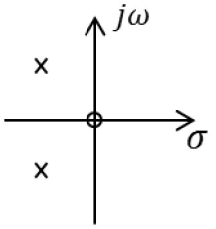
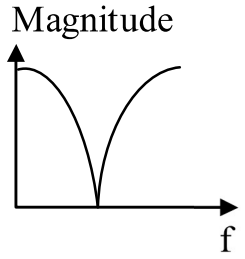
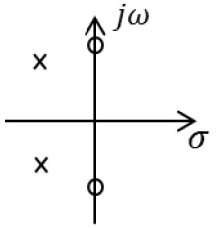
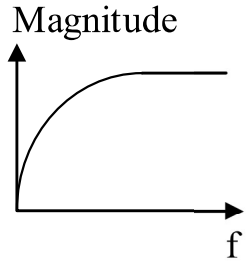
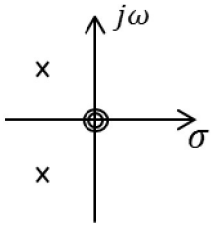
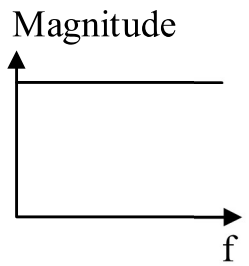
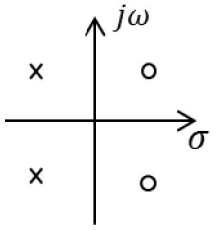
Q ← Q - factor
Q - Quality



- Passive 2nd order high-pass filter

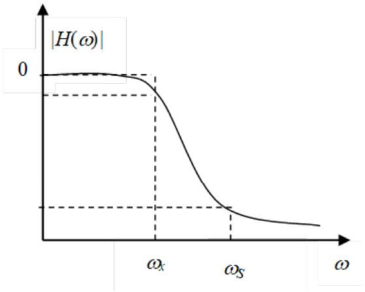
$$H(s) = \frac{s^2}{s^2 + b_o s + {\omega_o}^2}$$



Filter Type	Magnitude	Pole Location	Transfer function
Low-pass			$\frac{{\omega_o}^2}{s^2+\frac{\omega_o}{Q}s+{\omega_o}^2}$
Band-pass			$\frac{\frac{\omega_o}{Q}s}{s^2+\frac{\omega_o}{Q}s+{\omega_o}^2}$
Notch (Band-reject)			$\frac{s^2+{\omega_z}^2}{s^2+\frac{\omega_o}{Q}s+{\omega_o}^2}$
High-pass			$\frac{s^2}{s^2+\frac{\omega_o}{Q}s+{\omega_o}^2}$
All-pass			$\frac{s^2-\frac{\omega_o}{Q}s+{\omega_o}^2}{s^2+\frac{\omega_o}{Q}s+{\omega_o}^2}$

Note: Lowpass Butterworth Filter

The magnitude response of a Butterworth filter is shown below:



- Lowpass Butterworth filter is a monotonically decreasing function of ω (i.e. throughout pass band and stop band).

The transfer function $H(s)$ is given by:

$$H(s) = \frac{1}{s+1} \quad 1^{st} \text{ order}$$

$$H(s) = \frac{1}{s^2 + \sqrt{2}s + 1} \quad 2^{nd} \text{ order}$$

$$H(s) = \frac{1}{(s+1)(s^2+s+1)} \quad 3^{rd} \text{ order}$$

7.3 Lowpass to High pass transformation

To transform analogue low-pass filter $H(s)$ with unity cut-off frequency to low-pass filter $H(s)$ with cut-off frequency ω_c, we substitute

$$s \rightarrow \frac{s}{\omega_c}$$

Example: First order Butterworth prototype filter is given by

$$H(s) = \frac{1}{s+1} \qquad \longleftarrow \text{ Normalised}$$

To transform to new cut-off frequency $\omega_c = 5$, we replace s with $\frac{s}{\omega_c}$

$$H(s) = \frac{1}{\frac{s}{\omega_c}+1} = \frac{\omega_c}{s+\omega_c} = \frac{5}{s+5}$$

Note: To transform an analogue low-pass filter $H(s)$ with unit cut-off frequency to high-pass filter $H(s)$ with cut-off frequency ω_c, we substitute

$$s \rightarrow \frac{\omega_c}{s}$$

Example: First order Butterworth prototype filter is given by

$$H(s) = \frac{1}{s+1} \quad \longleftarrow \quad \text{Normalised}$$

To transfer to a high-pass filter with cut-off frequency $\omega_c = 2$

We replace $s \to \frac{\omega_c}{s}$; $\qquad H(s) = \frac{1}{\frac{\omega_c}{s}+1} = \frac{s}{s+2}$

Note: *High-pass filter contains zeros as well as poles*

Example:

$$H(s) = \frac{1}{(s+1)(s^2+s+1)} \quad \longleftarrow \quad 3^{rd} \text{ order Butterworth filter.}$$

Determine the transfer function of the corresponding high-pass filter with cut off frequency $\omega_c = 1$ (normalised)

$$s \to \frac{\omega_c}{s} \quad \text{i.e.} \quad s \to \frac{1}{s}$$

normalised

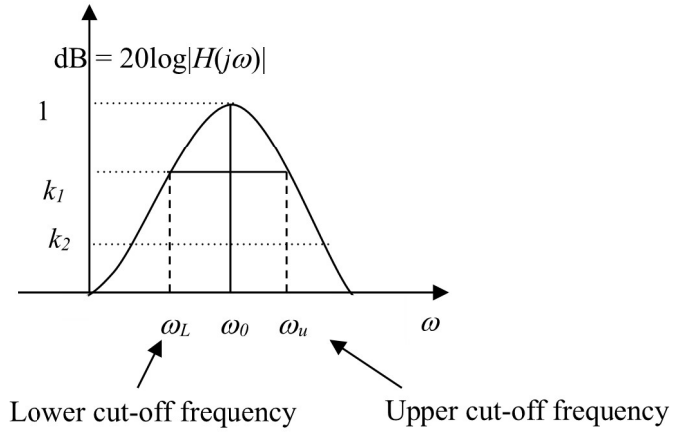
$$H(s) = \frac{1}{\left(\frac{1}{s}+1\right)\left(\left(\frac{1}{s}\right)^2 + \frac{1}{s}+1\right)} = \frac{s^3}{(s+1)\left(s^2+s+1\right)}$$

7.4 Low-pass to Band-pass Transformations

By definition, a band pass filter rejects both low and high frequency components and passes a certain band of frequencies some where between them. Thus the frequency response $H(j\omega)$, of a band-pass filter has the following properties.

1. $|H(j\omega)| = 0$ at both $\omega = 0$ & $\omega = \infty$

2. $|H(j\omega)| = 1$ for a frequency band centered on ω_0, where ω_0 is the mid frequency of the filter



$$B = \omega_u - \omega_L$$

Bandwidth of the band-pass filter

Method 1: To convert unity cut off low-pass filter $H(s)$ into a Band-Pass filter $H(s)$ with lower cut-off frequency ω_L and the upper cut-off frequency ω_u, we replace

$$s \rightarrow \frac{s^2 + \omega_L \omega_u}{s(\omega_u - \omega_L)}$$

Method 2: A lowpass to bandpass transformation can be performed by

$$s \rightarrow \frac{s^2 + {\omega_o}^2}{Bs}$$

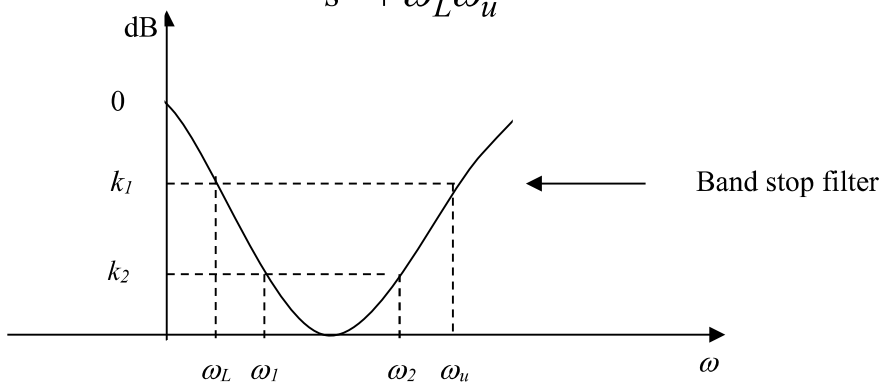
B - Bandwidth of the band-pass filter $(\omega_u - \omega_L)$

ω_0 - centre frequency.

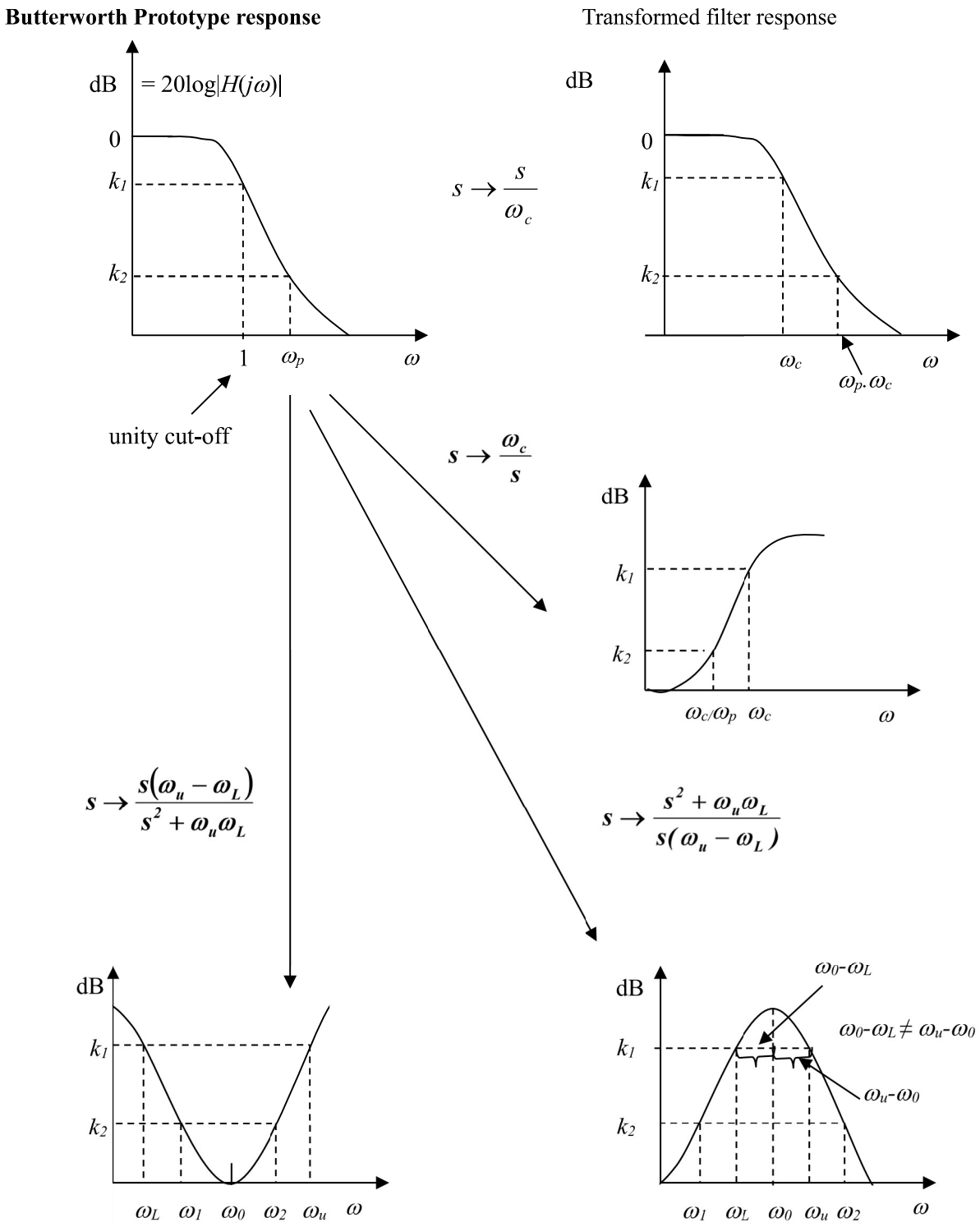
7.5 Low-pass to Band-stop Transformations

Similarly to convert unity cut-off lowpass filter $H(s)$ into a bandstop filter $H(s)$ with lower cutoff frequency ω_L and upper cut-off frequency ω_u, we replace:

$$s \rightarrow \frac{s(\omega_u - \omega_L)}{s^2 + \omega_L \omega_u}$$



Summary: Analogue to Analogue Transformation



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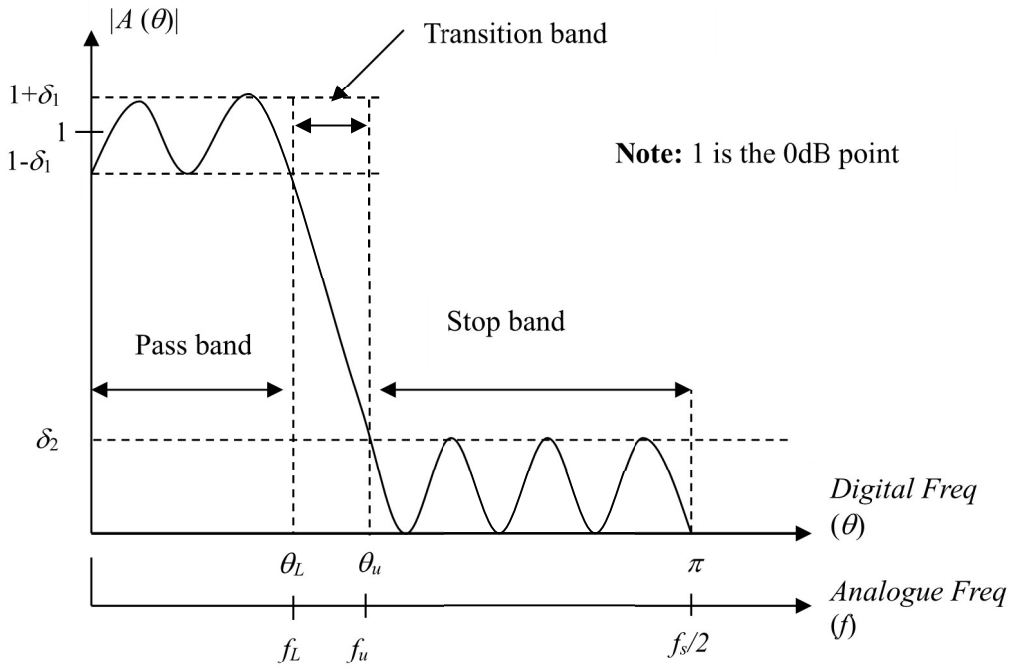
Chapter 8: Digital Filter Design

8.1 Introduction

The design of a digital filter is the task of determining a transfer function which is a rational function of z^{-1} (e.g. $\frac{a_0 + a_1 z^{-1} + a_2 z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}}$) in the case of a recursive filter (IIR) or a polynomial in z^{-1}

($a_0 + a_1 z^{-1} + a_2 z^{-2} + a_3 z^{-3}$) in the case of a non-recursive filter.

Performance specifications: A typical amplitude characteristic of a low-pass filter is shown below.



The goal of the design is to determine a transfer function H(z) so that its amplitude characteristic |H(θ)| satisfies the conditions.

$$1 - \delta_1 \le |H(\theta)| \le 1 + \delta_1 \quad \text{for } 0 \le \theta \leqslant \theta_L$$

$$|H(\theta)| \le \delta_2 \quad \text{for } \theta_u \le \theta \le \pi$$

8.2 Choosing between FIR and IIR filters

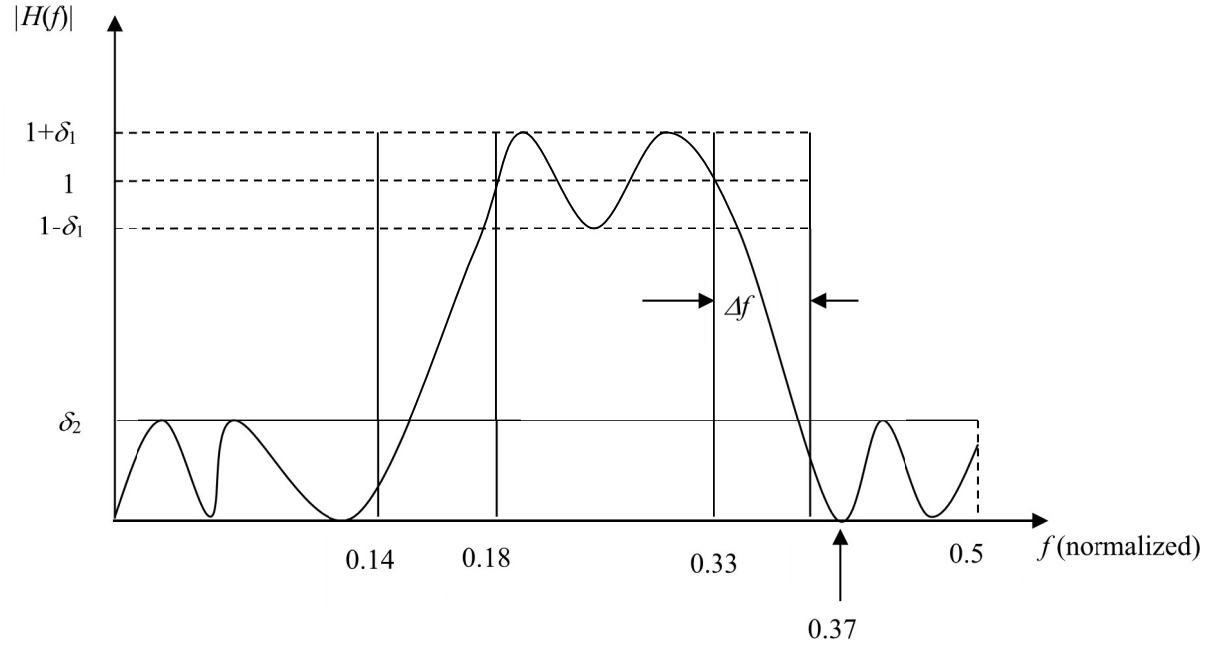
FIR Filters	IIR Filters
Difference equation $y[n] = b_0 x[n] + b_1 x[n-1] + ... + b_M x[n-M]$	Difference equation $y[n] = b_0 x[n] + b_1 x[n-1] + ... + b_M x[n-M]$ $- a_1 y[n-1] - ... - a_N y[n-N]$
Transfer function $H(z) = b_0 + b_1 z^{-1} + ... + b_M z^{-M}$	Transfer function $H(z) = \frac{b_0 + b_1 z^{-1} + ... + b_M z^{-M}}{1 + a_1 z^{-1} + ... + a_N z^{-N}}$
System function contain only zeros	Contain poles and zeros (normally)
Non-recursive or recursive structures are both possible; the best known is the non-recursive (transversal) structure.	Only recursive structure is possible; the most widely used form is the cascade connection of first-order and second order sections.
FIR Filters can have an exactly linear phase response. The implication of this is that no phase distortion is introduced into the signal by the filter.	The phase responses of IIR filters are nonlinear, especially at the band edges.
The effects of using a limited number of bits to implement filters such as round off noise and quantization errors are much less severe in FIR than in IIR.	Because of quantization of the filter coefficients, a pole can in principle move from a position inside the unit circle to a position outside the unit circle and hence cause instability.
FIR requires more coefficients for sharp cut-off filters than IIR. Thus for a given amplitude response specification, more processing time and storage will be required for FIR implementation.	IIR requires fewer coefficients for sharp cut off filters than FIR.
Complexity is proportional to the length of the impulse response.	No direct relation between the complexity and the length of the impulse response (which is infinite by definition) Filters with high selectivity can be realized with relatively low complexity.
FIR filters have no analogue counterpart. FIR design procedures are normally iterative procedures. Design equations do not exist.	Analogue filters can be readily transformed into equivalent IIR digital filters meeting similar specifications. IIR filters can be designed using design formulae.

Example: An FIR filter is to be designed to meet the following frequency response specifications.

- Pass band 0.18-0.33 (normalized)
- Transition band 0.04 (normalized)
- Stop-band deviation 0.001
- passband deviation 0.05

(i) Sketch the tolerance scheme for the filter.
(ii) Express the filter band edge frequencies in the standard unit of kHz, assuming a sampling frequency of 10 kHz and the stop band and pass band deviation in dBs.

The tolerance diagram for the filter is shown below:



f_s=10kHz , therefore

Pass band: 1.8-3.3 kHz
Stop band: 0-1.4 kHz and 3.7-5kHz
Stop band attenuation: $-20\log_{10}(0.001) = -60$dB
Pass band ripple: $20\log_{10}(1+0.05) = 0.42$ dB

Example: The following transfer functions represent two different filters meeting identical amplitude–frequency response specifications.

$$H_1(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}} \longleftarrow \text{Filter 1}$$

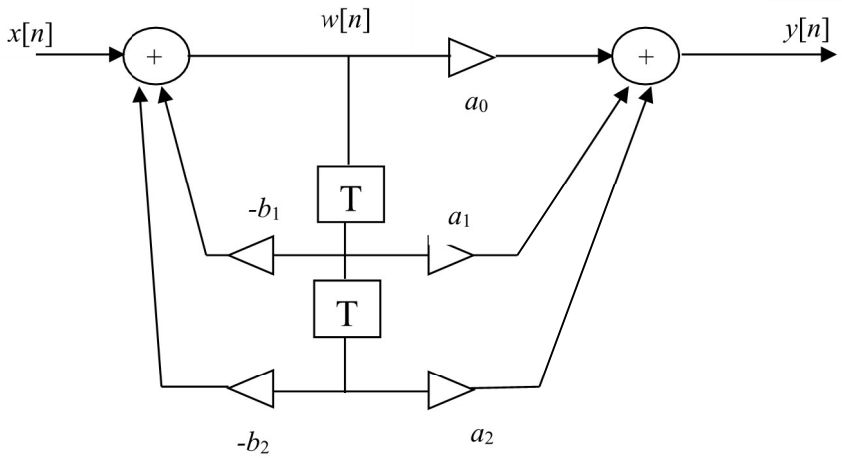
where $a_0 = 0.4981819$; $a_1 = 0.9274777$;
$a_2 = 0.4981819$; $b_1 = -0.6744878$; $b_2 = -0.3633482$;

$$H(z) = \sum_{k=0}^{11} h[k] z^{-k} \longleftarrow \text{Filter 2}$$

where

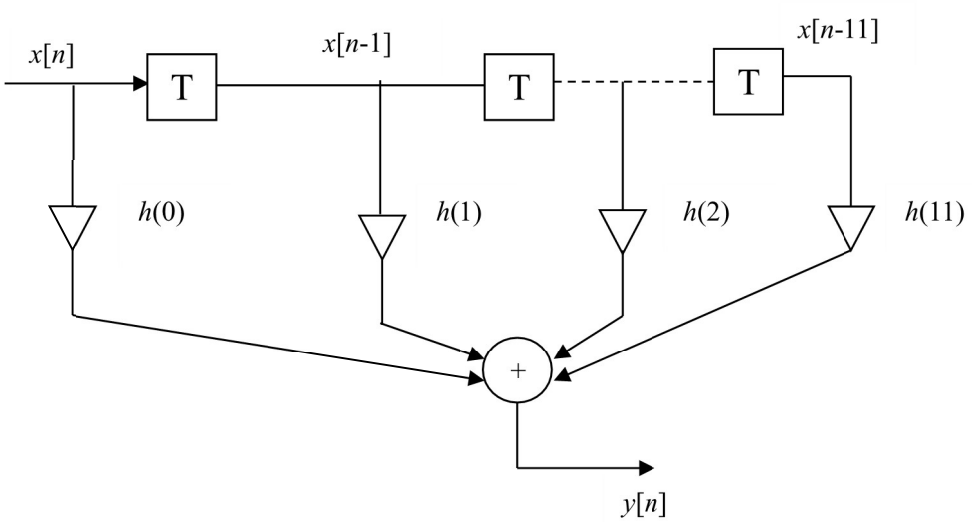
$$h[0] = 0.54603280 \times 10^{2} = h[11]$$
$$h[1] = -0.45068750 \times 10^{-1} = h)[10]$$
$$h[2] = 0.69169420 \times 10^{-1} = h[9]$$
$$h[3] = -0.55384370 \times 10^{-1} = h[8]$$
$$h[4] = -0.63428410 \times 10^{-1} = h[7]$$
$$h[5] = 0.57892400 \times 10^{0} = h[6]$$

Determine and comment on the computational and storage requirements.



Filter 1

$$w(n) = x(n) - b_1 w(n-1) - b_2 w(n-2)$$
$$y(n) = a_o w(n) + a_1 w(n-1) + a_2 w(n-2)$$



Filter 2

Computation and storage requirement of FIR and IIR

	Filter 1 (IIR)	**Filter 2 (FIR)**
Number of Multiplications	5	12
Number of Additions	4	11
Storage locations for coefficients and data	8	24

It is evident that the IIR filter is more economical in both computations and storage requirements than the FIR filter.

Exercise:

Determine the computional and storage requirements of the following filter

$$H(z) = \frac{0.8 - 0.5z^{-2} + 0.8z^{-4}}{1 - 0.45z^{-1} + 0.7z^{-4}}$$

8.3 Design Techniques

The method used to calculate the filter coefficients (h_k for FIR, a_k and b_k for IIR) depends on whether the filter is IIR or FIR type.

There are several methods of calculating filter coefficients of which the following are the most widely used.

FIR digital filters	IIR digital filters
• window	• Impulse invariant transformation
• Frequency sampling	• Bilinear transformation
• Optimisation method (e.g Remez Algorithm)	• Pole-zero placement method
$\left\{\frac{Y(z)}{X(z)} = h_0 + h_1 z^{-1} + h_2 z^{-2} + ... + h_{N-1} z^{-(N-1)}\right\}$	$\left\{\frac{Y(z)}{X(z)} = \frac{a_0 + a_1 z^{-1} + ... + a_n z^{-(N-1)}}{1 + b_1 z^{-1} + ... + b_L z^{-L}}\right\}$

We choose the method that best suits our particular applications.

In most cases, if the FIR properties are vital then a good candidate is the optimization method, where as, if IIR properties are desirable, then the bilinear method will in most cases suffice.

8.4 IIR Filter Design

In transforming an analogue filter to digital filter, we must obtain either $H(z)$ or $h[n]$ from the analogue filter design. In such transformations, we generally require that the essential properties of the analogue frequency response be preserved in the frequency response of the resulting digital filter. This implies that we want the imaginary axis of the s-plane to map into the unit circle of z-plane.

A second condition is that a stable analogue filter should be transformed to a stable digital filter. That is if the analogue

system has two poles only in the left half s-plane, then the digital filter must have poles inside the unit circle.

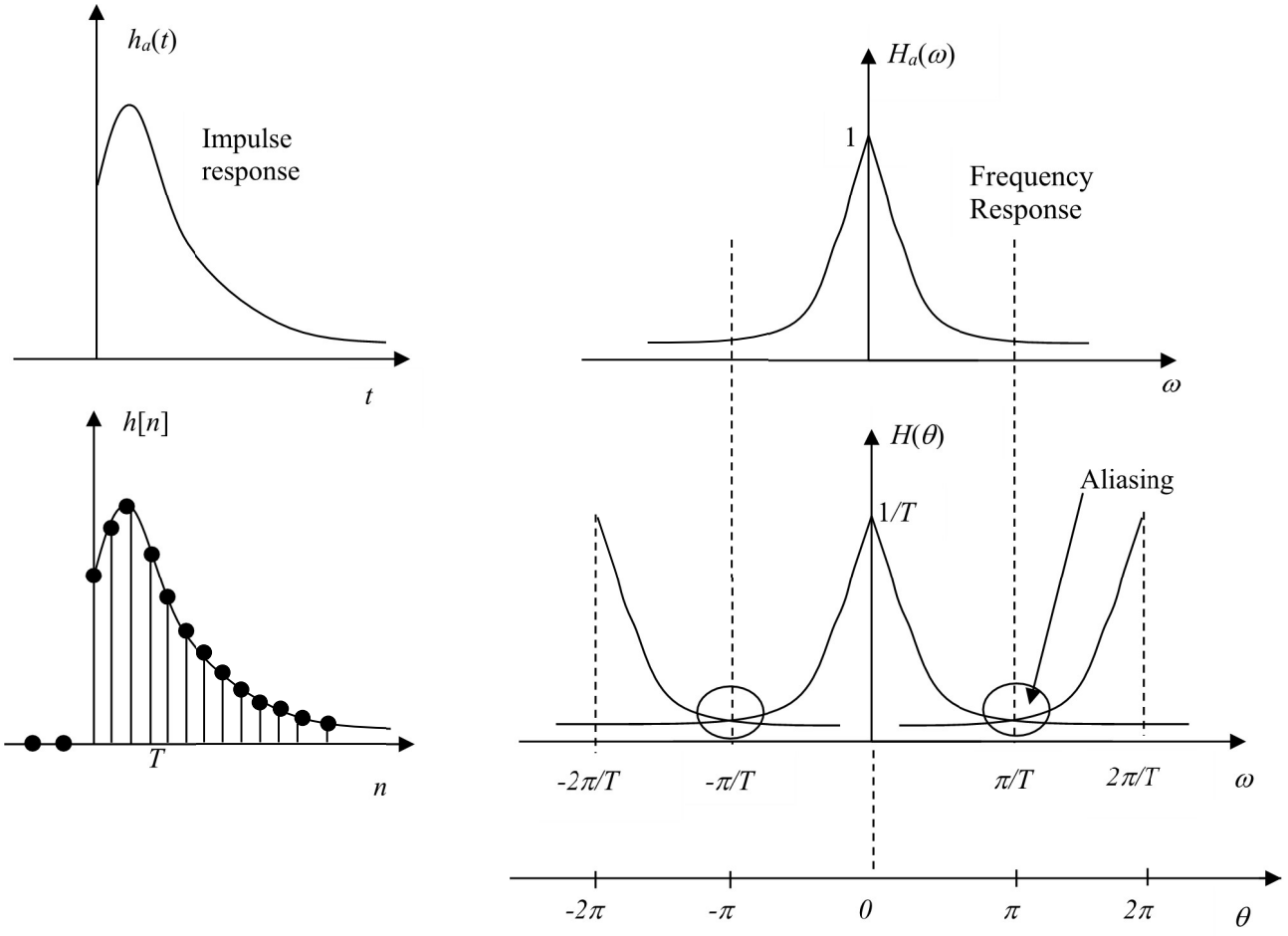
8.4.1 Impulse invariant method

In this method we start from an analogue filter of impulse response $h_a(t)$ and the system function $H_a(s)$.

The objective of our design is to realize an IIR filter with an impulse response $h[n]$ which satisfies :

$h[n] = h_a(nT)$ where T-sampling frequency

The characteristic property preserved by this transformation is that the impulse response of the resulting digital filter is a sampled version of the impulse response of the analogue filter.



We see that with this method there are problems to a greater or lesser extent depending on the choice of T.

The sampling frequency affects the frequency response of the impulse invariant discrete filter. A sufficient high sampling frequency is necessary for the frequency response to be close to that of the equivalent analogue filter. Thus due to aliasing, the frequency response of the digital filter will not be identical to that of the analogue filter. So how do we find the filter coefficients of the IIR filter in this design method?

To obtain the mapping let,

$$H_a(s) = \frac{1}{s+b}, \quad b > 0$$

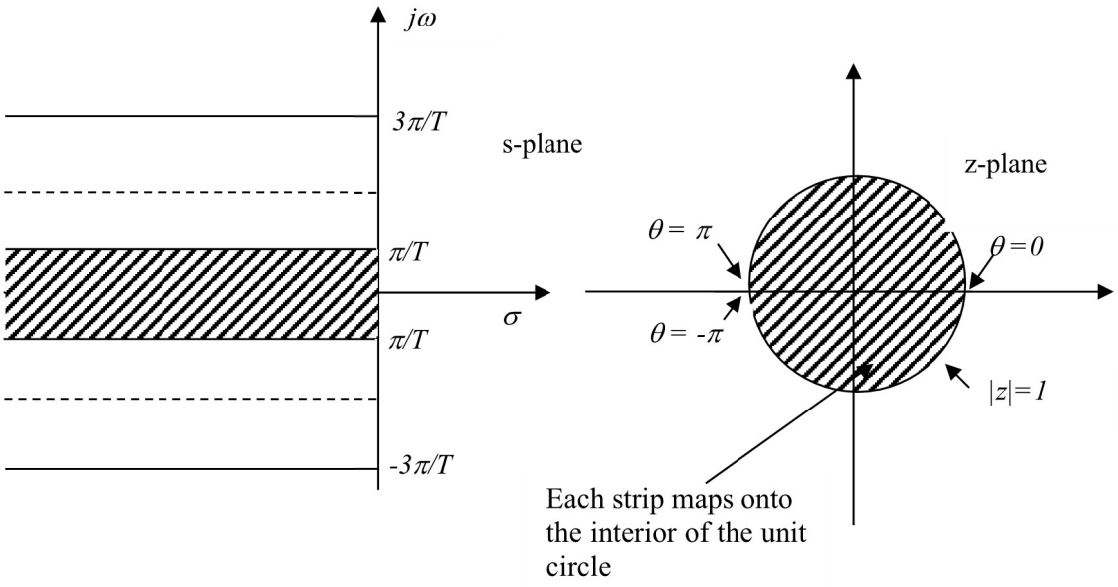
Inverse Laplace transform, $h_a(t) = e^{-bt}$

Sampled sequence, $h_a(nT) = e^{-bnT}$

Usually written as $h[n] = \begin{cases} e^{-bnT} & n \geq 0 \\ 0 & n < 0 \end{cases}$

z-transform, $H(z) = \dfrac{1}{1 - e^{-bT} z^{-1}}$

It is seen that $H(z)$ is obtained from $H_a(s)$ by using the mapping relationship, $\dfrac{1}{s+b} \rightarrow \dfrac{1}{1 - e^{-bT} z^{-1}}$, $b > 0$, T-sampling period



In this kind of mapping, the perimeter is the imaginary axis.
Note: Mapping does not exist for the zeros.

Example:

$$H(s)=\frac{2}{(s+1)(s+3)}$$

Using partial fractions this can be written as

$$H(s)=\frac{1}{s+1}-\frac{1}{s+3}$$

$$\downarrow \quad \frac{1}{s+b} \rightarrow \frac{1}{1-e^{-bT}z^{-1}}$$

$$H(z)=\frac{1}{1-e^{-T}z^{-1}}-\frac{1}{1-e^{-3T}z^{-1}}$$

$$=\frac{\left(e^{-T}-e^{-3T}\right)z^{-1}}{1-(e^{-T}-e^{-3T})z^{-1}+e^{-4T}z^{-2}}$$

Example:

Use the impulse invariant method to design digital filter from an analogue prototype that has a system function.

$$H(s)=\frac{s+a}{(s+a)^2+b^2}$$

To design a filter using the impulse invariant method, expand *H(s)* in a partial fraction form:

$$H(s)=\frac{s+a}{\left[s+\left(a+jb\right)\right]\left[s+\left(a-jb\right)\right]}=\frac{A}{s+(a+jb)}+\frac{B}{s+(a-jb)}$$

$$=\frac{1/2}{s+(a+jb)}+\frac{1/2}{s+(a-jb)}$$

$$=\frac{1}{2}\left[\frac{1}{s+(a+jb)}\right]+\frac{1}{2}\left[\frac{1}{s+(a-jb)}\right]$$

Substituting $\frac{1}{s+c} \rightarrow \frac{1}{1-e^{-cT}z^{-1}}$ (Impulse invariant transform)

$$H(z) = \frac{1}{2}\left[\frac{1}{1-e^{-(a+jb)T}z^{-1}}\right] + \frac{1}{2}\left[\frac{1}{1-e^{-(a-jb)T}z^{-1}}\right]$$
$$= \frac{\frac{1}{2}\left[1-e^{-(a-jb)T}z^{-1}\right] + \frac{1}{2}\left[1-e^{-(a+jb)T}z^{-1}\right]}{(1-e^{-(a+jb)T}z^{-1})(1-e^{-(a-jb)T}z^{-1})}$$
$$= \frac{1-e^{-aT}\cos(bT)z^{-1}}{1-2e^{-aT}\cos(bT)z^{-1}+e^{-2aT}z^{-2}}$$

Hence,

$$H(z) = \frac{1+a_1z^{-1}}{1+b_1z^{-1}+b_2z^{-2}}$$

where

$$a_1 = e^{-aT}\cos(bT), \quad b_1 = -2e^{-aT}\cos(bT) \quad \text{and} \quad b_2 = e^{-2aT}$$

$$\frac{s+a}{(s+a)^2+b^2} \rightarrow \frac{1-e^{-aT}\cos(bT)z^{-1}}{1-2e^{-aT}\cos(bT)z^{-1}+e^{-2aT}z^{-2}}$$

Note that the zero at $s = -a$ is mapped to a zero at $z=e^{-aT}\cos(bT)$.

Thus, the location of the zero in the discrete time filter depends on the position of the poles as well as the zero in the analogue filter.

Example:

Using impulse invariant method design a digital filter to approximate the following normalized analogue transfer function:

$$H(s)=\frac{1}{s^2+\sqrt{2}s+1}$$

Assume that the 3dB cut-off frequency of the digital filter is 150Hz and the sampling frequency is 1.28 kHz

Solution: Before applying the impulse invariant method, we need to de-normalize the transfer function.

$$H_1(s)=H(s)\Big|_{s\to\frac{s}{\omega_c}}=\frac{1}{\left(\frac{s}{\omega_s}\right)^2+\sqrt{2}\left(\frac{s}{\omega_s}\right)+1}$$

$$H(s)=\frac{{\omega_c}^2}{s^2+\sqrt{2}\omega_c s+{\omega_c}^2} \quad \text{where} \quad \omega_c=2\pi\times 150=942.4778$$

$$H_1(s)=\frac{\omega_c^2}{s+\sqrt{2}\omega_c s+{\omega_c}^2}=\frac{A}{s-p_1}+\frac{B}{s-p_2}$$

$$p_1,p_2=\frac{-\sqrt{2}\omega_c\pm\sqrt{2\omega_c^2+4\omega_c^2}}{2}$$

$$=\frac{-\sqrt{2}\omega_c\pm j\sqrt{2}\omega_c}{2}$$

$$=\frac{-\sqrt{2}\omega_c[1\pm j]}{2}$$

$$\therefore\ p_1=\frac{-\sqrt{2}\omega_c}{2}[1+j]$$

$$p_2=\frac{-\sqrt{2}\omega_c}{2}[1-j]$$

$$p_1=-666.4324(1+j)$$

$$p_2=-666.4324(1-j)$$

$$A=-\frac{\omega_c}{\sqrt{2}}j$$

$$B=\frac{+\omega_c}{\sqrt{2}}j$$

$$H_1(z) = \frac{A}{1 - e^{p_1 T} z^{-1}} + \frac{B}{1 - e^{p_2 T} z^{-1}} = \frac{A(1 - e^{-p_2 T} z^{-1}) + B(1 - e^{p_1 T} z^{-1})}{(1 - e^{p_1 T} z^{-1})(1 - e^{p_2 T} z^{-1})}$$

$$H(z) = \frac{(A + B) - (Ae^{p_2 T} + Be^{p_1 T})z^{-1}}{1 - (e^{p_1 T} + e^{p_2 T})z^{-1} + e^{p_1 T} z^{-2}}$$

$$H(z) = \frac{393.9264 z^{-1}}{1 - 1.0308 z^{-1} + 0.3530 z^{-2}}$$

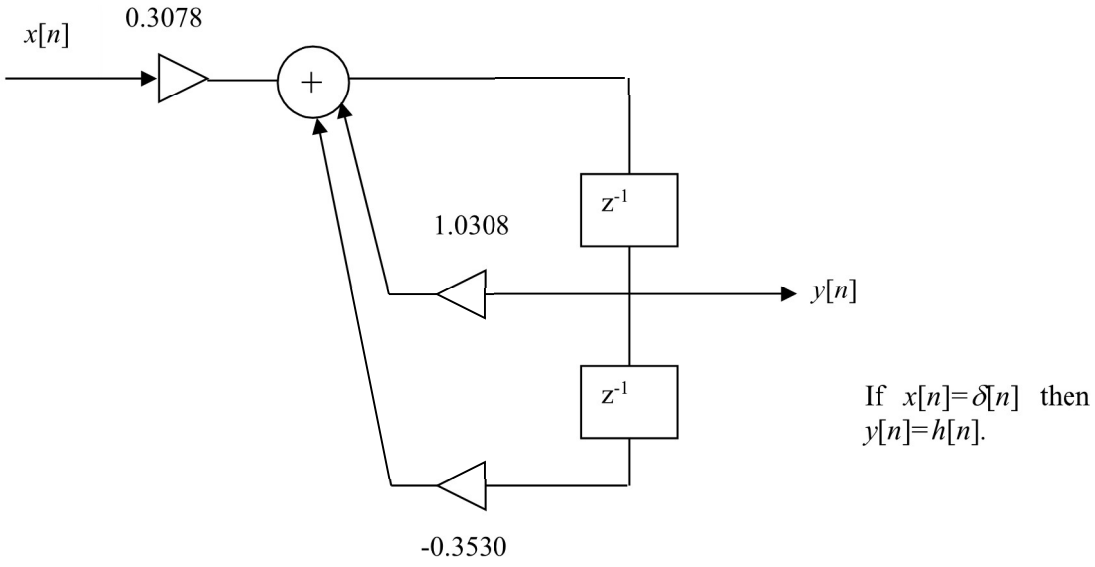
$$H(\theta) = \frac{393.9264 e^{-j\theta}}{1 - 1.0308 e^{-j\theta} + 0.3530 e^{-j2\theta}}$$

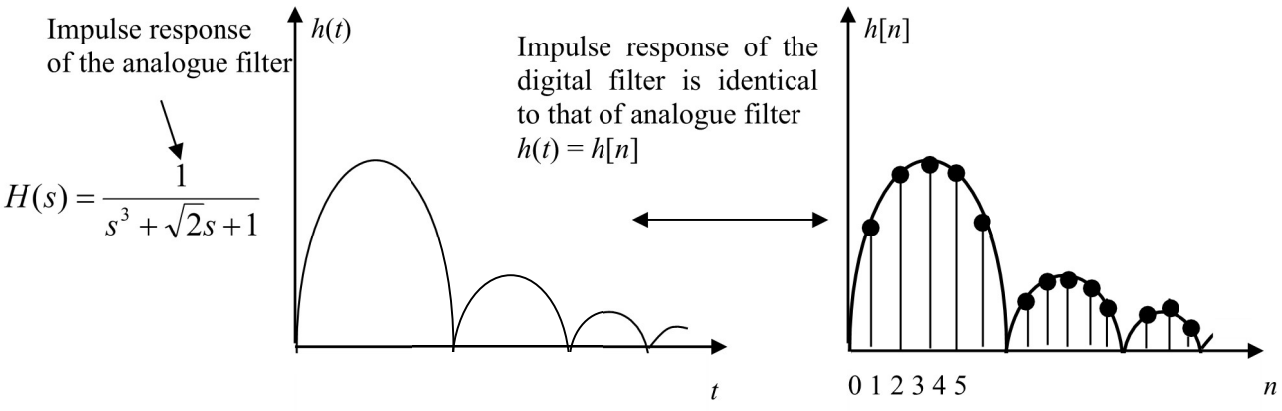
$$H(\theta)\Big|_{\theta=0} = \frac{393.9264}{1 - 1.0308 + 0.3530} \approx 1223$$

This is approximately equal to the sampling frequency

Such a large gain is characteristic of impulse invariant filters.To keep the gain down (and to avoid overflows when the filter is implemented). It is common practice to divide the gain by f_s. Thus the new transfer function becomes

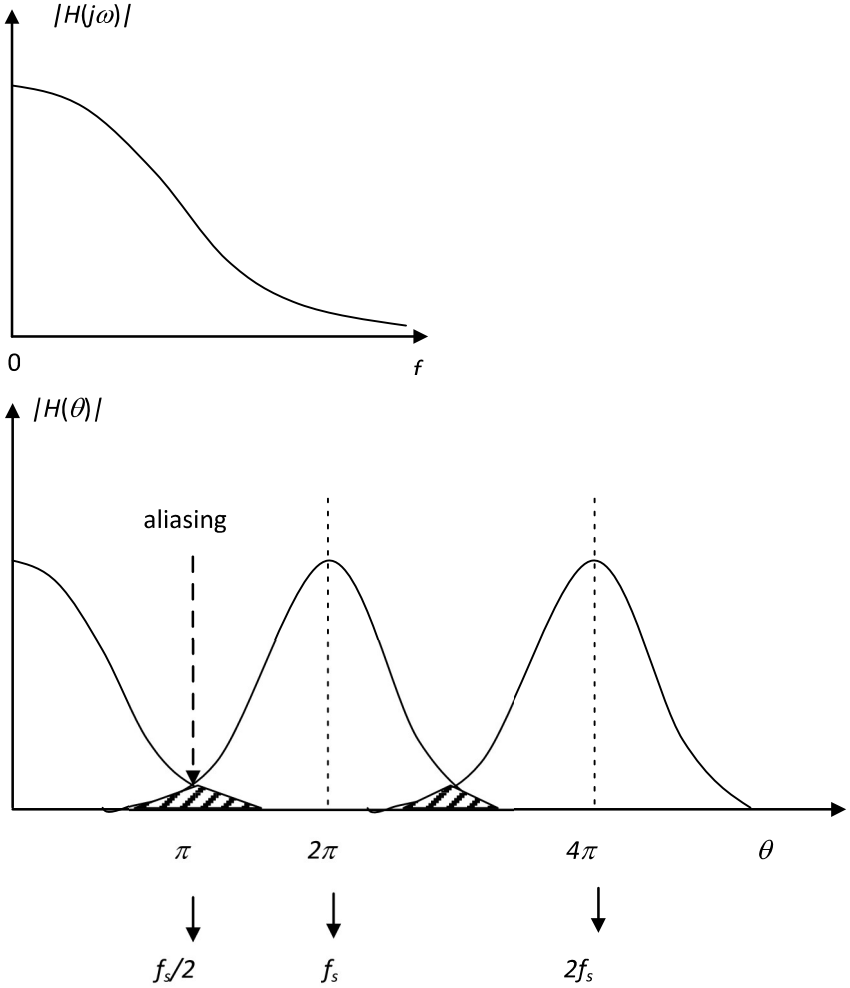
$$H(z) = \frac{0.3078 z^{-1}}{1 - 1.0308 z^{-1} + 0.3530 z^{-2}}$$





Note: The sampling frequency affects the frequency response of the digital filter obtained using impulse invariant transformation.

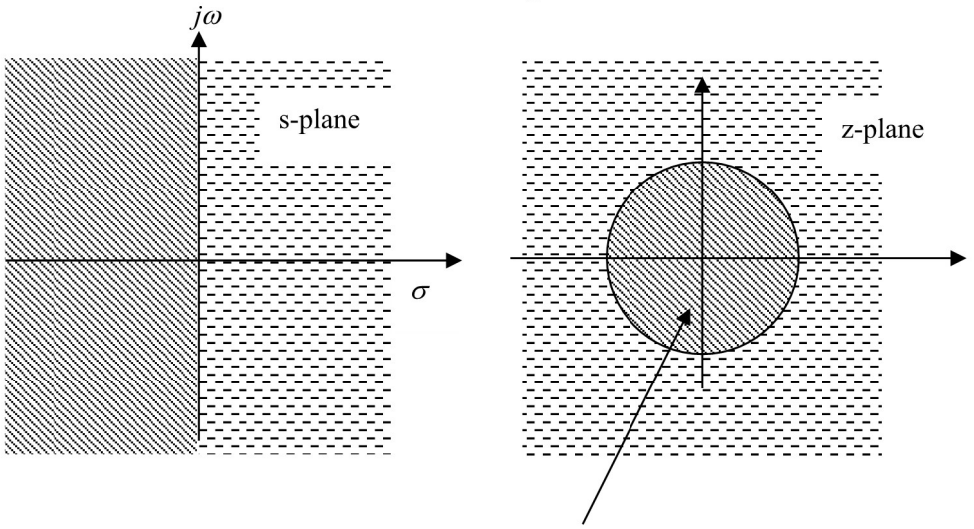
A sufficient high samples frequency is necessary for the frequency response to be closer to that of the equivalent analogue filter (see below)



Low degree of aliasing can be achieved by making the sampling frequency high.

8.4.2 Bilinear Transformation

The bilinear transformation yields stable digital filters from stable analogue filters (the impulse invariant technique may not). Also the bilinear transformation avoids the problem of aliasing encountered with the use of the impulse invariant transformation, because it maps the entire imaginary axis in the s-plane on to the unit circle in the z-plane.



$$z = e^{sT} = \frac{e^{\frac{sT}{2}}}{e^{\frac{-sT}{2}}} = \frac{1+\frac{sT}{2}+\left(\frac{sT}{2}\right)^2\cdot\frac{1}{2}+\ldots}{1-\frac{sT}{2}+\left(\frac{sT}{2}\right)^2\cdot\frac{1}{2}+\ldots}$$

$$z \approx \frac{1+\frac{sT}{2}}{1-\frac{sT}{2}}$$

(drop out higher order terms)

$$\therefore\ s = \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}$$

The price paid for the avoidance of aliasing is an introduction of distortion in the frequency axis. Consequently, the design of digital filters using the bilinear transformation is only useful when the distortion can be compensated.

Example: We have a system function $H_a(s)$ such that

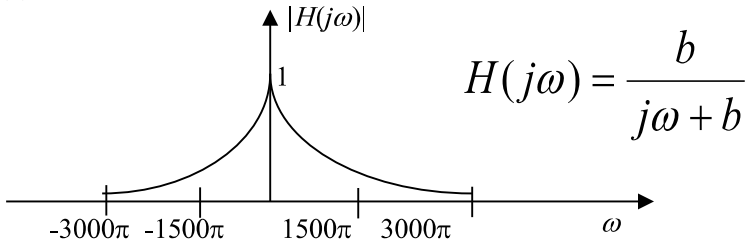
$$H_a(s) = \frac{b}{s+b}$$

Applying bilinear transformation,

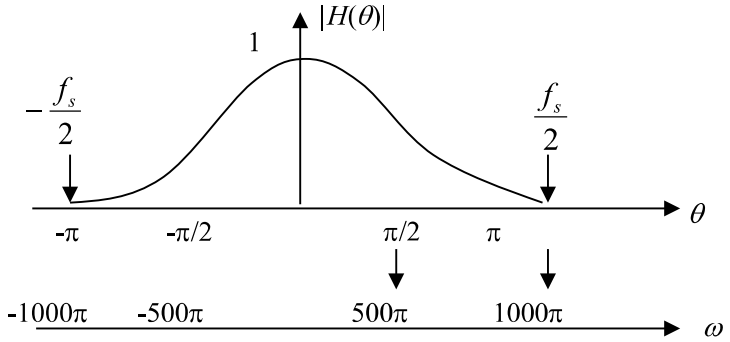
$$H(z) = \frac{b}{\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}} + b} = \frac{bT(1+z^{-1})}{bT+2+(bT-2)z^{-1}}$$

Let $b = 1000$ & $T = \frac{1}{1000}$

The modulus of the frequency response $H(\omega)$ and $H(\theta)$ are shown below:



$$H(j\omega) = \frac{b}{j\omega + b}$$



There is a very important property of the bilinear transformation that can be seen in the above example. The entire frequency range ($-\infty \le \omega_a \le \infty$) of the continuous system maps into the fundamental interval ($-\pi \le \theta \le \pi$) of the discrete system, where $\omega = 0$ corresponds to $\theta = 0$, $\omega = \infty$ to $\theta = \pi$ and $\omega = -\infty$ to $\theta = -\pi$.

To demonstrate that this mapping has the property that the imaginary axis in the s-plane maps onto the unit circle, let $z = e^{j\theta}$ and s = $j\omega$. Then since $s = \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}$,

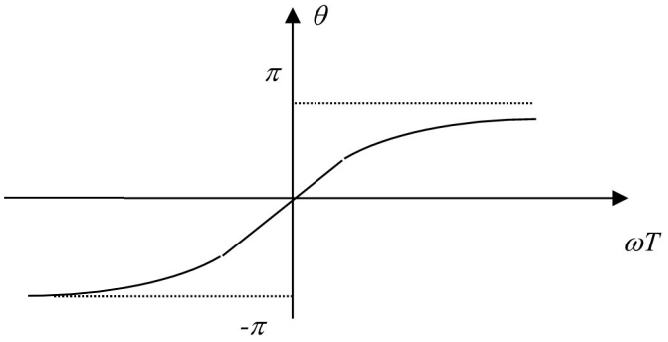
$$j\omega = \frac{2}{T}\frac{1-e^{-j\theta}}{1+e^{-j\theta}}$$

$$j\omega = \frac{2}{T} j \tan\left(\frac{\theta}{2}\right)$$

Hence,

$$\omega = \frac{2}{T}\tan\left(\frac{\theta}{2}\right) \text{ or } \theta = 2\tan^{-1}\left(\frac{\omega T}{2}\right)$$

We see that a nonlinear relation exists between ω and θ. This effect is called 'Warping' and is shown below.



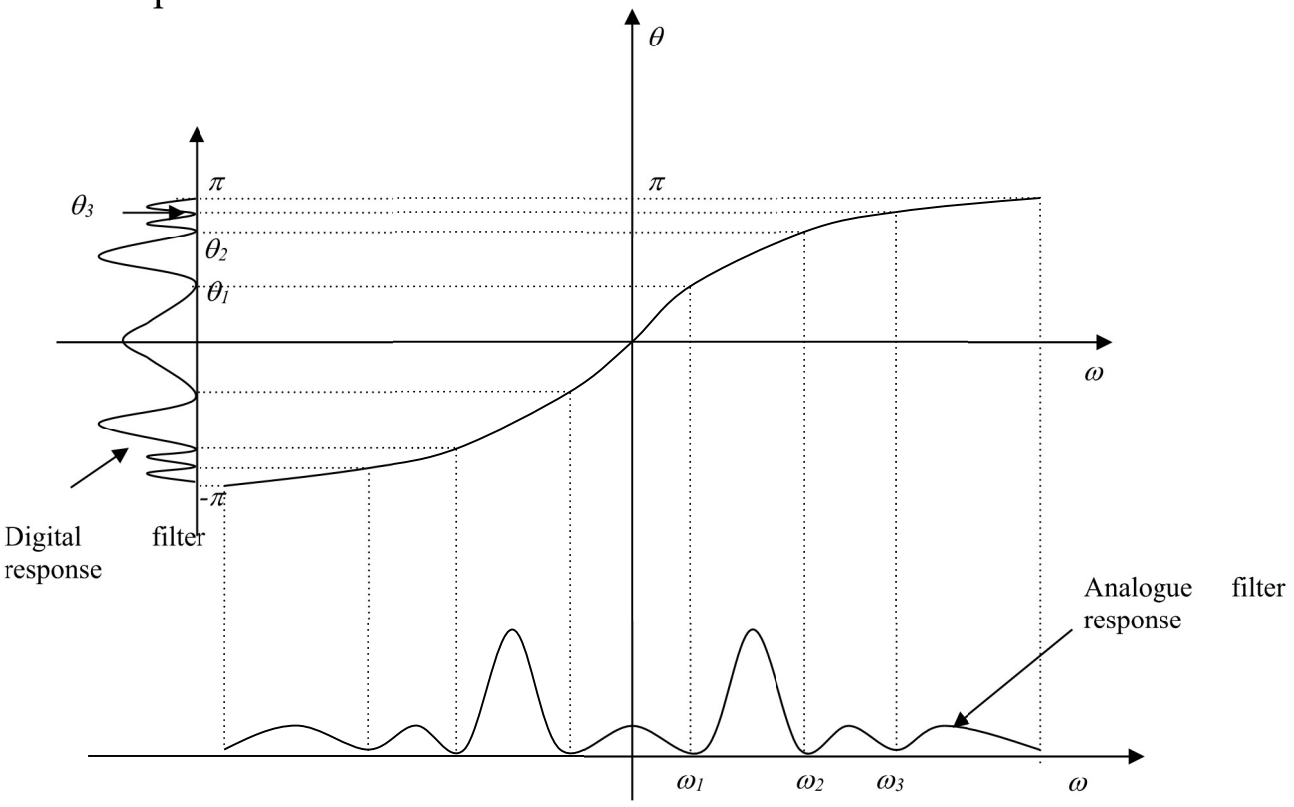
At low frequencies $\tan^{-1}\left(\frac{\omega T}{2}\right) = \frac{\omega T}{2}$.

$$\therefore \theta = 2 \times \frac{\omega T}{2} = \omega T$$

The great advantage of warping is that no aliasing of the frequency characteristic can occur in the transformation of an analogue filter to a discrete filter, which we encountered in the impulse-invariant method.

We must however check carefully just how the various characteristic frequencies of the continuous characteristic frequencies of the discrete filter.

We can illustrate this with the aid of a diagram (below) for a band-pass filter.



The effect of "warping" in the conversion of $|H(\omega)| \rightarrow |H(\theta)|$ is seen from the above diagram.

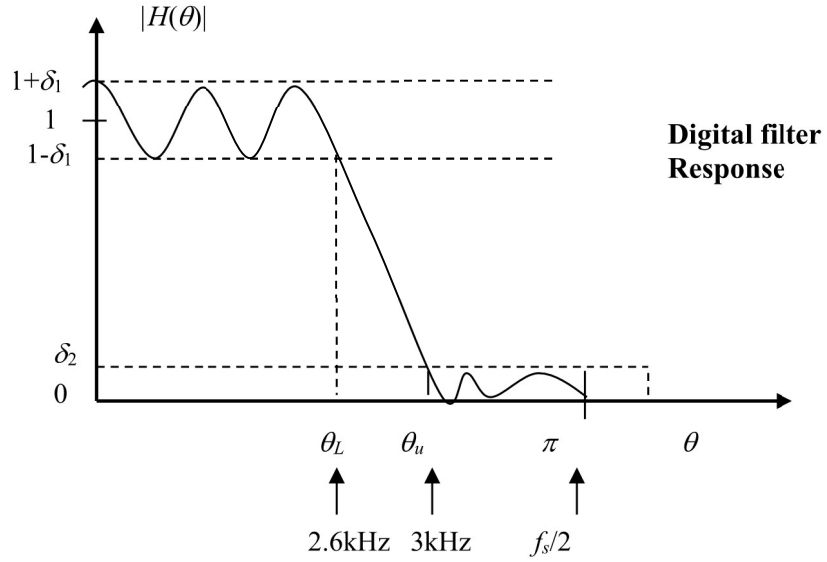
In designing a digital filter by this method we must first **pre-wrap** the given filter specifications to find the continuous filter to which we are going to apply the bilinear transformation.

Example: The specification of a desired digital low pass filter is shown below.

Sampling frequency: $f_s = 8\text{kHz}$ ($T = 1/f_s = 125\mu\text{s}$)

A pass band up to $f_L = 2.6$ kHz ($\theta_L = \omega_L T = 2\pi \frac{f_L}{f_s} = 0.65\pi$)

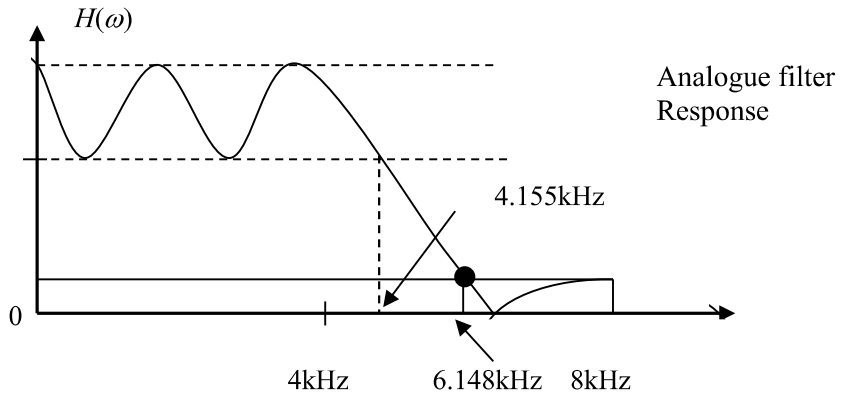
and a stop band, f_u, above 3kHz ($\theta_u = \omega_L T = 2\pi \frac{f_u}{f_s} = 0.75\pi$).



We must start from an analogue filter with

$$\omega_L = 2\pi f_l = \frac{2}{T}\tan\frac{\theta_L}{2} = 2\pi(4155) \Rightarrow f_L = 4.155\text{kHz}$$

$$\omega_u = 2\pi f_h = \frac{2}{T}\tan(\frac{\theta_u}{2}) = 2\pi(6148) \Rightarrow f_u = 6.148\text{kHz}$$



Example: Determine, using bilinear transformation method, the transfer function and difference equation for the digital equivalent of the RC filter. The normalized transfer function for the RC filter is

$$H(s) = \frac{1}{s+1}$$

Assume a sampling frequency of 150Hz and a cut-off frequency of 30Hz.

Solution: $\theta_c = \omega_c T = 2\pi f_c \cdot \frac{1}{f_s}$ before prewarping

$$\theta_c = 2\pi(30) \cdot \frac{1}{150} = 0.4\pi$$

The analogue frequency after prewarping is

$$\omega_c' = \frac{2}{T}\tan\left(\frac{0.4\pi}{2}\right) = 217.95 \text{ rad/sec}$$

Pre-warped frequency

f_c' is always $> f_c$ (un-warped freq= 30 Hz) and after warping, $f_c' = 34.68$Hz.

The de-normalized analogue filter transfer function (this is achieved by replacing s with $\frac{s}{\omega_c}$) is obtained from $H(s)$ as

$$H_1(s) = H(s)\Big|_{s=\frac{s}{\omega_c'}} = \frac{\omega_c'}{s+\omega_s'} = \frac{\left(\frac{2}{T}\right)0.7265}{s+\frac{2}{T}\times 0.7265}$$

$$H(z) = H_1(s)\Big|_{s=\frac{2}{T}\frac{z-1}{z+1}\cdot} = \frac{0.7265(1+z)}{(1+7265)z+0.7265-1} = \frac{0.4208(1+z^{-1})}{1-0.1584z^{-1}}$$

Example:

It is required to design a digital filter to approximate the following analogue transfer function

$$H(s) = \frac{1}{s^2 + \sqrt{2}s + 1}$$

using the Bi-linear transformation method obtain the transfer function, *H(s)* of the digital filter assuming a 3dB cut-off frequency of 150Hz and a sampling frequency of 1.28kHz.

Solution:

$$f_c = 150\text{Hz} \qquad f_s = 1280\text{Hz} \quad T = \frac{1}{f_s}$$

$$\downarrow$$

$$\theta_c = 2\pi\left(\frac{f_c}{f_s}\right) = 2\pi\left(\frac{150}{1280}\right) = \frac{15}{64}\pi$$

The analogue frequency after pre-warping

$$\omega_c' = \frac{2}{T}\tan\left(\frac{\theta_c}{2}\right) = \frac{2}{T} \times 1280 \tan\left(\frac{\frac{15\pi}{64}}{2}\right) = 987.5009 = \frac{2}{T} \times 0.3859 rad/\sec$$

$$f_c' = \frac{987.5009}{2\pi} = 157.1656 Hz$$

f_c' is always greater than f_c.

Prewarped analogue filter is given by,

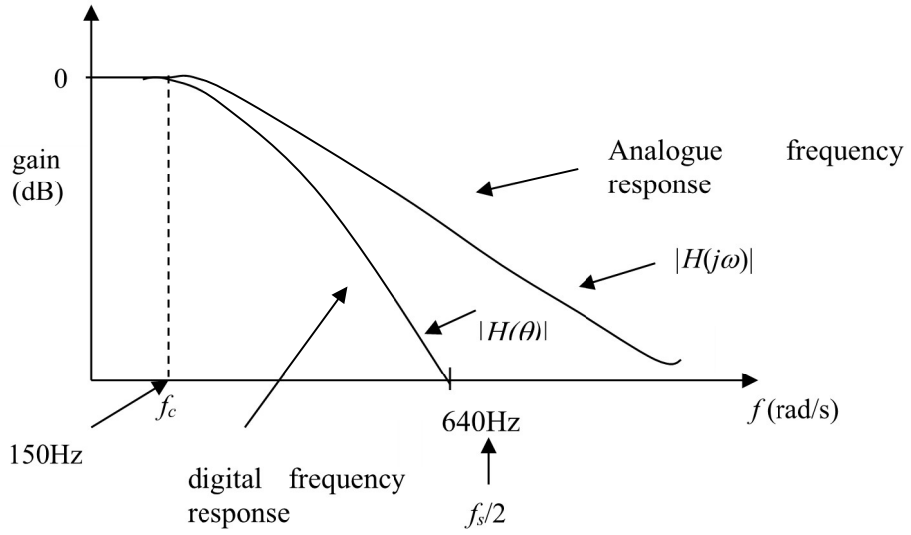
$$H_1(s) = \frac{\left(\frac{2}{T}\right)^2 (0.3857)^2}{s^2 + \frac{2}{T}(0.3857)\sqrt{2}s + \left(\frac{2}{T}\right)^2 (0.3857)^2}$$

$$H(z) = H_1(s)\Big|_{s=\frac{2}{T}\frac{z-1}{z+1}} = \frac{\left(\frac{2}{T}\right)^2 (0.3857)^2}{\left(\frac{2}{T}\frac{z-1}{z+1}\right)^2 + \frac{2}{T}\times 0.3857\times\sqrt{2}\left(\frac{2}{T}\frac{z-1}{z+1}\right)+\left(\frac{2}{T}\right)^2(0.3857)^2}$$

$$H(z) = \frac{0.0878\left(1+2z^{-1}+z^{-2}\right)}{1-1.0048z^{-1}+0.3561z^{-2}}$$

$$\frac{\left(\omega_c{}'\right)^2}{s^2+\sqrt{2}\omega_c' s+\left(\omega_{c'}\right)^2} \xrightarrow{\textit{Bilinear transformation}} \frac{0.0878(1-2z^{-1}+z^{-2})}{1-1.0048z^{-1}+0.3561z^{-2}}$$

All pole — Poles & zeros



Note:

- Same cut-off frequency
- increased roll off and attenuation in stop band.

Example:

(a) An analogue transfer function can be converted to a digital transformation using the bilinear transformation. Derive this transform relationship using the following equation.

$$y[n]-y[n-1]=\frac{T}{2}[x[n]+x[n-1]] \quad \longleftarrow \text{Digital Integrator}$$

$$H(s)=\frac{1}{s} \quad \longleftarrow \text{Analogue Integrator}$$

T-sampling period, $x[n]$ - input , $y[n]$ - output

Solution:

$$Y(z)-Y(z)z^{-1}=\frac{T}{2}\left\{X(z)-X(z)z^{-1}\right\}$$

$$H(z)=\frac{T}{2}\frac{1+z^{-1}}{1-z^{-1}} \qquad (1)$$

$$H(s)=\frac{1}{s} \qquad (2)$$

Setting (1) = (2),

$$\frac{1}{s}=\frac{T}{2}\frac{1+z^{-1}}{1-z^{-1}} \quad \Rightarrow s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}$$

(b) Convert the analogue filter $H(s)$

$$H(s)=\frac{s+0.1}{(s+0.1)^2+16}$$

into a digital IIR filter by means of the bilinear transformation.

The digital filter is to have a resonant frequency $\theta_0 = \frac{\pi}{2}$.

$$H(s) = \frac{s + 0.1}{s^2 + 0.2s + 16.01} \quad \leftarrow \omega_0^2$$

The analogue filter has a resonant frequency $\omega_0 \approx 4$ rad/sec.

The frequency is to be mapped into $\theta_0 = \frac{\pi}{2}$ by selecting the value of the parameter T.

$$\omega_0 = \frac{2}{T}\tan\frac{\theta_0}{2}$$

$$4 = \frac{2}{T}\tan\frac{\frac{\pi}{2}}{2} = \frac{2}{T}\tan\frac{\pi}{4} \quad \Rightarrow 4 = \frac{2}{T}$$

$$\therefore T = \frac{1}{2} \text{ in order to have } \theta_0 = \frac{\pi}{2} \text{ rad/sec}$$

Thus the desired mapping is

$$s = 4\left(\frac{1 - z^{-1}}{1 + z^{-1}}\right)$$

$$H(z) = H(s)\Big|_{s=4\left(\frac{1-z^{-1}}{1+z^{-1}}\right)} = \frac{4\left(\frac{1 - z^{-1}}{1 + z^{-1}}\right) + 0.1}{4\left(\frac{1 - z^{-1}}{1 + z^{-1}}\right)^2 + 0.2 \times 4\left(\frac{1 - z^{-1}}{1 + z^{-1}}\right) + 16.01}$$

$$= \frac{0.128 + 0.006z^{-1} - 0.122z^{-2}}{1 + 0.0006z^{-1} + 0.975z^{-2}} = \frac{0.128 + 0.006z^{-1} - 0.122z^{-2}}{1 + 0.975z^{-2}}$$

This filter has poles at

$$p_1 = 0.987e^{+j\frac{\pi}{2}} \quad \leftarrow \text{resonant frequency}$$

$$p_2 = 0.987e^{-j\frac{\pi}{2}} \quad \leftarrow \text{resonant frequency}$$

and zeros at $z_1 = -1$ and $z_2 = 0.95$.

Example: Convert the simple low pass filter $H(s)=\frac{1}{s+1}$ into an equivalent high pass discrete filter. Assume f_s = 150Hz, analogue cut-off f_c = 30Hz.

$$\theta_c = 2\pi\left(\frac{30}{150}\right)= 0.4\pi$$

$$\omega_c^{'} = \frac{2}{T}\tan(\frac{0.4\pi}{2}) = \frac{2}{T}\times 0.7265$$

LPF to HPF transformation

$$H^{'}(s) = H(s)\Big|_{s\to\frac{\omega_c^{'}}{s}} = \frac{1}{\frac{\omega_c^{'}}{s}+1} = \frac{s}{s+\omega_c^{'}}$$

$$H(z) = H^{'}(s)\Big|_{z=\frac{2}{T}\left(\frac{z-1}{z+1}\right)} = \frac{\frac{2}{T}\left(\frac{z-1}{z+1}\right)}{\frac{2}{T}\left(\frac{z-1}{z+1}\right)+\frac{2}{T}0.7265} = \frac{z-1}{z-1+0.7265(z+1)}$$

$$H(z) = 0.5792\frac{1-z^{-1}}{1+0.1584z^{-1}}$$

Exercise:

Using the second order normalised Butterworth lowpass filter G(s), design an IIR digital filter G(z) to approximate the analogue transfer function G(s). The digital filter is to be designed to operate at a sampling rate of 8 kHz with a 3 dB cut-off frequency at 2 kHz. Using the bilinear transformation method, obtain the transfer function G(z) of the digital filter and show that G(z)

$$G(z) = \frac{(z+1)^2}{(2+\sqrt{2})(z^2+\frac{2-\sqrt{2}}{2+\sqrt{2}})}$$

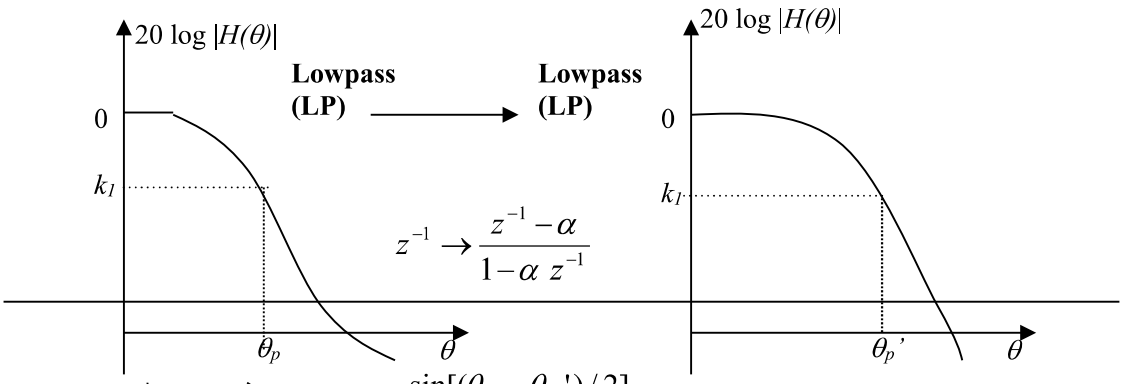
8.4.3 Digital-To-Digital Transformations

We have seen in the lecture notes that one method for the design of analogue filters relied on applying a transformation to an analogue low-pass filter with a unit bandwidth. It was shown that we could obtain low-pass, high-pass, band-pass and band-stop filters by selecting the appropriate transformation.

Similarly, a set of transformations can be formed that take a low-pass digital filter and turn it into high-pass, band-pass, and band-stop or another low-pass digital filter.

The transformations are given below:

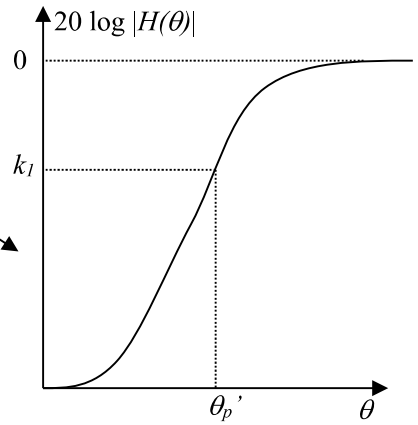
Digital to Digital Transformations



$$z^{-1} \rightarrow \frac{z^{-1}-\alpha}{1-\alpha\, z^{-1}}$$

$$\alpha = \frac{\sin[(\theta_p - \theta_p')/2]}{\sin[(\theta_p + \theta_p')/2]}$$

Lowpass to Highpass



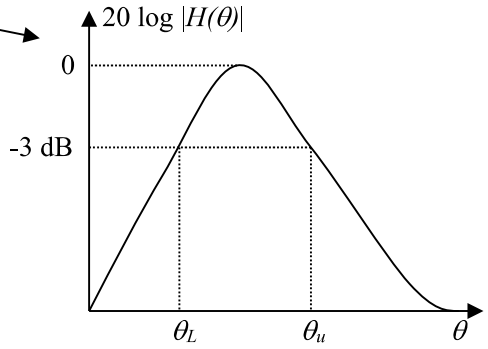
$$z^{-1} \rightarrow \frac{z^{-1}+\alpha}{1+\alpha\, z^{-1}}$$

Simplest transformation is to change the signs of poles and zeros of the LP → LP transformation

Lowpass to Bandpass

$$\alpha = \frac{\cos[(\theta_p + \theta_p')/2]}{\cos[(\theta_p - \theta_p')/2]}$$

$$z^{-1} \rightarrow -\frac{z^{-2} - \frac{2\alpha k}{k+1} z^{-1} + \frac{k-1}{k+1}}{\frac{k-1}{k+1} z^{-2} - \frac{2\alpha k}{k+1} z^{-1} + 1}$$

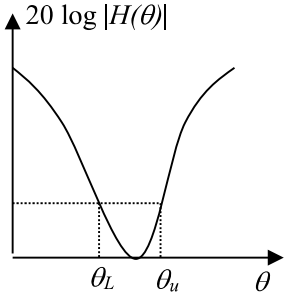


$$\alpha = \frac{\cos[(\theta_u + \theta_L)/2]}{\cos[(\theta_u - \theta_L)/2]}$$

$$k = \cot((\theta_u - \theta_L)/2) \cdot \tan\left(\frac{\theta_p}{2}\right)$$

Lowpass to Bandstop

$$z^{-1} = \frac{z^{-2} - \frac{2\alpha}{k+1} z^{-1} + \frac{k-1}{k+1}}{\frac{k-1}{k+1} z^{-2} - \frac{2\alpha}{k+1} z^{-1} + 1}$$



8.5 Notch filters

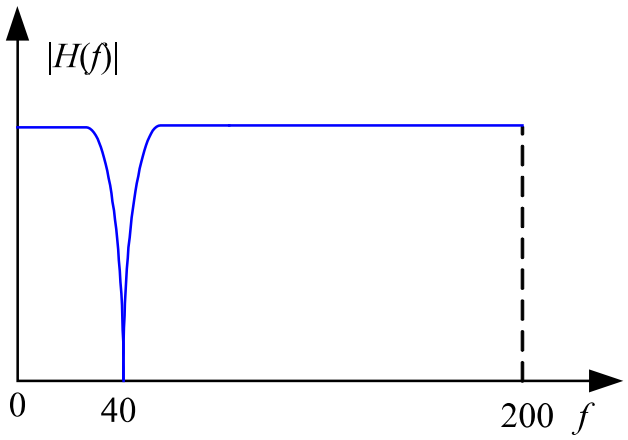
When a zero is placed at a given point on the z-plane, the frequency response will be zero at the corresponding point. A pole on the other hand produces a peak at the corresponding frequency point.

Poles that are close to the unit circle give rise large peaks, where as zeros close to or on the unit circle produces troughs or minima. Thus, by strategically placing poles and zeros on the z-plane, we can obtain sample low pass or other frequency selective filters (**notch filters**).

Example:
Obtain, by the pole-zero placement method, the transfer function of a sample digital notch filter (see figure below) that meets the following specifications:

- Notch Frequency: 40Hz
- 3db width of the Notch: ±4Hz
- Sampling frequency: 400 Hz

The radius, r of the poles is determined by: $r = 1 - \left(\frac{\Delta f}{f_s}\right)\pi$

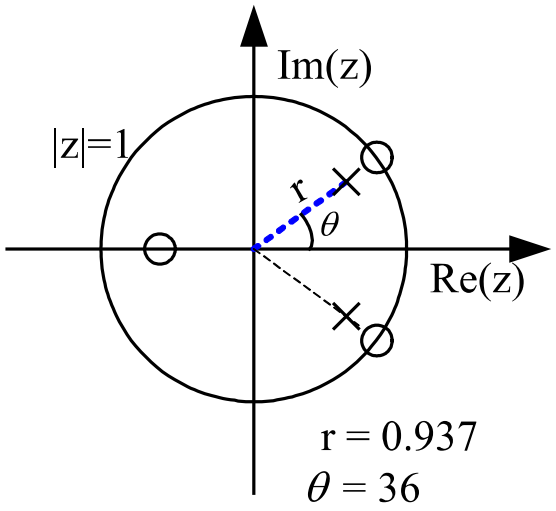


To reject the component at 40 Hz, place a pair of complex zeros at the points on the unit circle corresponding to ±40 Hz.

$$\text{i.e. at angles of } 360^{\bullet} \times \frac{40}{400} = \pm 36^{\bullet} = \pm 0.2\pi$$

To achieve a sharp notch filter and improved magnitude response either side of the notch frequency, a pair of complex conjugate poles are placed at a radius $r < 1$, at the same frequency as the zeroes, as shown below.

$$r = 1 - \left(\frac{\Delta f}{f_s}\right)\pi = 1 - \left(\frac{8}{400}\right)\pi = 0.937$$



$$H(z) = \frac{\left(z - e^{-j0.2\pi}\right)\left(z - e^{j0.2\pi}\right)}{\left(z - 0.937e^{-j0.2\pi}\right)\left(z - 0.937e^{j0.2\pi}\right)}$$

$$= \frac{z^2 + 1 - (e^{j0.2\pi} + e^{-j0.2\pi})}{z^2 + 0.878 - 0.937(e^{j0.2\pi} + e^{-j0.2\pi})z}$$

$$= \frac{z^2 + 1 - 2\cos(0.2\pi)}{z^2 + 0.878 - 2 \times 0.937\cos(0.2\pi)}$$

$$= \frac{1 - 1.6180z^{-1} + z^{-2}}{1 - 1.5161z^{-1} + 0.878z^{-2}}$$

Exercise: A bandpass digital filter is required to meet the following specifications:

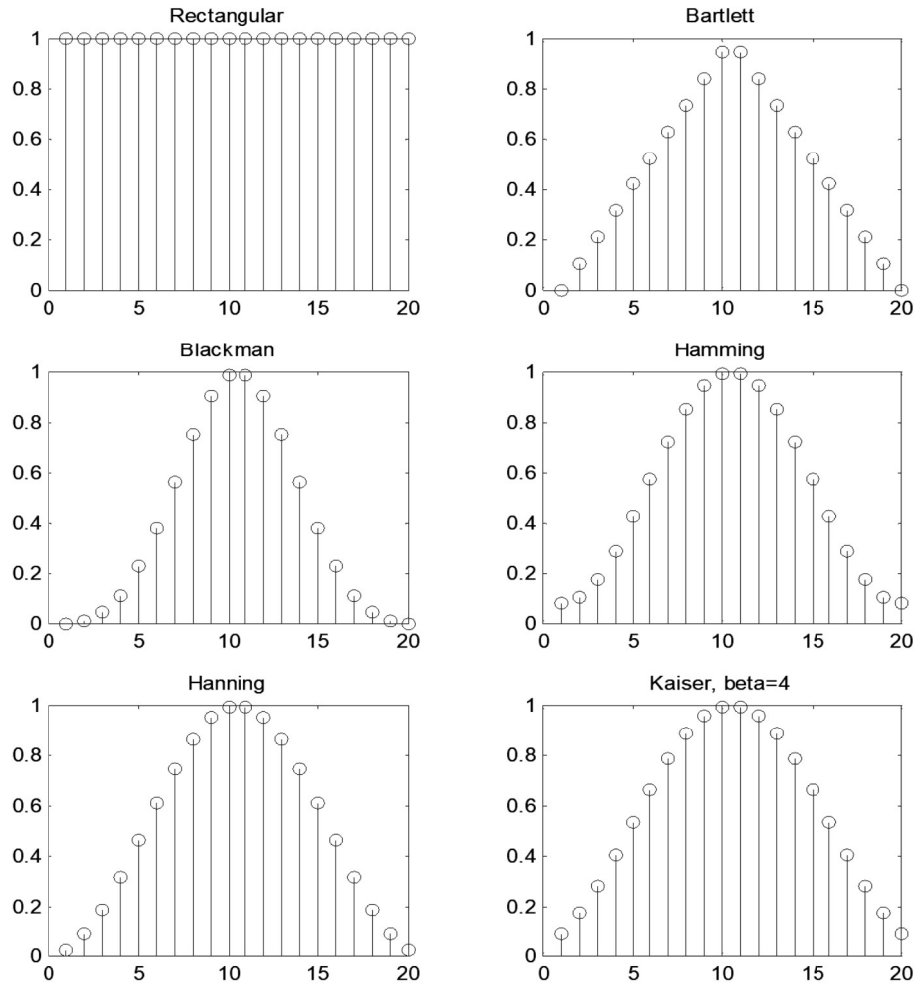
(1) Complete signal rejection at dc and 250Hz;
(2) A narrow passband centred at 125Hz;
(3) A 3dB bandwidth of 10Hz

Assuming a sampling frequency (f_s) of 500Hz, obtain the transfer function of the filter, by suitably placing z-plane poles and zeros, and it difference equations.

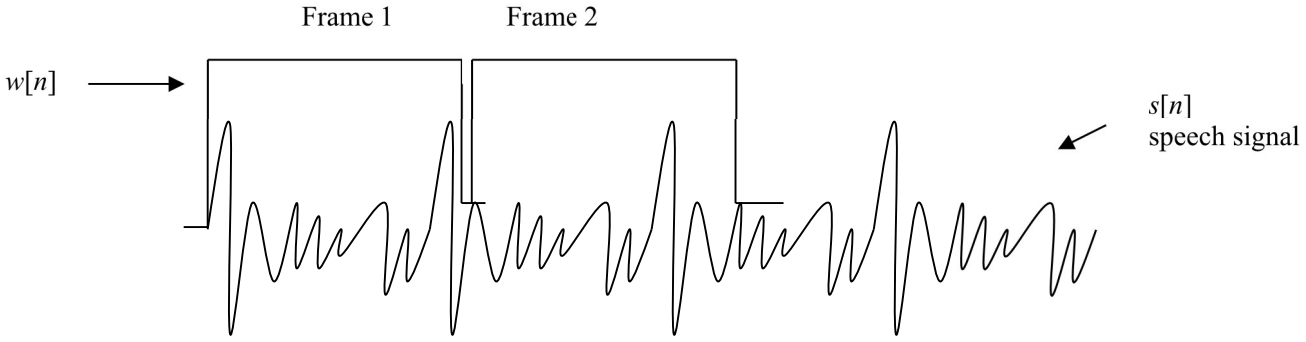
The radius, r of the poles is determined by the desired bandwidth. An approximate relationship between r, for r>0.9, and bandwidth, BW, is given by $r \approx 1 - \left(\frac{BW}{f_s}\right)\pi$

8.6 Window Functions

Window functions are used to truncate a signal or impulse response to produce a signal or impulse response of finite duration. Some of the most commonly used window functions are:



To analyse a **truncation** process we model it as a multiplication of the desired sequence by finite duration window sequence denote by $w[n]$. Truncation of a sequence $s[n]$ is equivalent to placing a rectangular time window around $s[n]$.



Vocal tract shape changes every 15ms. When using a sampling frequency of 8 kHz (T = 125μs) with 100 samples in each frame (12.5ms),

$$y[n] = s[n] \cdot w[n]$$

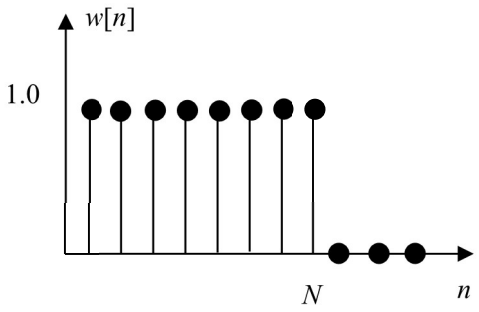
(Multiplication becomes convolution in the frequency domain)

In frequency domain,

$$Y(\theta) = S(\theta) * W(\theta)$$

Thus when window is applied, the frequency domain convolution causes distortion in the spectrum $Y(\theta)$. It can be shown that the rectangular window creates ripples in the spectrum $Y(\theta)$. To reduce the distortion, use Hamming or Kaiser Window.

Rectangular Window

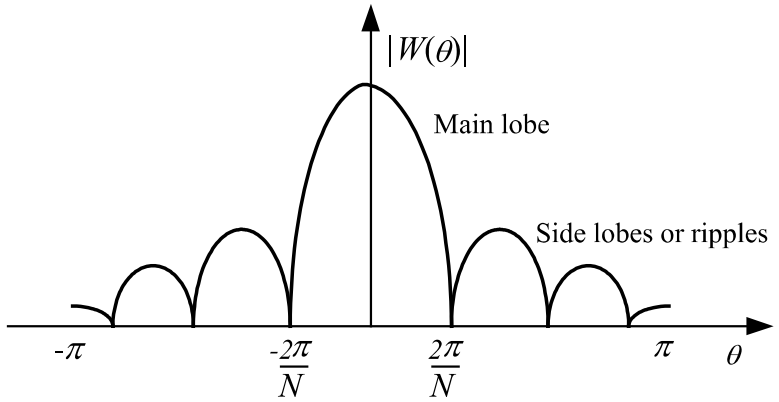


$$w[n] = \begin{cases} 1 & 0 \le n \le N-1 \\ 0 & \text{otherwise} \end{cases}$$

$$W(z) = \sum_{n=0}^{N-1} w(n) z^{-n} = 1 + z^{-1} + z^{-2} + \ldots\ldots + z^{-(N-1)} = \frac{1 - z^{-N}}{1 - z^{-1}}$$

$$W(\theta) = W(z)\Big|_{z=e^{j\theta}} = \frac{1 - e^{-jN\theta}}{1 - e^{-j\theta}}$$

$W(\theta) = e^{-j\frac{N-1}{2}\theta} \sin\frac{\frac{N\theta}{2}}{\frac{\theta}{2}}$	$\phi(\theta) = -\left(\frac{N-1}{2}\right)\theta$	$\lvert W(\theta)\rvert = \left\lvert \frac{\sin\frac{N\theta}{2}}{\sin\frac{\theta}{2}} \right\rvert$



If *N* increases the width of the main lobe decreases but the peak amplitude of the side lobes grows in a manner such that the area under each lobe is constant while the width of each lobe decreases with *N*.

8.7 Design Methods for FIR filters

The most essential feature of FIR filters is, by definition, the finite length of the impulse response. Another important point is that it can be seem directly from the impulse response of an FIR filter whether we have a linear phase characteristic or not.

A filter is said to have a linear phase response of its response satisfies one of the following relationships.

$$\phi(\theta) = -a\theta \qquad (1)$$

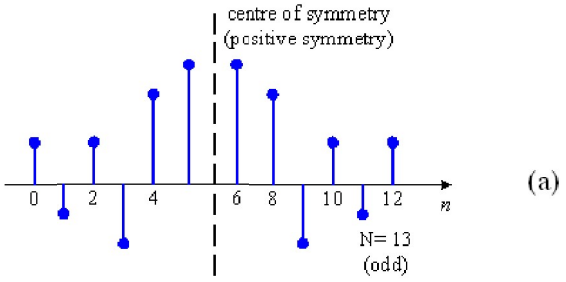
$$\phi(\theta) = b - a\theta \qquad (2)$$

where a and b are constants.
It can be shown that for condition (1) above to be satisfied the impulse response of the filter must have positive symmetry.

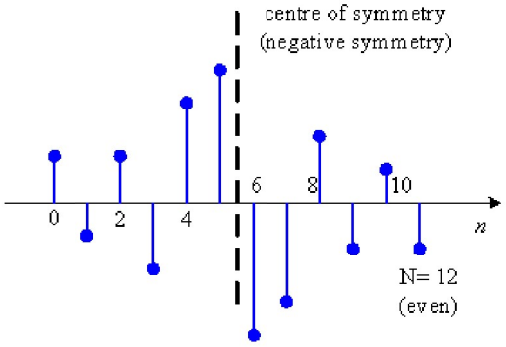
$$h[n] = h[N-n-1], \quad a = \frac{N-1}{2}$$

where N denotes the filter length. When the condition given in (2) is satisfied, the impulse response of the filter has negative symmetry.

$$h[n] = -h[N-n-1], \quad a = \frac{N-1}{2} \text{ and } b = \frac{\pi}{2}$$



(a)



(b)

8.7.1 Design of FIR filters using Windows.

The easiest way to obtain an FIR filter is to simply truncate the impulse response of an IIR filter. If $h_d[n]$ represents the impulse response of a desired IIR filter, then an FIR filter with impulse response $h[n]$ can be obtained as follows:

$$h[n] = \begin{cases} h_d[n] & N_1 \le n \le N_2 \\ 0 & \text{otherwise} \end{cases}$$

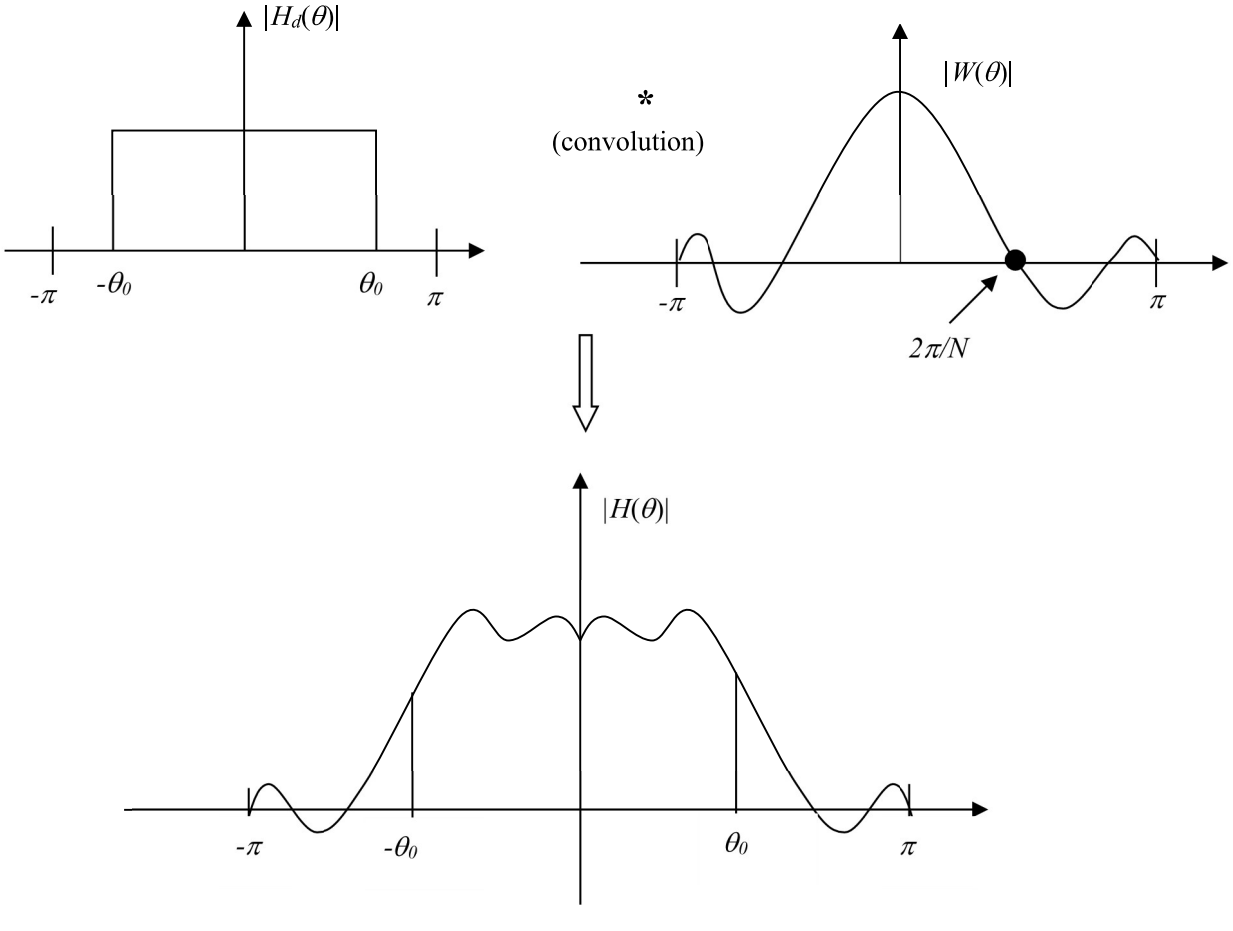
In general $h[n]$ can be thought of as being formed by the product $h_d[n]$ and a window function $w[n]$ as follows:

$$h[n] = h_d[n] \cdot w[n]$$

let it be, for example a rectangular window

$$H(\theta) = H_d(\theta) * W(\theta)$$

let it be an ideal low pass filter with cut off frequency θ_0



Therefore it is seen that the convolution produces a smeared version of the ideal low pass frequency response $H_d(\theta)$.

In general, the wider the main lobe of $W(\theta)$, the more spreading, where as the narrower the main lobe (larger N) the closer $|H(\theta)|$ comes to $|H_d(\theta)|$.

In general we are left with a trade off of making N large enough so that smearing is minimized, yet small enough to allow reasonable implementation.

8.7.2 Design Procedure

An ideal low pass filter with linear phase of slope $-\beta$ and cut-off ω_c can be characterized in the frequency domain by

$$H_d(\theta) = \begin{cases} e^{-j\theta\beta} & |w| \le \omega_c \\ 0 & w_c < |\omega| \le \pi \end{cases}$$

The corresponding impulse response $h_d[n]$ can be obtained by taking the inverse Fourier transform of $\boldsymbol{H}_d(\theta)$ and easily shown to be

$$h_d[n] = \frac{\sin[\omega_c(n-\beta)]}{\pi(n-\beta)}$$

A causal FIR filter with impulse response $h[n]$ can be obtained by multiplying $h_d[n]$ by a window beginning at the origin and ending at N-1 as follows :

$$h[n] = \frac{\sin[\omega_c(n-\beta)]}{\pi(n-\beta)}\omega[n]$$

For $h[n]$ to be a linear phase, β must be selected so that the resulting $h[n]$ is symmetric.

As $\frac{\sin[\omega_c(n-\beta)]}{\pi(n-\beta)}$ is symmetric about $n=\beta$ and the window symmetric is about $n=\frac{N-1}{2}$

$$\therefore \quad \beta = \frac{N-1}{2} \quad \longleftarrow \text{Symmetric about } \beta$$

Example:

(a) Determine the impulse response $h_d[n]$ of the lowpass filter whose frequency response is given by

$$H(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{3} \\ 0 & \frac{\pi}{3} < |\theta| \le \pi \end{cases}$$

(b) To obtain a finite impulse response from $h_d[n]$ a rectangular window of length $N = 9$ is used. Compute the coefficients of the FIR filter with a linear phase characteristic ands with this finite impulse response.

Solution:

(a)

$$h_d[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\theta) e^{j\theta n} d\theta$$

$$= \frac{1}{2\pi}\left[\frac{e^{j\theta n}}{jn}\right]_{\frac{-\pi}{3}}^{\frac{\pi}{3}} = \frac{1}{\pi}\left(\frac{\sin n\frac{\pi}{3}}{n}\right)$$

$$h_d[n] = \begin{cases} 1/3 & n = 0 \\ \dfrac{\sin(n\pi/3)}{n\pi} & n \ne 0 \end{cases}$$

(b) For a linear phase filter, $h_d[n]$ is symmetric about $n = 0$ and the window is symmetric about $n = \frac{(N-1)}{2} = \frac{9-1}{2} = 4$.

$$h[n] = h_d[n] \cdot \omega[n]$$

$$h[0] = 0.333$$

$$h[1] = \frac{\sqrt{3}}{2\pi}$$

$$h[2] = \frac{\sqrt{3}}{4\pi}$$

$$h[3] = 0$$

$$h[4] = -\frac{\sqrt{3}}{8\pi}$$

The coefficients are

	$\frac{-\sqrt{3}}{8\pi}$	0	$\frac{\sqrt{3}}{4\pi}$	$\frac{\sqrt{3}}{2\pi}$	0.333	$\frac{\sqrt{3}}{2\pi}$	$\frac{\sqrt{3}}{4\pi}$	0	$\frac{-\sqrt{3}}{8\pi}$
n	0	1	2	3	4	5	6	7	8

Symmetry about $n = 4$

Exercise: The filter $G(\theta)$ need not be very sharp in practice. It only serves for attenuating higher spectral components resulting from the increase in the sampling rate. You are requested to design a linear phase FIR filter for this purpose. The length of the impulse response is set to N=7 and the coefficients are determined using a windowing technique. To obtain a linear phase FIR filter after windowing a phase factor, $f(\theta)$ must be first introduced.

$$G_d(\theta) = \begin{cases} e^{j\phi(\theta)} & 0 \le |\theta| \le \pi/4 \\ 0 & \pi/4 \le |\theta| \le \pi \end{cases}$$

(i) Determine $\phi(\theta)$.

(ii) Using part (i) above, Determine the impulse response $g_d[n]$ using a rectangular window.

CHAPTER 8: PROBLEM SHEET 8

Q1)

a) A second-order analogue band pass filter with an *s*-domain transfer function is given by

$$H(s) = \frac{b_p s}{s^2 + b_p s + \omega_p{}^2} \quad (1)$$

Where ω_p and b_p are the centre frequency and bandwidth of the filter; respectively, both expressed in rad/s. By applying the bilinear transformation to equation (1) a digital filter with the following transfer function can be obtained:

$$H(z) = \frac{a_0 - a_1 z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}} \quad (2)$$

Show that the digital filter coefficients are given by

$$a_0 = a_1 = \frac{2b_p T}{4 + 2b_p T + \omega_p{}^2 T^2}; \quad b_1 = \frac{2\omega_p{}^2 T^2 - 8}{4 + 2b_p T + \omega_p{}^2 T^2}; \quad b_2 = \frac{4 + \omega_p^2 T^2 - 2b_p T}{4 + 2b_p T + \omega_p^2 T^2}$$

b) A digital filter with a centre frequency of 1000 Hz and a bandwidth of 150 Hz is required. Assuming a sampling frequency of 10kHz, compute the digital filter coefficients a_0, a_1, b_1 & b_2 and show that $a_0 = a_1 = 0.04409115$; $b_1 = -1.551$ and $b_2 = 0.918176$

c) Comment on the stability of the digital filter *H(z)* (see equation 2 above) which you have obtained .

Ans: *Filter is stable.*

Q2)

a) Determine the impulse response $h_d[n]$ of the bandpass filter whose frequency response is given by

$$H_d(\theta) = \begin{cases} e^{-j3\theta} & \frac{\pi}{4} \le |\theta| \le \frac{3\pi}{4} \\ 0 & \text{otherwise} \end{cases}$$

b) To obtain a finite impulse response from $h_d[n]$ a Bartlett window of length $N = 7$ is used. Compute the coefficients of the FIR filter with this impulse response.

Note: The Bartlett Window function is given by

$$w_B[n] = \begin{cases} \frac{2n}{N-1} & 0 \le n < \frac{(N-1)}{2} \\ 2 - \frac{2n}{N-1} & \frac{N-1}{2} \le n \le N-1 \\ 0 & \text{otherwise} \end{cases}$$

where N is the length of the window.

Ans:

$$h_d[n] = \begin{cases} \frac{1}{\pi(n-3)}\left[\sin(n-3)\frac{3\pi}{4} - \sin(n-3)\frac{\pi}{4}\right] & n \ne 3 \\ \frac{1}{2} & n = 3 \end{cases}$$

h[n]	0	-1/(3π)	0	1/2	0	-1/(3π)	0
n	0	1	2	3	4	5	6

Q3) Design a digital filter which meets the following specifications:

Low-pass filter:	0 to 10 kHz passband
Sampling frequency:	f_s = 100 kHz
Transition band:	10 kHz to 20 kHz
Stopband attenuation:	-10 dB(Starting at 20 kHz)

The filter must be monotonic in the pass and stop bands, (i.e. no ripple).

Ans: $H(z) = \dfrac{0.0675 + 0.1349z^{-1} + 0.0675z^{-2}}{(1 - 1.143z^{-1} + 0.4123z^{-2})}$

Q4) Consider the following analogue system with a transfer function

$$H(s) = \frac{\alpha}{s+\alpha} \Rightarrow h(t) = e^{-\alpha t}$$

where α (= 10^4 rad/sec) is the analogue cut-off frequency.

(a) Using bilinear transformation, show that the transfer function *H(z)* is

$$H(z) = \frac{1-a}{2}\left[1 + \frac{(1+a)z^{-1}}{1-az^{-1}}\right] \quad \text{where } a = \frac{2-\alpha T}{2+\alpha T}$$

where a = 10^4 rad/sec and the sampling period T is 100 μs.

(i) What is the dc gain of *H(z)*?

(ii) At what frequency is the *H(θ)* equal to zero? (θ - digital frequency).

(iii) Calculate the impulse response *h[n]*.

(iv) Assuming that the impulse response decays to *1/e* of its initial value at $n = N$ samples, show that:

$$N = \frac{\ln\left\{\dfrac{a}{a+1}\right\} - 1}{\ln(a)}$$

End of Chapter 8

Chapter 9

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Chapter 9: Multirate Digital Signal Processing

The increasing need in modern digital systems to process data at more than one sampling rate has lead the development of a new sub-area in DSP known as multirate processing. The two primary operations in multirate proceeding are

- **Decimation**
  (We can decrease the sampling rate f_s of a given signal $x[n]$: This is sometimes called **down sampling**).

- **Interpolation**
  (We can increase the sampling rate f_s of a given signal $x[n]$: This is sometimes called up **sampling**).

The process of decimation and interpolation are the fundamental operations in multirate signal processing.

9.1 Decimation

A sampling rate decreaser is shown in Figure 9.1 below. We should confine our attention to a decrease by an integer factor M. (e.g. $M = 3$)

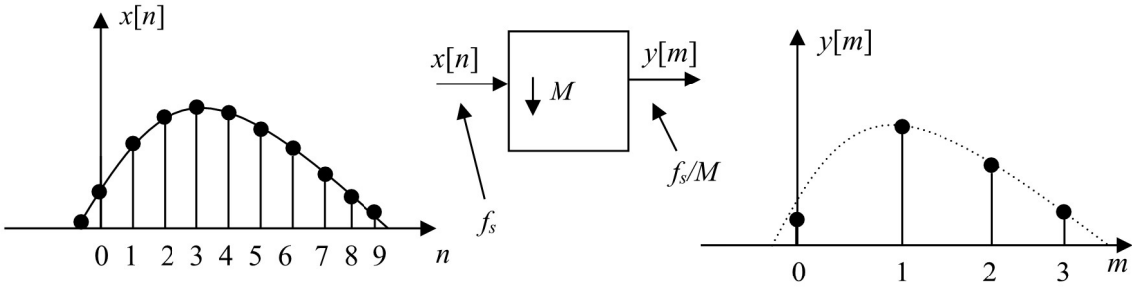


Figure 9.1: *A discrete-time signal, x[n], is down-sampled by a factor of M=3, yielding another discrete-time signal, y[n].*

Example: $x[n] = \{1, 2, 4, 3, 5, -6, -8, 2, -3, 2\}$

$\downarrow$ Down sample by 2

$y[m] = \{1,4,5,-8,-3\}$

The output signal $y[n]$ is obtained by taking every M^{th} sample of the input signal. If $M = 3$, we should just take every third sample of $x[n]$ to form the desired signal $y[m]$.

Obviously, it only makes sense to reduce the sampling rate if the information constant of the signal we wish to preserve is band limited to $\frac{f_s}{6}$; half the desired sampling rate since the spectral components above this frequency will be aliased into frequencies below $\frac{f_s}{6}$ according to the sampling rule.

Figure 9.2 shows a block representation of a times M decimator. We see that the signal $x[n]$ is first passed through a low pass filter that attenuates the band from $\frac{f_s/2}{M} = \frac{f_s}{2M}$ to $\frac{f_s}{2}$ to prevent aliasing

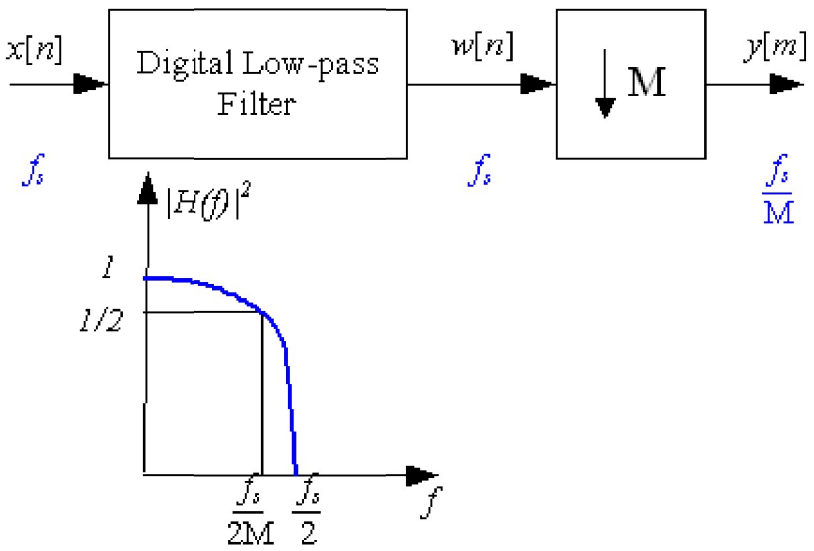


Figure 9.2: *Prior to down-sampling, a discrete-time signal must first go through a low-pass filter to remove its high-frequency components. This prevents aliasing.*

The spectral description of the decimation process is shown in Figure 9.3.

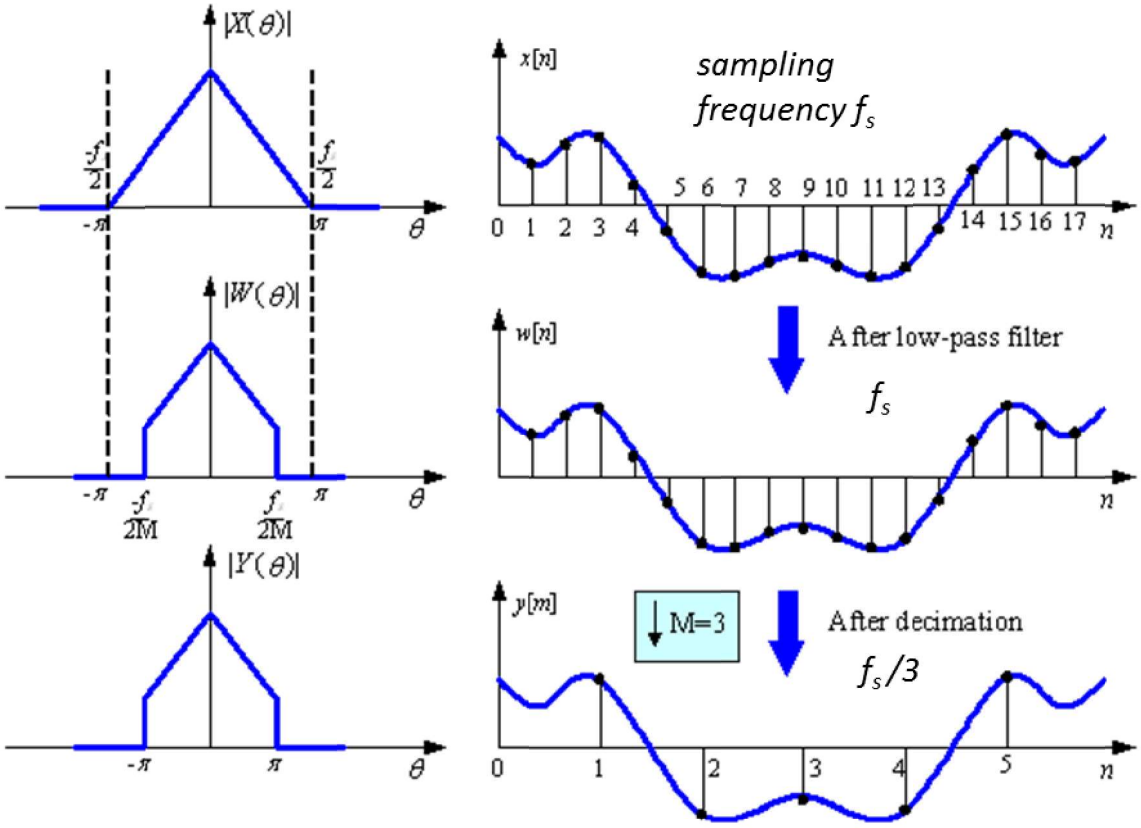
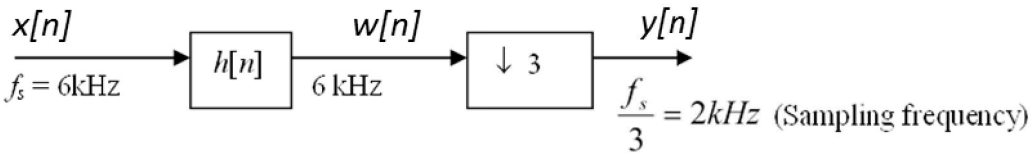
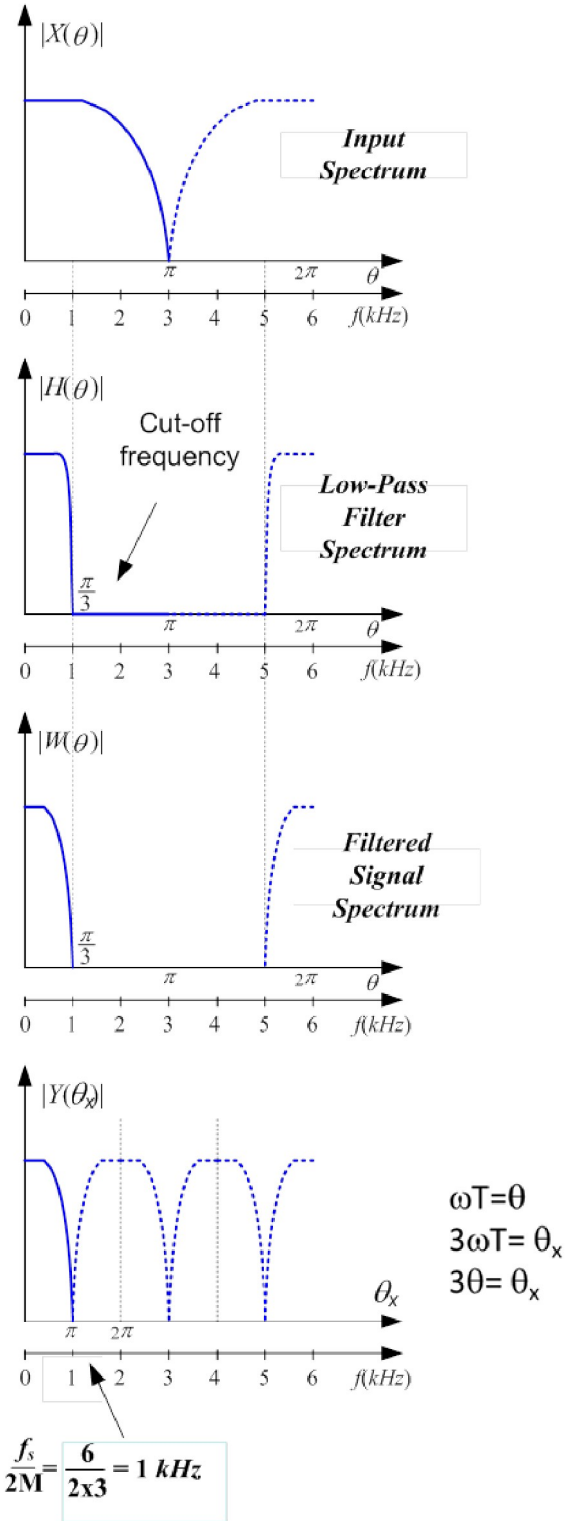


Figure 9.3: Spectral description of decimation process





Spectral interpretation of decimation of a signal from 6kHz to 2kHz

Note: Such a processing requirement may arise, for example when a speech signal is over sampled at f_s = 32 kHz. Since we are interested only in a band of 0-4 kHz, sampling rate can be reduced to 8 kHz, so the first step in the decimation process has to be the digital filtering of the signal $x[n]$ is band limited to $f_s/8$. Should we use IIR (recursive) or FIR (non-recursive) for the low pass filtering required?

Using an IIR filter in this case has an obvious shortcoming. We cannot take advantage of the fact that we only have to compute every N^{th} output, since previous outputs are required to compute the M^{th} output. Thus no saving realized. On the other hand, *using an FIR filter*, in this case implies that we can do our computations at the rate $\frac{f_s}{M}$. Thus, using an FIR filter in the decimation process will lead to a significantly lower computation rate.

Another advantage of using an FIR filter is the fact that we can easily design linear phase filters and this is desirable in many applications.

Exercise: Let x[n]={1 0 -1 2 4 2 5 1 3 2 8 7 -2}

a) x[n] is down-sampled by 3 to obtain y[m]. Determine y[m].

b) x[n] is down-sampled by 4 to obtain z[k]. Determine z[k].

9.2 Interpolation

The process of interpolation involves a sampling rate increase (see Figure 9.5)

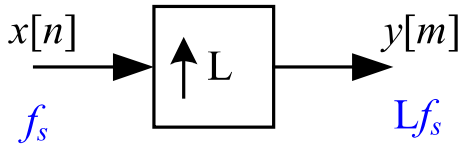


Figure 9.3: *A general block diagram representation of up-sampling process.*

Example:

$x[n]=\{1,2,4,3,-5,6,-7,2,4,3\}$

$\uparrow 2$

$y[m]=\{1,0,2,0,4,0,3,0,-5,0,6,0,-7,0,2,0,4,0,3,0\}$

The sequence $x[n]$ was derived by sampling $x(t)$ at a sampling rate of f_s and we want to obtain a sequence $y[m]$ that approximates as closely as possible the sequence that would have been obtained had we sampled $x(t)$ at the rate Lf_s.

This involves inserting between any two samples $x[n]$ and $x[n\text{-}1]$ an additional L-1 samples.

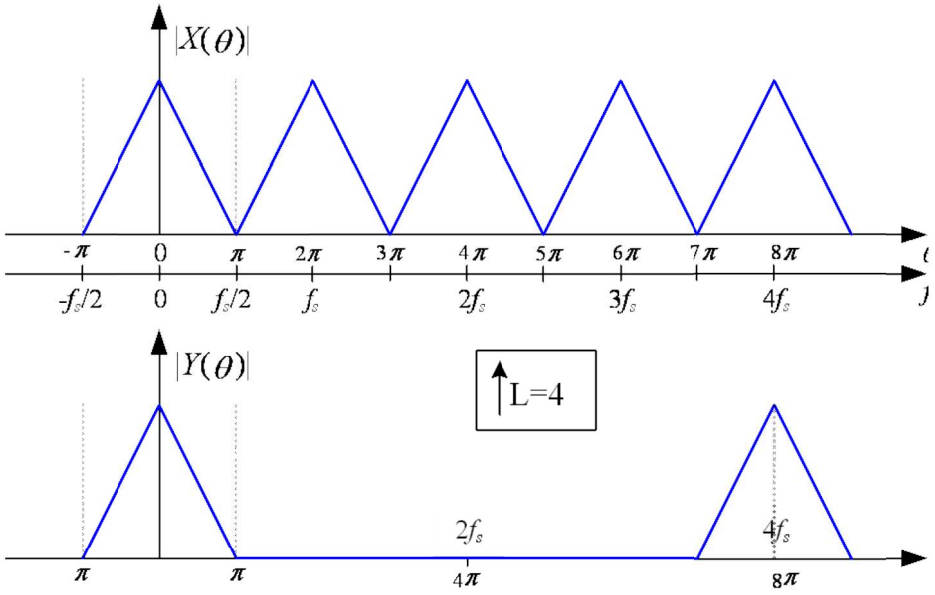


Figure 9.4: Spectral description of the interpolation process. Sampling frequency of $y(n) = 4f_s$.

We observe from Figure 9.6 that to go from $X(\theta)$ to $Y(\theta)$, we have to pass $x[n]$ through a lowpass digital filter designated at the Lf_s sampling rate that attenuates sufficiently any frequency components above $\frac{f_s}{2}$.

The frequency domain interpretation of the process is given in Figure 9.8 below.

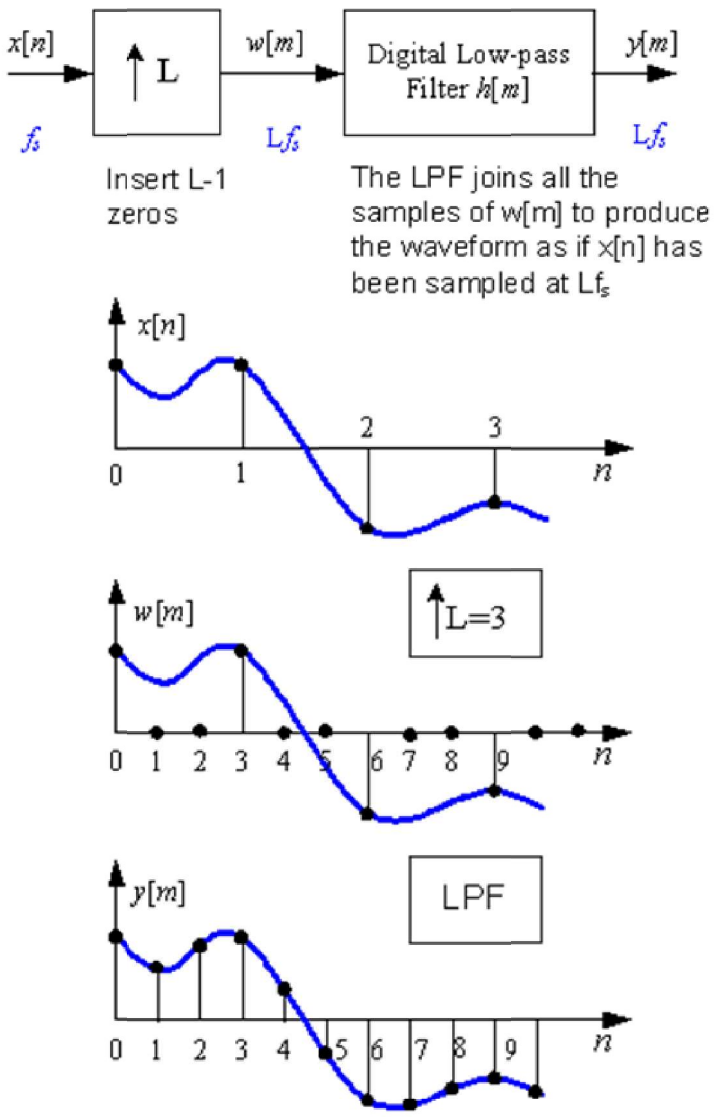


Figure 9.5: *A general description of the interpolation process. A discrete-time signal is up-sampled by inserting zeros and then filtered by a low-pass filter. Note that for each sample of x[n], three output samples y[m] are obtained.*

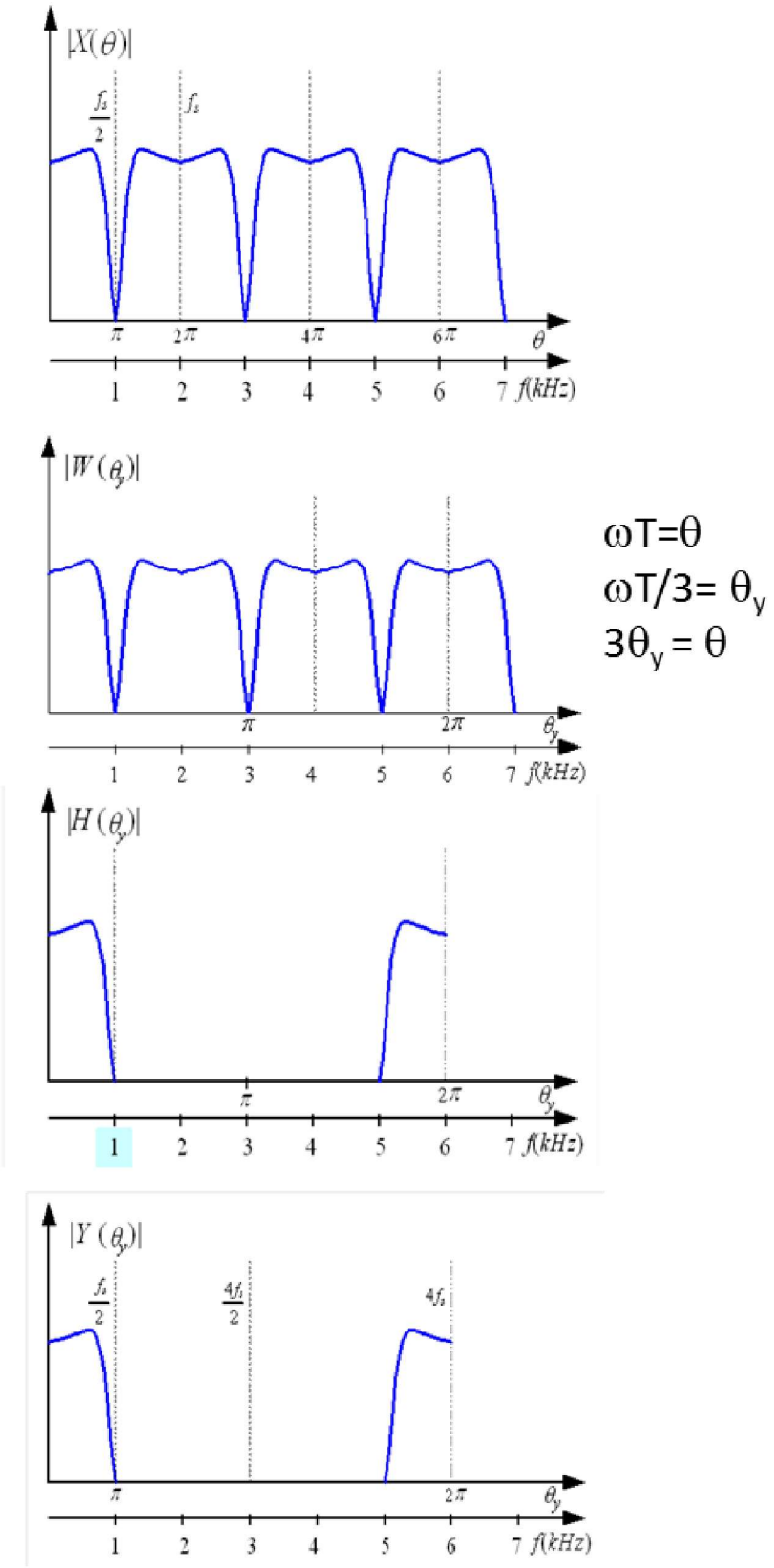
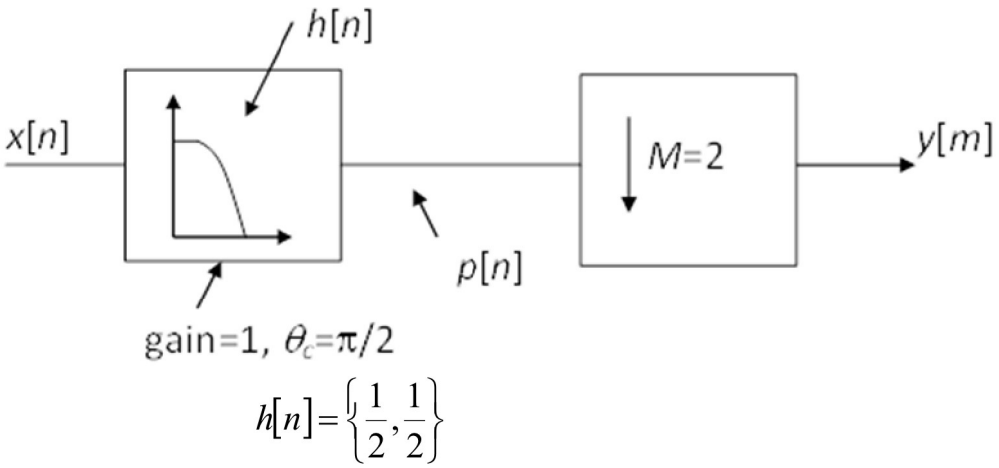


Figure 9.6: *Spectral interpretation of interpolation of a signal from 2kHz to 6 kHz.*

Note: Obviously the same reasoning that led us to believe that FIR filters are preferable in the decimation holds here also.

Example: Down sampling, $x[n] = \{2, 6, 4, 2, 6, 8, 4, 2, 4, 4\}$

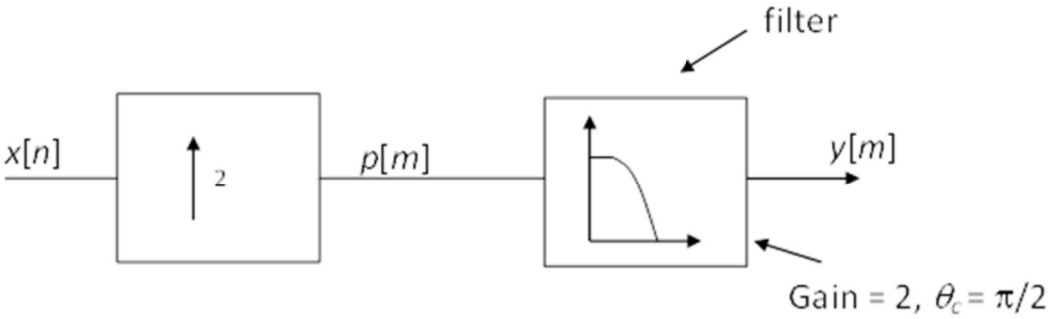


$$h[n] = \left\{\frac{1}{2}, \frac{1}{2}\right\}$$

$$p[n] = x[n] * h[n] = \{4,5,3,4,7,6,3,3,4,2\}$$

When p[n] is downsampled by 2, we obtain $y[n] = \{4,3,7,3,4\}$

9.2.1 Linear Interpolation



$x[n] = \{1,3,5,3,7\}$

$p[n] = \{1,0,3,0,5,0,3,0,7,0\}$ (insert zeros)

$$h[n] = \left\{\frac{1}{2}, 1, \frac{1}{2}\right\}$$

$y[n] = p[n] * h[n] = \{1,2,3,4,5,4,3,5,7,3.5\}$

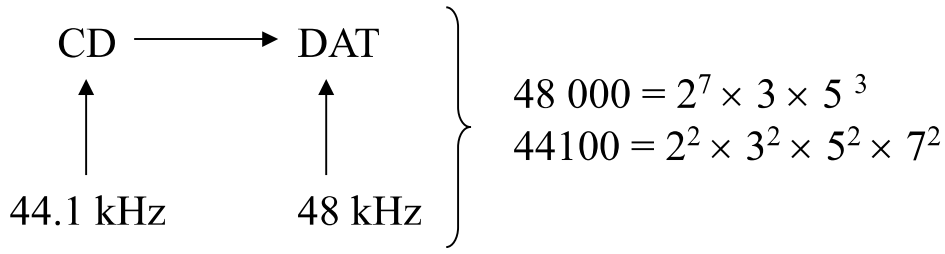
9.2.2 Sampling rate conversion by Non-integer factors

In some applications, the need often arises to change the sampling rate by a non-integer factor. An example is in digital audio applications where it may be necessary to transfer data from one storage system to another, where both systems employ different rates.

An example is transferring data from the compact disk system at a rate of 44.1 kHz to a digital audio Tape (DAT) at 48 kHz. This can be achieved by increasing the data rate of the CD by a factor of $\frac{48}{44.1}$, a non-integer.

In practice, such a non-integer factor is represented by a rational number that is a ratio of two integers, say L and M. The sampling frequency change is then achieved by first interpolating the data by L and then decimating by M.

Note: It is necessary that the interpolation process precede decimation otherwise the decimation process would remove some of the desired frequency components.



To change the sampling rate we require:

$$\frac{L}{M} = \frac{2^7 \times 3 \times 5^3}{2^2 \times 3^2 \times 5^2 \times 7^2} = \frac{160}{147}$$

Therefore if we up-sample by $L = 160$ and then down-sample by $M = 147$, we achieve the desired sampling rate conversion.

Figure 9.9 shows that the sampling frequency change is achieved by first interpolating the data by L and then decimating by M.

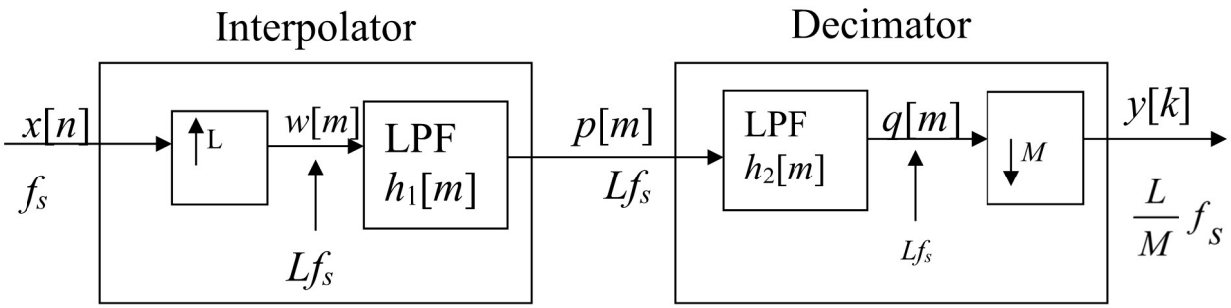


Figure 9.7: A discrete-time signal is interpolated and then decimated in order to achieve a non-integer conversion of sampling rate.

The two LPFs, $h_1[m] \& h_2[m]$ can be combined into a single filter since they are in cascade and have a common sampling frequency.

If $M > L$, the resulting operation is a decimation process by a non-integer.

If $M < L$, it is interpolation

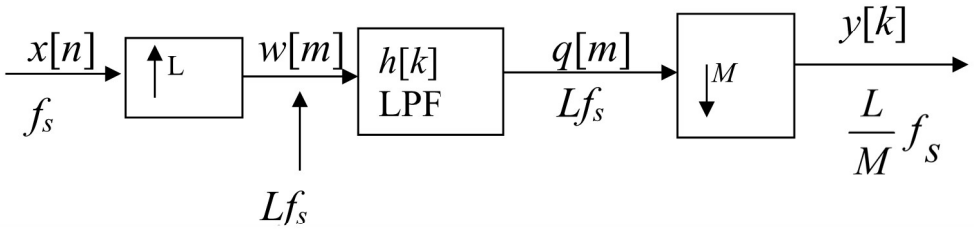
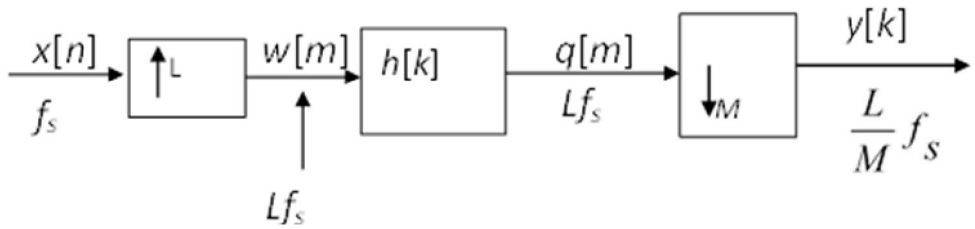


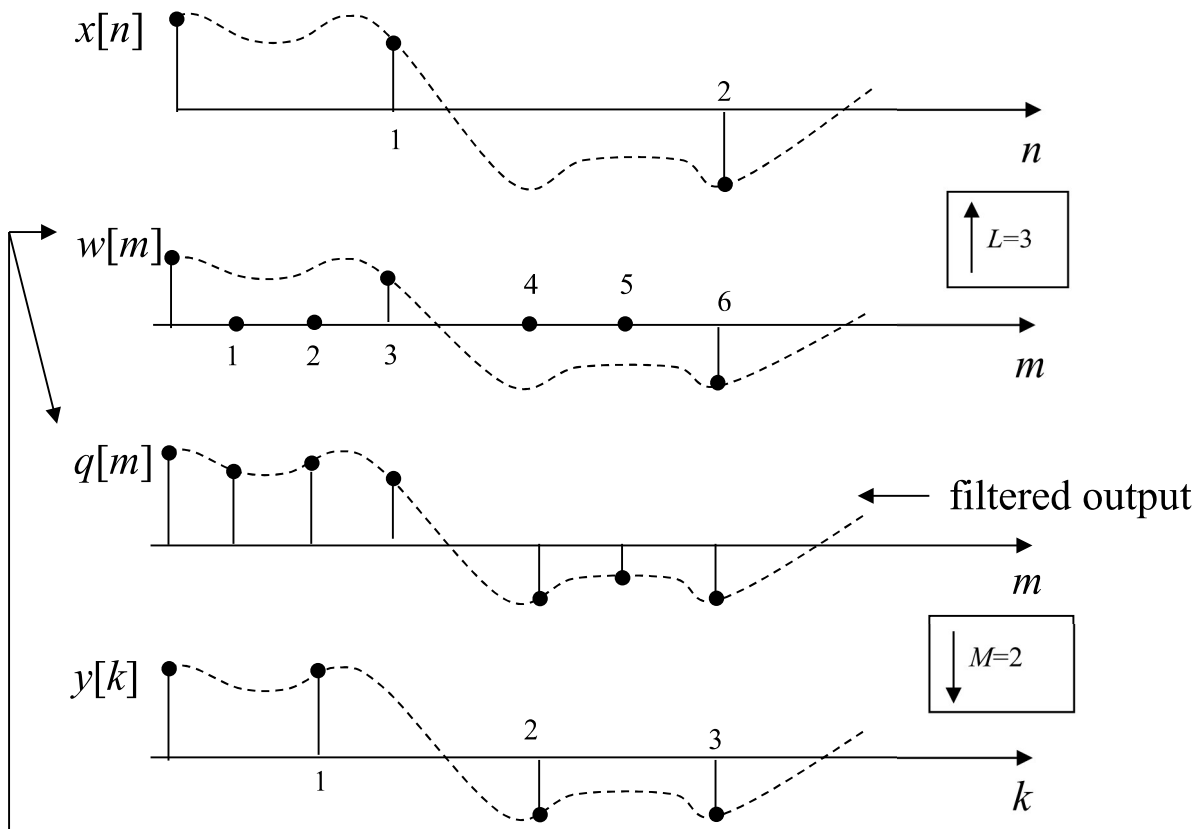
Figure 9.8: A block diagram representation of sampling rate conversion with combined low-pass filter.

The lowpass filter that we require is the one that has a cut-off frequency, $\boxed{\theta_c = \min\left(\frac{\pi}{L}, \frac{\pi}{M}\right)}$

9.2.3 Illustration of Interpolation by a factor 3/2



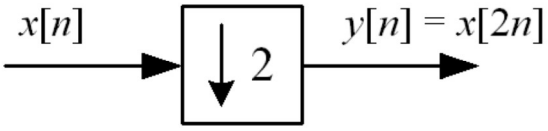
$$\frac{L}{M} = \frac{3}{2}$$



The sample rate is first increased by 3, by inserting two zero-value samples for each sample of $x[n]$ and low-pass filtered to yield $q[m]$.

The filtered data is then reduced by a factor of 2 by retaining only one sample for every two samples of $q[m]$.

9.3 Decimation by 2

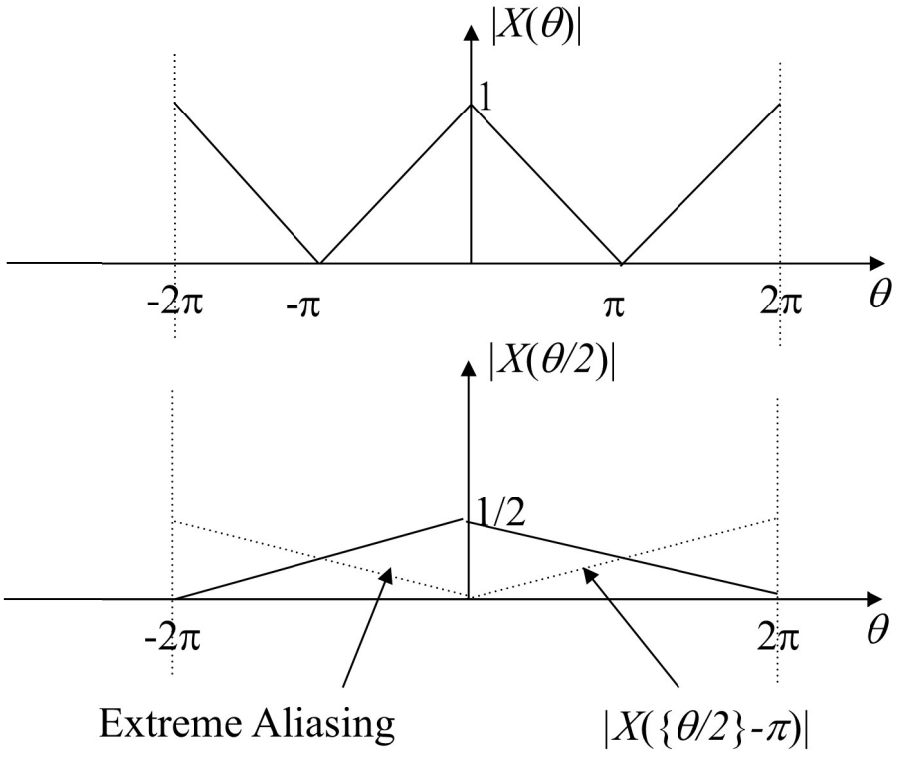


$$Y(z) = \frac{1}{2}\left[X(z^{\frac{1}{2}}) + X(-z^{\frac{1}{2}}) \right]$$ Aliasing term

$$Y(\theta) = \frac{1}{2}\left[X(e^{\frac{j\theta}{2}}) + X(-e^{\frac{j\theta}{2}}) \right] = \frac{1}{2}\left[X(e^{\frac{j\theta}{2}}) + X(e^{j(\frac{\theta}{2}-\pi)}) \right]$$

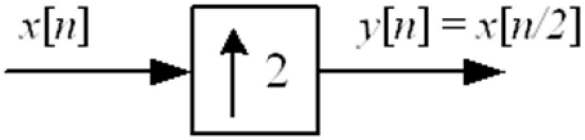
$$Y(\theta) = \frac{1}{2}\left[X(\frac{\theta}{2}) + X(\frac{\theta}{2}-\pi) \right]$$ Aliasing term

Stretch $X(\theta)$ by a factor 2 to obtain $X(\theta/2)$



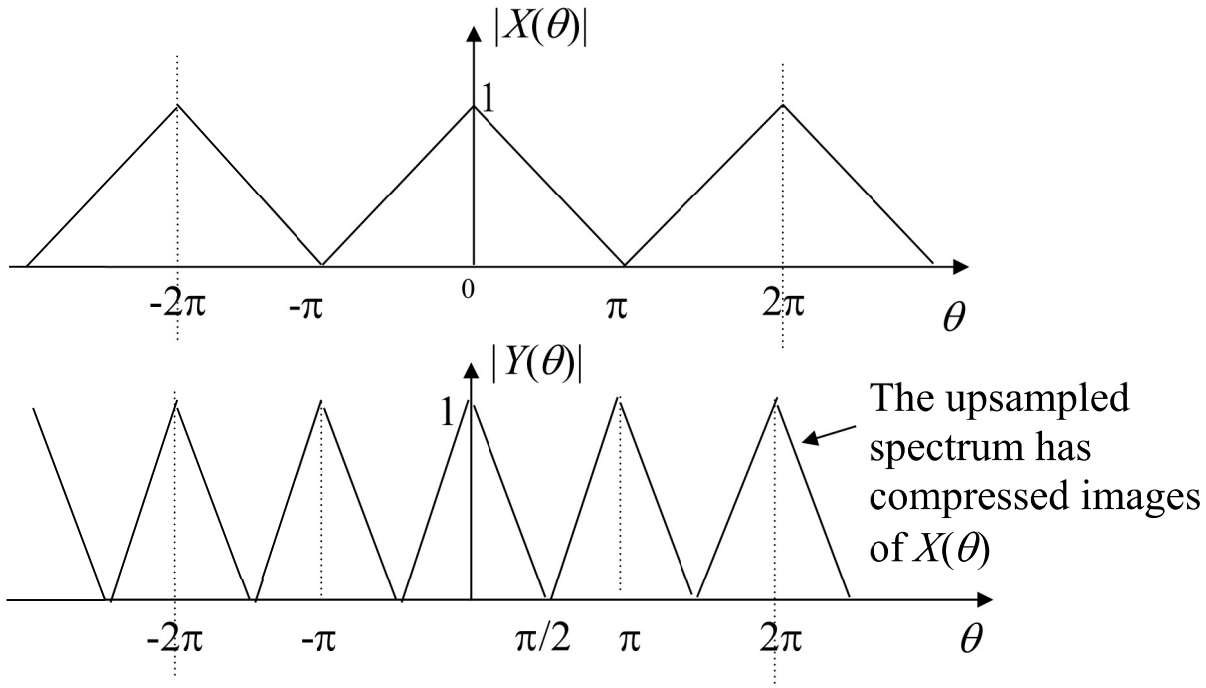
Note: The spectrum is stretched by down sampling.

9.4 Interpolation by 2



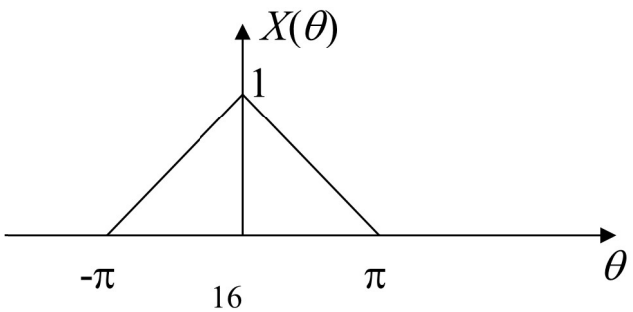
$$y[n] = x\left[\frac{n}{2}\right]; \quad Y(z) = X(z^2)$$

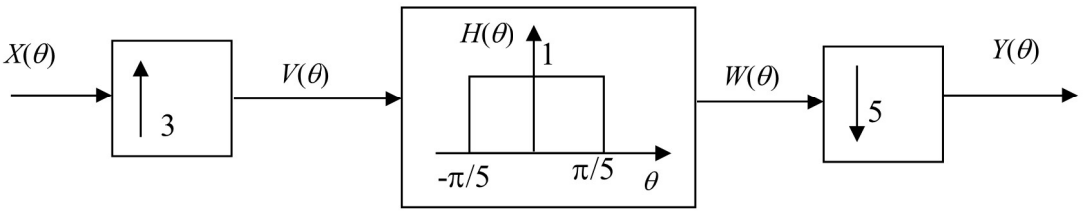
$$Y(\theta) = \left[X(e^{j2\theta}) \right] = \left[X(2\theta) \right]$$

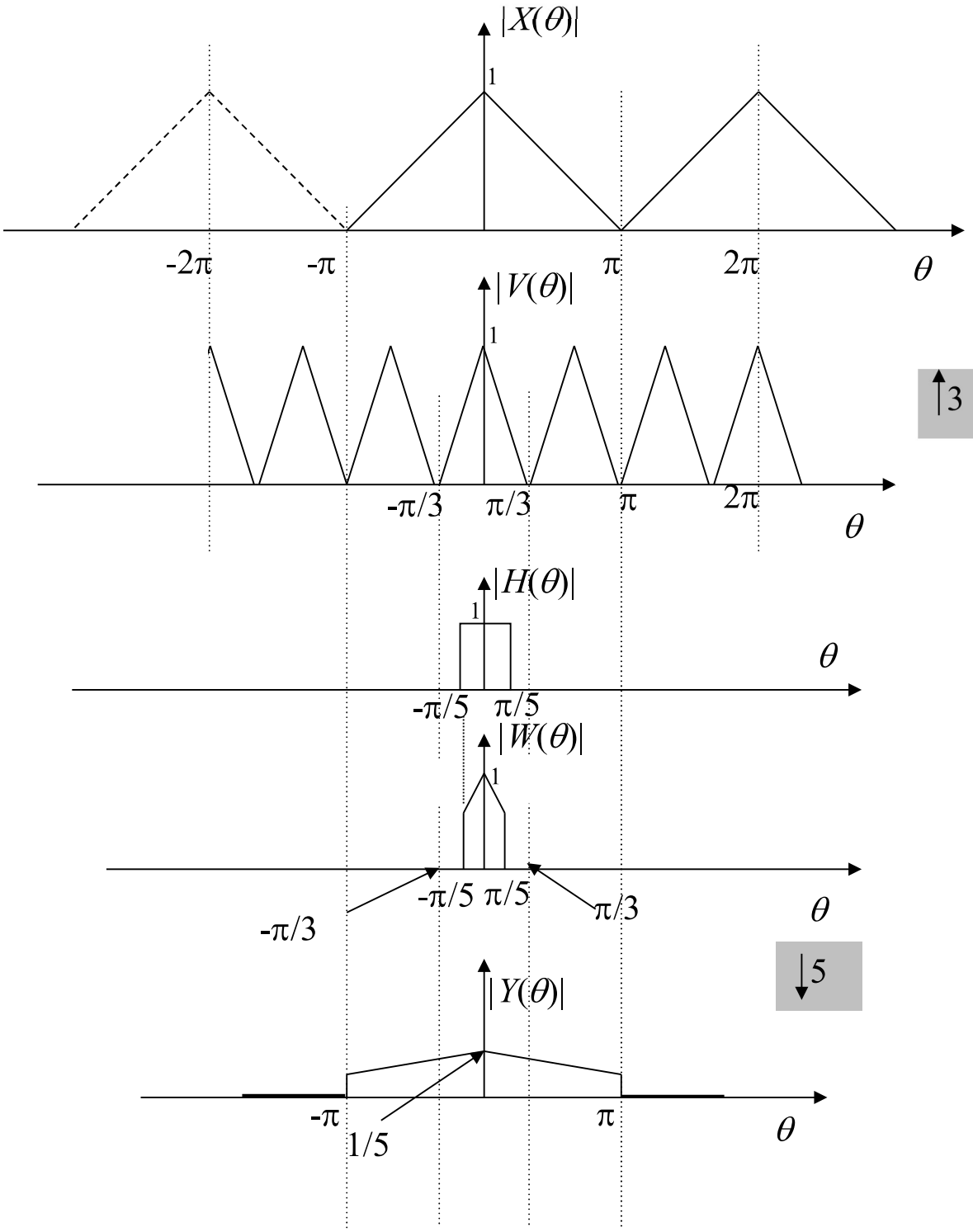


Note: The spectrum is compressed by upsampling.

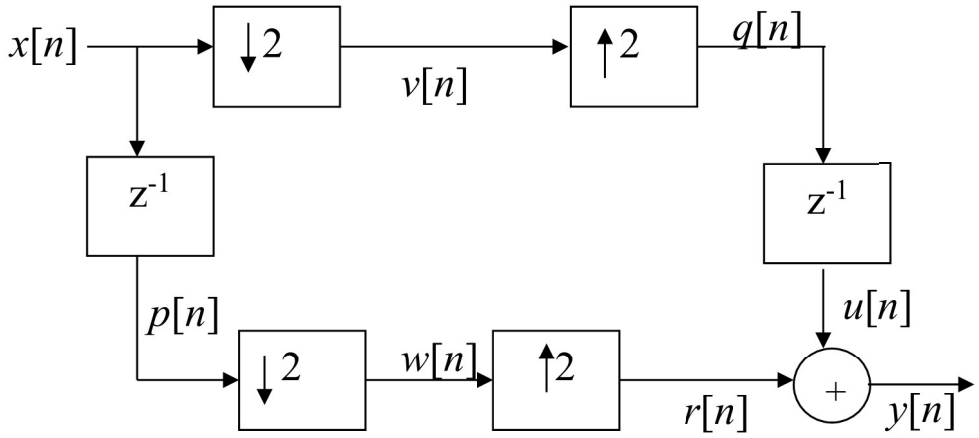
Example: An input signal $x[n]$ with the spectrum $X(\theta)$ is applied to the system shown below. Sketch, one above another, $|X(\theta)|$, $|V(\theta)|$, $|W(\theta)|$, $|Y(\theta)|$ against θ.




X(θ)
3
V(θ)
H(θ)
1
-π/5
π/5
θ
W(θ)
5
Y(θ)


|X(θ)|
1
-2π
-π
π
2π
θ
|V(θ)|
1
-π/3
π/3
π
2π
θ
3
|H(θ)|
1
θ
-π/5
π/5
|W(θ)|
1
-π/5
π/5
π/3
-π/3
θ
5
|Y(θ)|
-π
1/5
π
θ

Example: Express the output $y[n]$ of the figure below as function of the input $x[n]$. By simplifying the expression derived show that $y[n] = x[n\text{-}1]$.



$$p[n] = x[n-1] \Rightarrow P(z) = X(z)z^{-1}$$

$$P(z^{\frac{1}{2}}) = X(z^{\frac{1}{2}})z^{\frac{-1}{2}} \Rightarrow P(-z)^{\frac{1}{2}} = X(-z^{\frac{1}{2}})(-1)z^{\frac{-1}{2}}$$

$$w[n] = p[2n] \Rightarrow W(z) = \frac{1}{2}[P(z^{\frac{1}{2}}) + P(-z^{\frac{1}{2}})] \qquad \Rightarrow W(z) = \frac{1}{2}[X(z^{\frac{1}{2}})z^{\frac{-1}{2}} - X(-z^{\frac{1}{2}})z^{\frac{-1}{2}}]$$

$$v[n] = x[2n] \Rightarrow V(z) = \frac{1}{2}[X(z^{\frac{1}{2}}) + X(-z^{\frac{1}{2}})]$$

$$q[n] = v[\tfrac{n}{2}] \Rightarrow Q(z) = v(z^2); \quad U(z) = Q(z)z^{-1}$$

Similarly $R(z) = W(z^2)$

$$Q(z) = V(z^2) = \frac{1}{2}[X(z) + X(-z)]$$

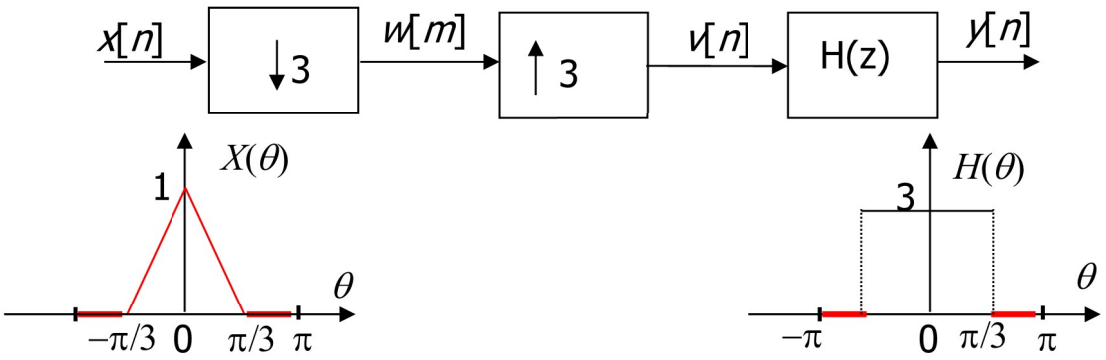
$$R(z) = W(z^2) = \frac{z^{-1}}{2}X(z) - \frac{z^{-1}}{2}X(-z)$$

$$U(z) = z^{-1}Q(z) = \frac{1}{2}z^{-1}[X(z) + X(-z)]$$

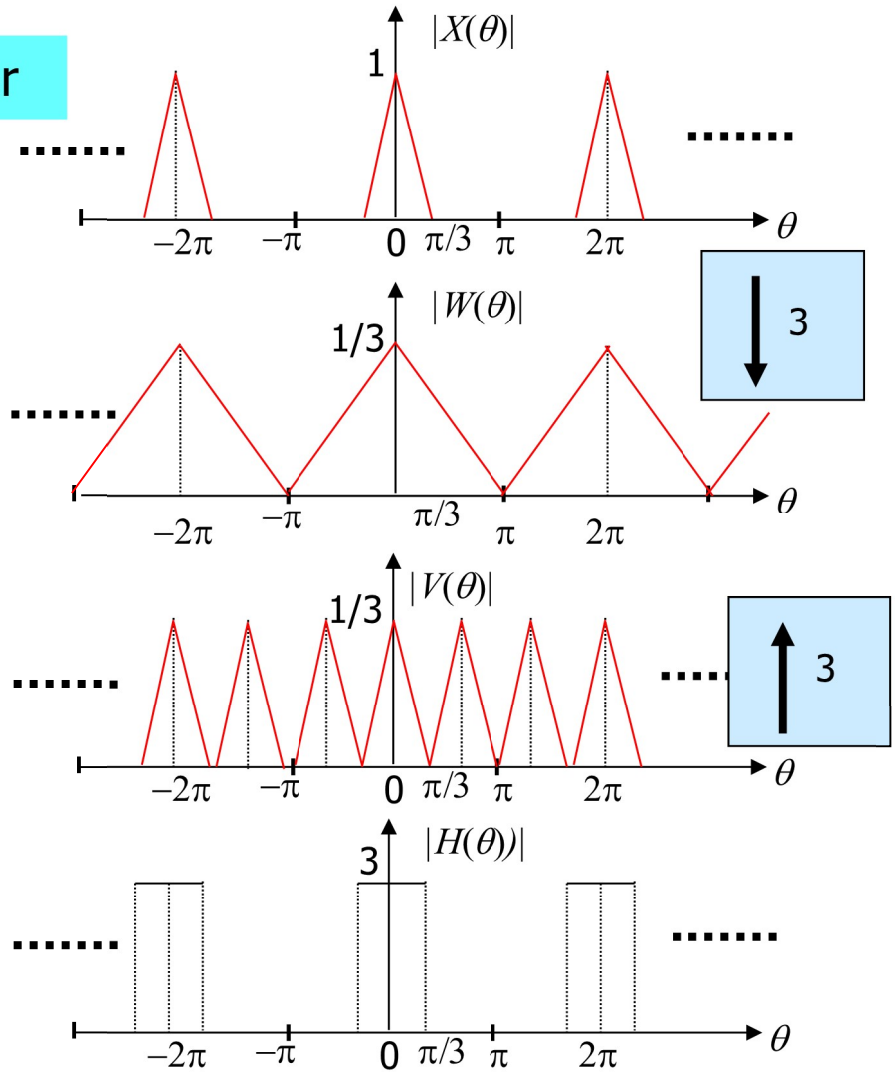
$$Y(z) = U(z) + R(z) = \frac{z^{-1}}{2}X(z) - \frac{z^{-1}}{2}X(-z) + \frac{1}{2}z^{-1}[X(z) + X(-z)]$$

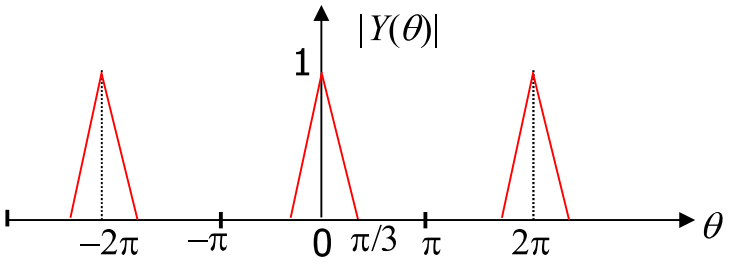
$Y(z) = X(z)z^{-1} \Rightarrow y[n] = x[n-1]$ ← The original signal $x[n]$ is reconstructed. The above structure is a perfect reconstruction system.

Example: An input signal $x[n]$ with spectrum $X(\theta)$ is shown below. The input signal is applied to the system shown below. Sketch $|X(\theta)|$, $|W(\theta)|$,$|V(\theta)|$, $|Y(\theta)|$ against θ.



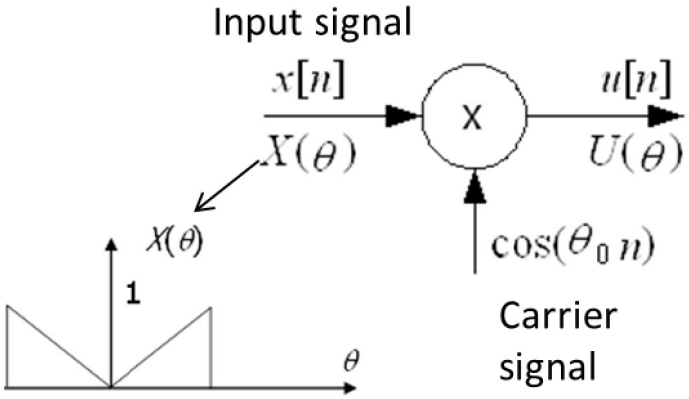
Answer





Modulation

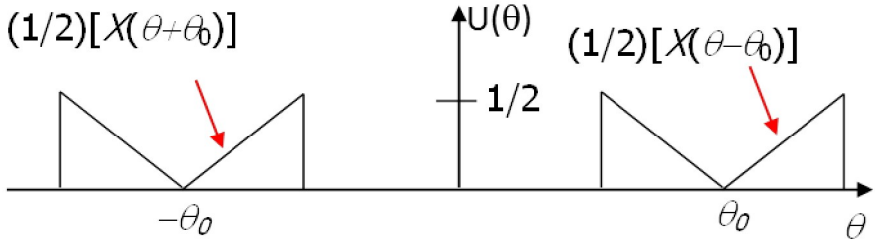
In the time domain, modulation is the process of multiplying an input signal $x[n]$ with a sinusoidal signal known as the carrier, illustrated in the diagram below.



According to the modulation property of the discrete-time Fourier transform:

$$x[n]\cos\theta_0 n \leftrightarrow \tfrac{1}{2}X(\theta+\theta_0)+\tfrac{1}{2}X(\theta-\theta_0),$$

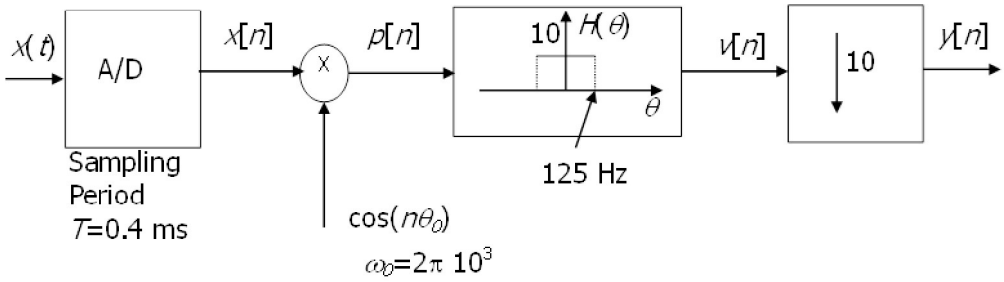
In the frequency domain, the modulated signal comprises two shifted versions of the original signal, translated by the carrier frequency.



Time Domain	Frequency Domain
$x(n)\cos(\theta_0 n)$	$\frac{1}{2}[X(\theta-\theta_0)+X(\theta+\theta_0)]$
$x(n)\sin(\theta_0 n)$	$\frac{1}{2j}[X(\theta-\theta_0)-X(\theta+\theta_0)]$
$x(n)e^{-jn\theta_0}$	$X(\theta+\theta_0)$

Example: $x(t)$ is the input signal for the system shown below. The analogue signal $x(t)$ has the spectrum $X(f)$ given by:

$$X(f)=\begin{cases}1 & 0.875kHz \le f \le 1.125kHz\\ 1 & -1.125kHz \le f \le -0.875kHz\\ 0 & \text{elsewhere}\end{cases}$$



$H(\theta)$ is an ideal lowpass filter (gain=10) with cut-off frequency f_c=125 Hz. Sketch, one above another, $|X(\theta)|$, $|P(\theta)|$,$|V(\theta)|$, $|Y(\theta)|$ against θ.

$$\text{Note}: \quad P(\theta)=\frac{1}{2}[X(\theta+\theta_0)+X(\theta-\theta_0)]$$

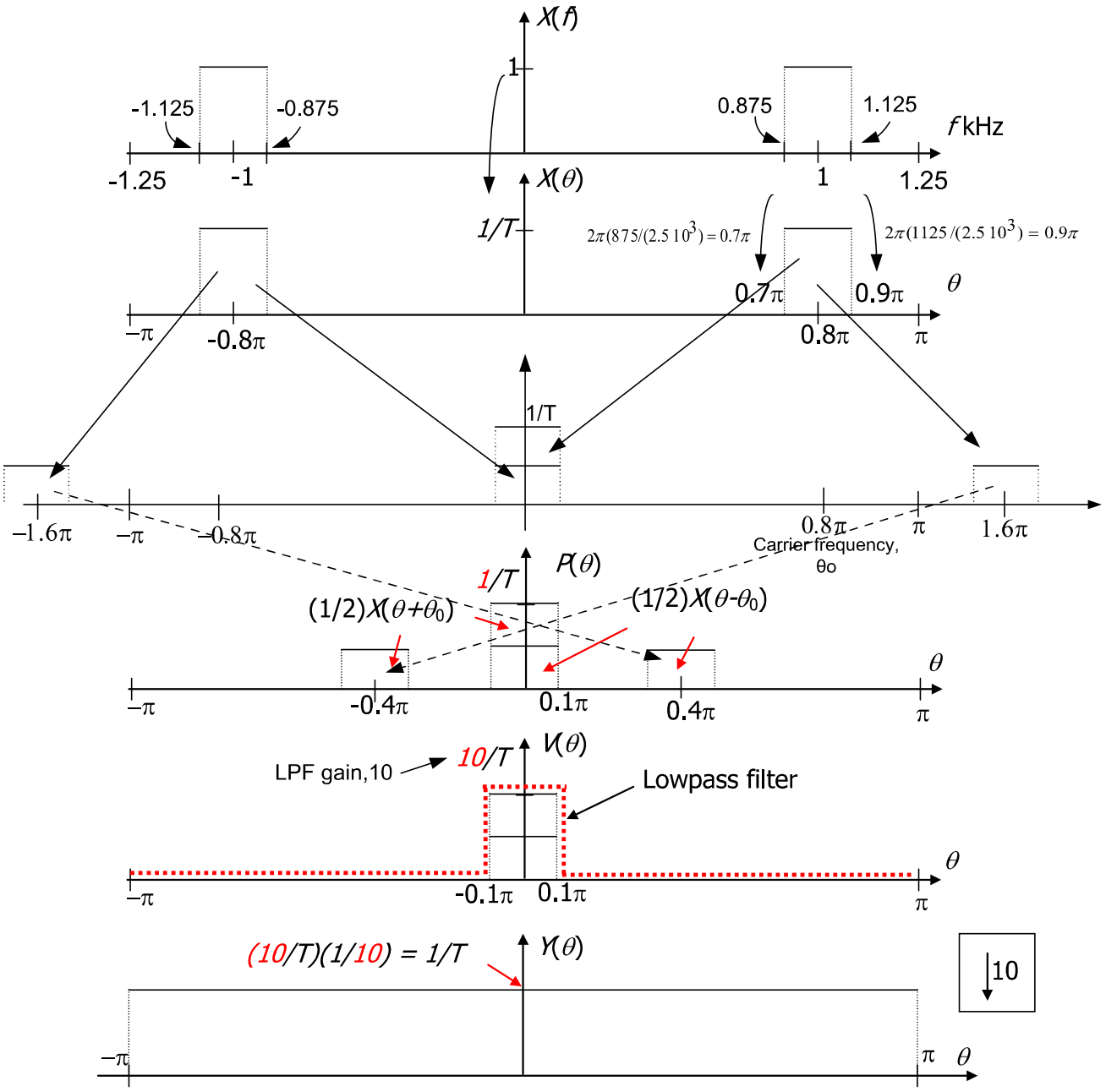
Solution:

Sampling Period, T = 0.4ms

$$f_S=\frac{1}{T}=2.5kHz;$$

Carrier Frequency: $\theta_0 = 2\pi 10^3 (0.4\ 10^{-3}) = 0.8\pi;$

LPF Cutoff Frequency: $\theta_c = 2\pi(125/(2.5 \quad 10^3) = 0.1\pi$


X(f)
1
-1.125
-0.875
0.875
1.125
f kHz
-1.25
-1
1
1.25
X(θ)
1/T
2π(875/(2.5 10^3) = 0.7π
2π(1125/(2.5 10^3) = 0.9π
0.7π
0.9π
θ
−π
-0.8π
0.8π
π
1/T
−1.6π
−π
−0.8π
0.8π
π
1.6π
Carrier frequency, θo
P(θ)
1/T
(1/2)X(θ+θ0)
(1/2)X(θ-θ0)
θ
−π
-0.4π
0.1π
0.4π
π
V(θ)
LPF gain,10
10/T
Lowpass filter
θ
−π
-0.1π
0.1π
π
(10/T)(1/10) = 1/T
Y(θ)
10
−π
π
θ

Example

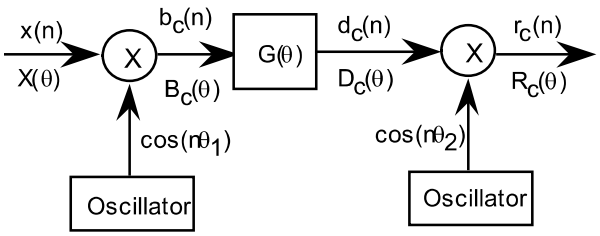
a. A signal has been sampled at 44 kHz and for a particular application this signal is required to be used at a reduced sampling rate of 8 kHz.
   (i) Draw a block diagram to show how you would achieve this.
   (ii) Indicate the type of digital filter which you would choose for part (i) above (give reasons).

b. The sampling period T of the input signal shown in the figure below is 125 μs. The relative frequency is θ = ωT. The first oscillator generates a carrier with a relative frequency $\theta_1 = \omega_1 T$, where $\omega_1 = 2\pi.2.10^3\ rad/\sec.$ The second oscillator generates a carrier with a relative frequency $\theta_2 = (\omega_C + \omega_1)T$, where $\omega_c = 2\pi.10^3\ rad/\sec.$ The low-pass filter has the following characteristics:

$$G(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \dfrac{\pi}{4} \\ 0 & \dfrac{\pi}{4} < |\theta| \le \pi \end{cases}$$

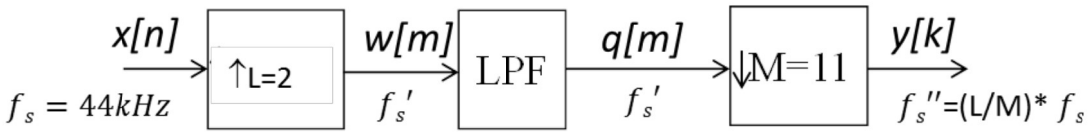
If the input spectrum is $X(\theta) = 1 + \cos(\theta) \quad -\pi \le \theta \le \pi$,

   (i) Write appropriate equations for $B_C(\theta)$, $D_C(\theta)$, and $R_C(\theta)$
   (ii) Sketch |X(θ)|, $|B_C(\theta)|$, $|D_C(\theta)|$, and $|R_C(\theta)|$ against θ,



Solution

(a)
(i)



$$\frac{L}{M} = \frac{8000}{44000} = \frac{2}{11}$$

(ii)
Using IIR filter, we cannot take advantage of the fact that we have to compute only every *N* th output, since previous outputs are needed to compute the *N* th output.

On the other hand, using FIR filter we can do our computation at the rate of fs'/N → a significantly lower computational rate.

(b)

$$\theta_1 = 2\pi \times 2\times 10^3 \times \frac{1}{8\times 10^3} = \frac{\pi}{2}$$

$$\theta_2 = \omega_C T + \omega_1 T = \frac{\pi}{2} + 2\pi \times 10^3 \times \frac{1}{8\times 10^3} = \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$$

$$X(\theta) = 1 + \cos\theta$$

$$B_C(\theta) = \frac{1}{2}\left[X(\theta-\theta_1) + X(\theta+\theta_1)\right] = \frac{1}{2}\left[1+\cos\left(\theta-\frac{\pi}{2}\right)+1+\cos\left(\theta+\frac{\pi}{2}\right)\right] = \frac{1}{2}\times 2 = 1$$

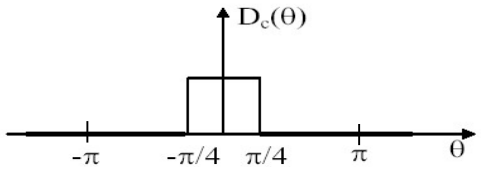
$$D_C(\theta) = B_C(\theta)\cdot G(\theta) = 1, \qquad 0 < |\theta| < \tfrac{\pi}{4}$$

$$R_C(\theta) = \frac{1}{2}\left[D_C(\theta-\theta_2) + D_C(\theta+\theta_2)\right] = \frac{1}{2}\left[D_C(\theta-\tfrac{3}{4}\pi) + D_C(\theta+\tfrac{3}{4}\pi)\right]$$

$$R_C(\theta) = \begin{cases} \frac{1}{2} & -\frac{\pi}{4} \le \theta - \frac{3\pi}{4} \le \frac{\pi}{4} \Rightarrow \frac{\pi}{2} \le \theta \le \pi \\ \frac{1}{2} & -\frac{\pi}{4} \le \theta + \frac{3\pi}{4} \le \frac{\pi}{4} \Rightarrow -\pi \le \theta \le -\frac{\pi}{2} \\ 0 & \text{elsewhere} \end{cases}$$

Then from the above calculation, sketch $X(\theta), B_C(\theta), D_C(\theta),$ and $R_C(\theta)$.

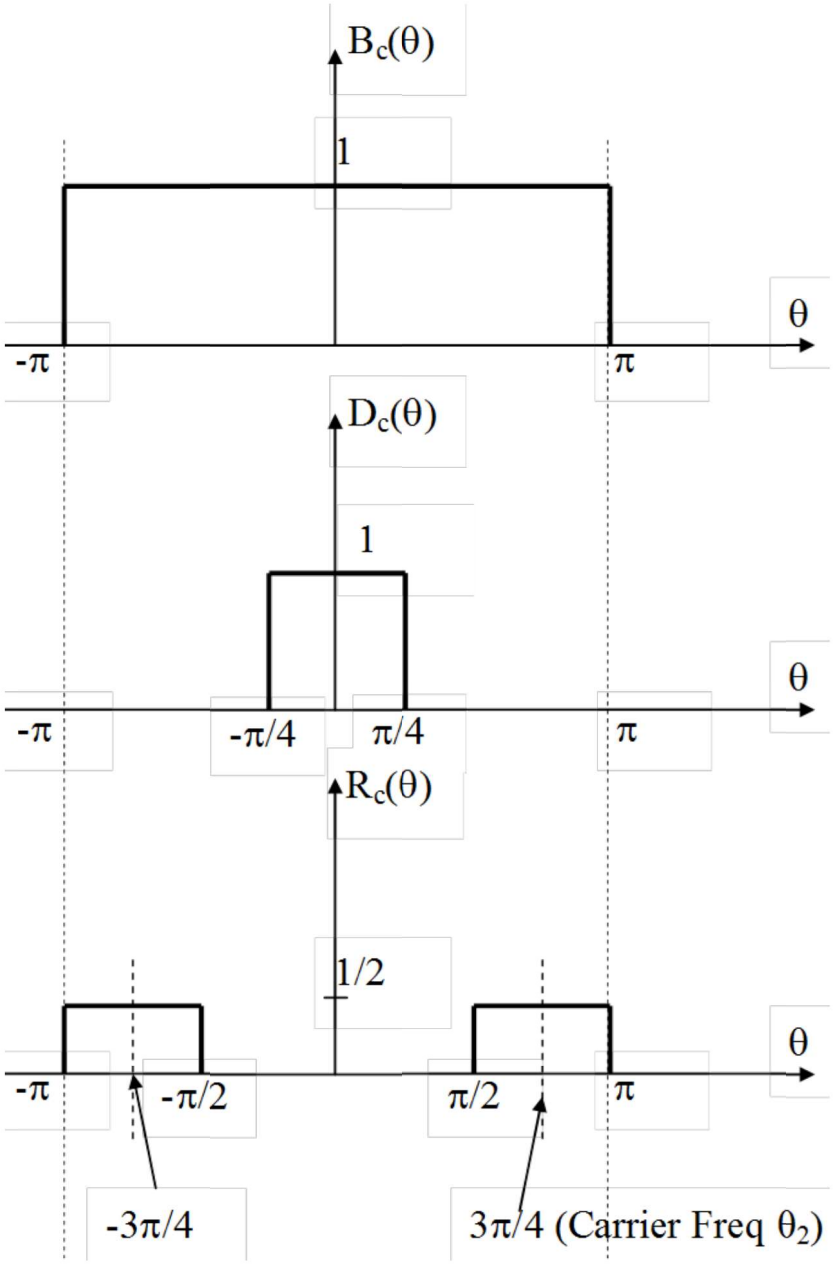
$$\mathrm{B_c}(\theta) = 1, \mathrm{D_c}(\theta) = 1$$



$$\mathrm{R_c}(\theta) = \frac{1}{2}\left\{\mathrm{D_c}(\theta - \theta_1) + \mathrm{D_c}(\theta + \theta_1)\right\}$$

$$= \frac{1}{2}\left\{\mathrm{D_C}(\theta - \frac{3\pi}{4}) + \mathrm{D_c}(\theta + \frac{3\pi}{4})\right\}$$

$$R_c(\theta)=\begin{cases}\frac{1}{2} & \frac{-\pi}{4}\le(\theta-\frac{3\pi}{4})\le\frac{\pi}{4}\\ \frac{1}{2} & \frac{-\pi}{4}\le\theta+\frac{3\pi}{4}\le\frac{\pi}{4}\\ 0 & \text{elsewhere}\end{cases}=\begin{cases}\frac{1}{2} & \frac{\pi}{2}\le\theta\le\pi\\ \frac{1}{2} & -\pi\le\theta\le\frac{-\pi}{2}\\ 0 & \text{elsewhere}\end{cases}$$

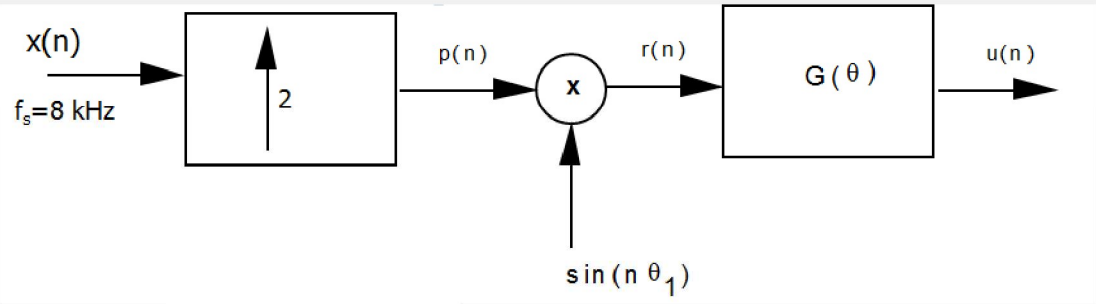


Exercise:

A section of a digital modulator is shown below. If the input signal x(n) is $x(n) = \frac{1}{2}\delta(n+1) + \delta(n) + \frac{1}{2}\delta(n-1)$

(a) Show $P(\theta) = 1+\cos(2\theta)$ for $-\pi \le \theta \le \pi$,

(b) By writing appropriate equations for $R(\theta)$ and $U(\theta)$, sketch $|R(\theta)|$ and $|U(\theta)|$ vs θ, and where $\theta_1 = \pi/4$.



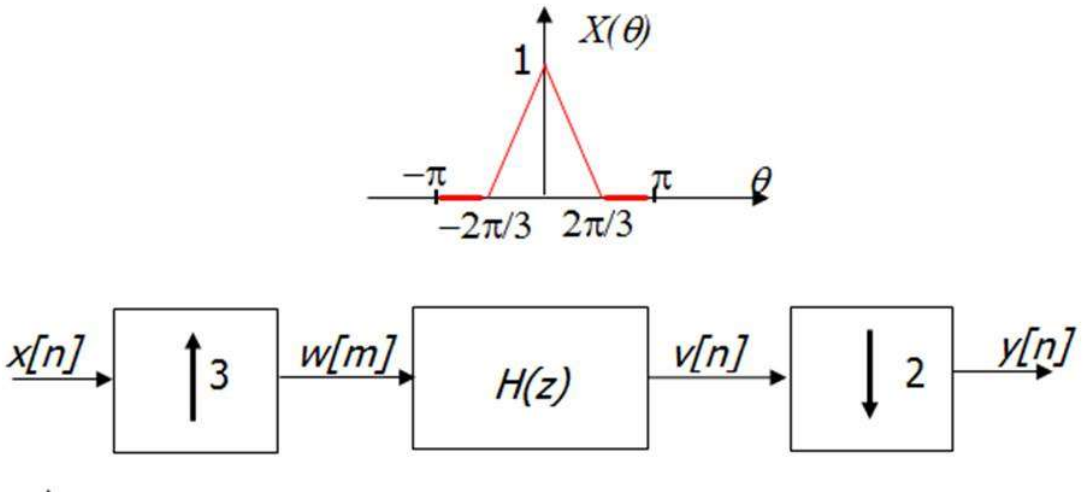
$$G(\theta) = \begin{cases} 1/2 & 0 \le |\theta| \le \frac{\pi}{4} \\ 0 & \frac{\pi}{4} < |\theta| \le \pi \end{cases}$$

Ans:

$$U(\theta) = R(\theta).G(\theta) = \begin{cases} \frac{1}{2j}\sin 2\theta & 0 \le |\theta| \le \frac{\pi}{4} \\ 0 & \frac{\pi}{4} \le \theta \le \pi \end{cases}$$

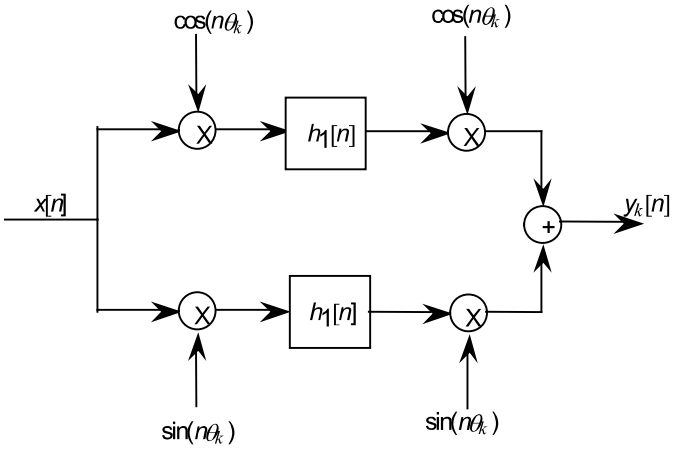
CHAPTER 9: PROBLEM SHEET 9

Q1) A signal *x[n]* has a spectrum $X(\theta)$ as shown below. The signal is applied to the system shown below. The ideal lowpass filter *H(z)* has a gain factor of 1 in the passband and a cut-off frequency $\theta = \frac{\pi}{3}$. Sketch $|X(\theta)|, |W(\theta)|, |V(\theta)|, |Y(\theta)|$ against θ



Q2) A signal has been sampled at 44 kHz. For a particular application, this signal is required to be used at a reduced sampling rate of 16 kHz. Explain how you would efficiently implement a system to achieve this, indicating the type of digital filter you would use (give reasons).

Q3) (a) Show that the impulse response of the system shown below for the k^{th} channel is $h_k[n] = h_1[n]\cos(n\theta_k)$.

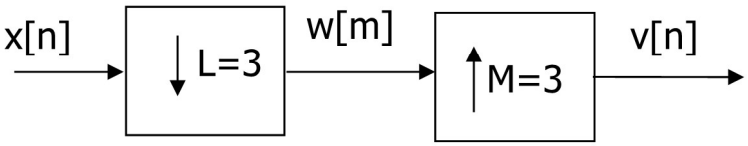


(b) Show that the frequency response $H_k(\theta)$ of the system is given by

$$H_k(\theta) = \tfrac{1}{2}H(\theta - \theta_k) + \tfrac{1}{2}H(\theta + \theta_k)$$

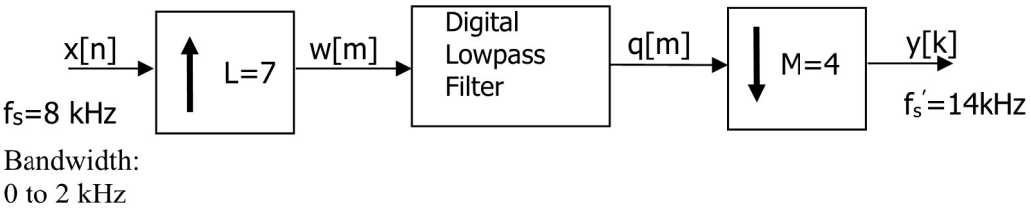
(c) If $x[n] = \cos(n\theta_0)$, then show using the figure, that the output signal $y_k[n]$ is essentially identical to the input signal *x[n]*.

Q4) (a) For the diagram shown below, tick the correct answer:



(A) w[m] = x[n]
(B) v[n]= w[m]
(C) v[n]=x[n/3]
(D) None of the above

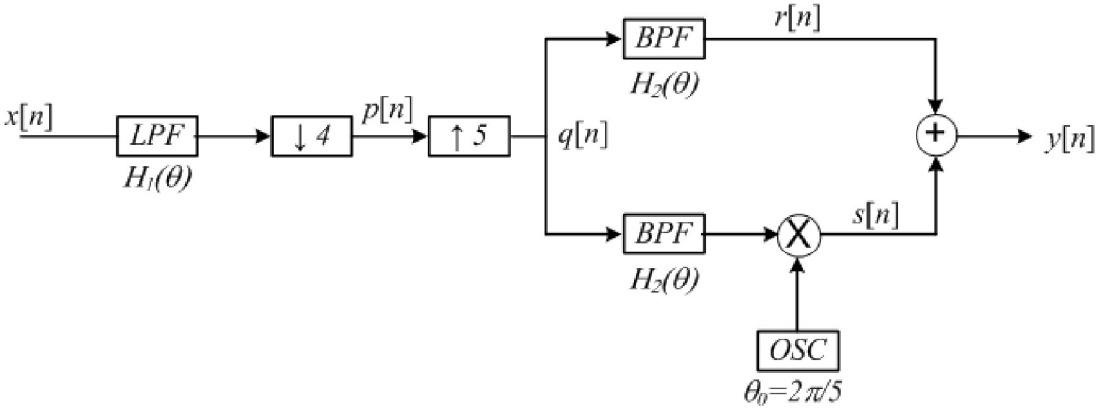
(b) Figure below shows sampling rate conversion by non-integer factors. Calculate the minimum cut-off frequency of the digital lowpass filter.



Bandwidth:
0 to 2 kHz

Ans: (a) Answer: (D)
(b) 2 kHz

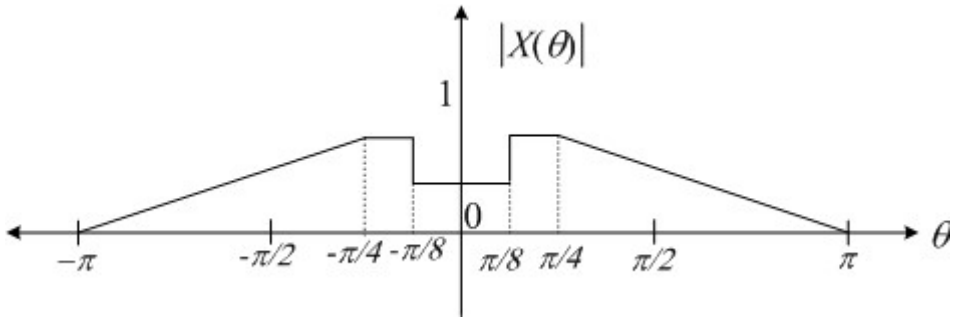
Q5) A multi-rate system is given below.



where,

$$H_1(\theta)=\begin{cases}1 & |\theta|\le \pi/4\\ 0 & otherwise\end{cases} \qquad H_2(\theta)=\begin{cases}1 & \pi/5\le|\theta|\le 2\pi/5\\ 0 & otherwise\end{cases} \qquad H_3(\theta)=\begin{cases}1 & 2\pi/5\le|\theta|\le 3\pi/5\\ 0 & otherwise\end{cases}$$

If the frequency spectrum of $x[n]$, $X(\theta)$ is given as follows, sketch $P(\theta)$, $Q(\theta)$, $R(\theta)$, $S(\theta)$ and $Y(\theta)$.



End of Chapter 9